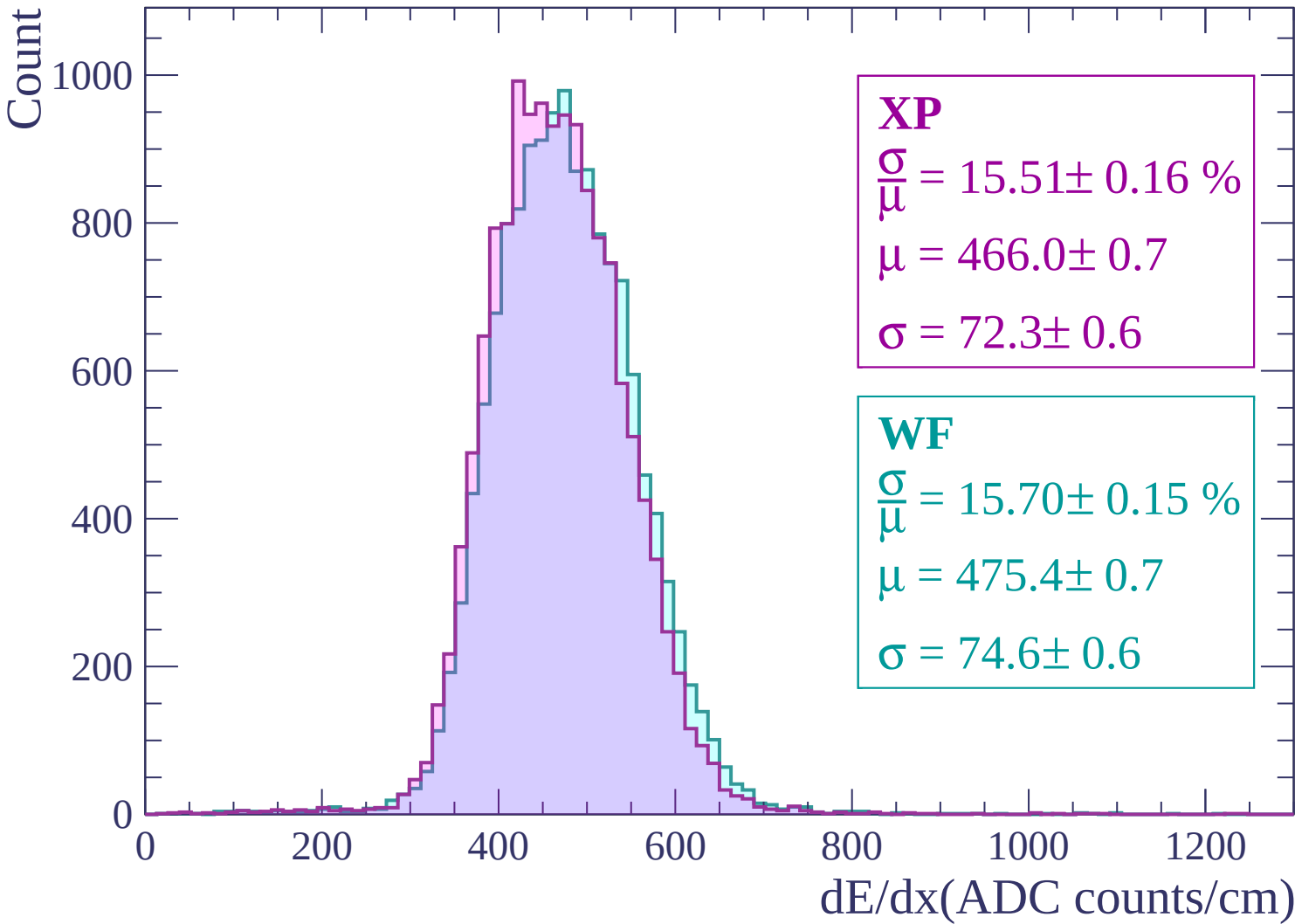
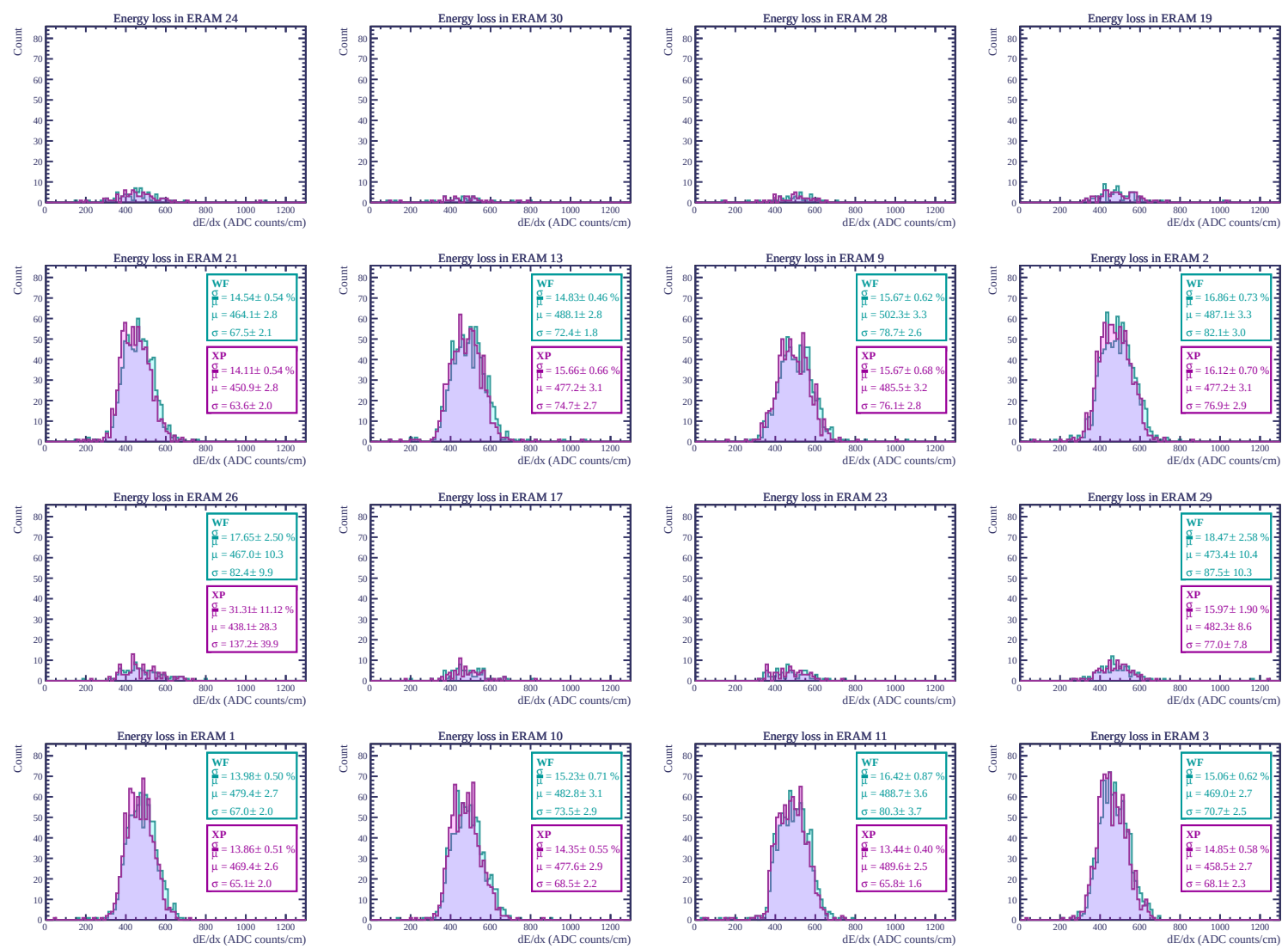
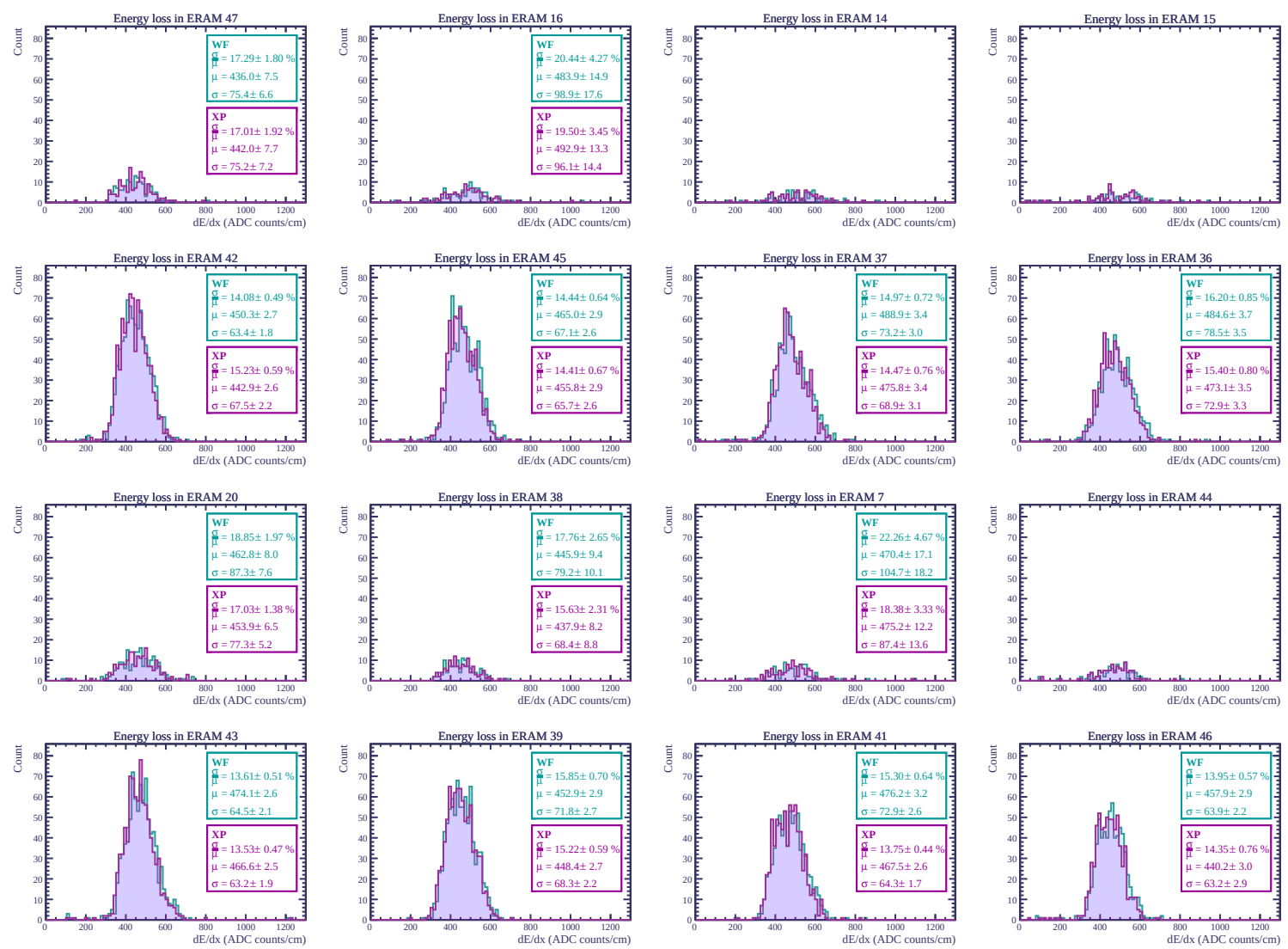
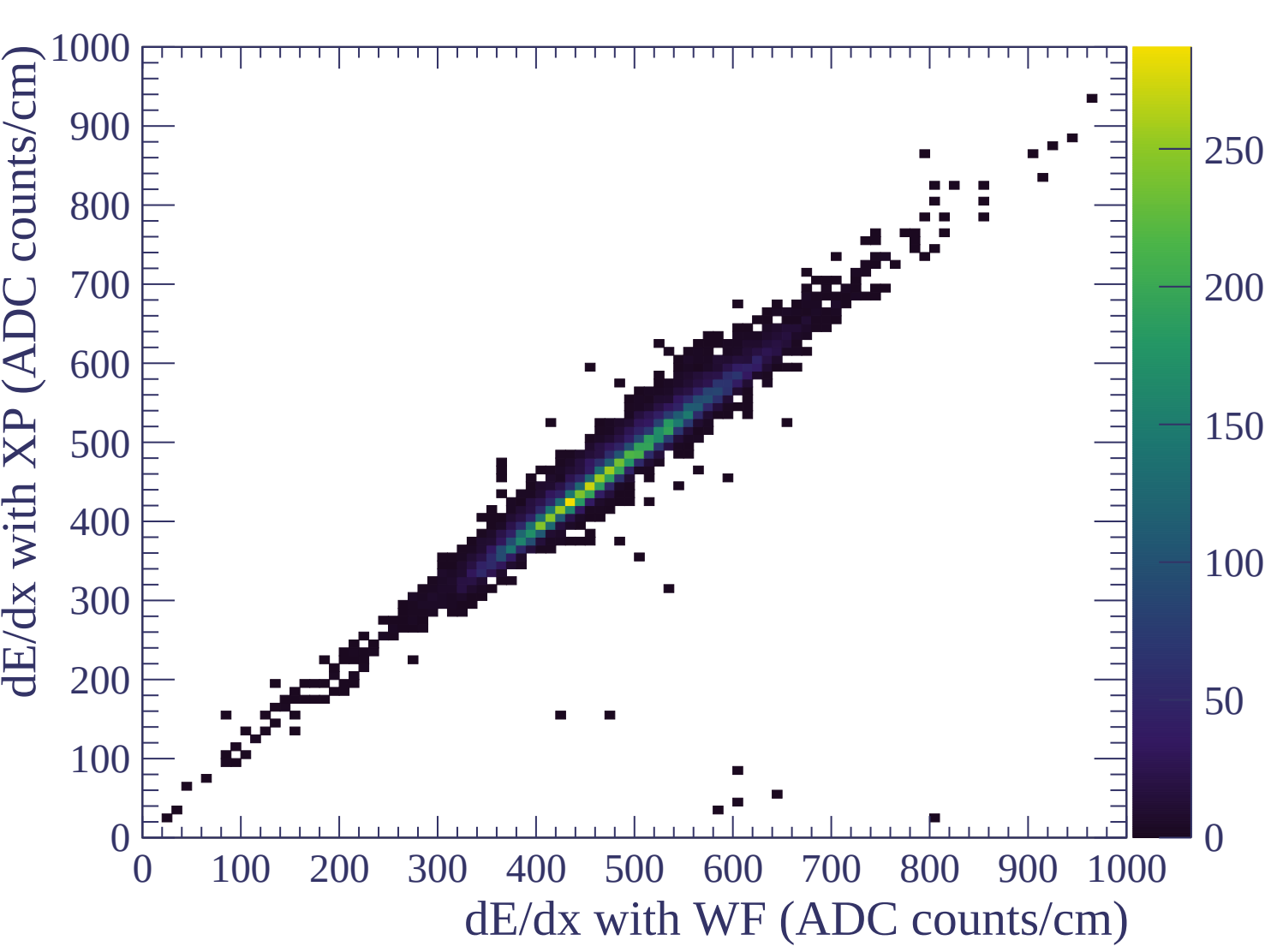
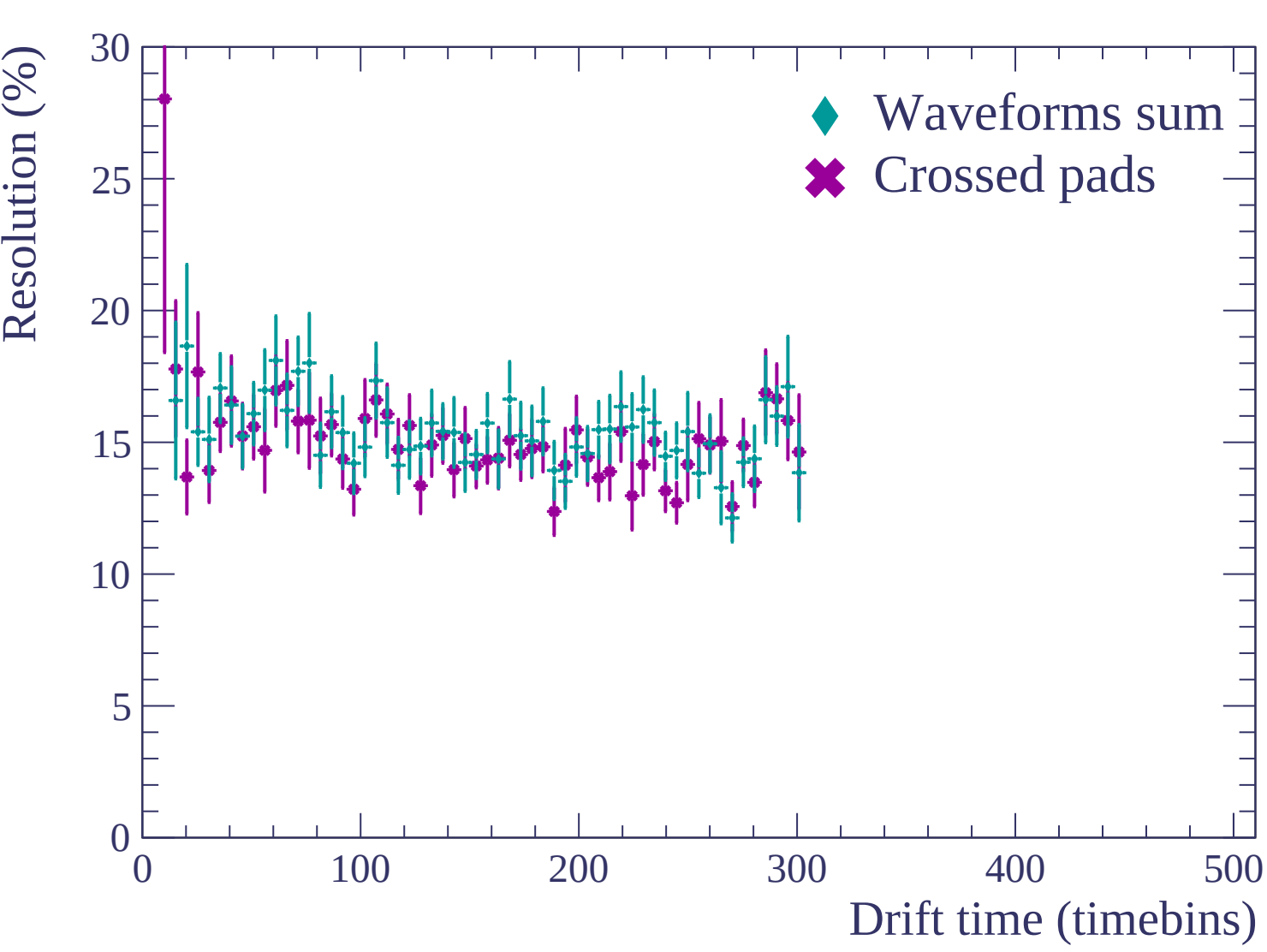


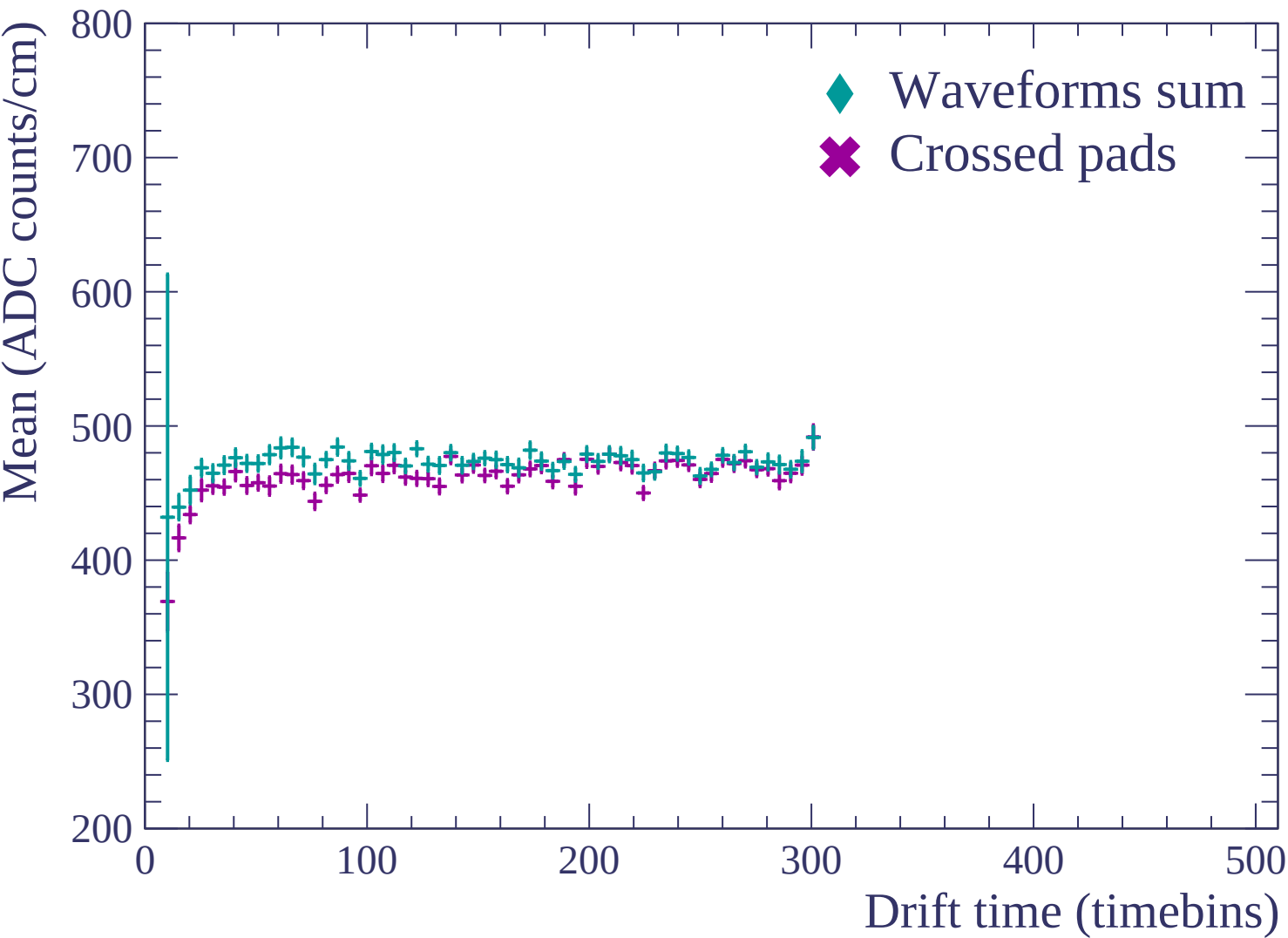


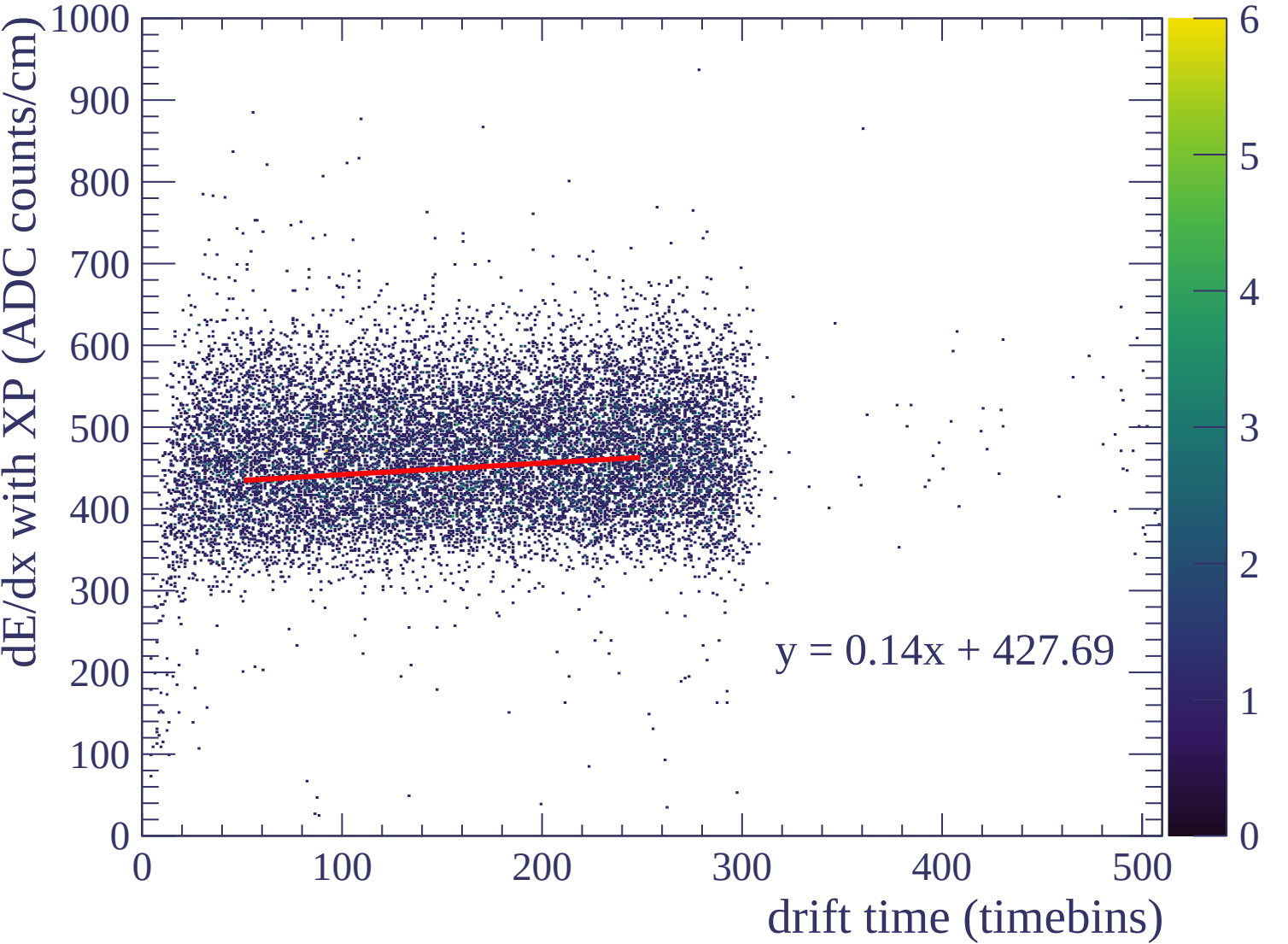


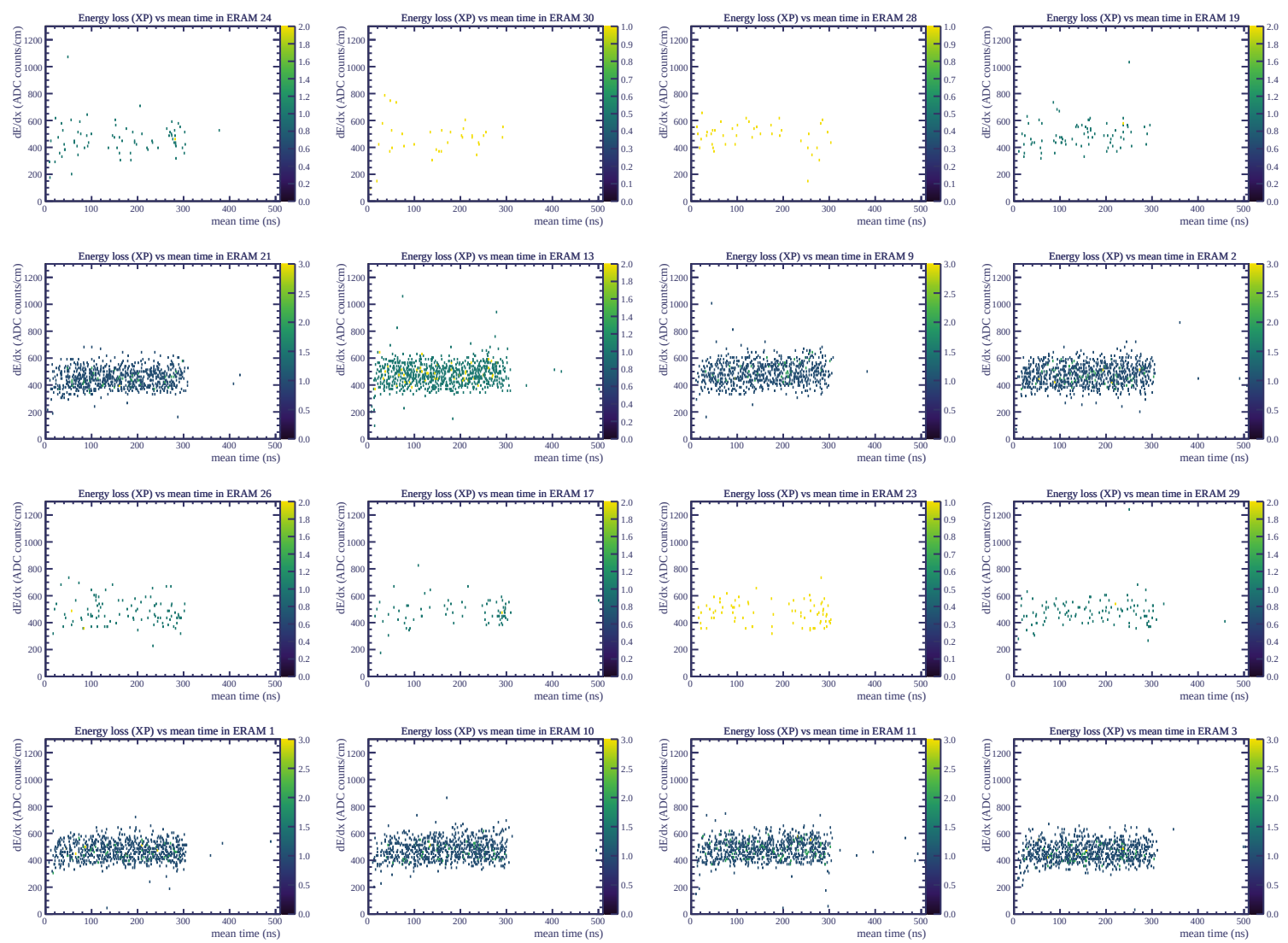


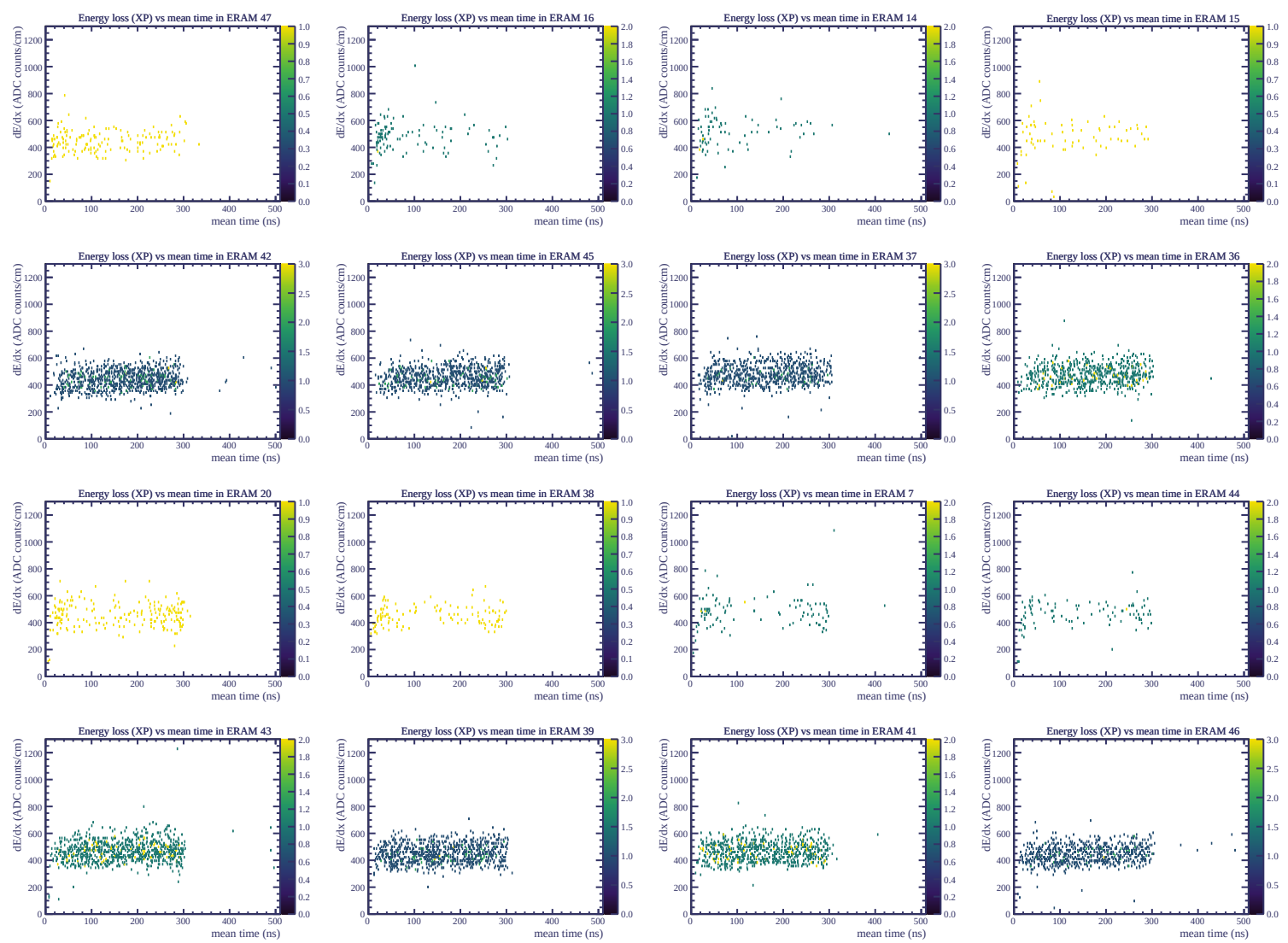


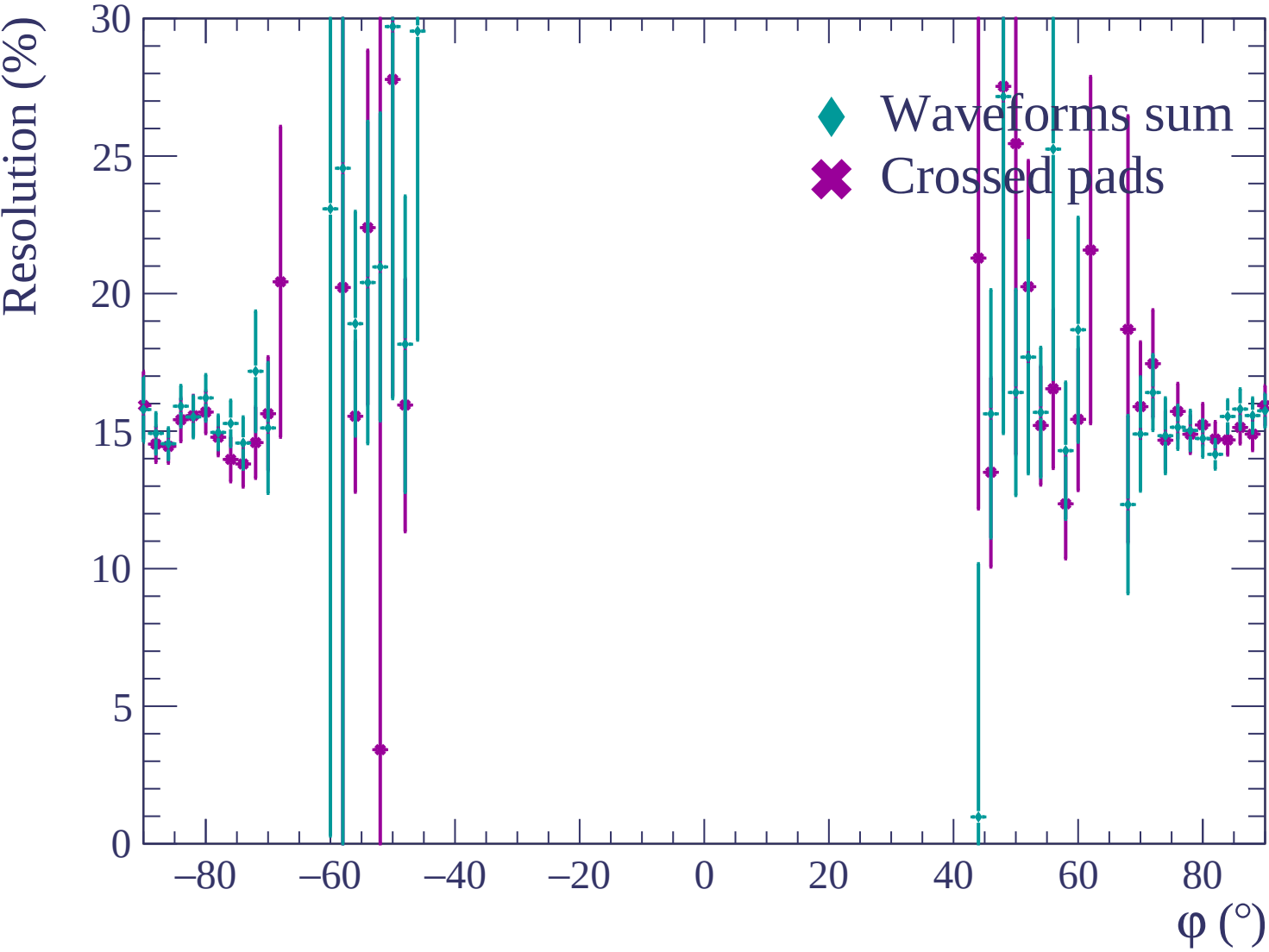


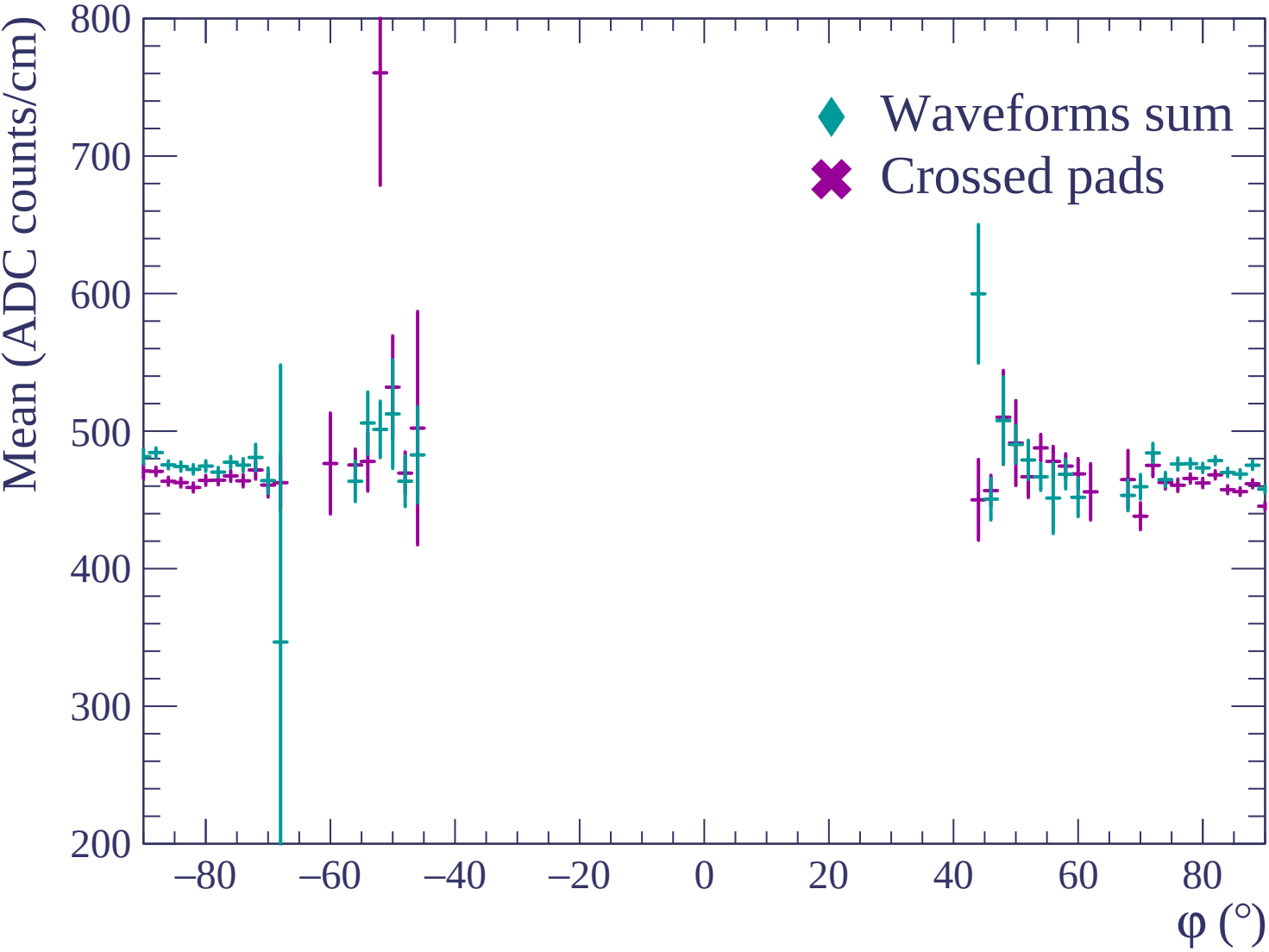


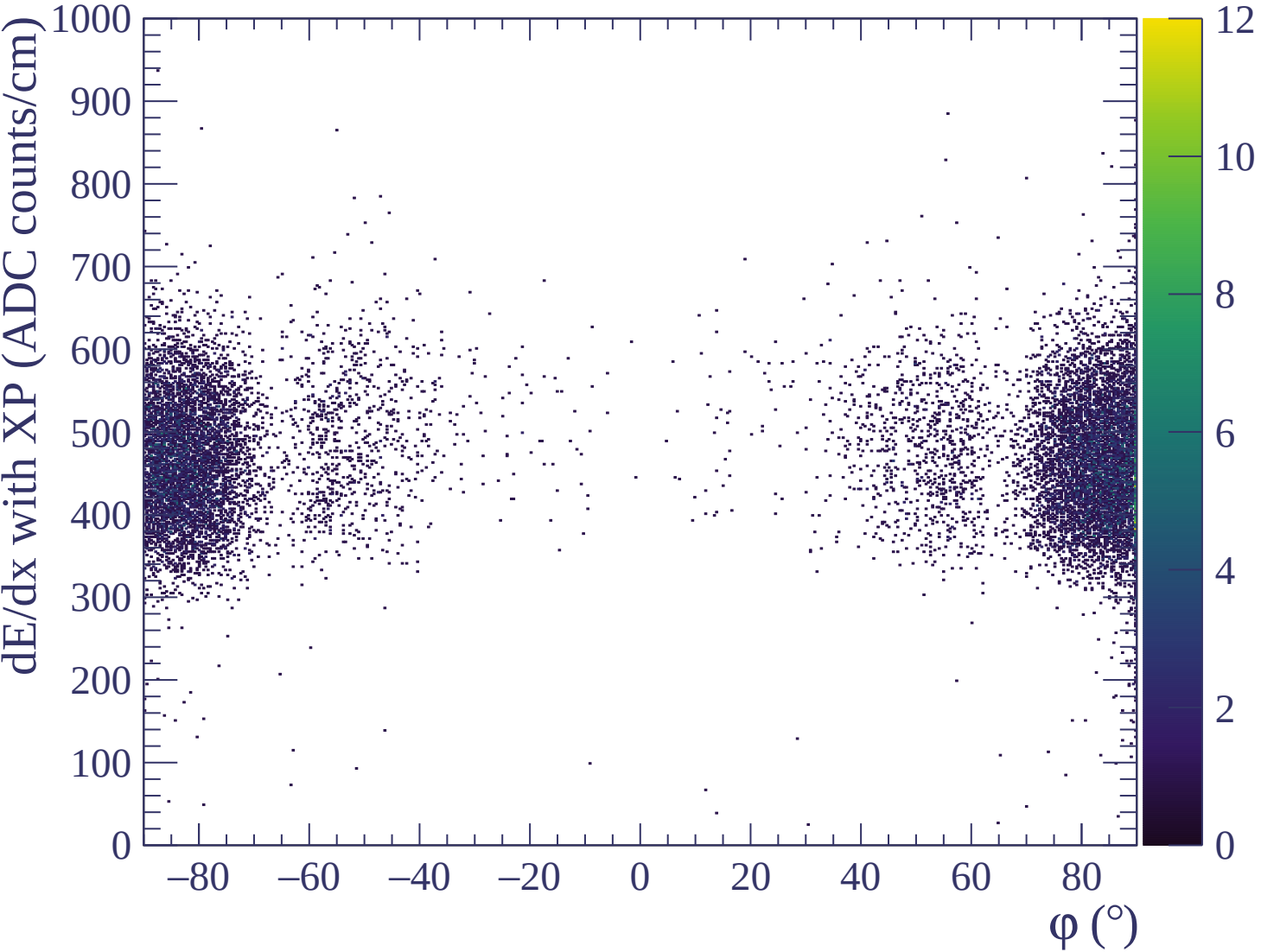


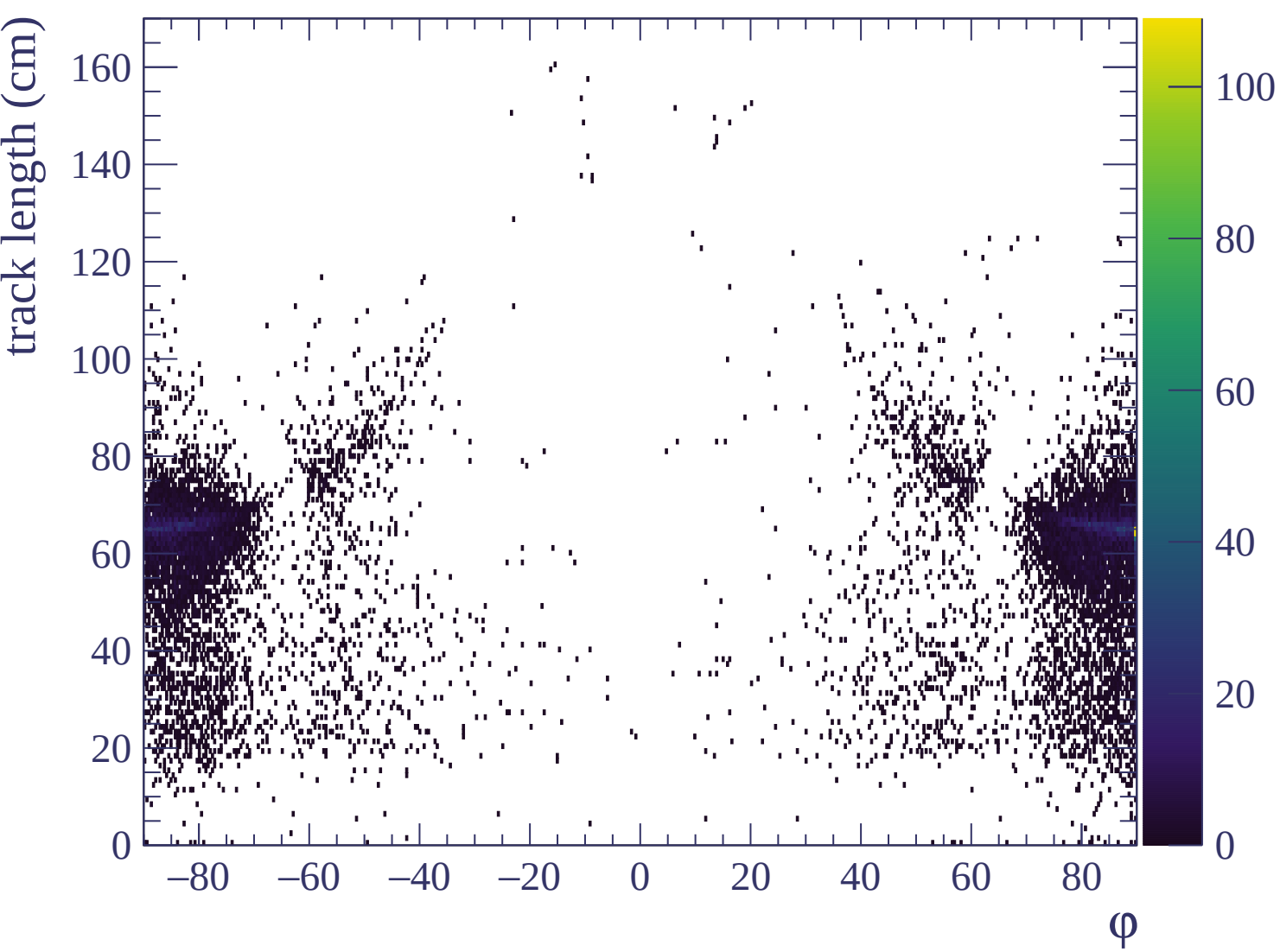


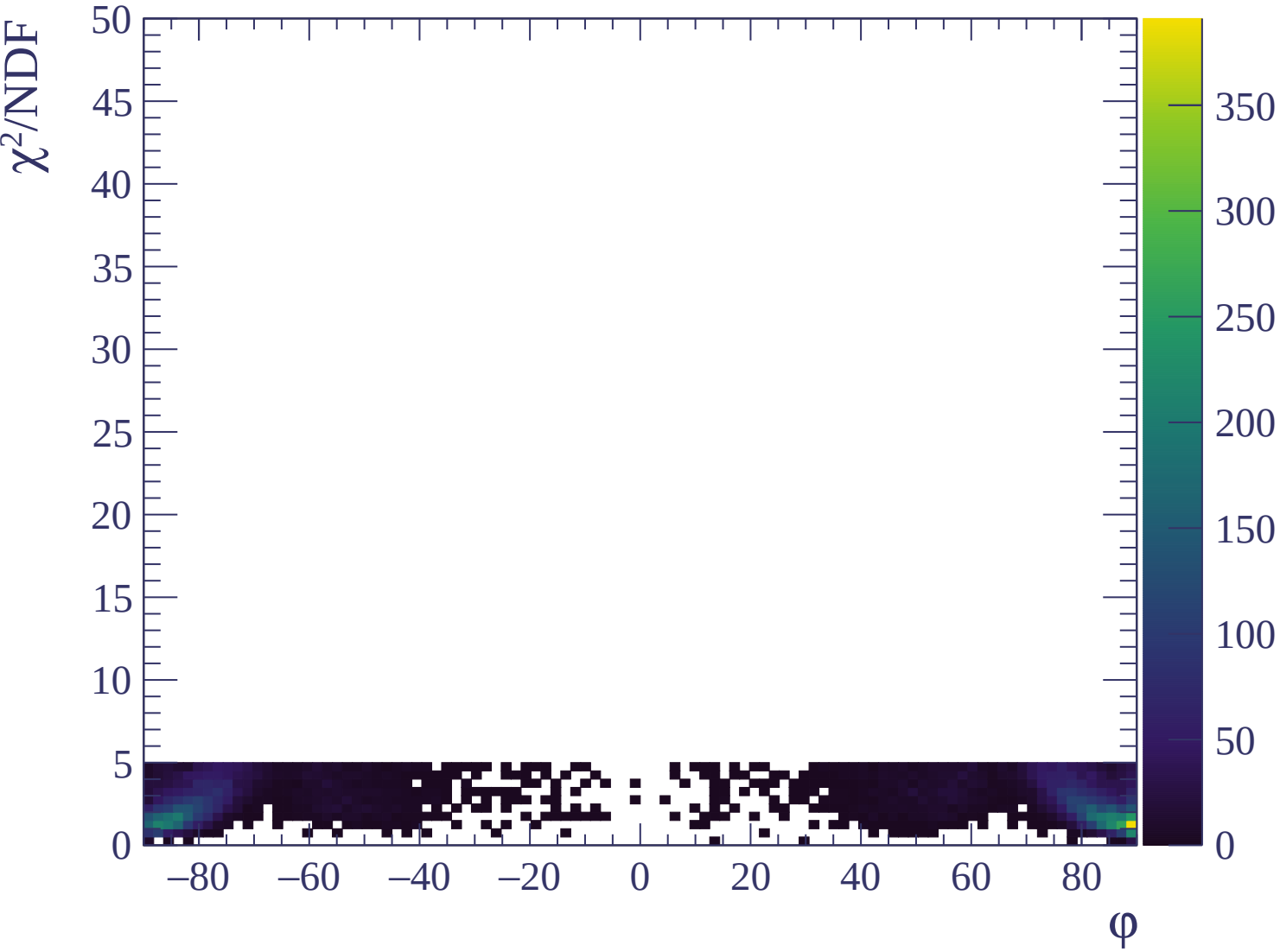


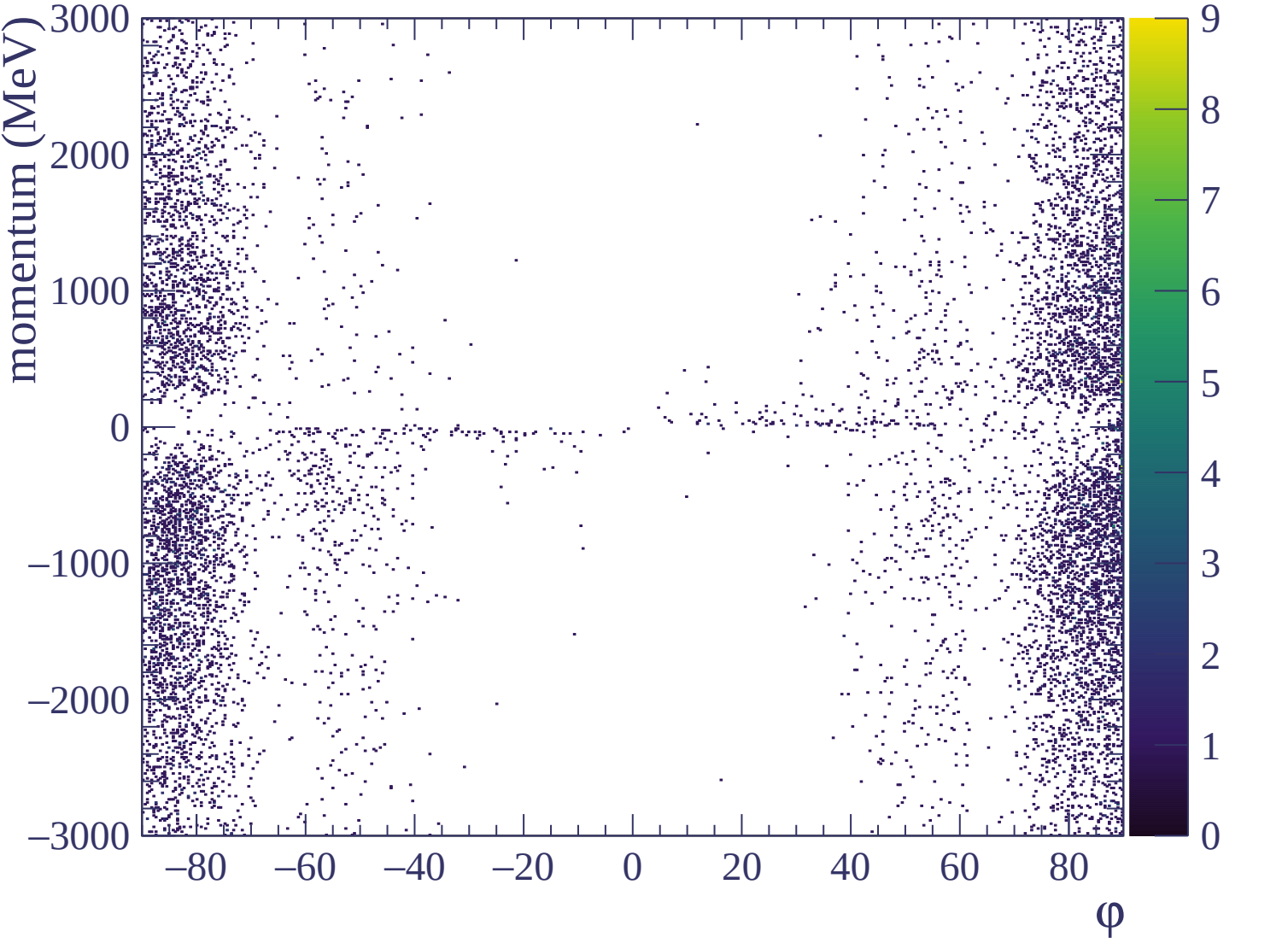


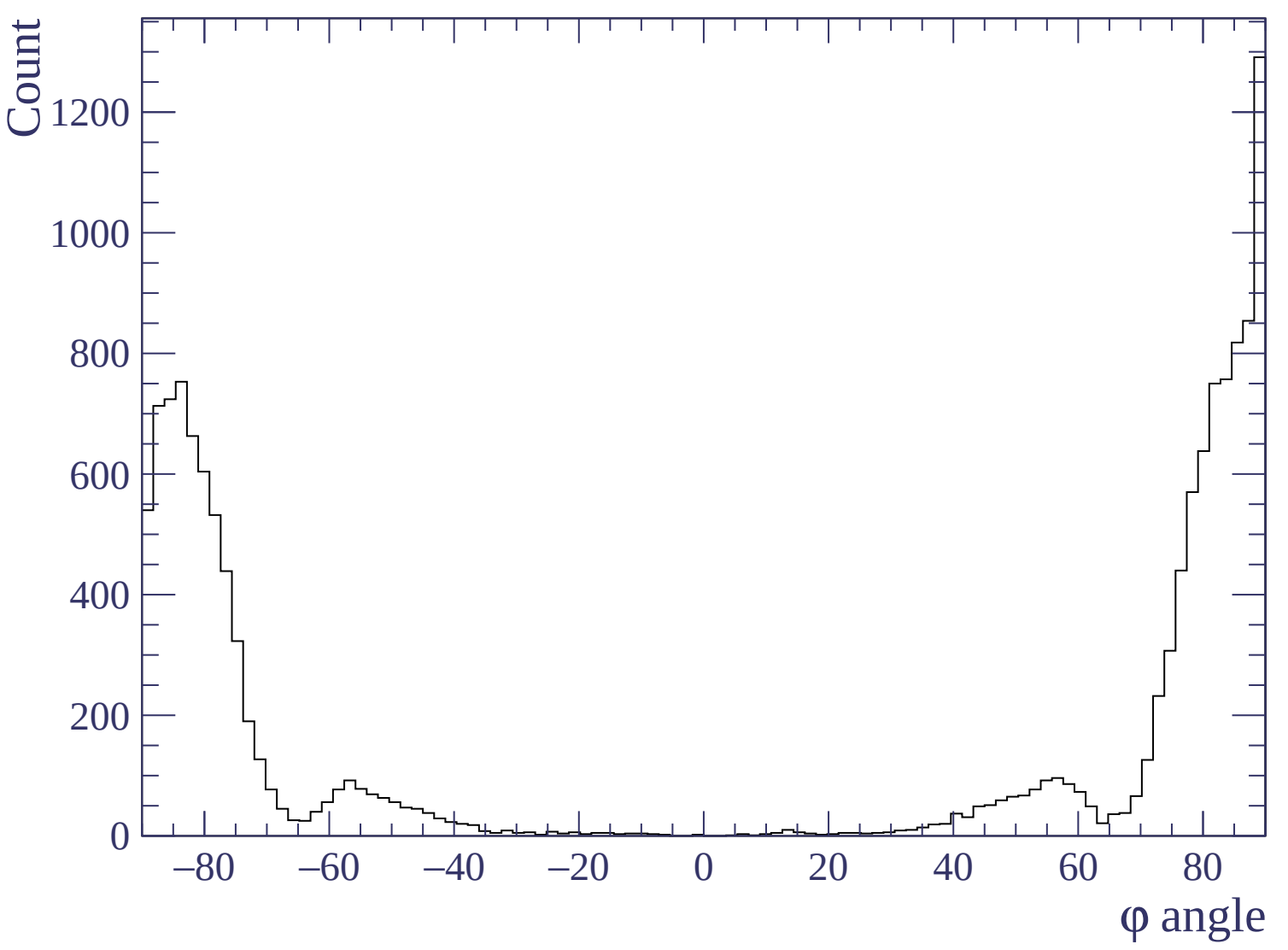


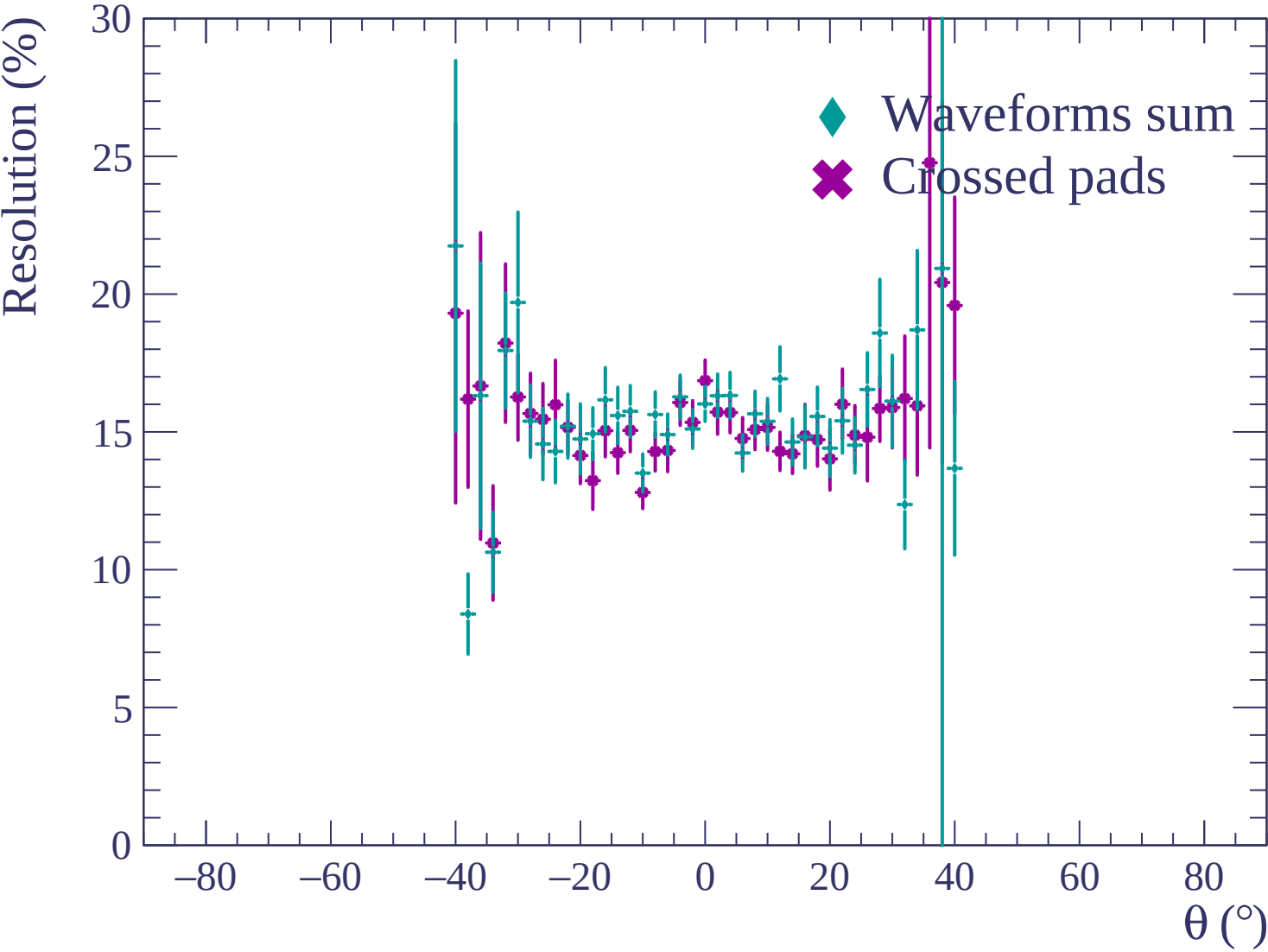


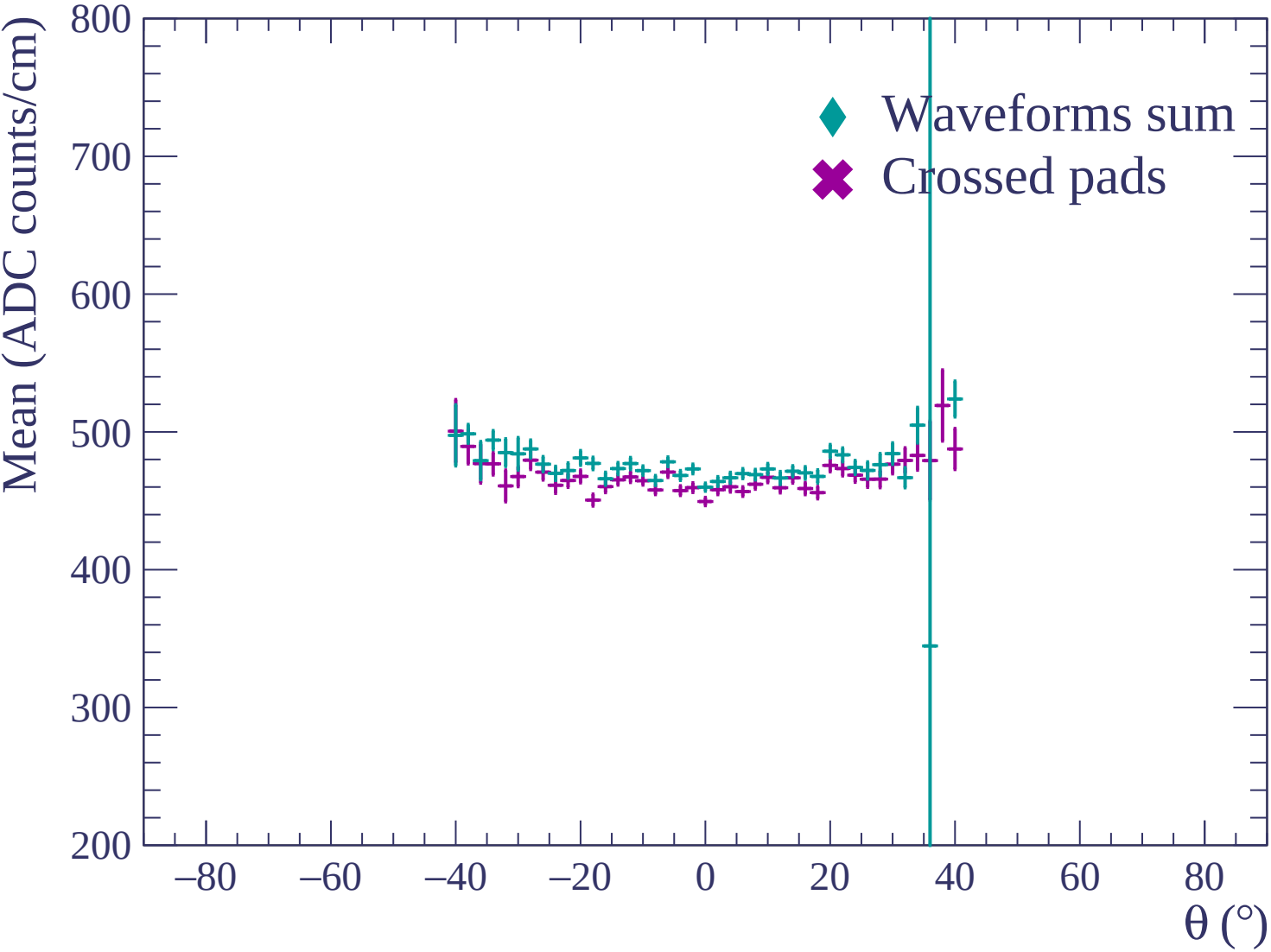


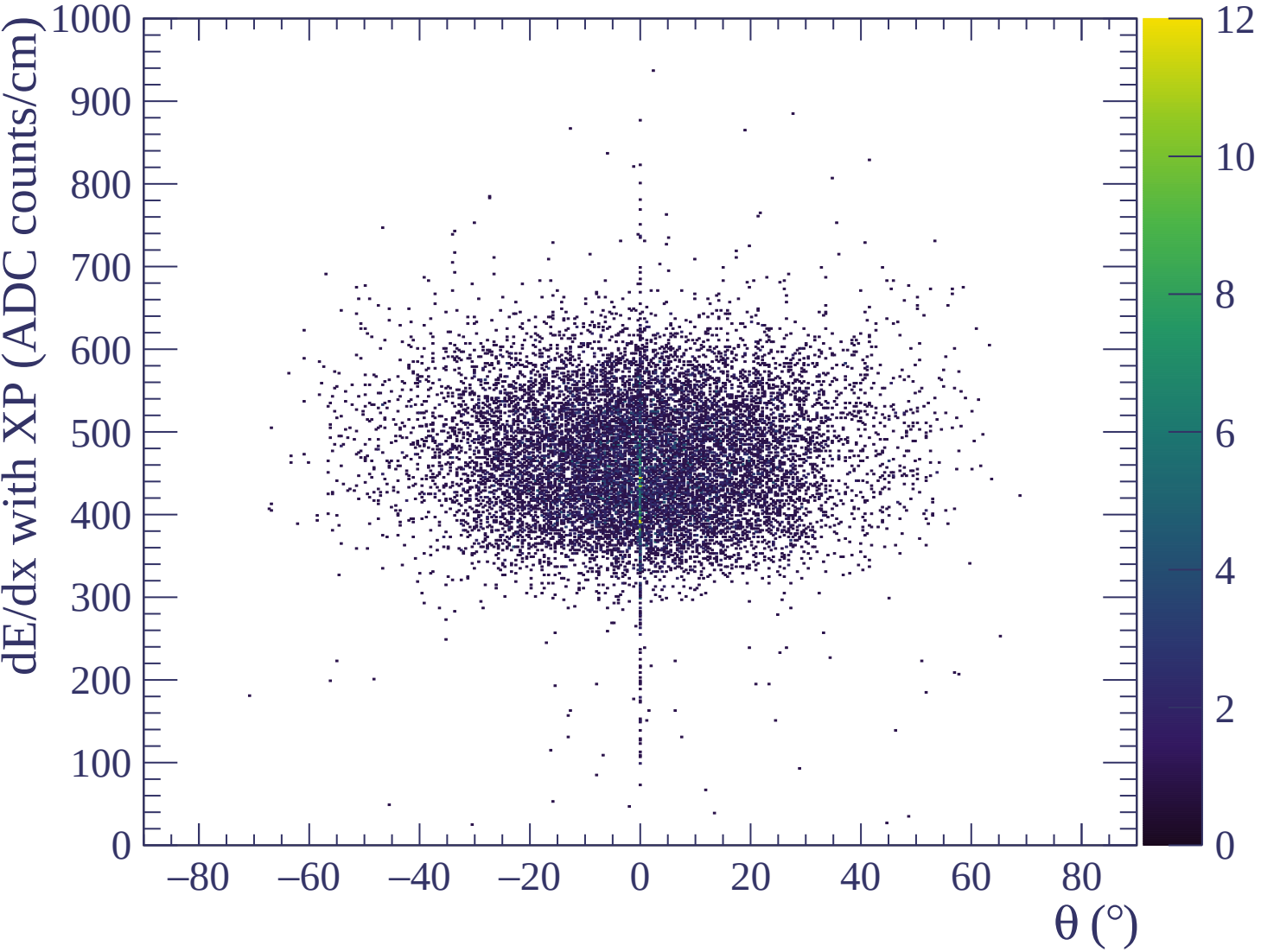


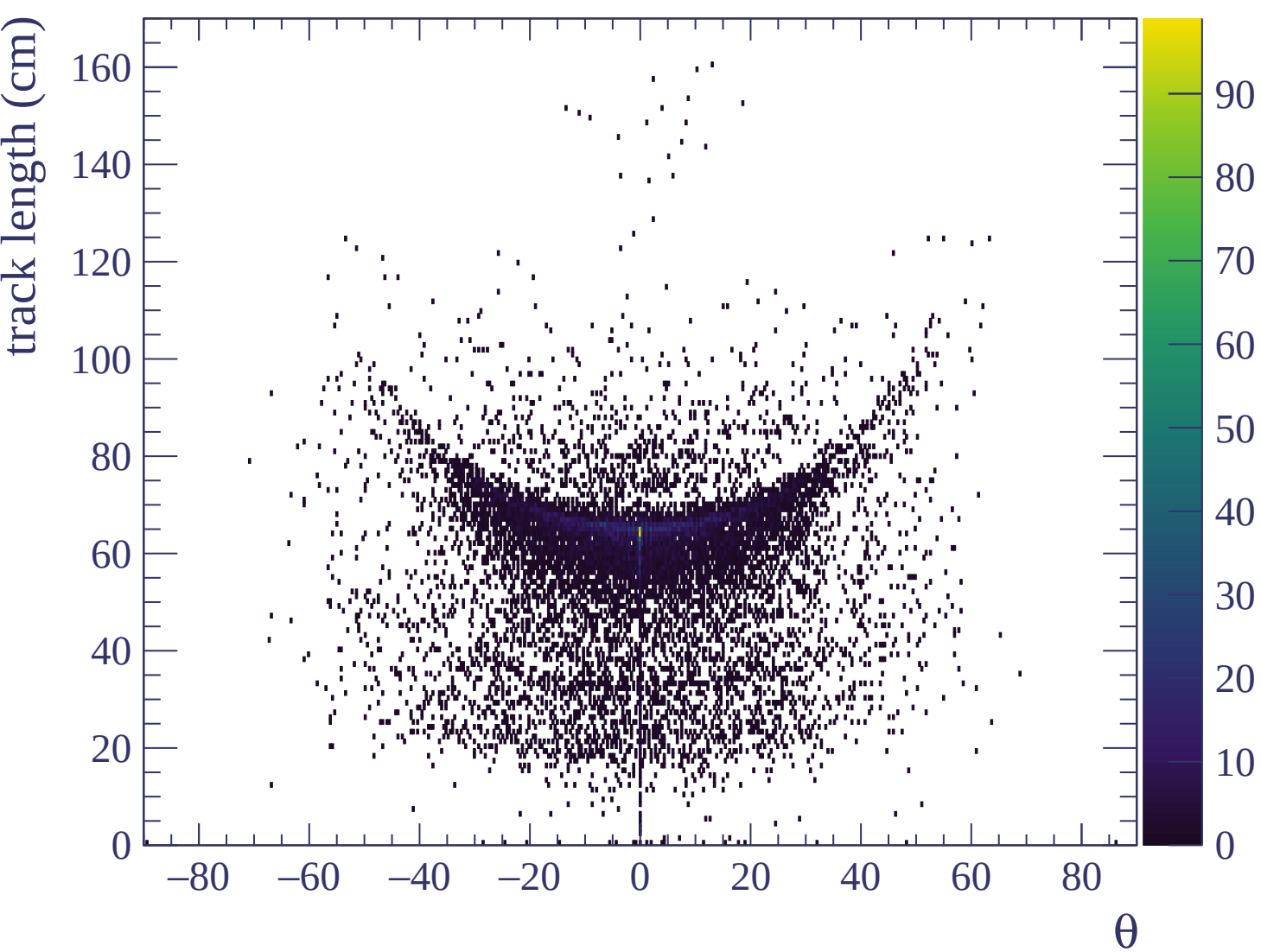


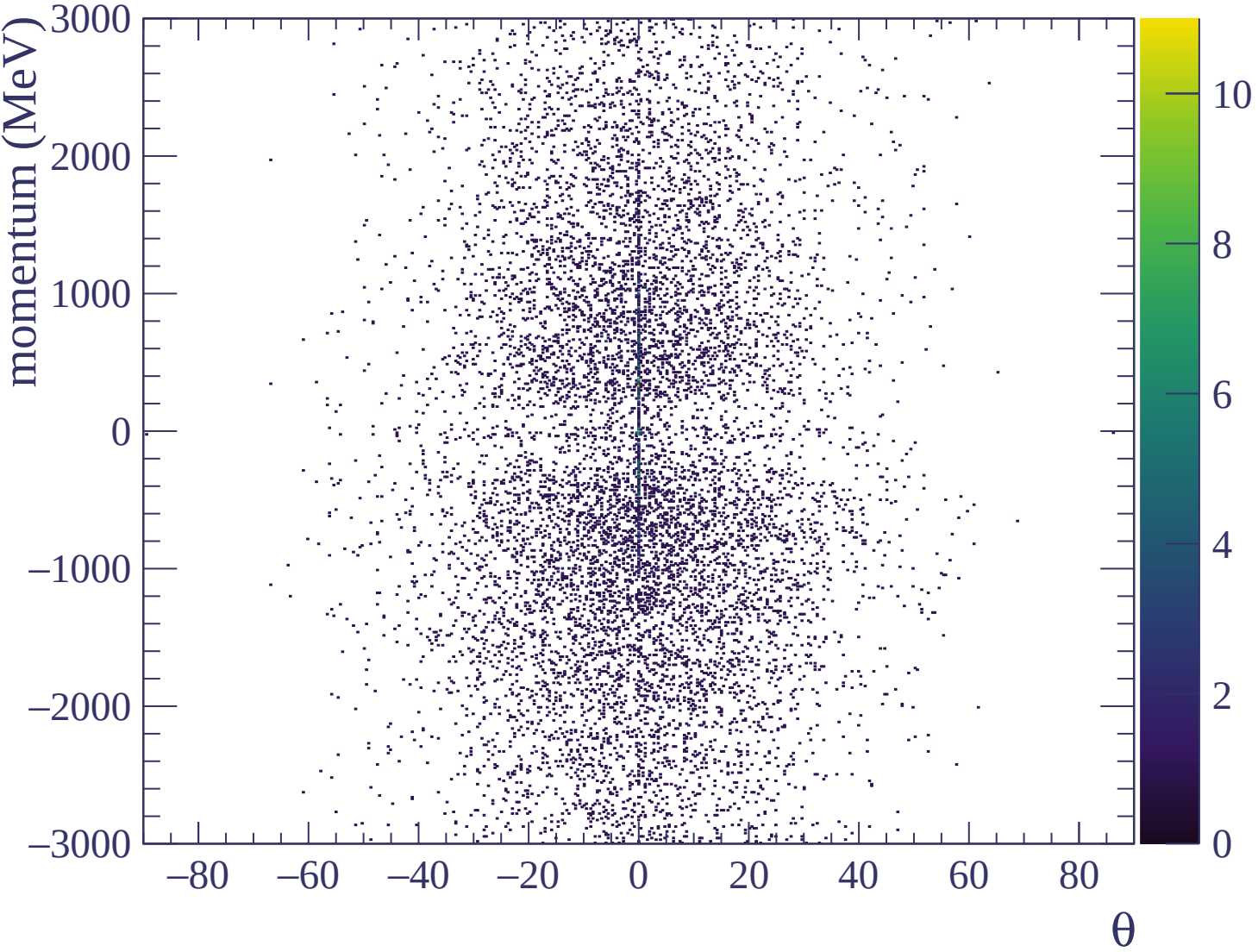


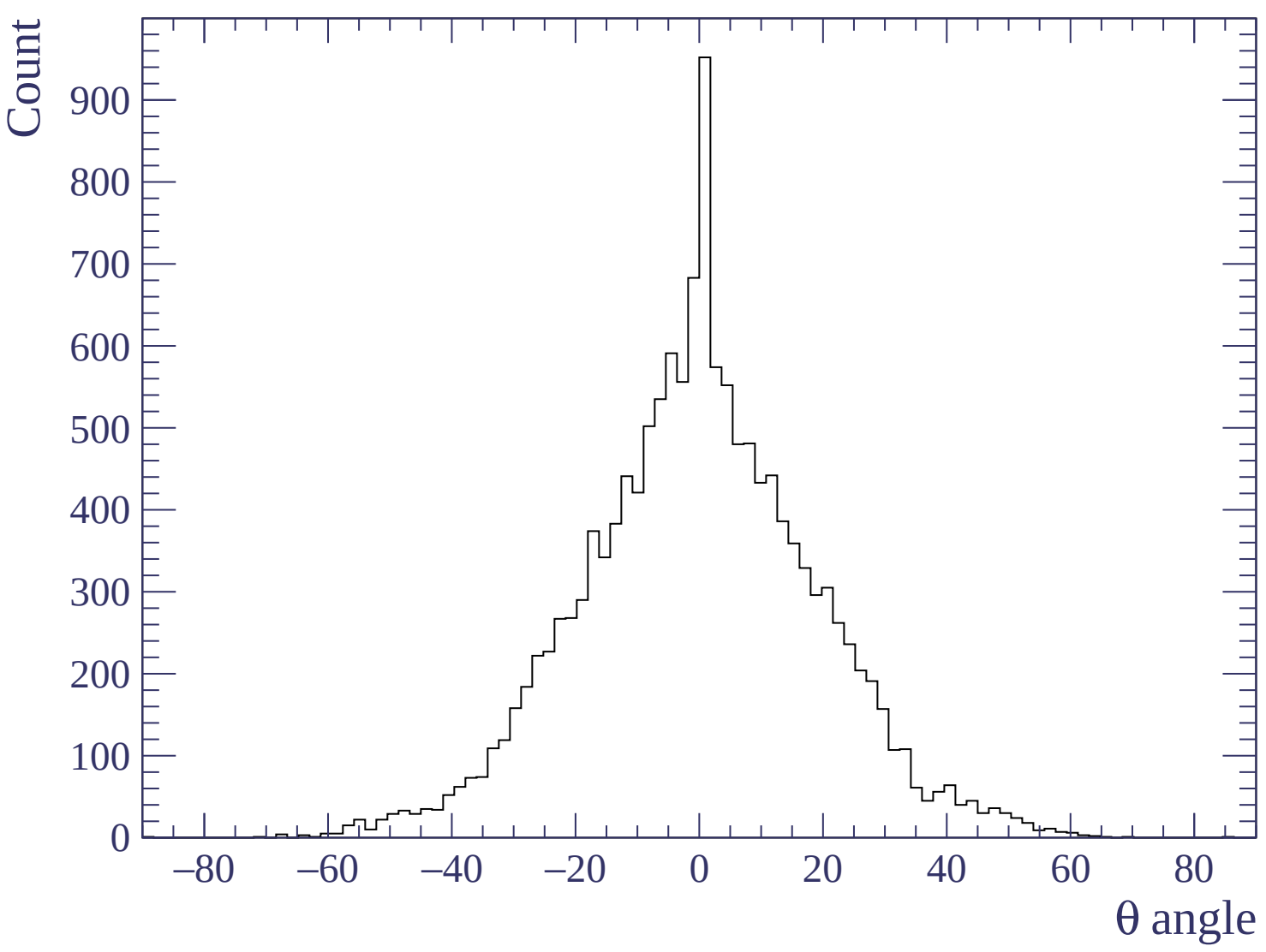


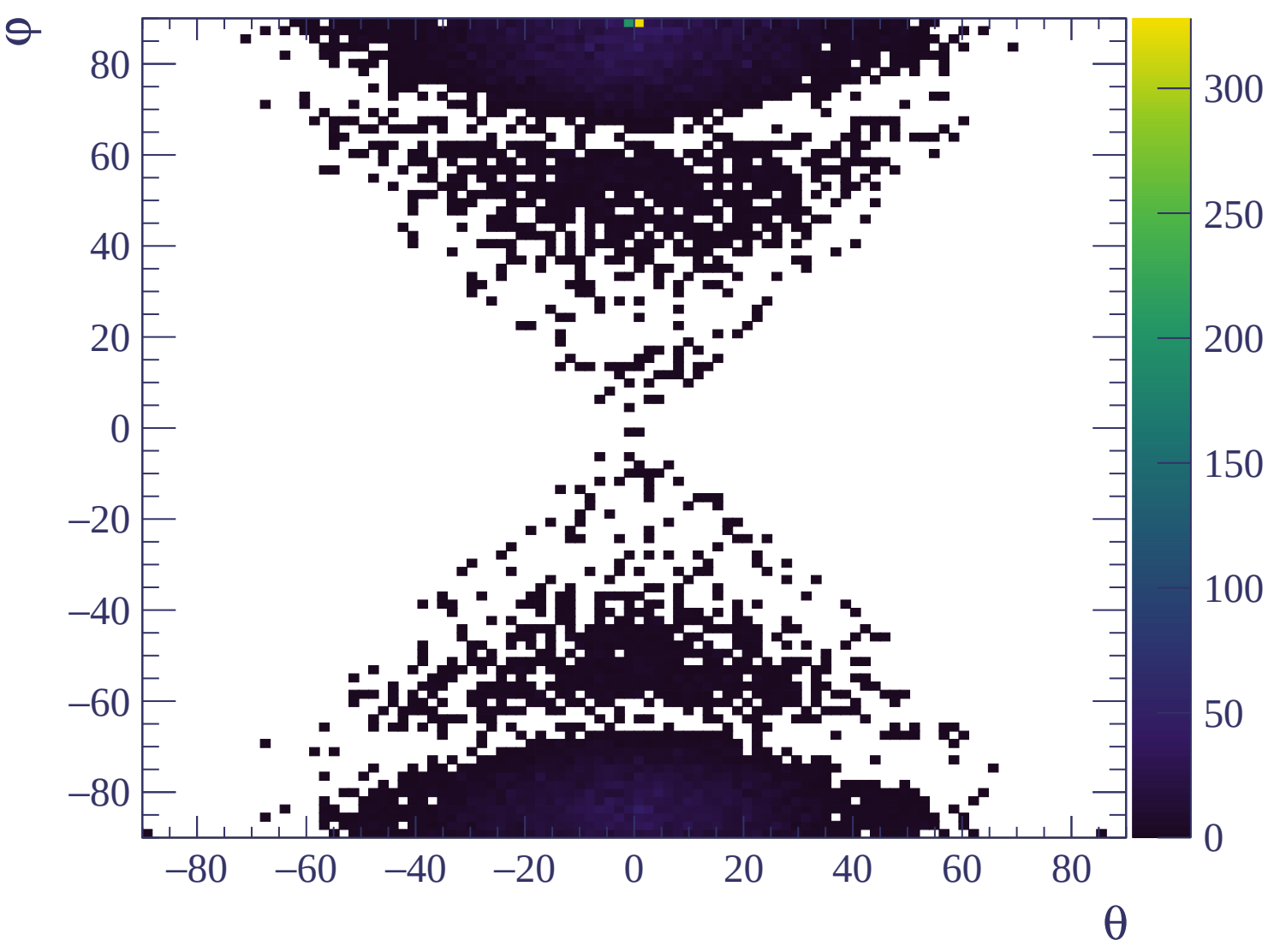


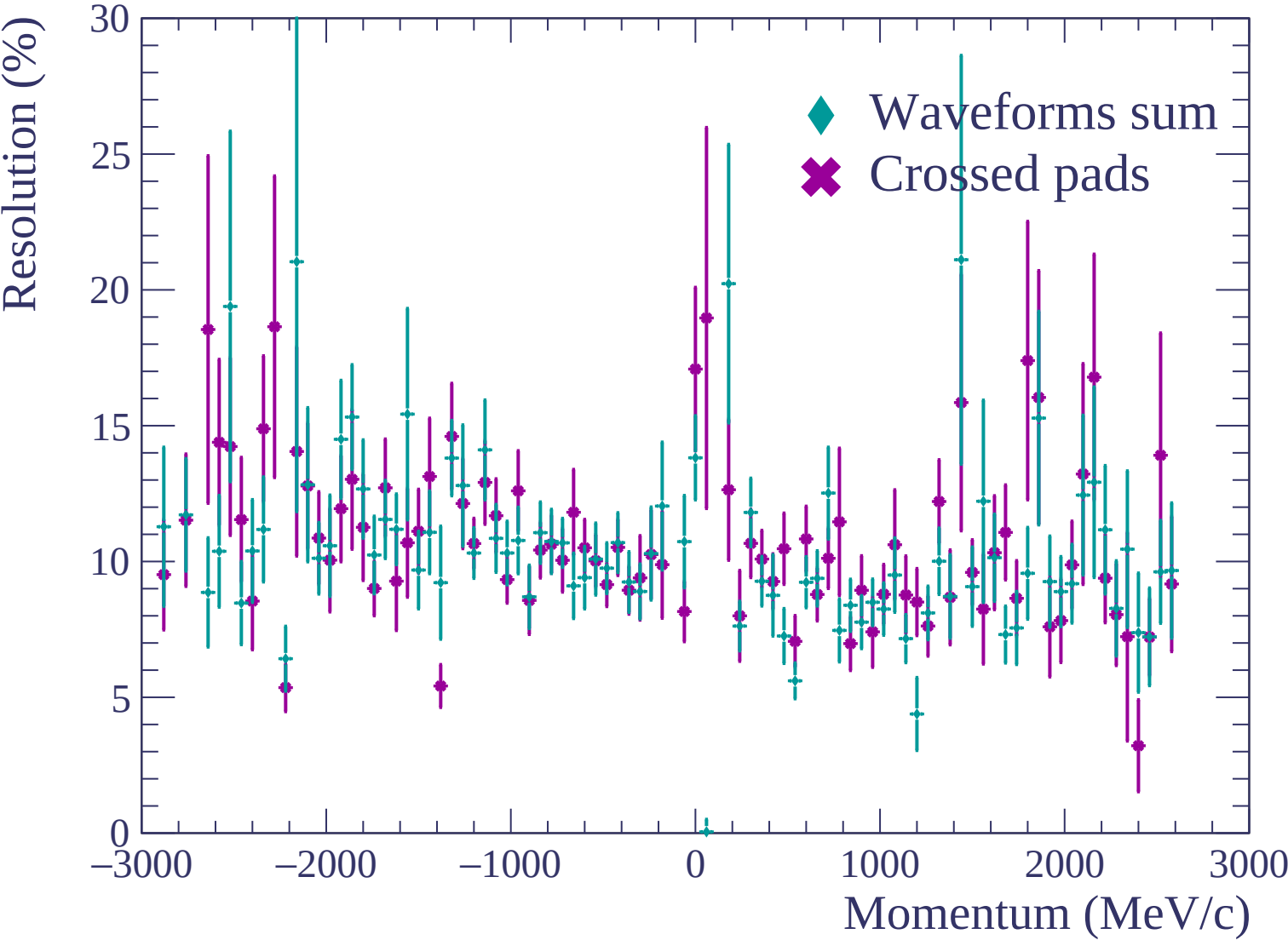


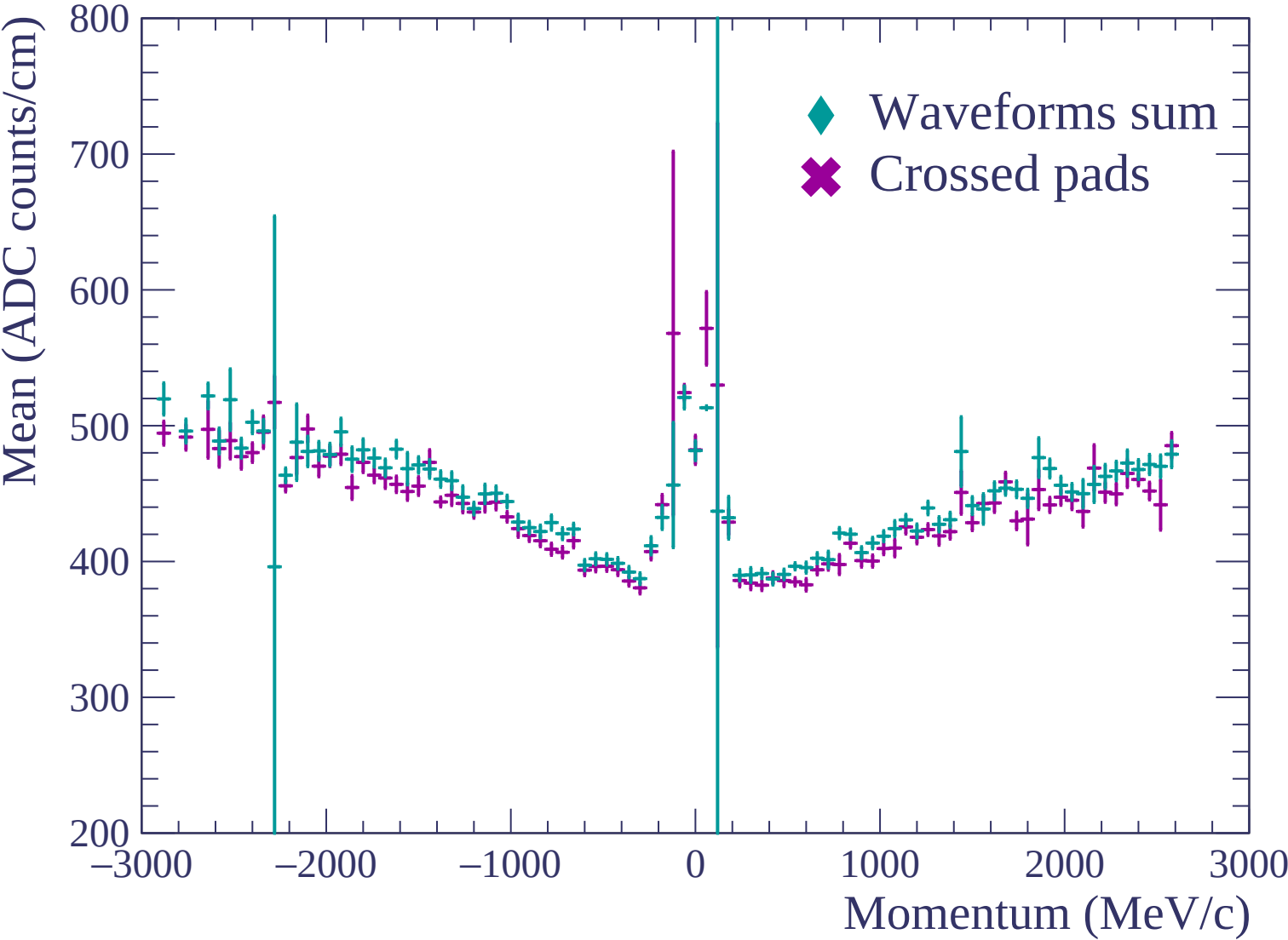


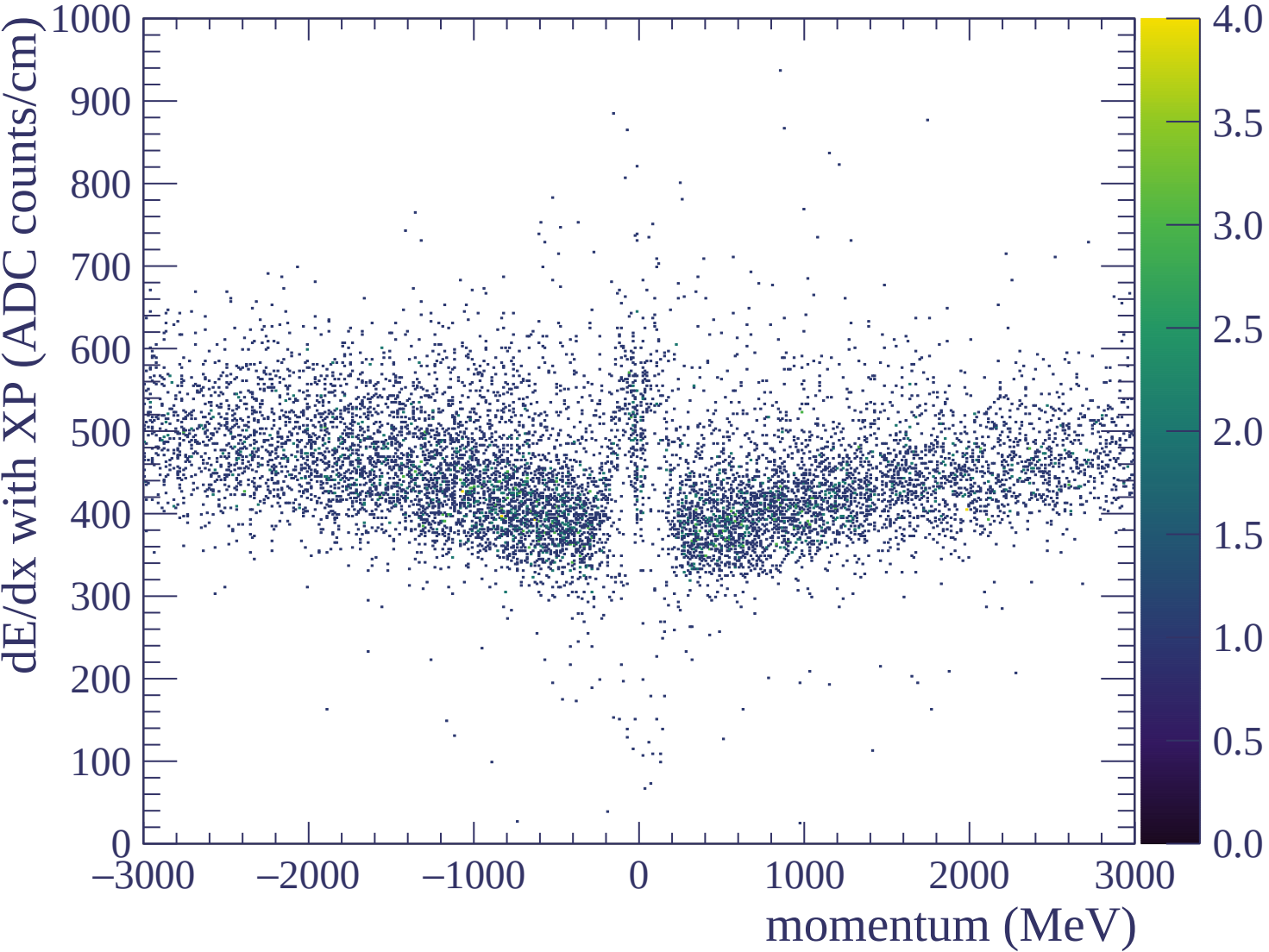


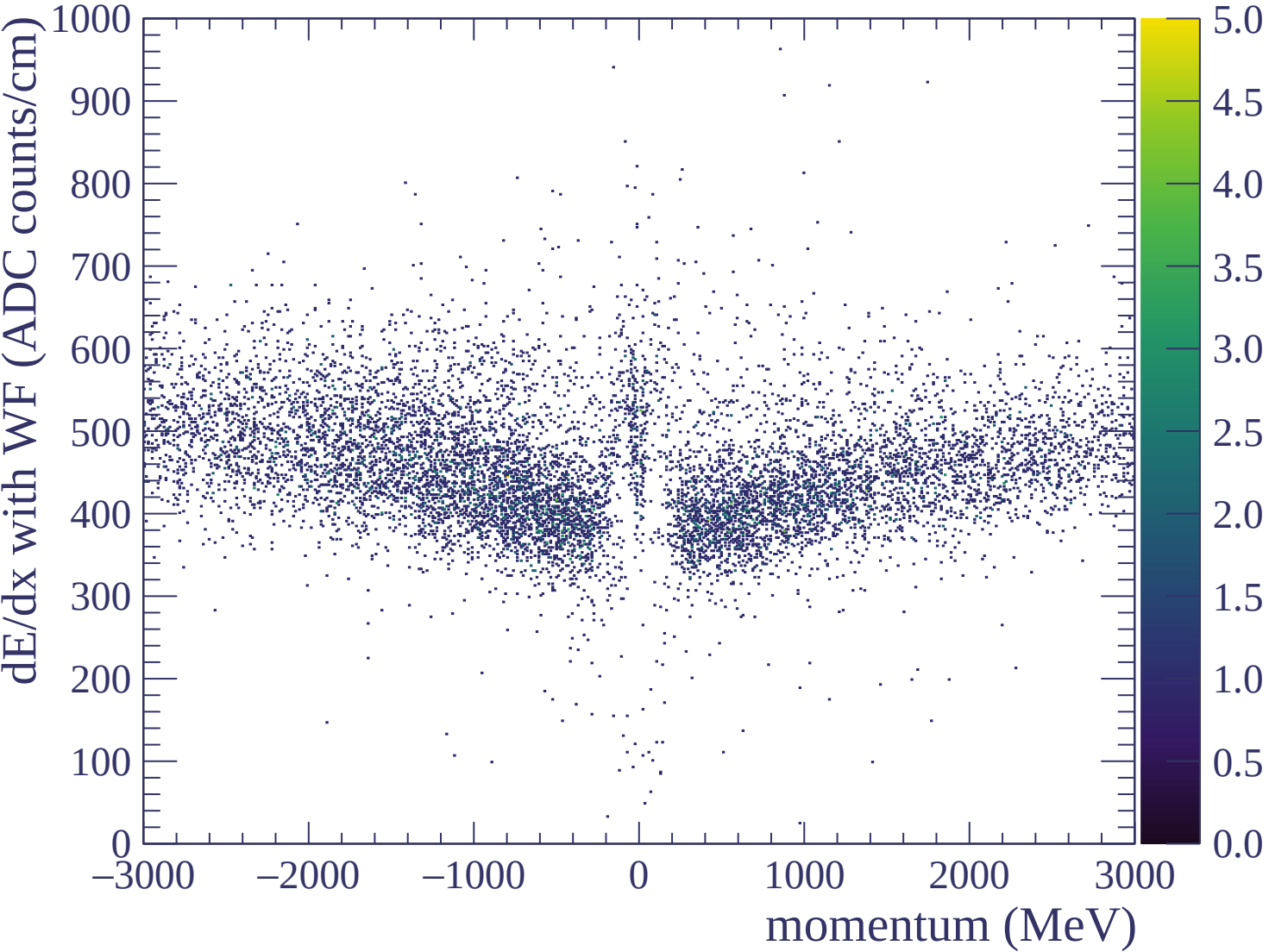


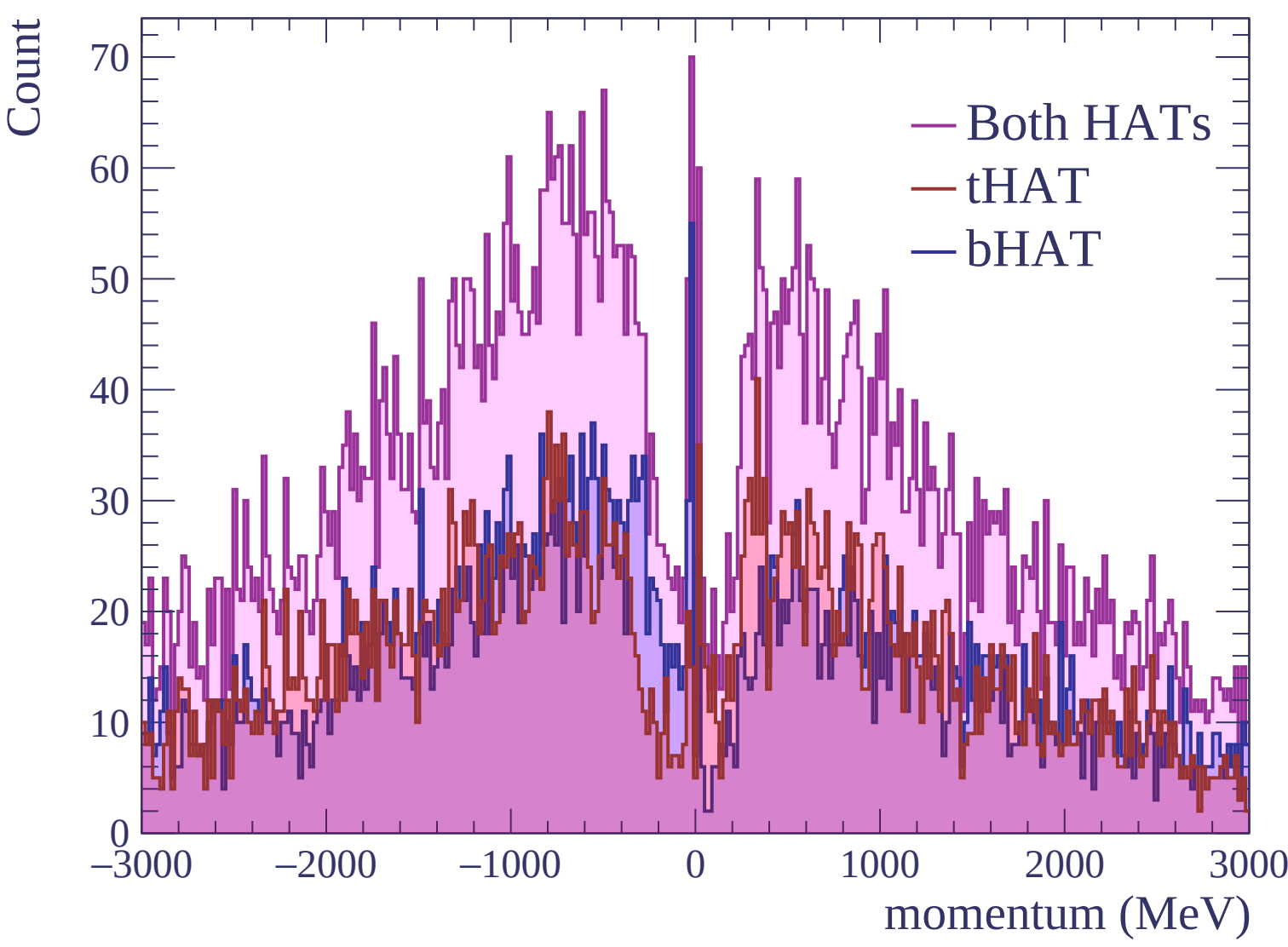


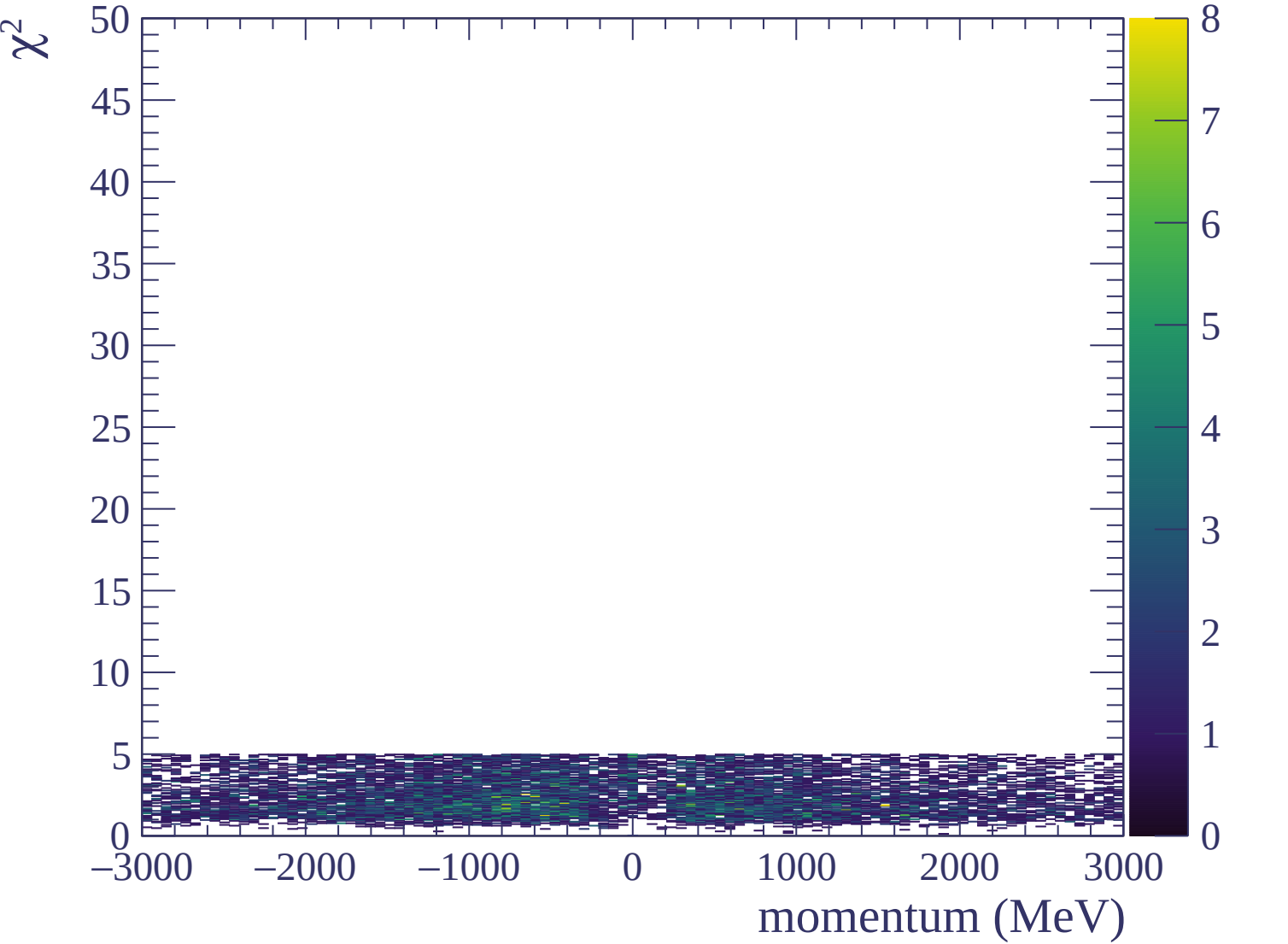


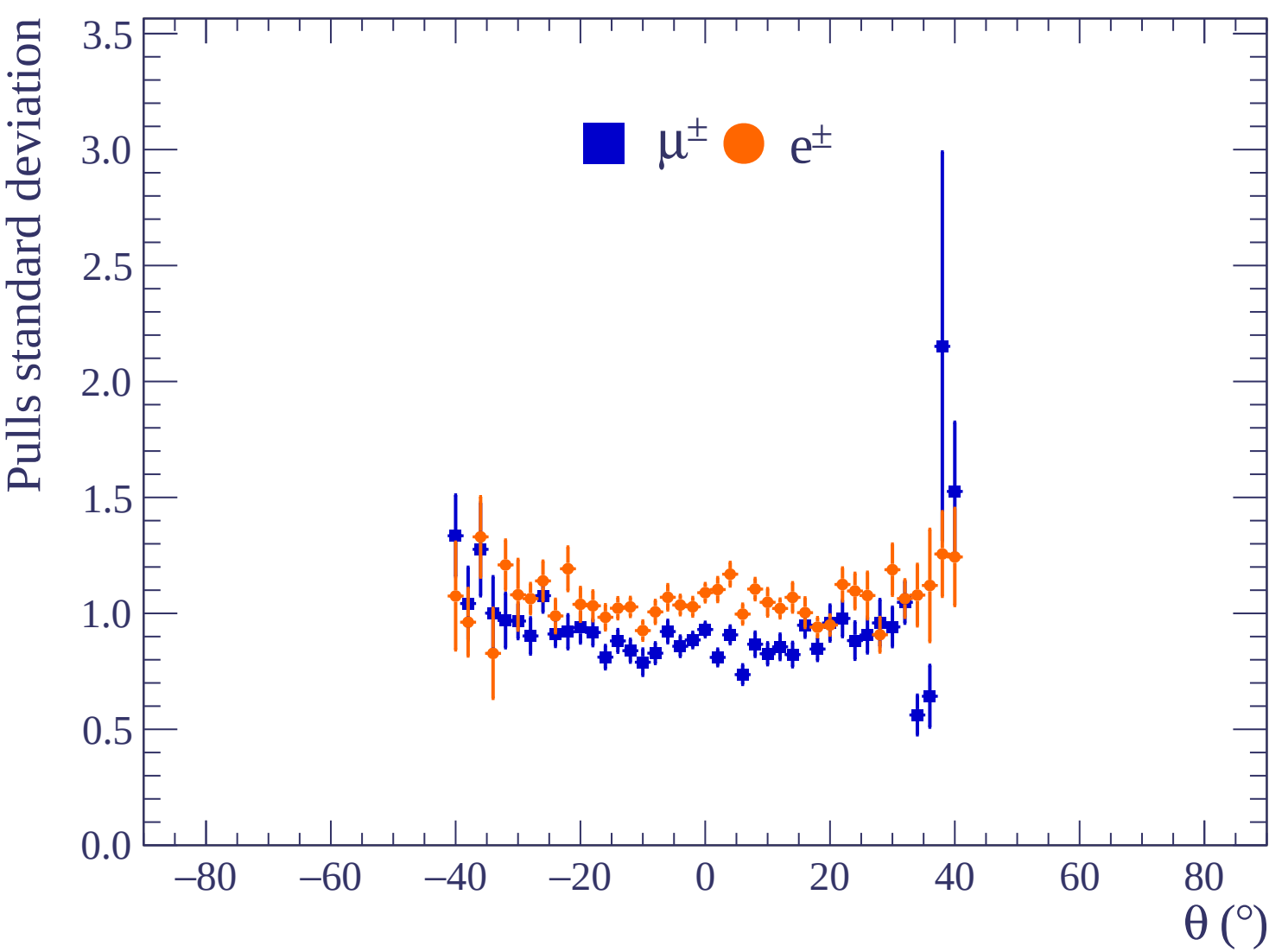


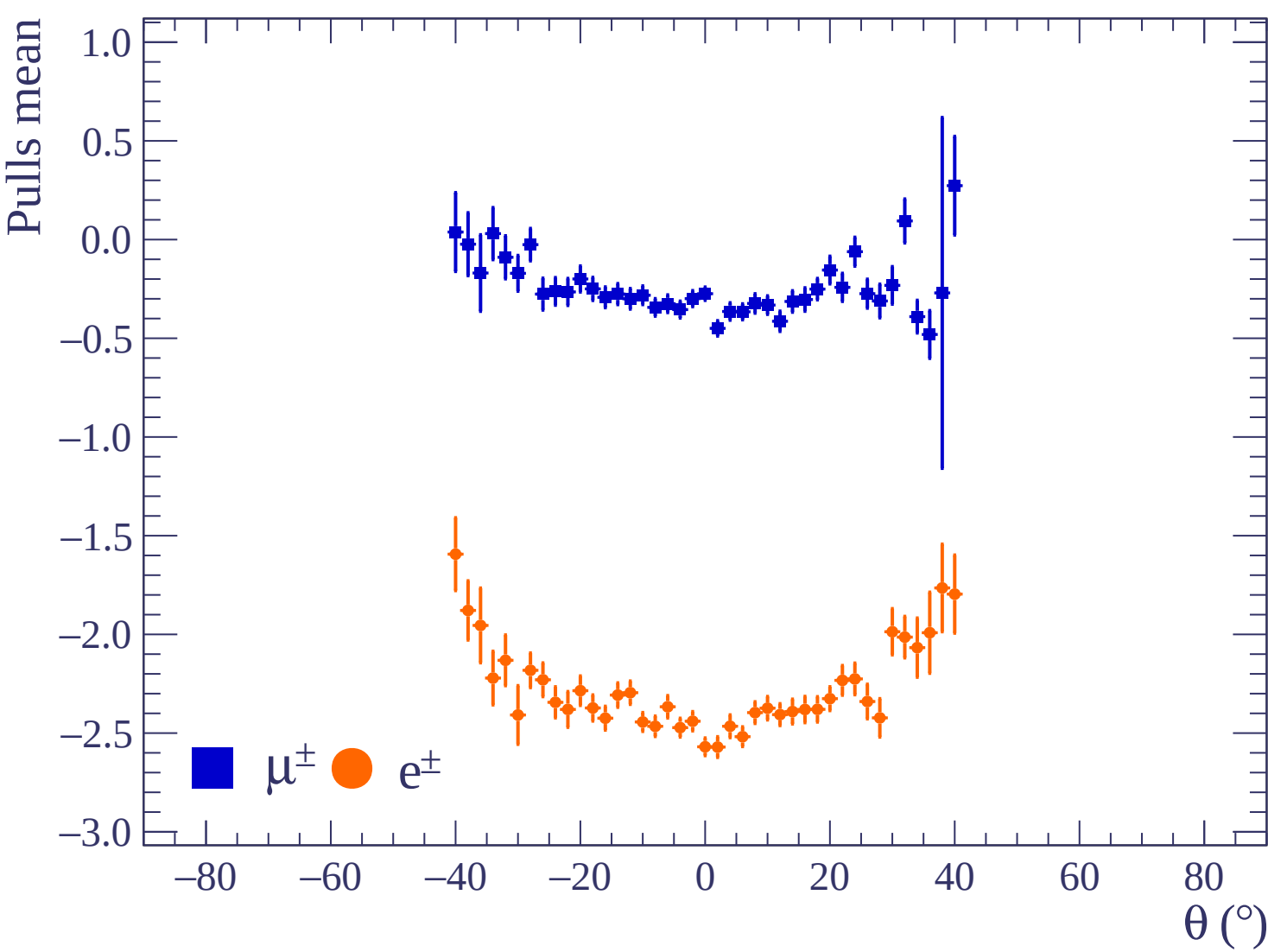


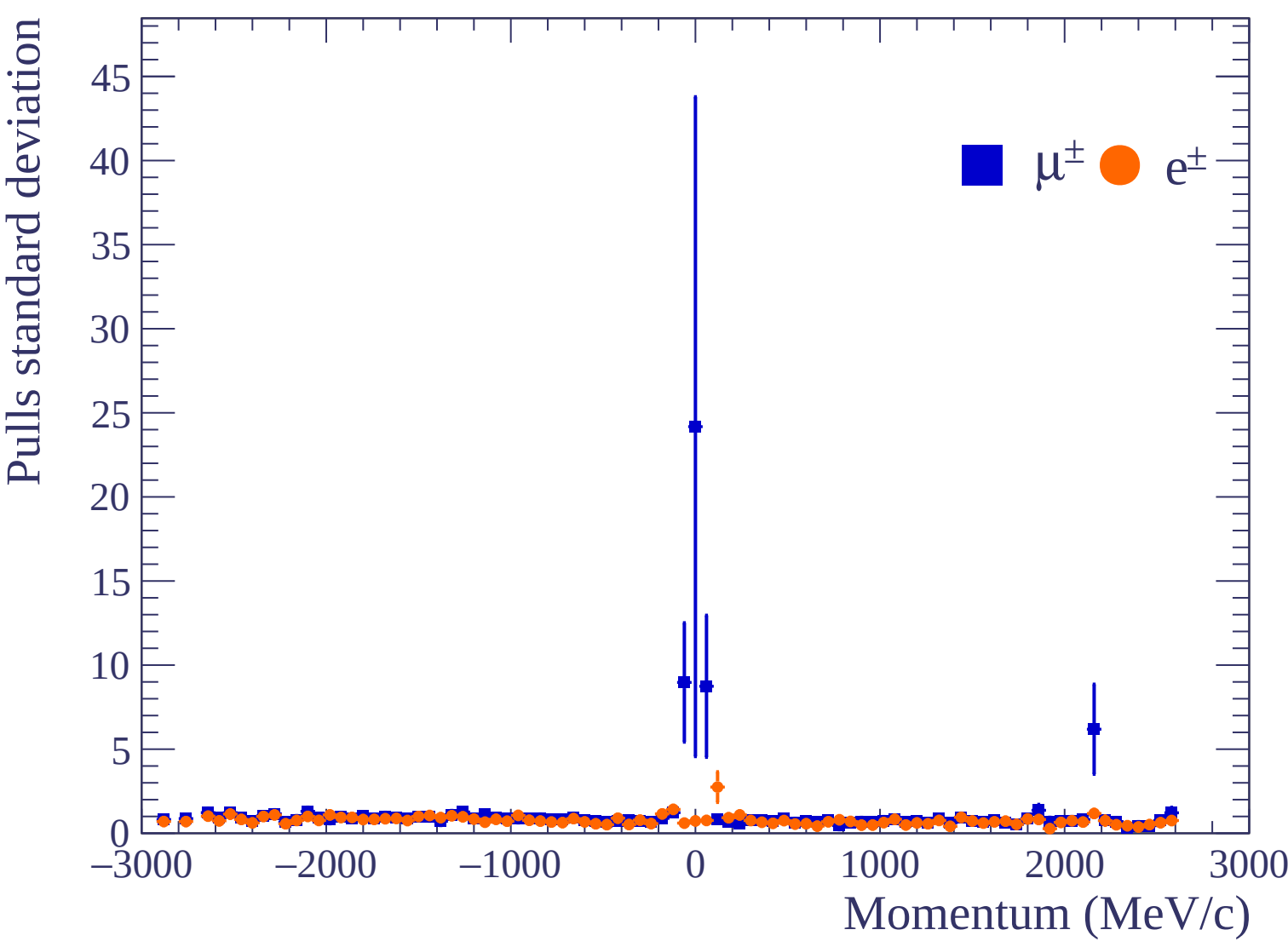


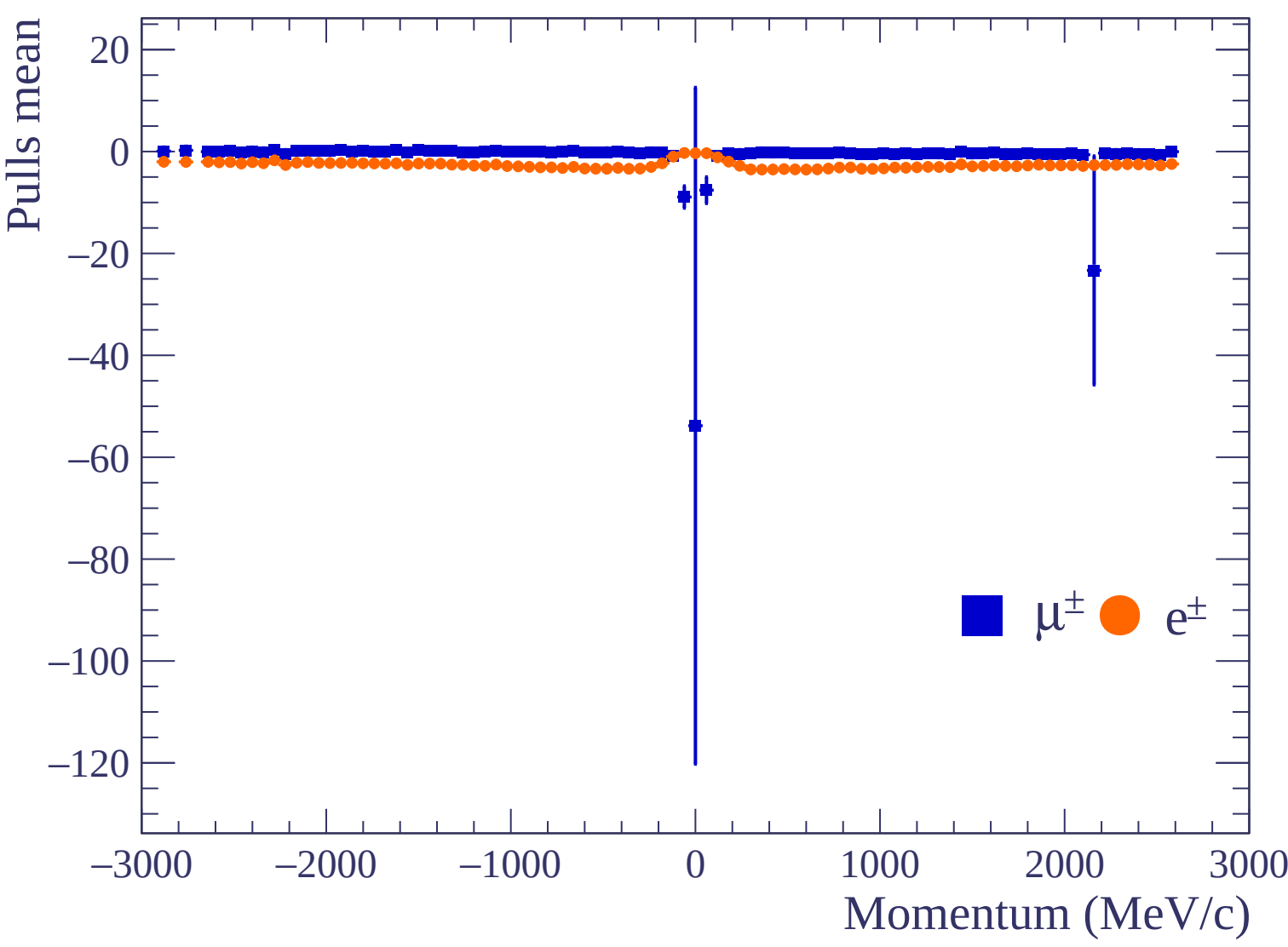


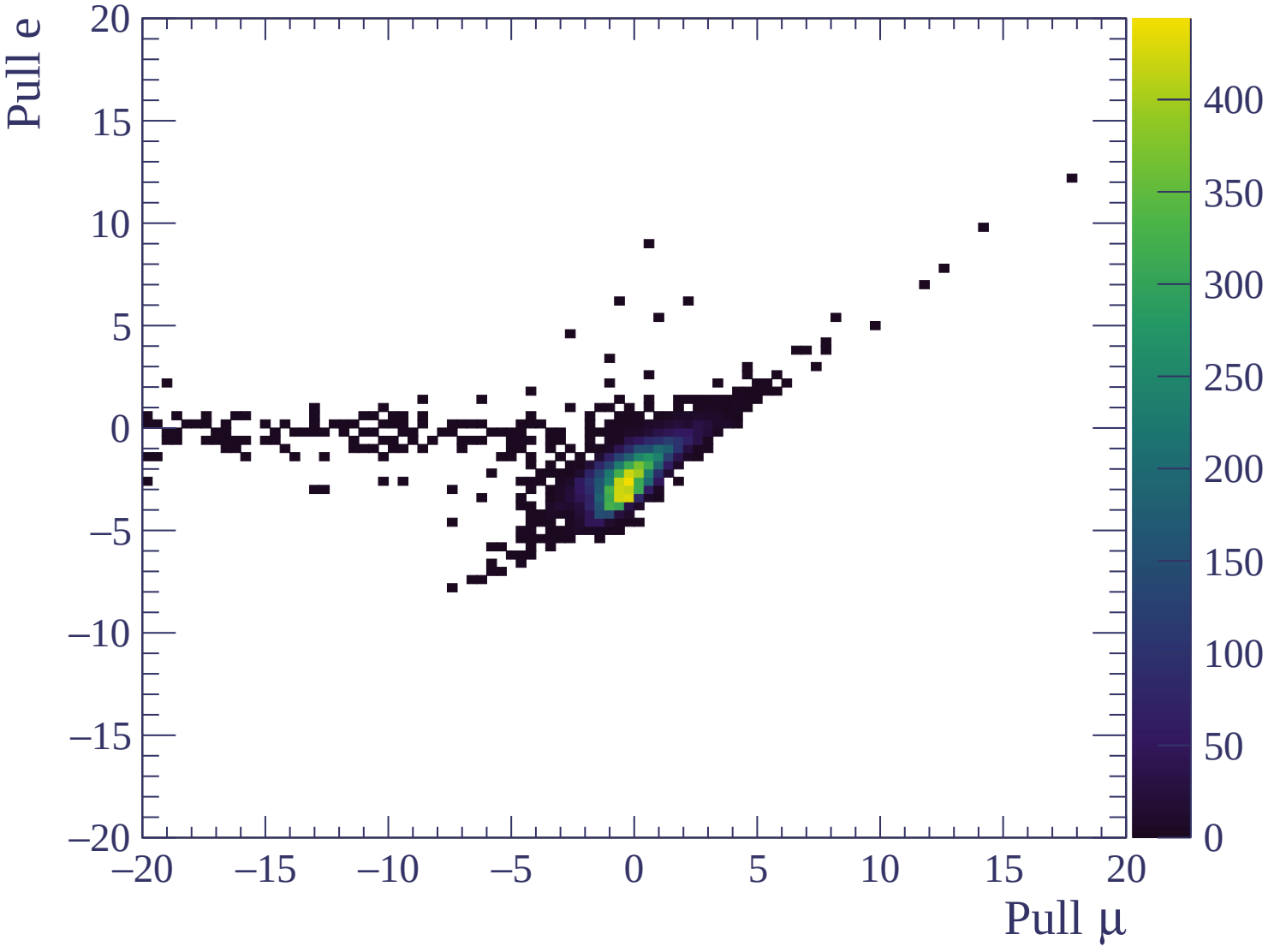


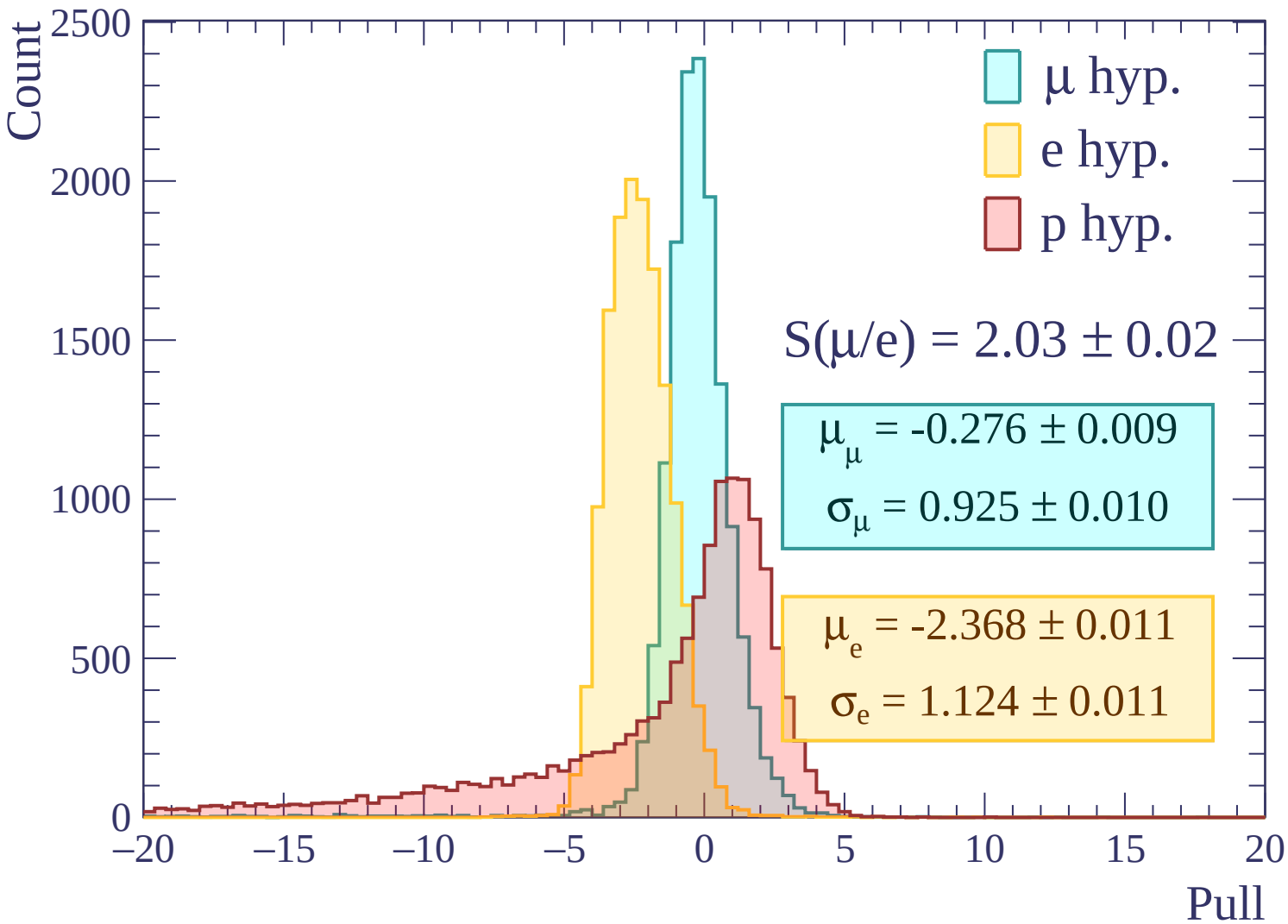


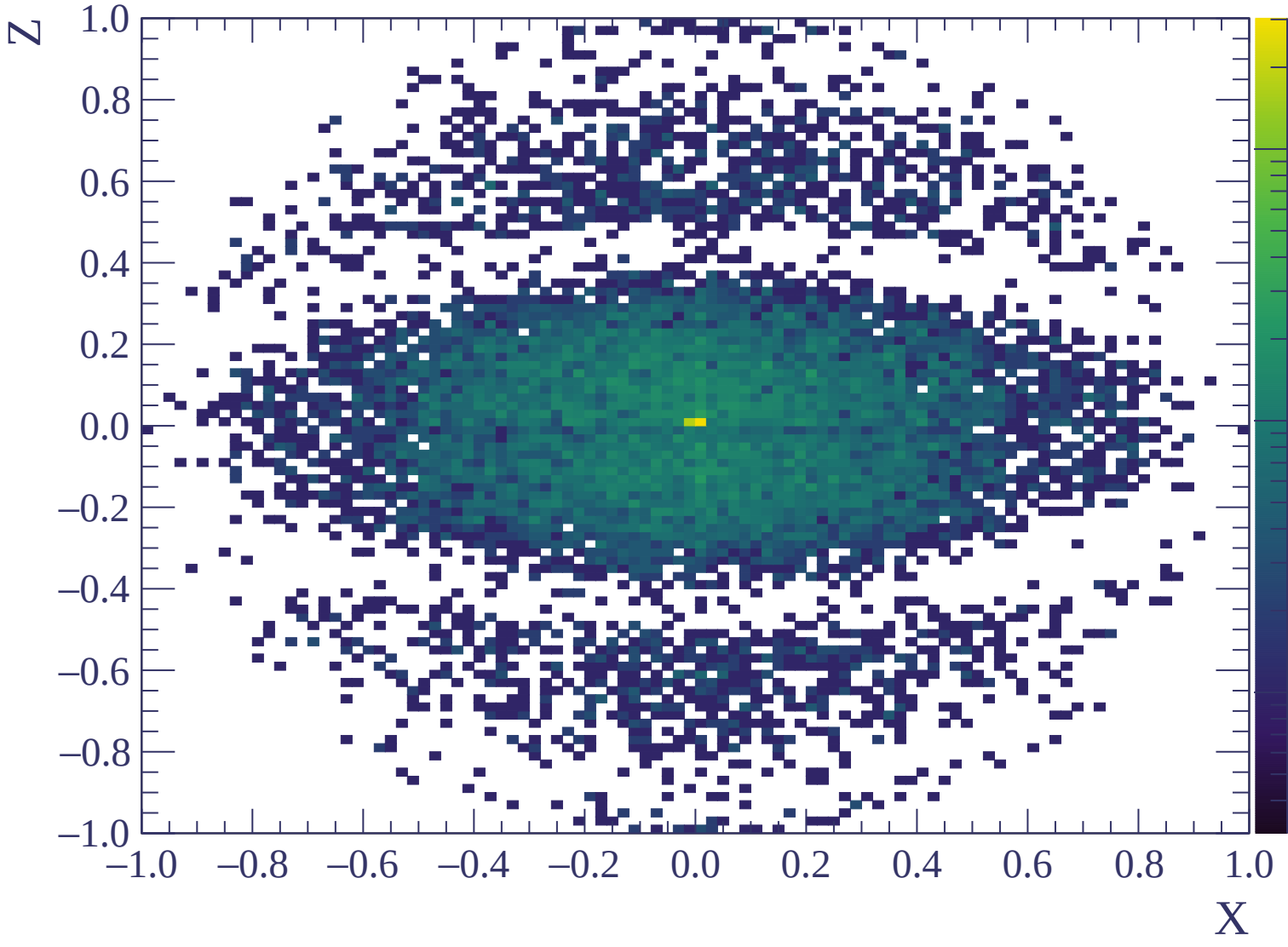


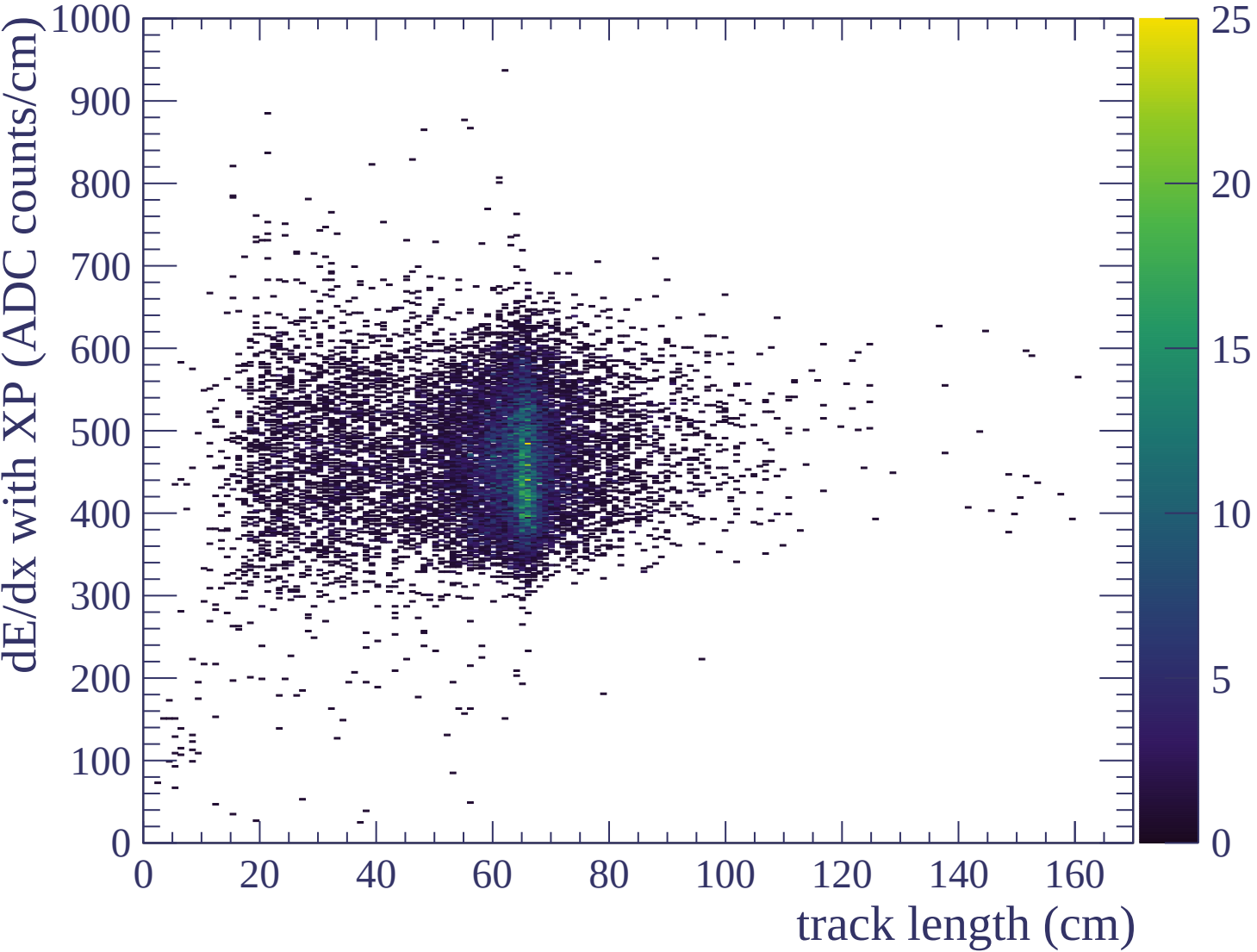


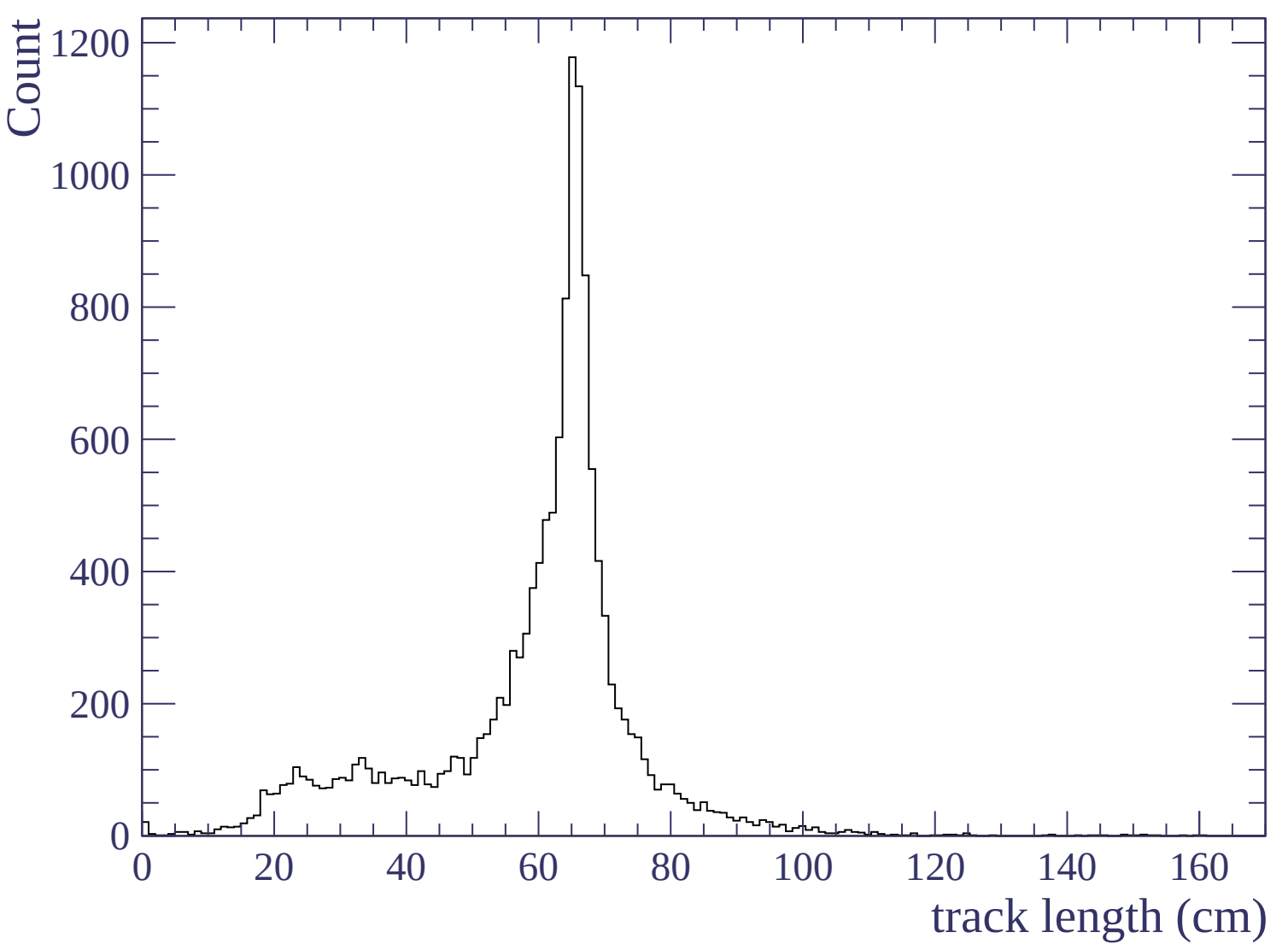


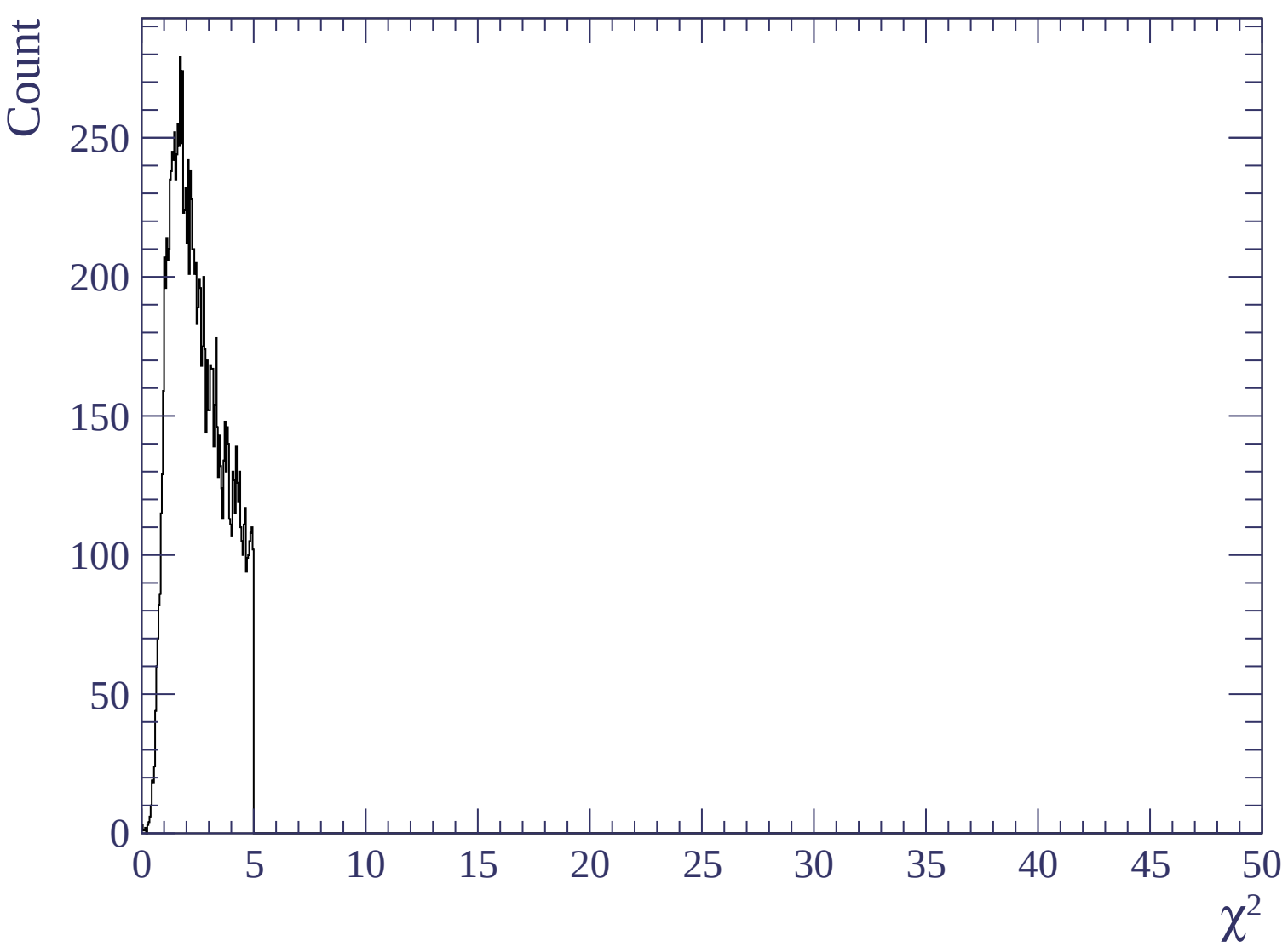


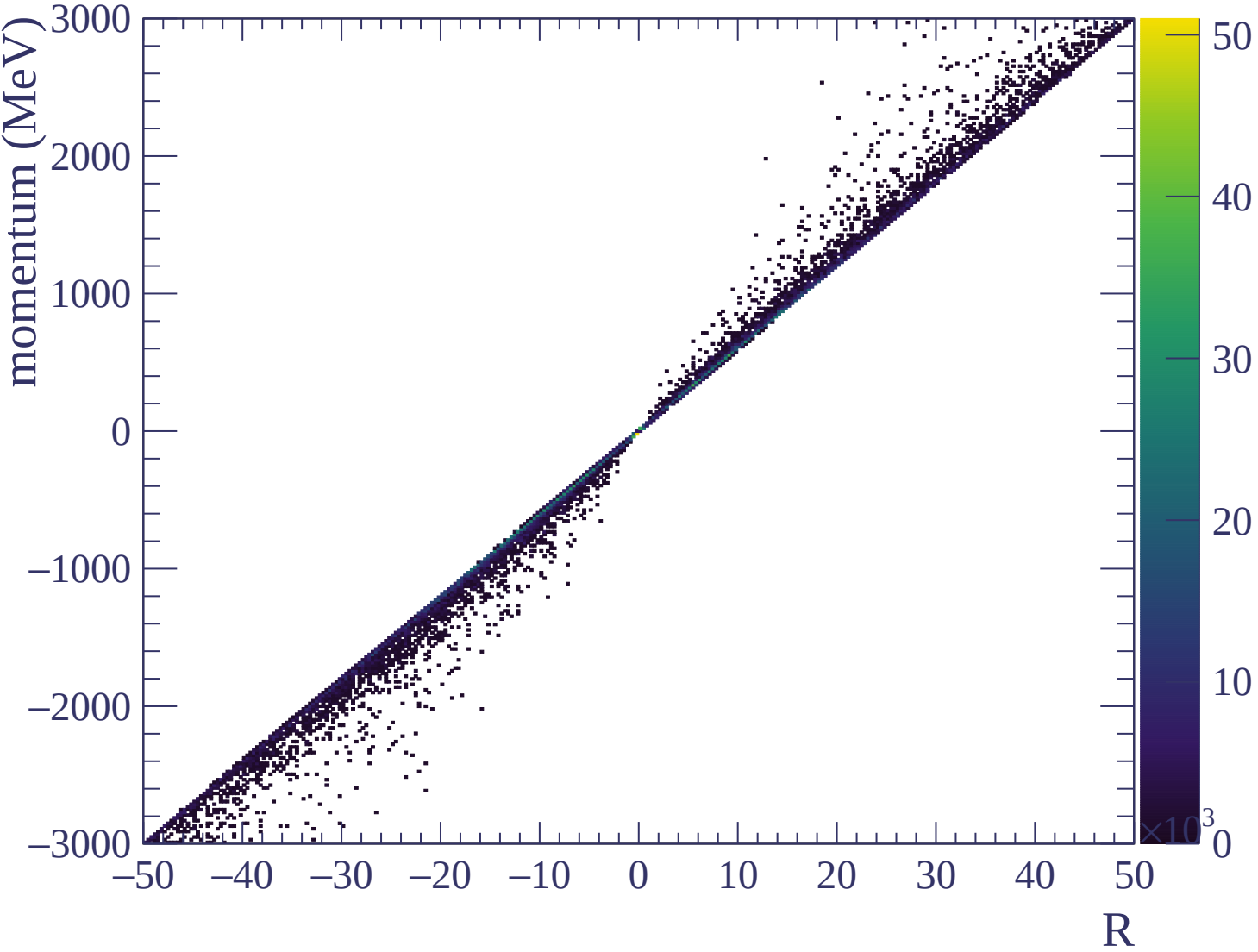


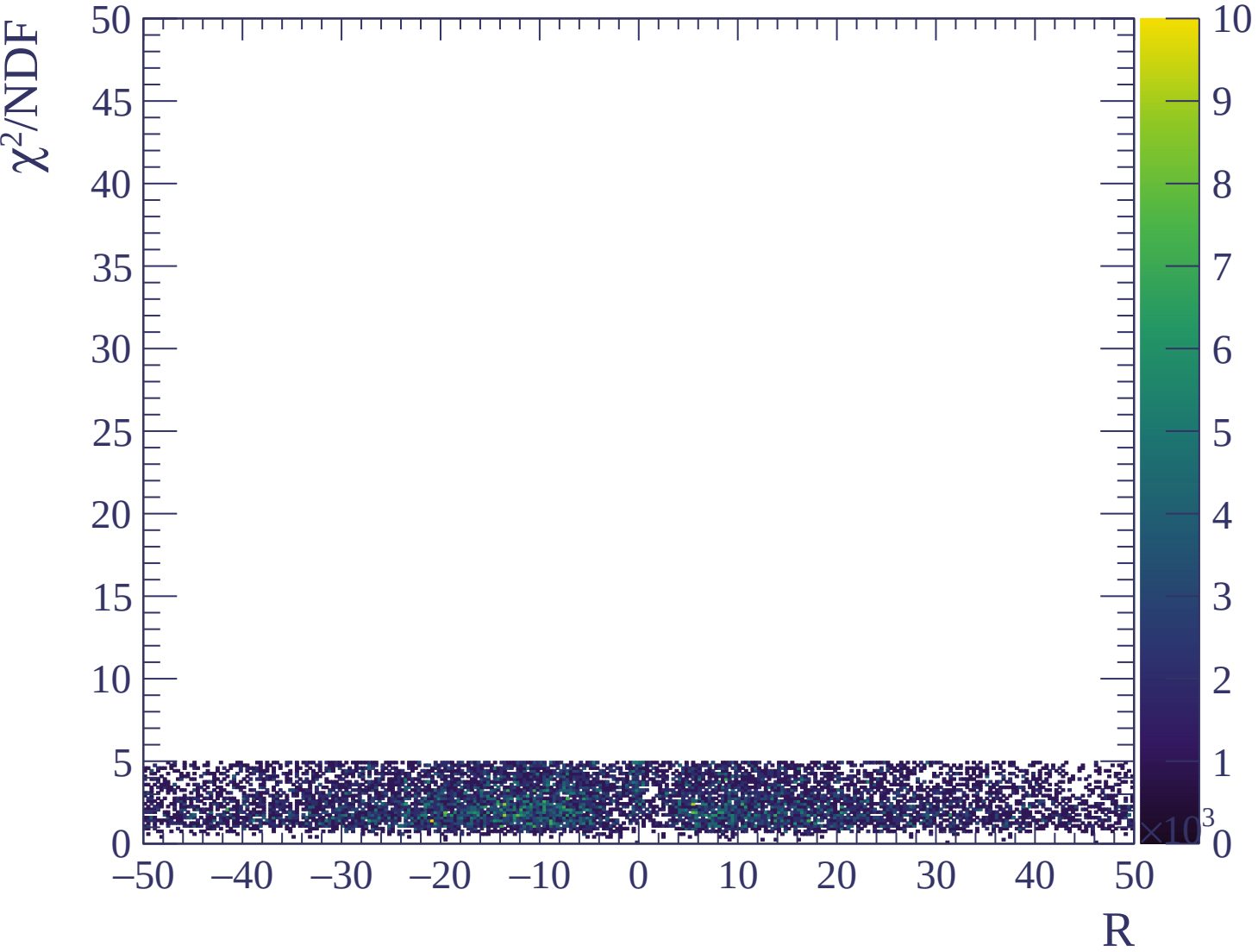






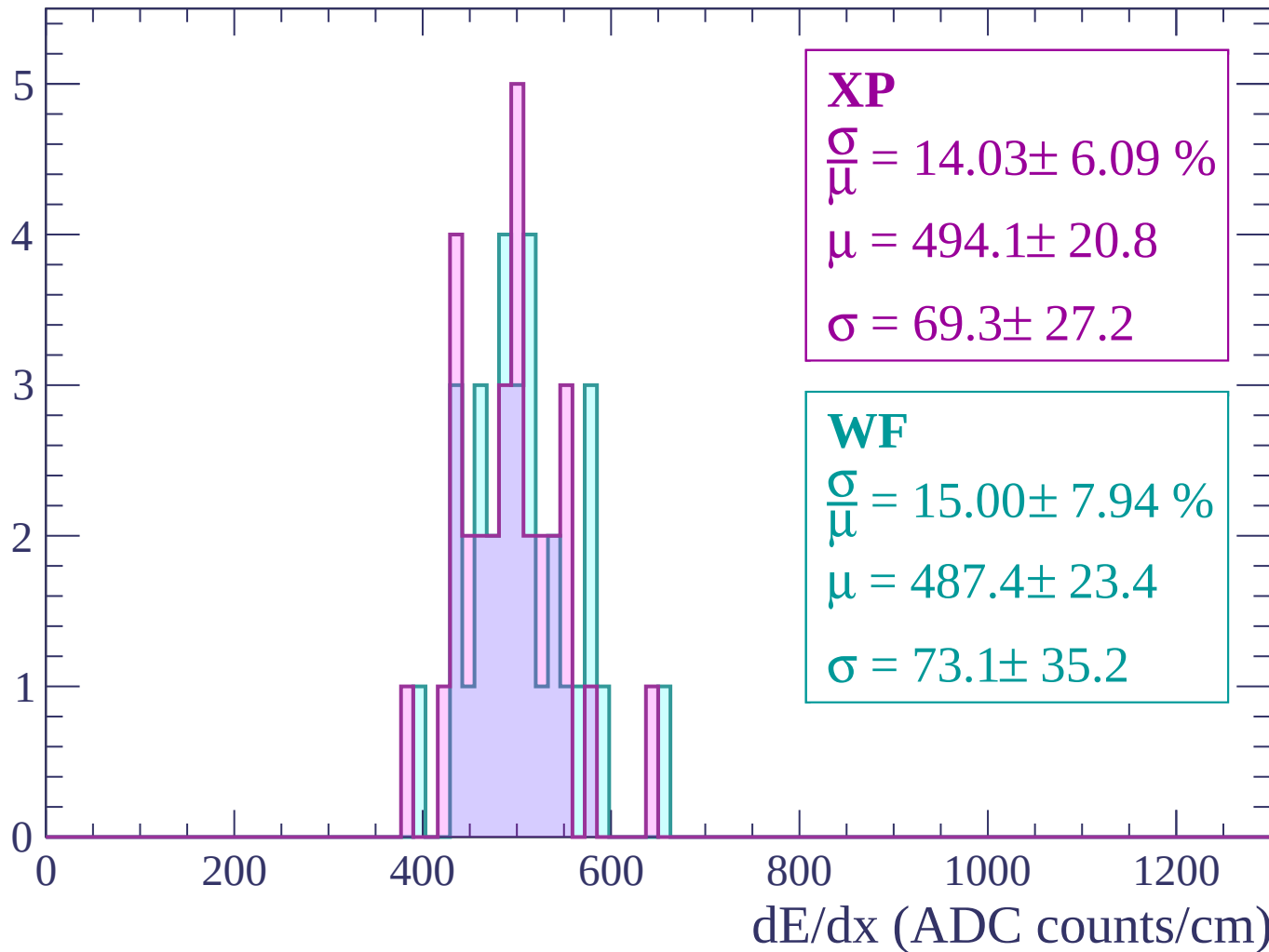






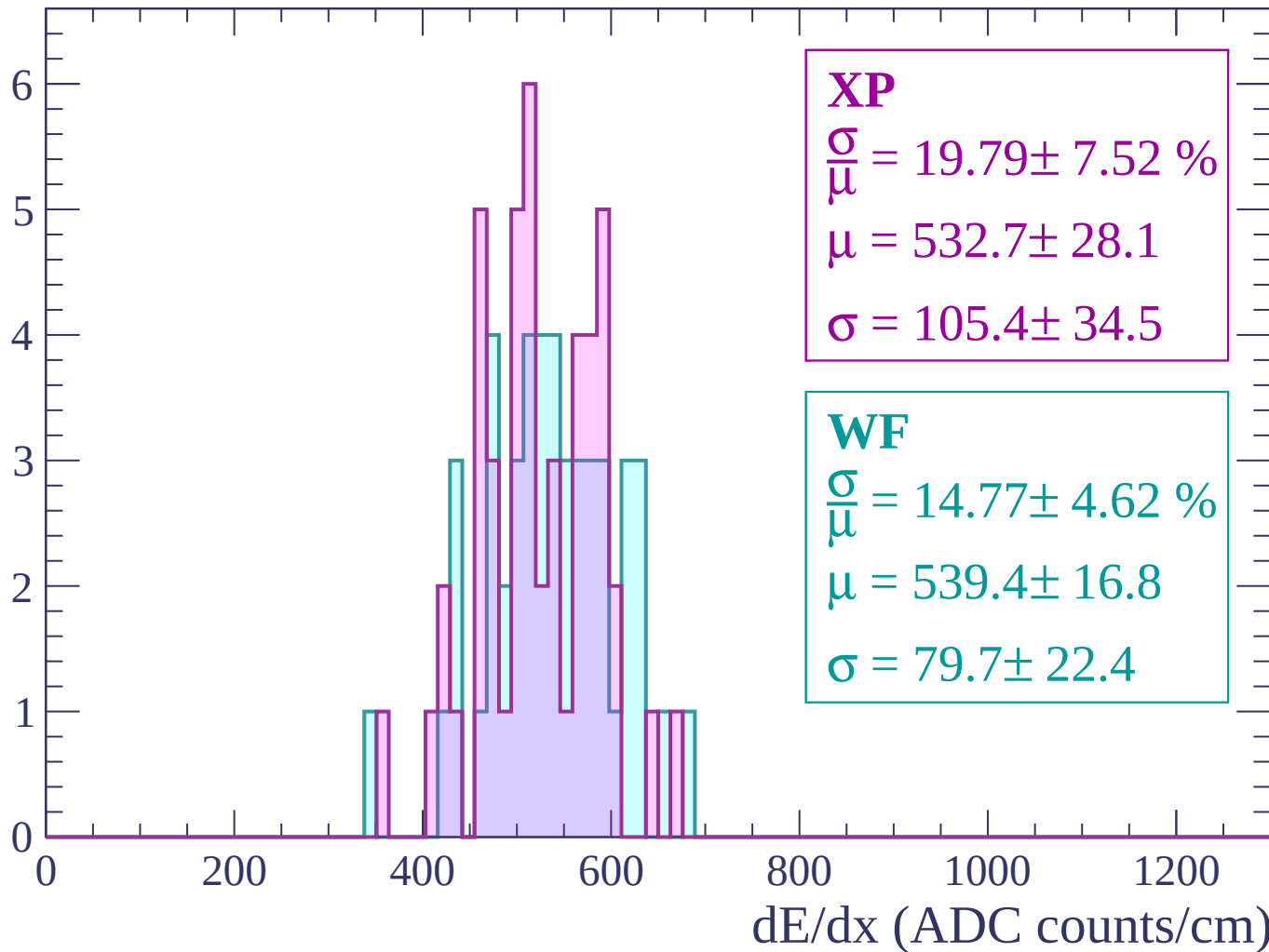
Energy loss | -3000 < p < -2940

Count



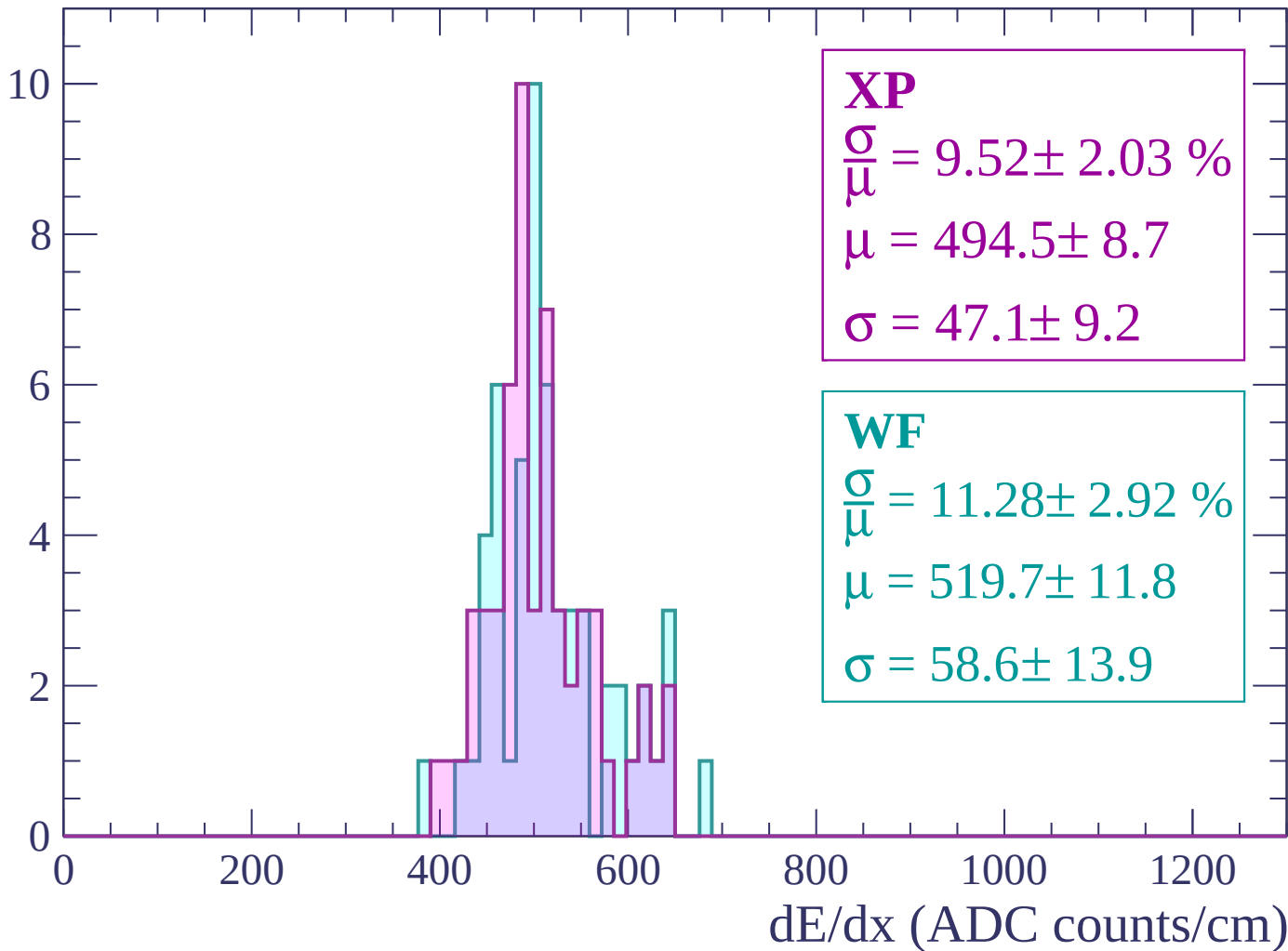
Energy loss | -2940 < p < -2880

Count



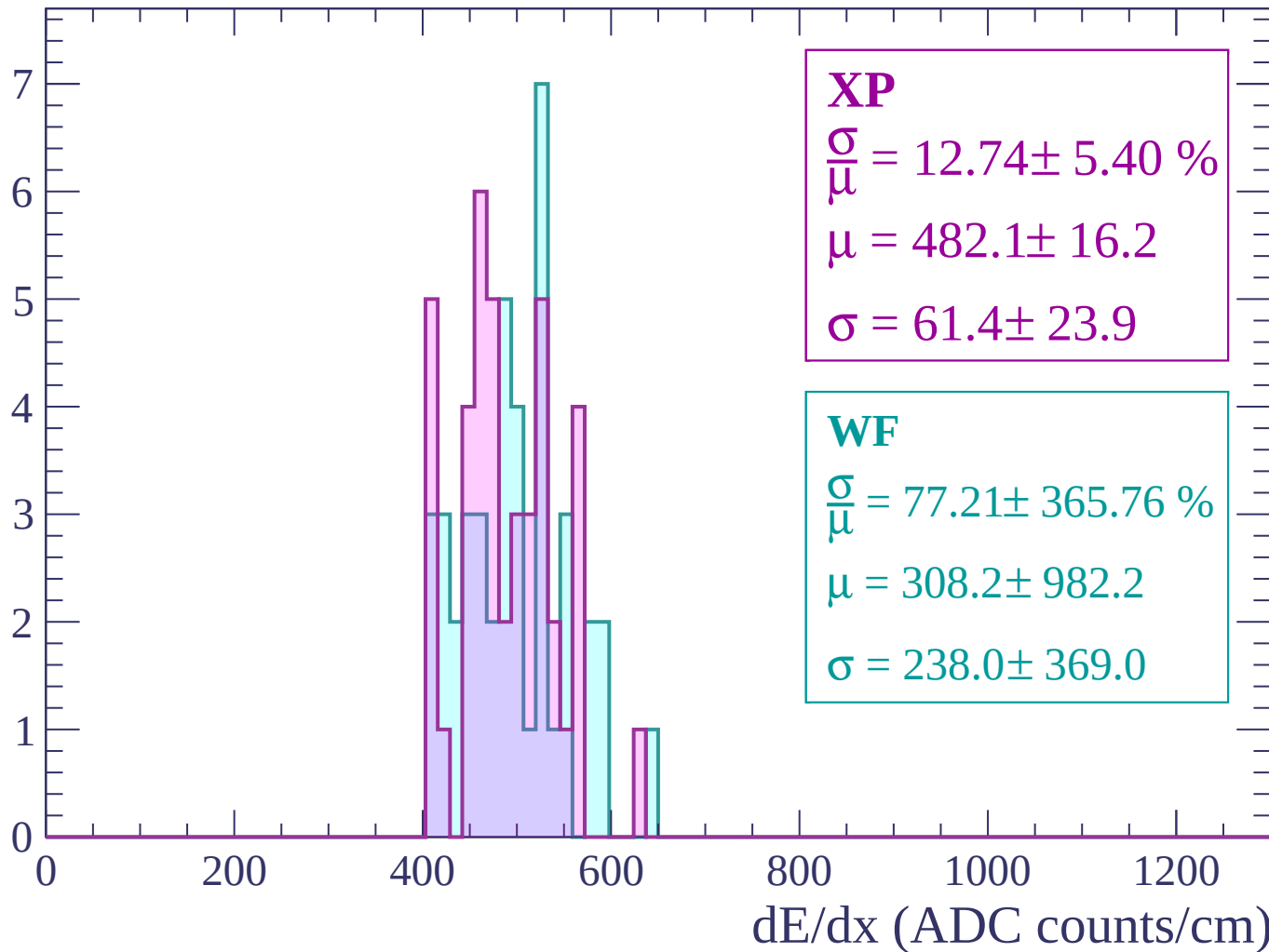
Energy loss | -2880 < p < -2820

Count



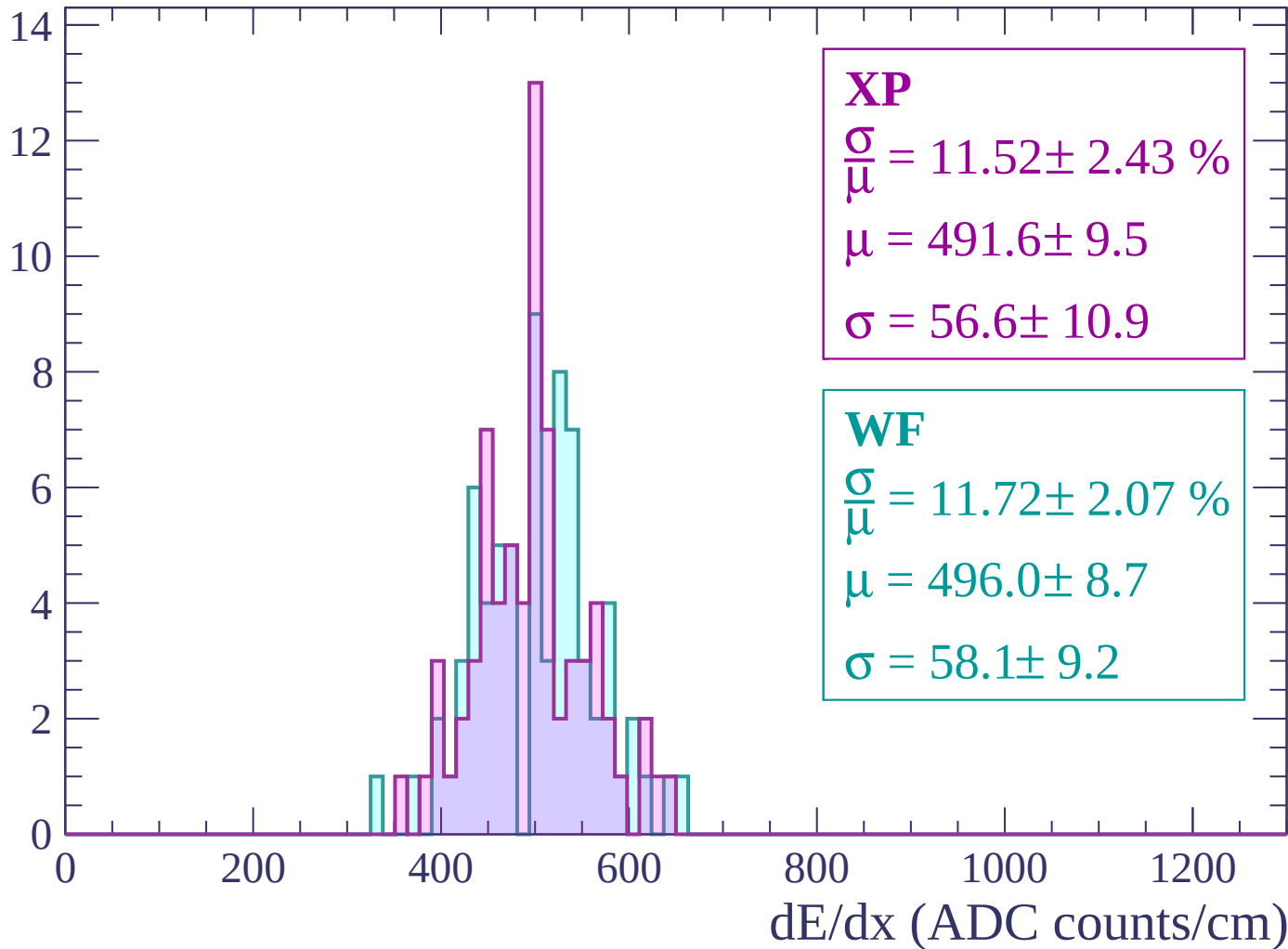
Energy loss | -2820 < p < -2760

Count



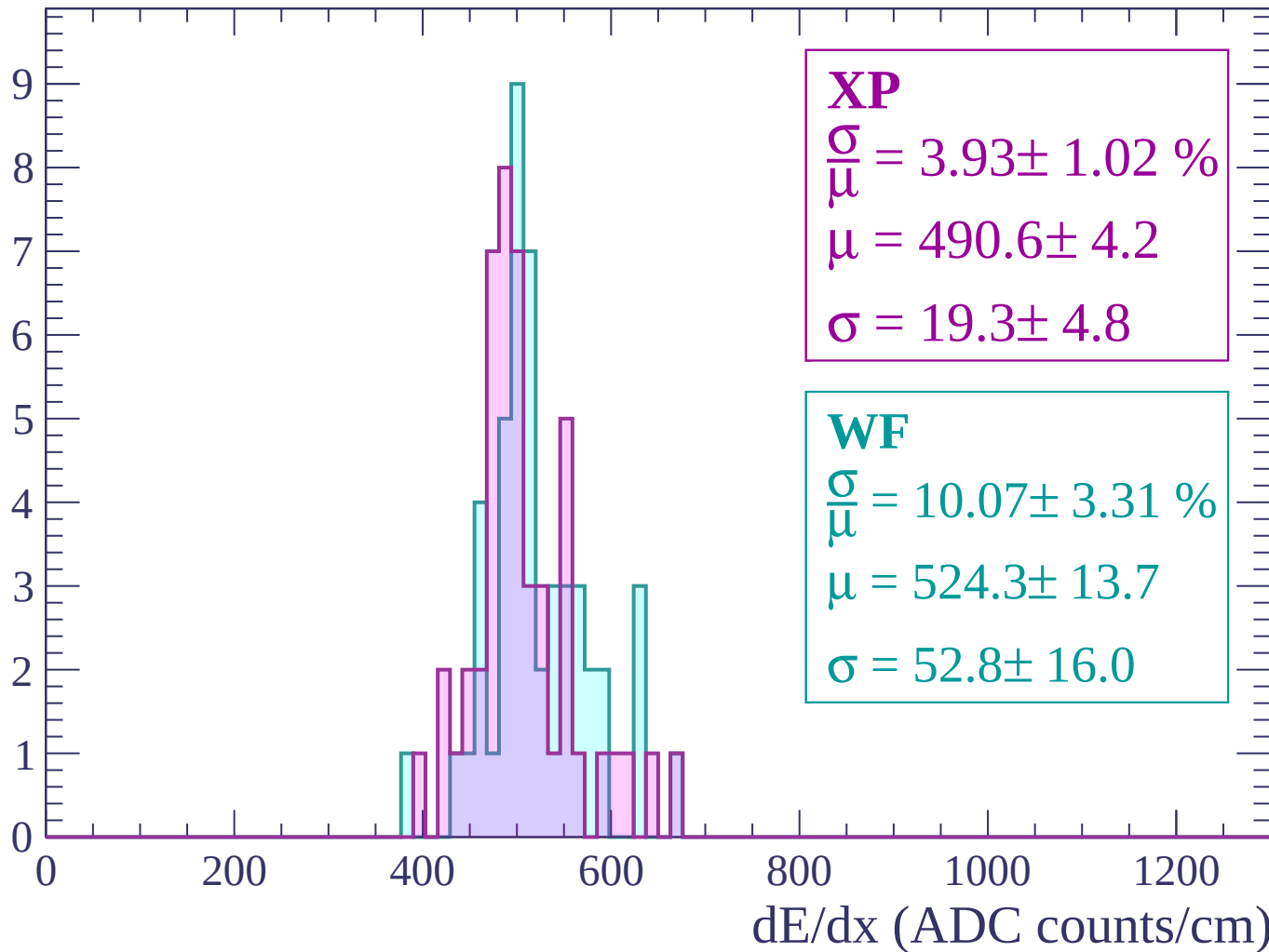
Energy loss | -2760 < p < -2700

Count



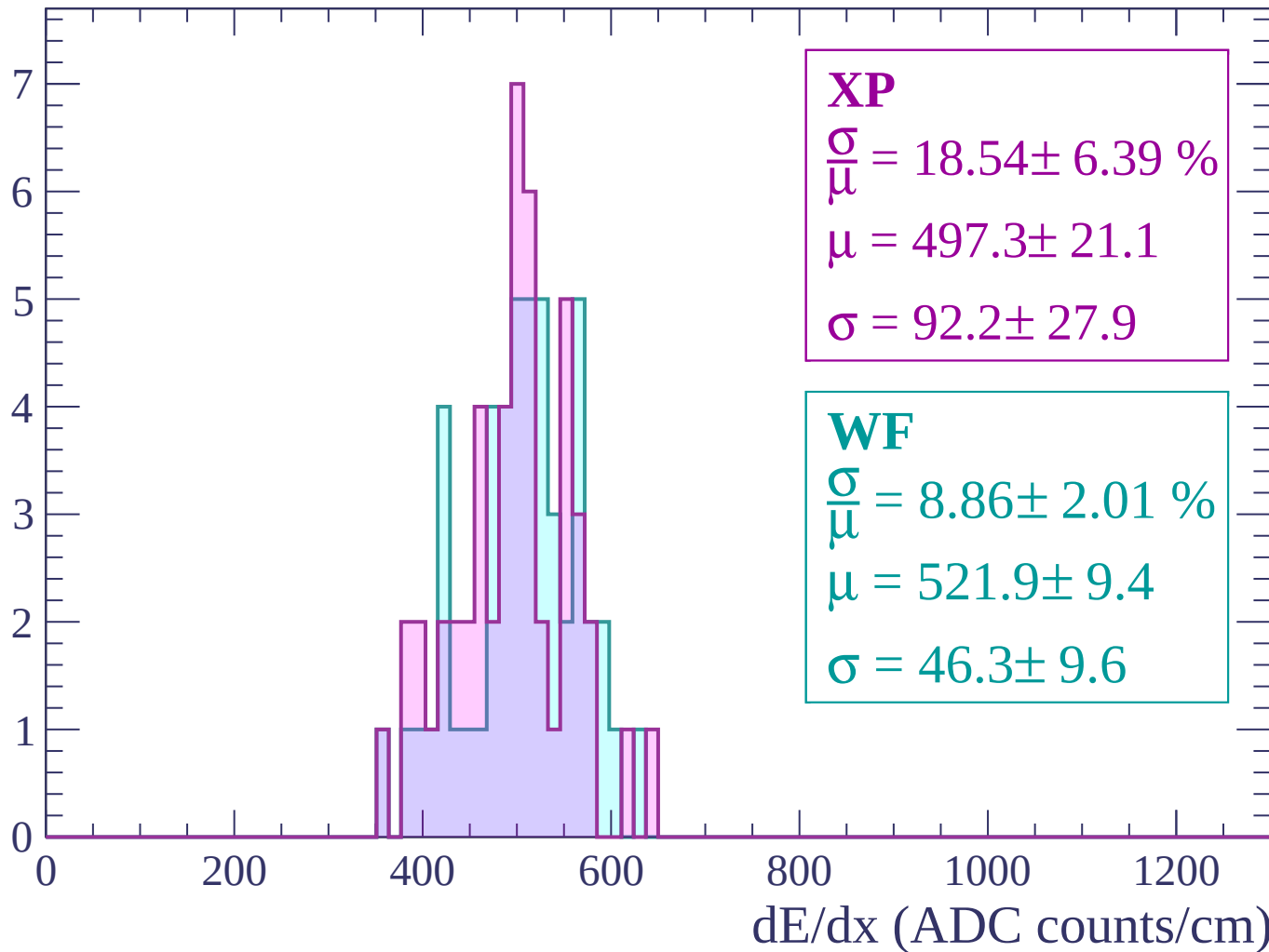
Energy loss | -2700 < p < -2640

Count



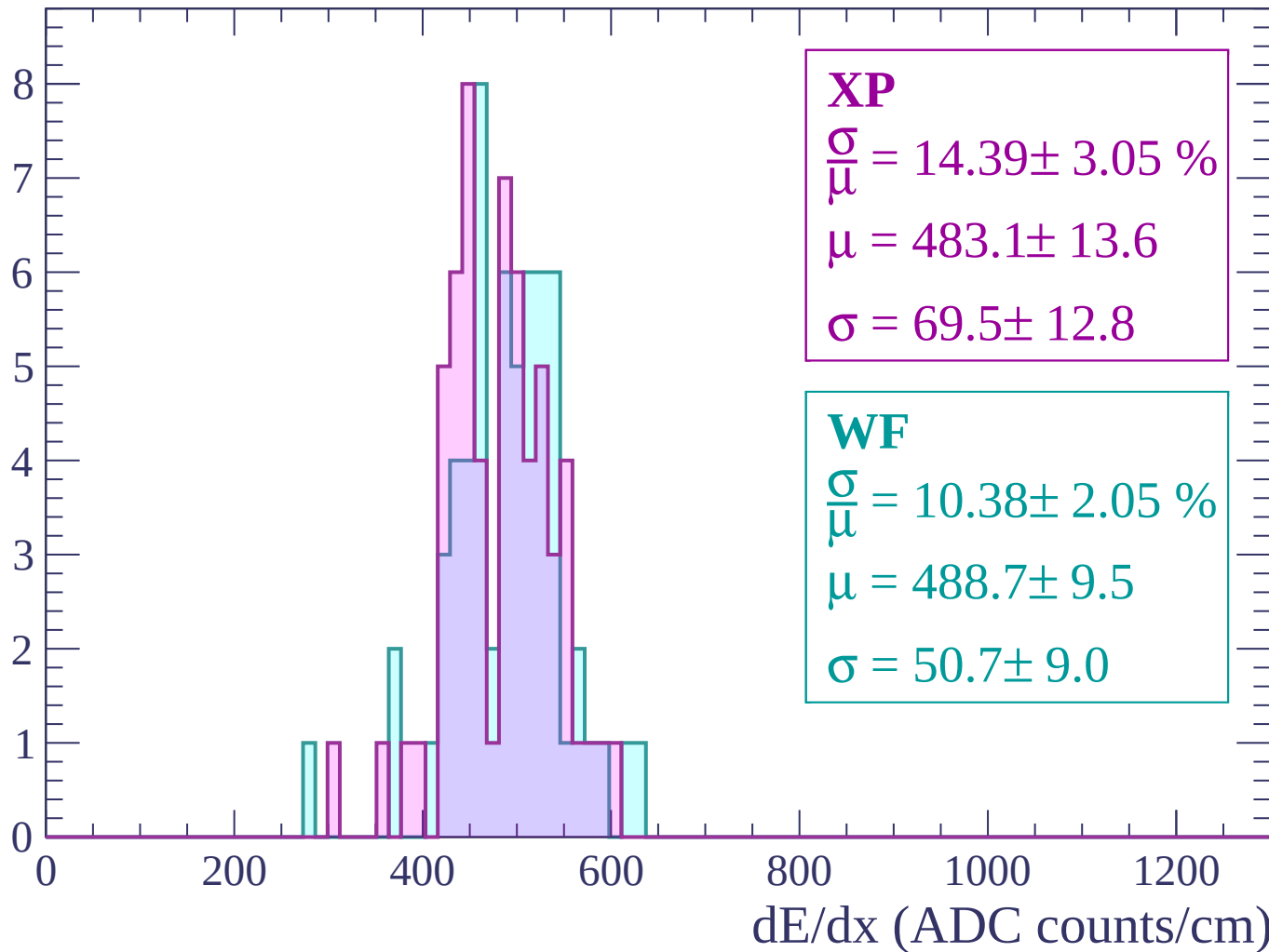
Energy loss | $-2640 < p < -2580$

Count



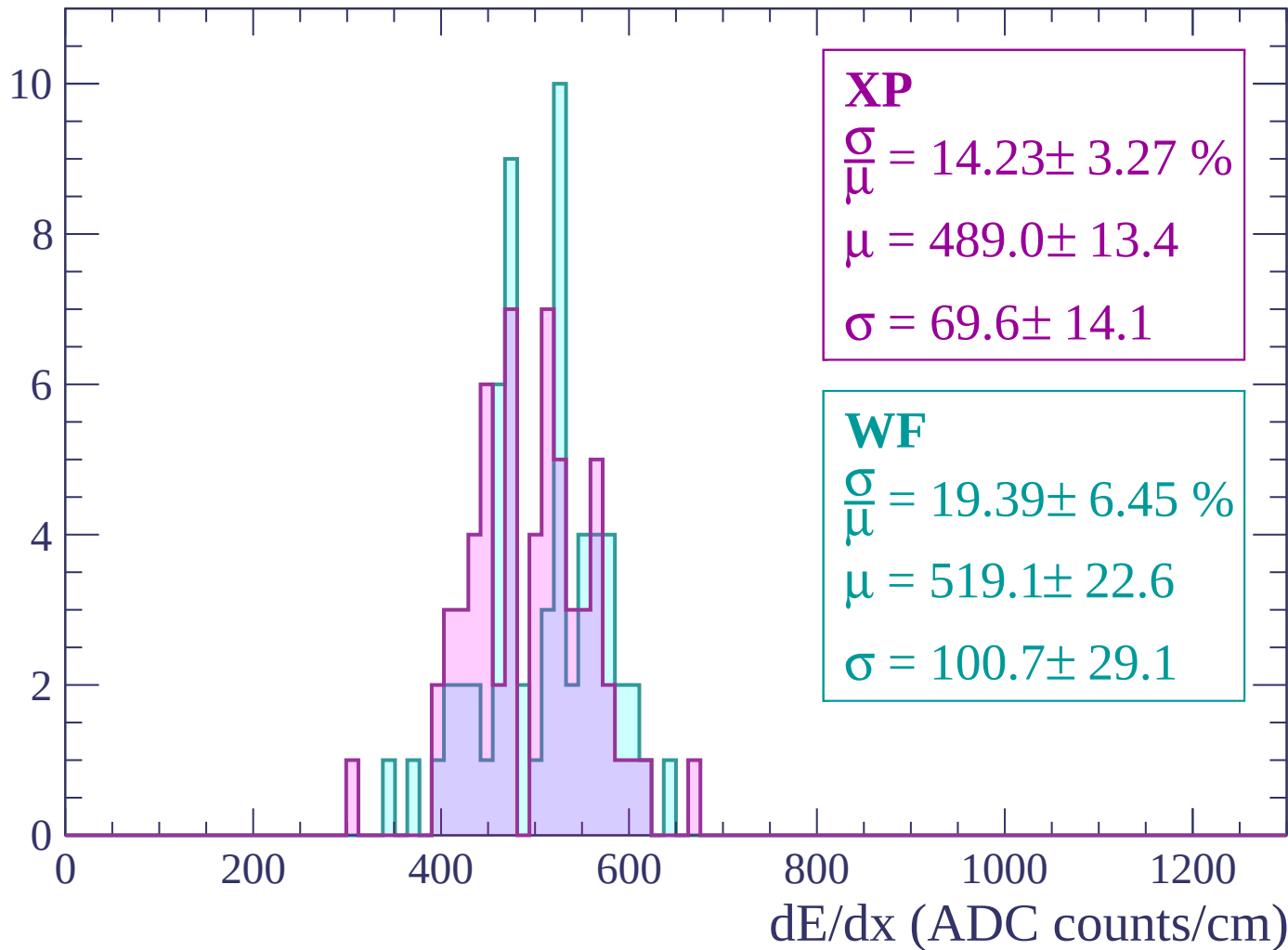
Energy loss | $-2580 < p < -2520$

Count



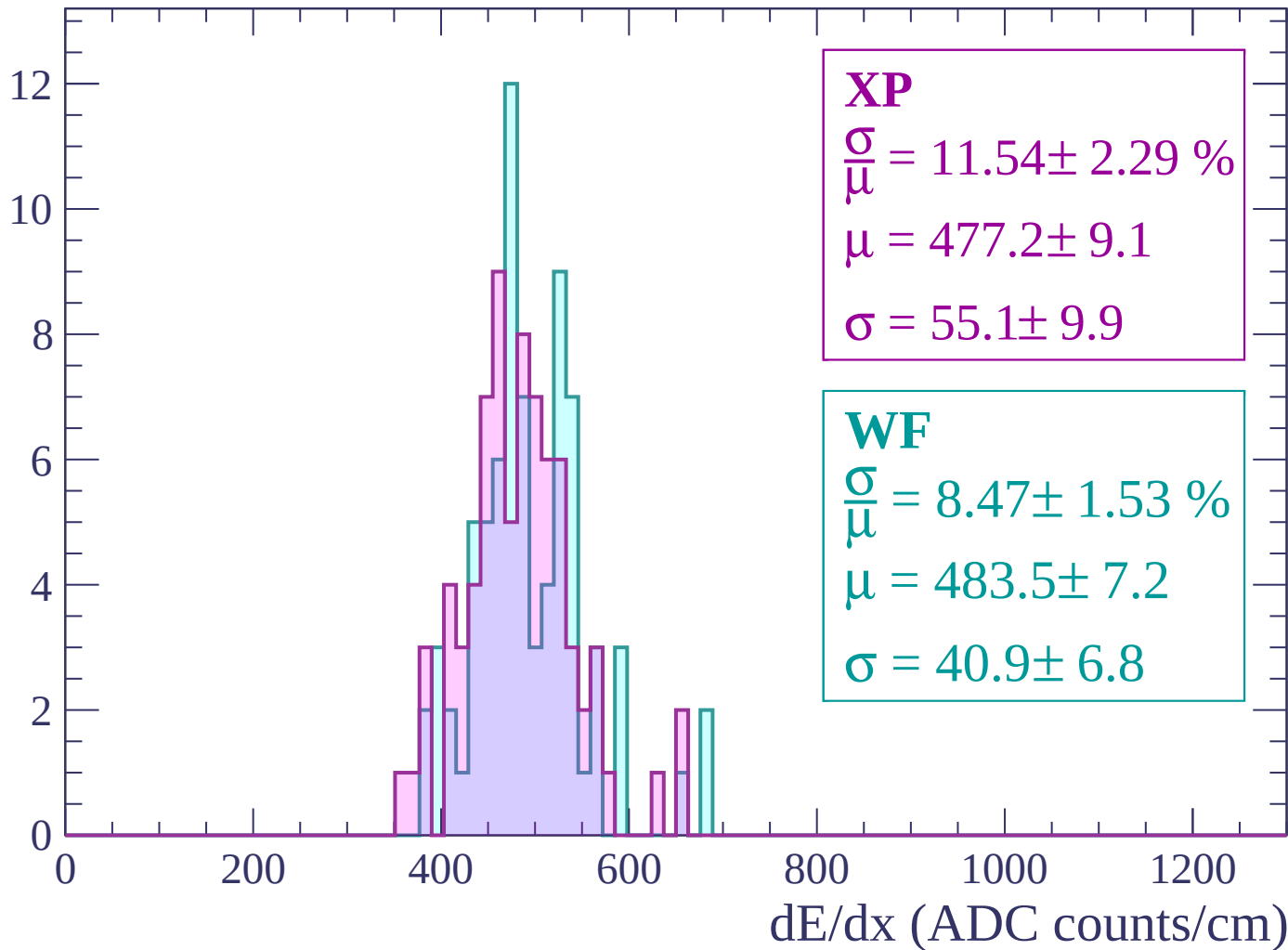
Energy loss | -2520 < p < -2460

Count



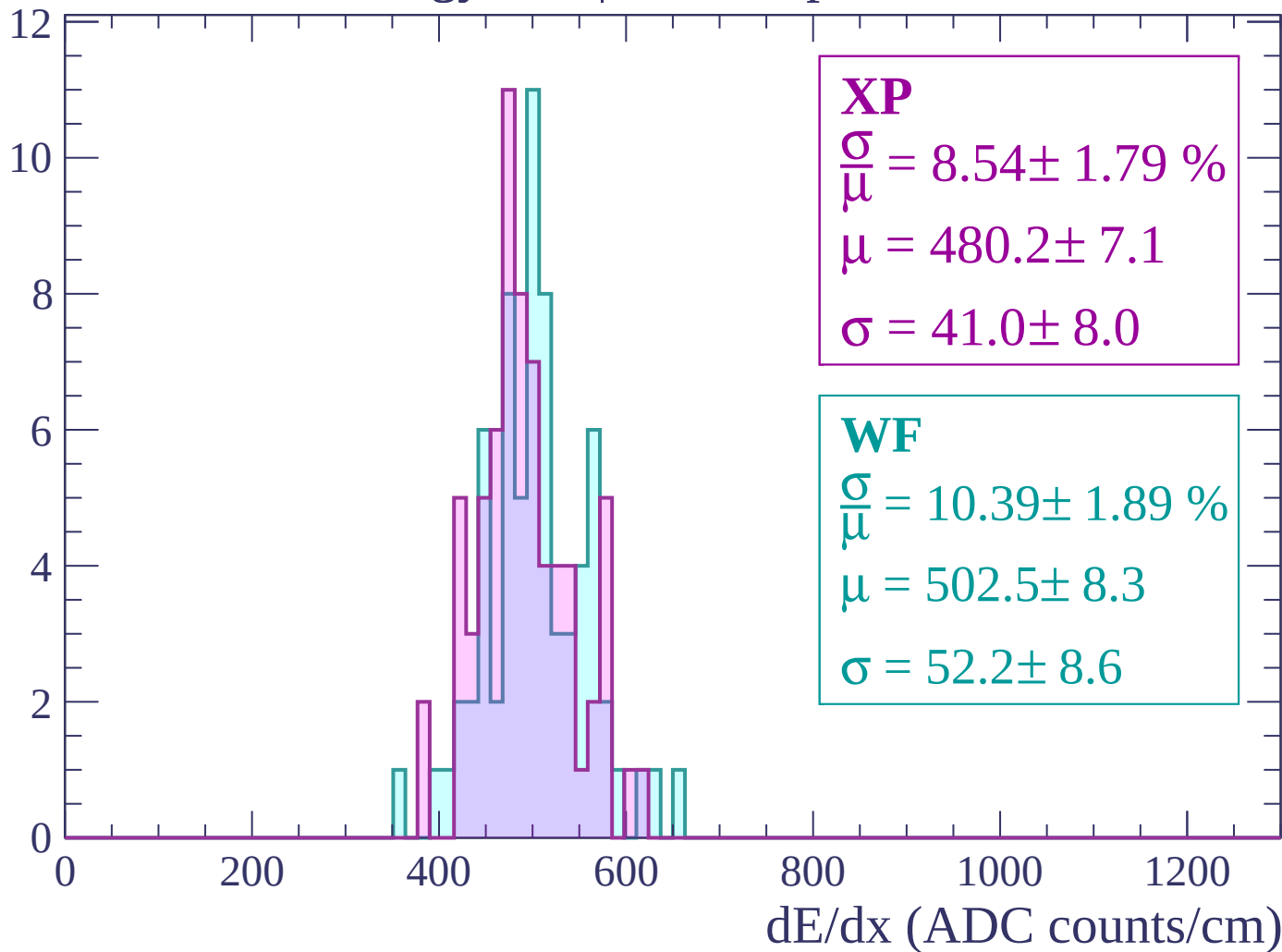
Energy loss | -2460 < p < -2400

Count



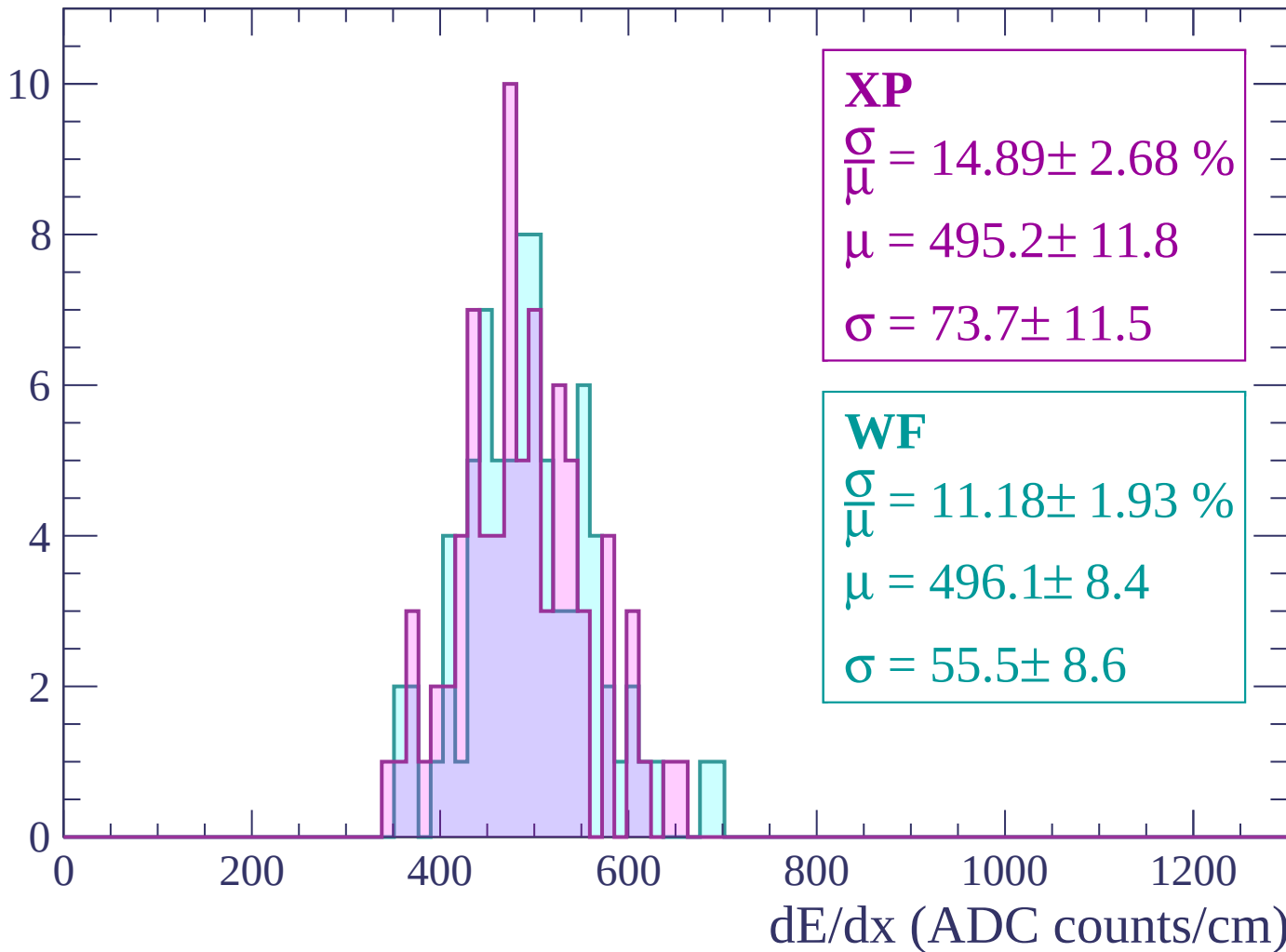
Energy loss | -2400 < p < -2340

Count



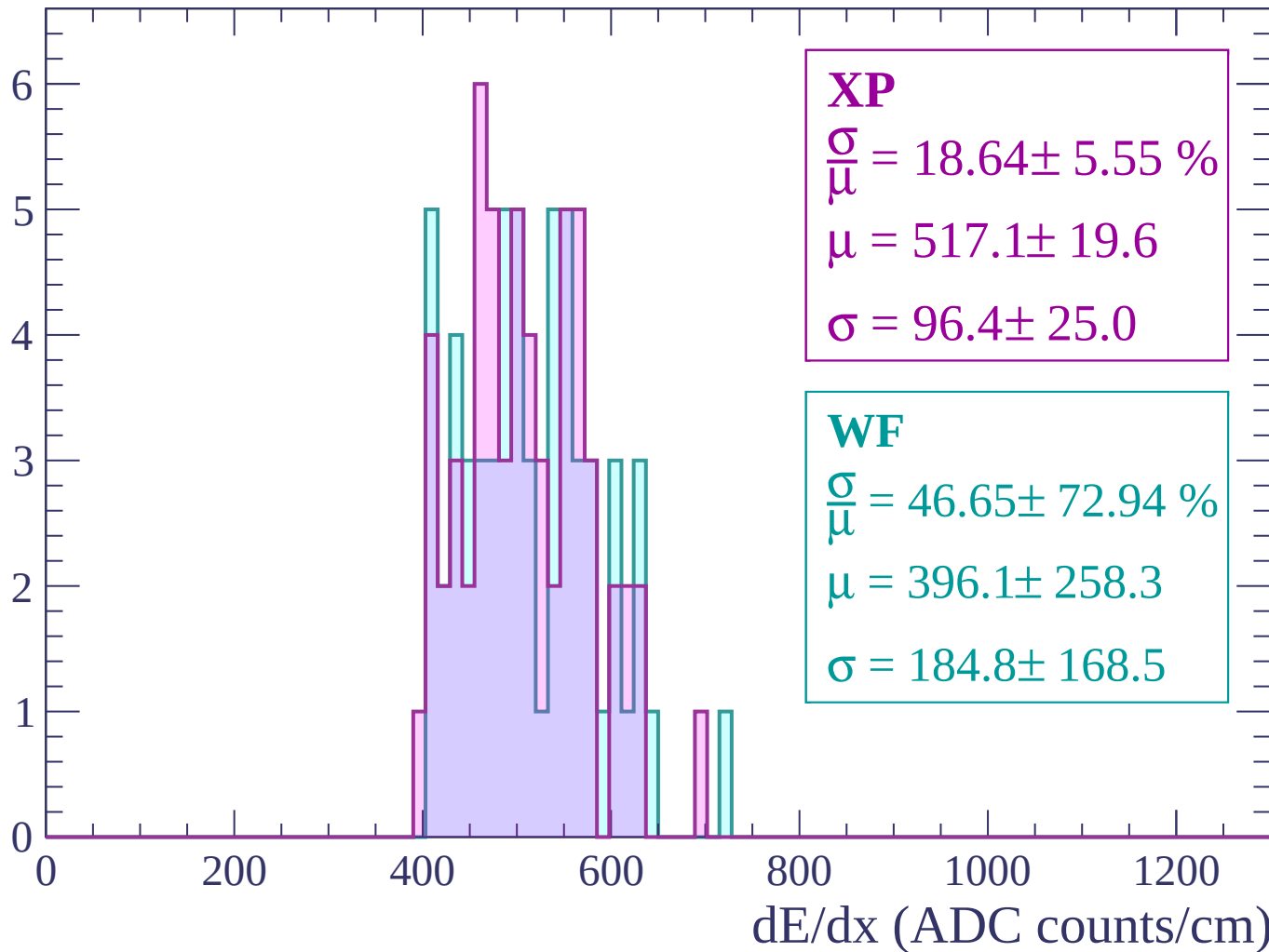
Energy loss | $-2340 < p < -2280$

Count



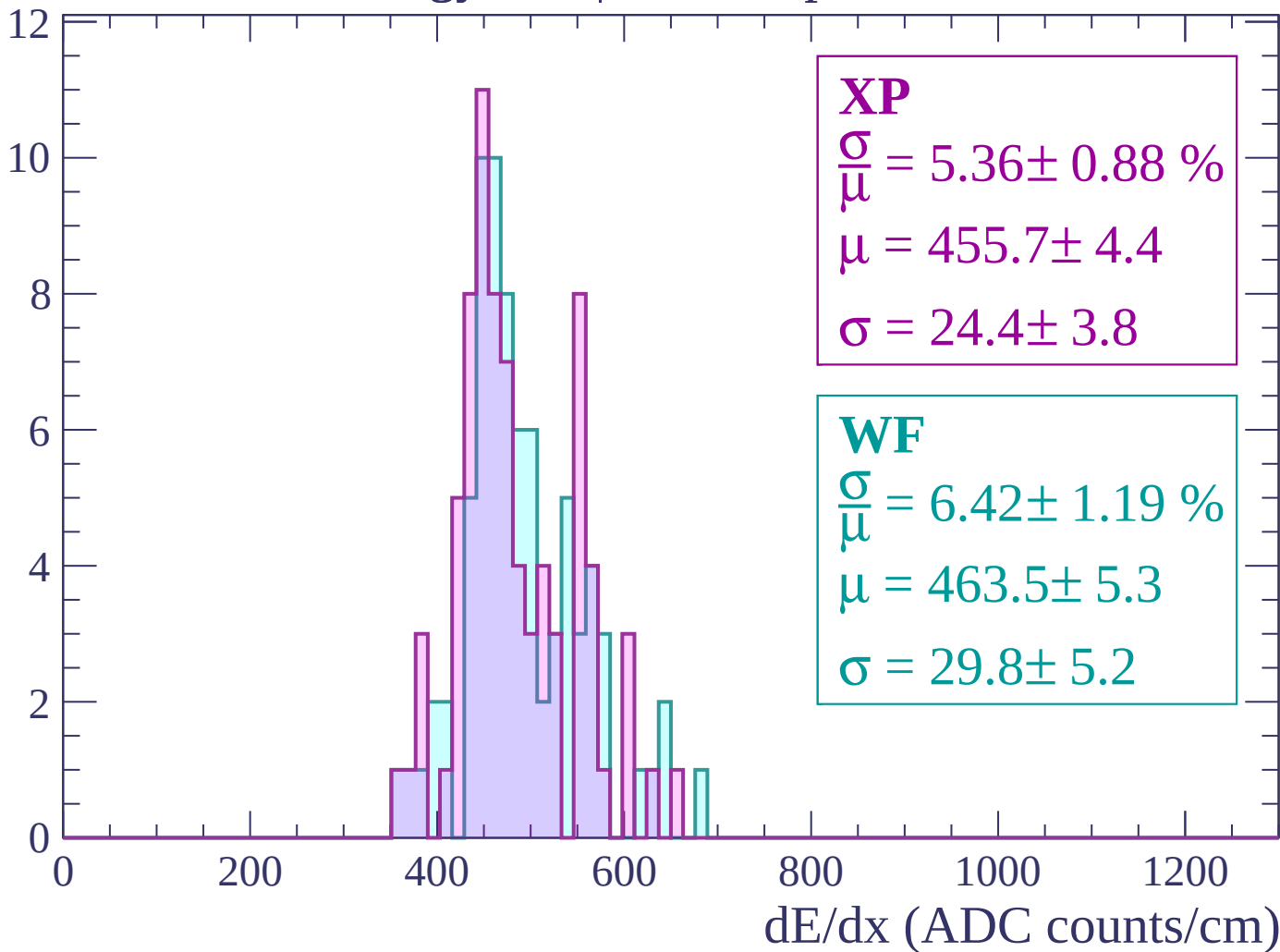
Energy loss | -2280 < p < -2220

Count



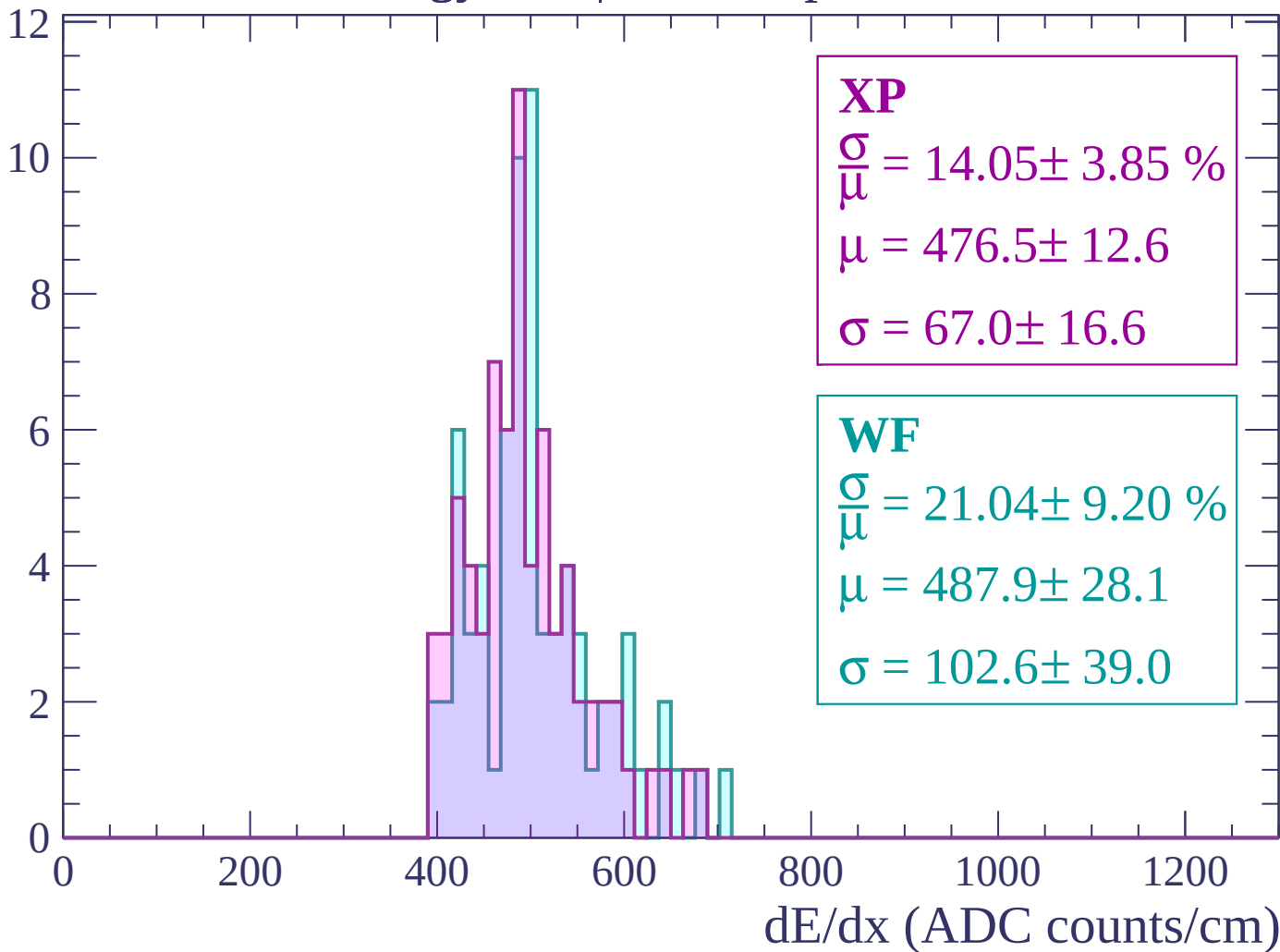
Energy loss | -2220 < p < -2160

Count



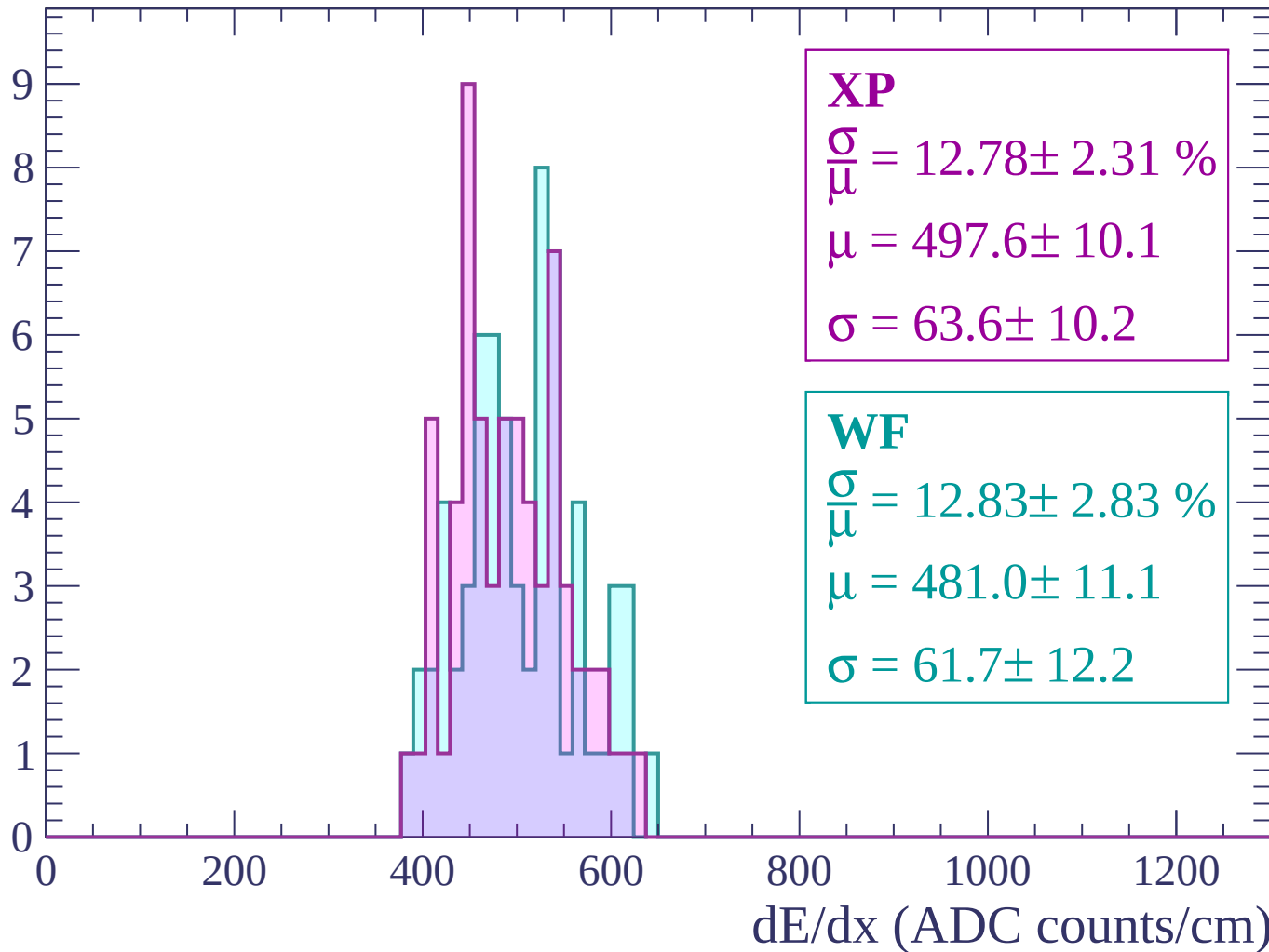
Energy loss | -2160 < p < -2100

Count



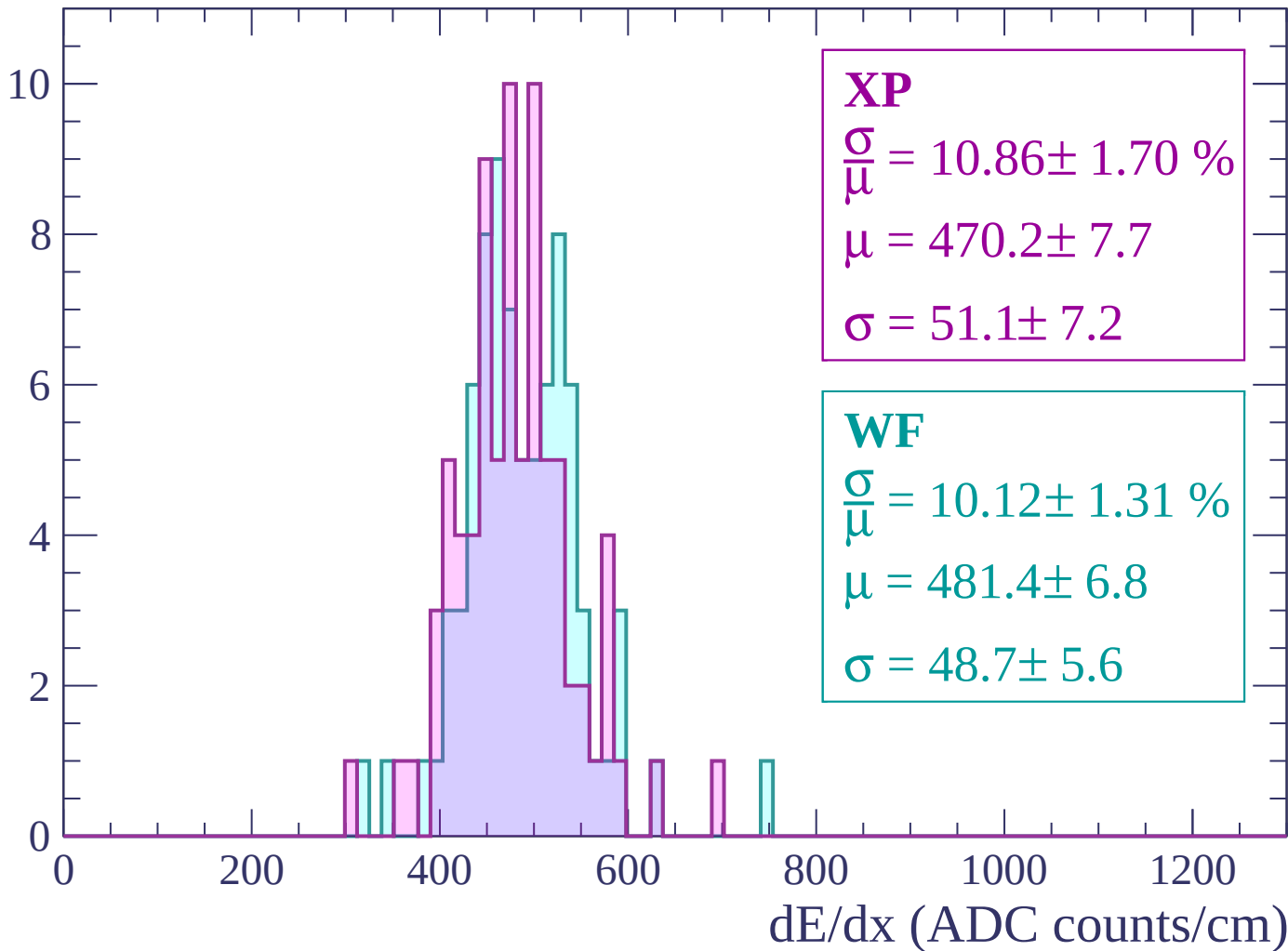
Energy loss | -2100 < p < -2040

Count



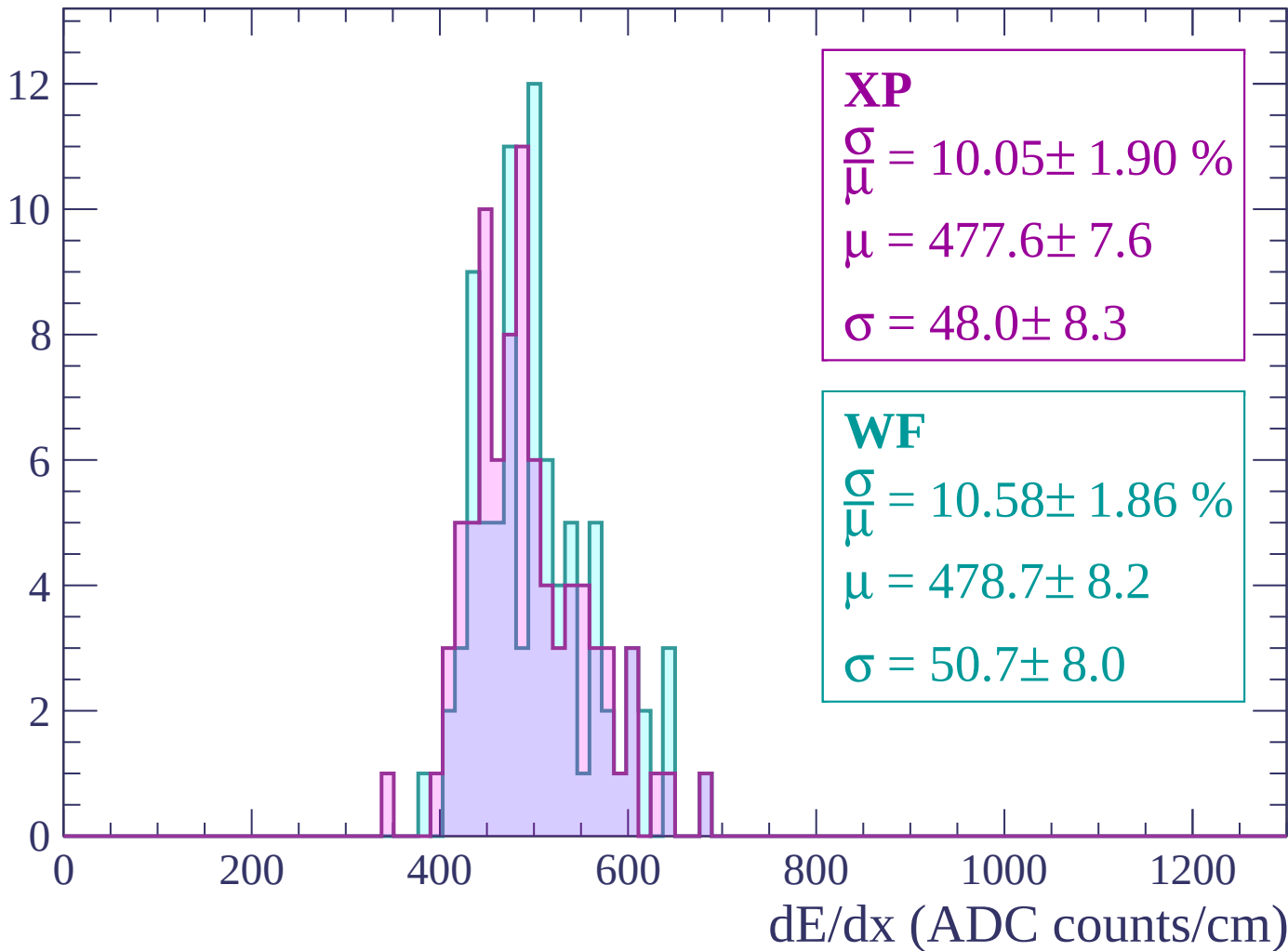
Energy loss | -2040 < p < -1980

Count



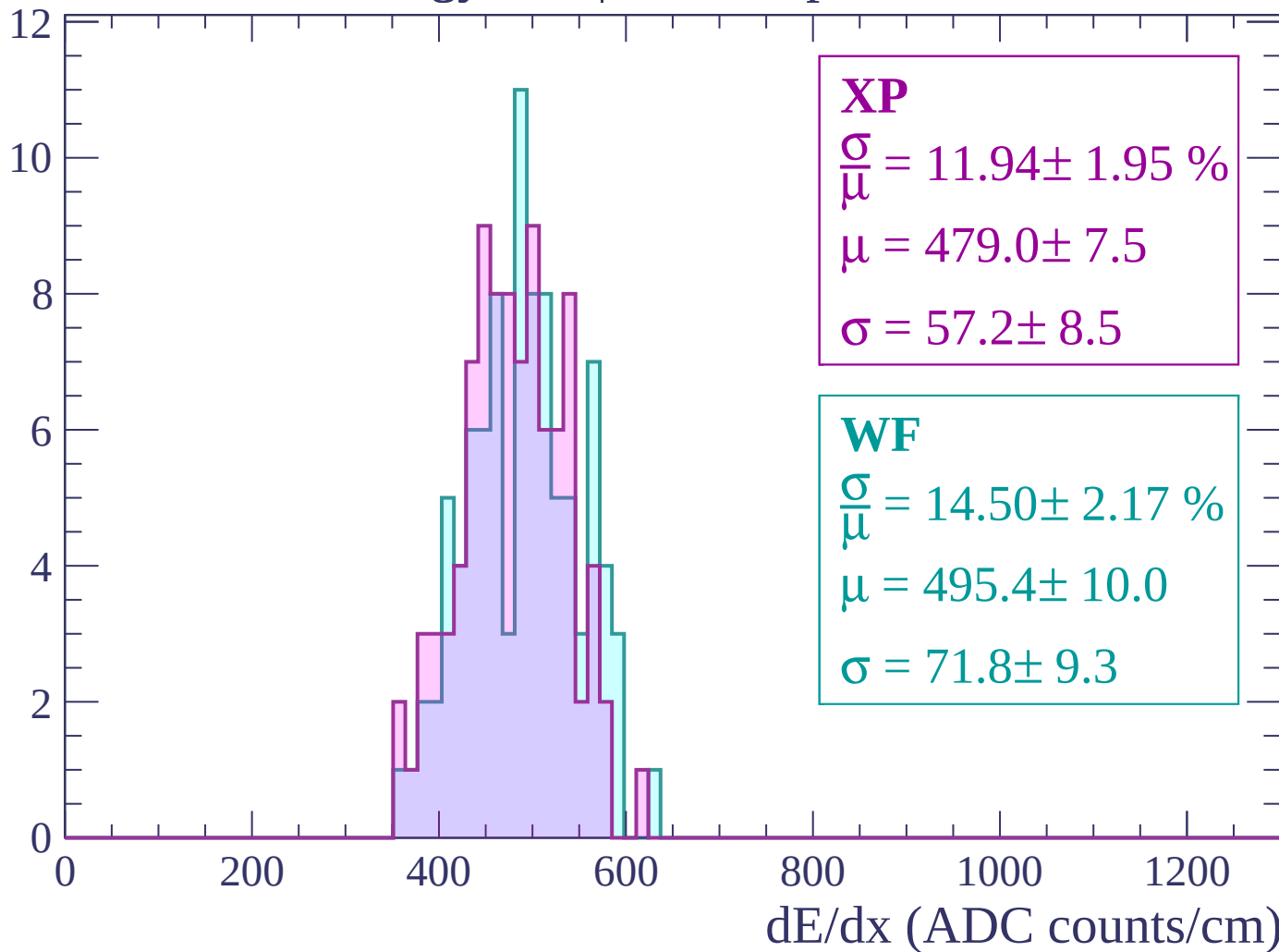
Energy loss | -1980 < p < -1920

Count



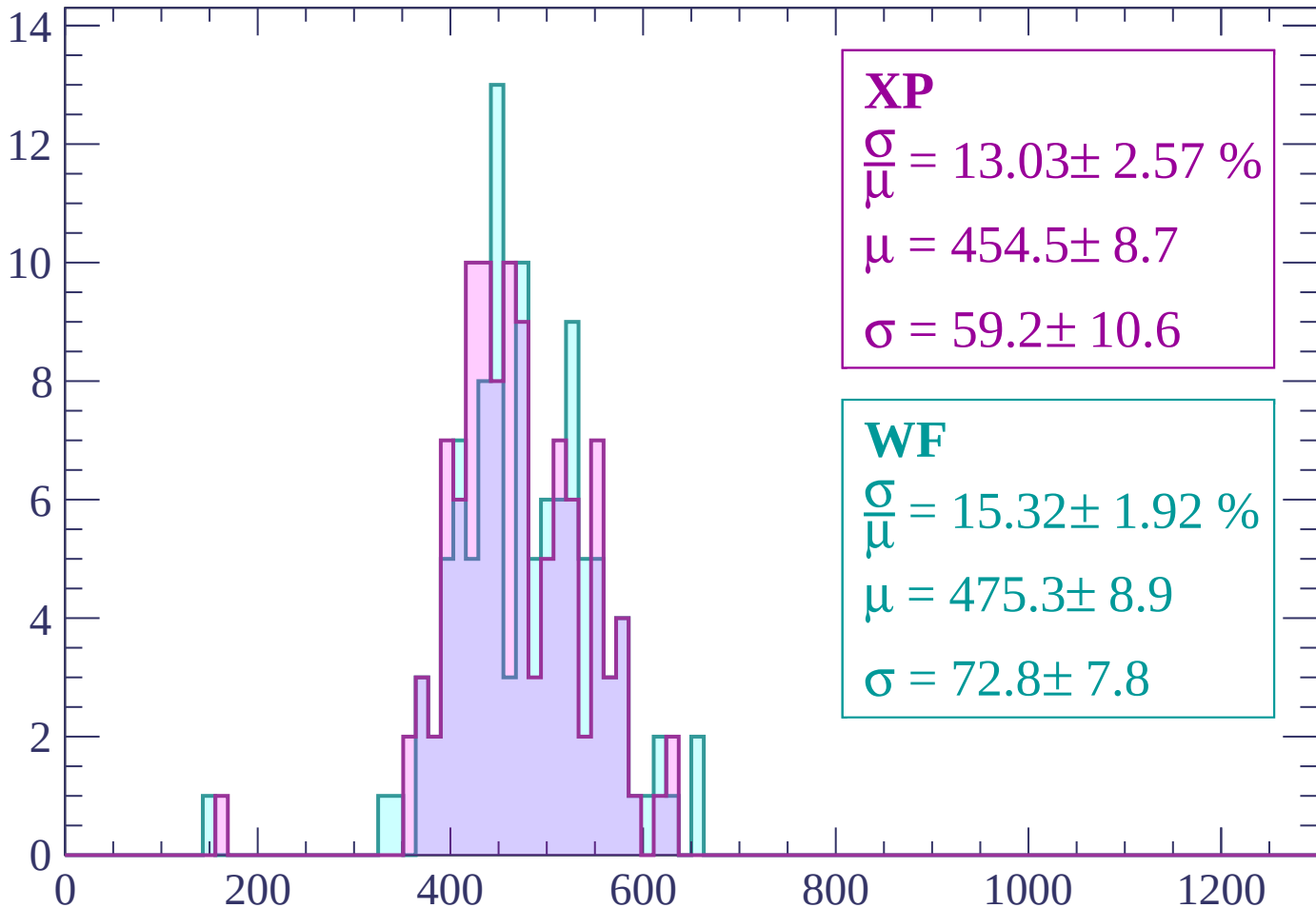
Energy loss | -1920 < p < -1860

Count



Energy loss | -1860 < p < -1800

Count



XP

$$\frac{\sigma}{\mu} = 13.03 \pm 2.57 \%$$

$$\mu = 454.5 \pm 8.7$$

$$\sigma = 59.2 \pm 10.6$$

WF

$$\frac{\sigma}{\mu} = 15.32 \pm 1.92 \%$$

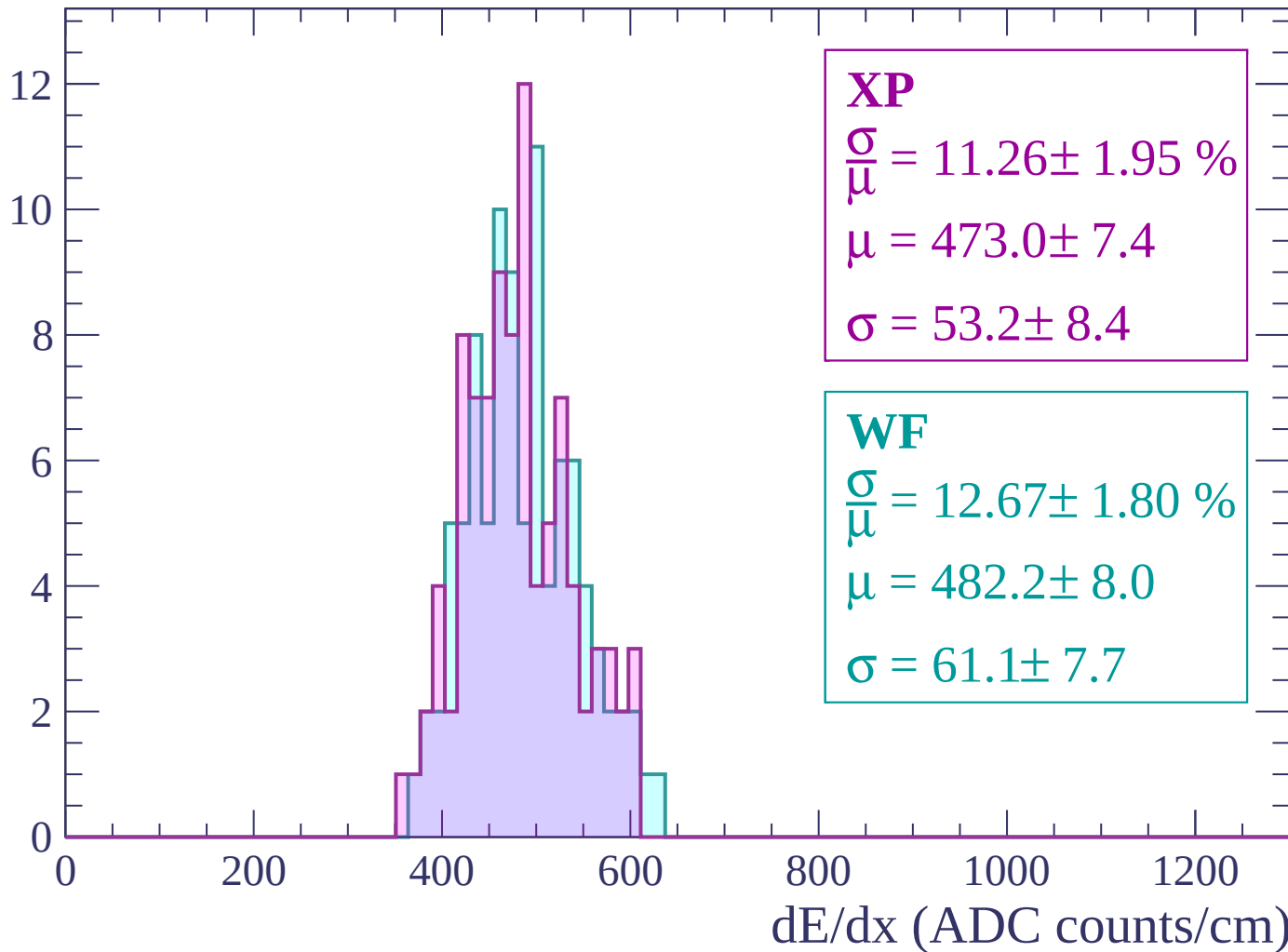
$$\mu = 475.3 \pm 8.9$$

$$\sigma = 72.8 \pm 7.8$$

dE/dx (ADC counts/cm)

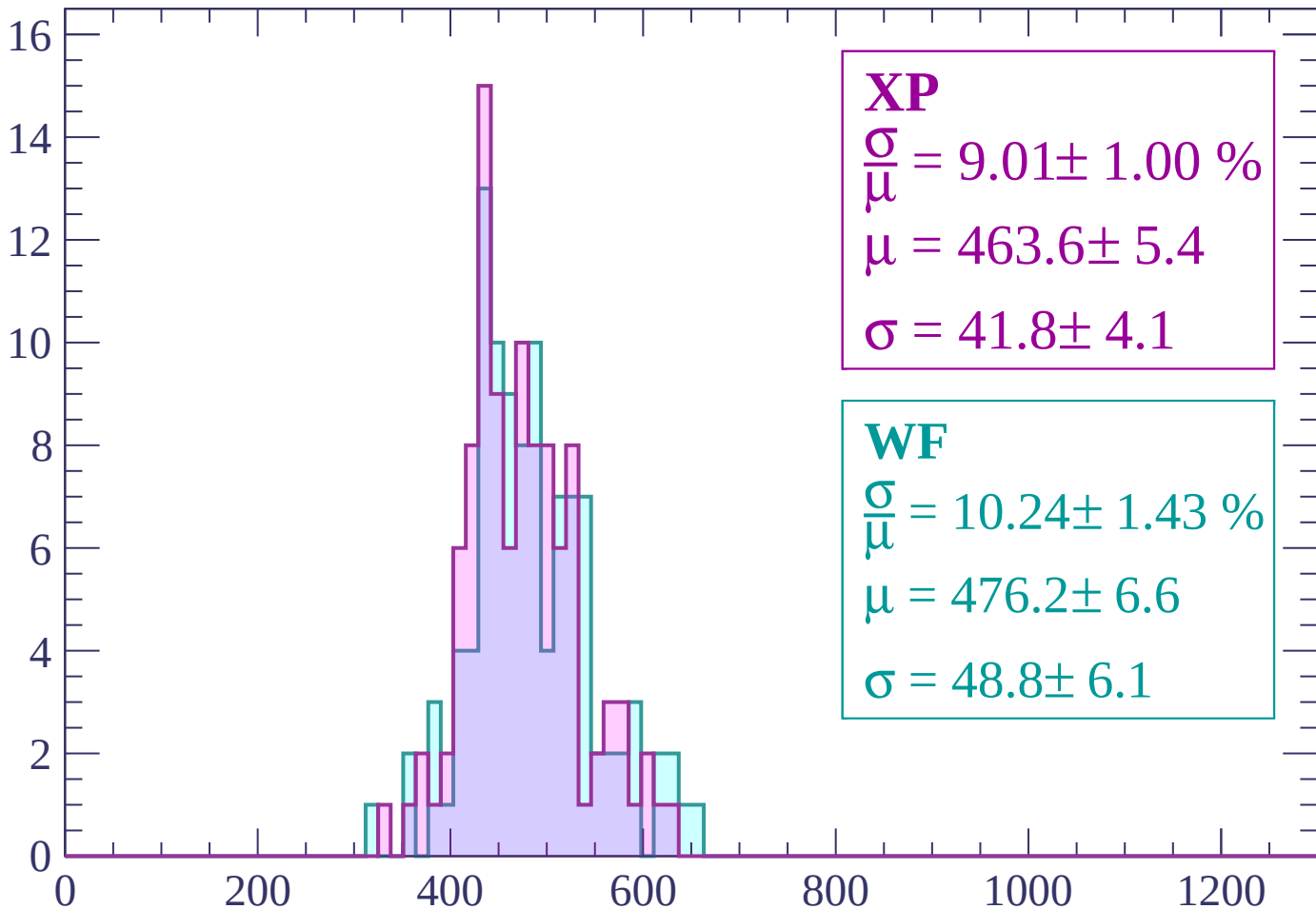
Energy loss | -1800 < p < -1740

Count



Energy loss | $-1740 < p < -1680$

Count



XP

$$\frac{\sigma}{\mu} = 9.01 \pm 1.00 \%$$

$$\mu = 463.6 \pm 5.4$$

$$\sigma = 41.8 \pm 4.1$$

WF

$$\frac{\sigma}{\mu} = 10.24 \pm 1.43 \%$$

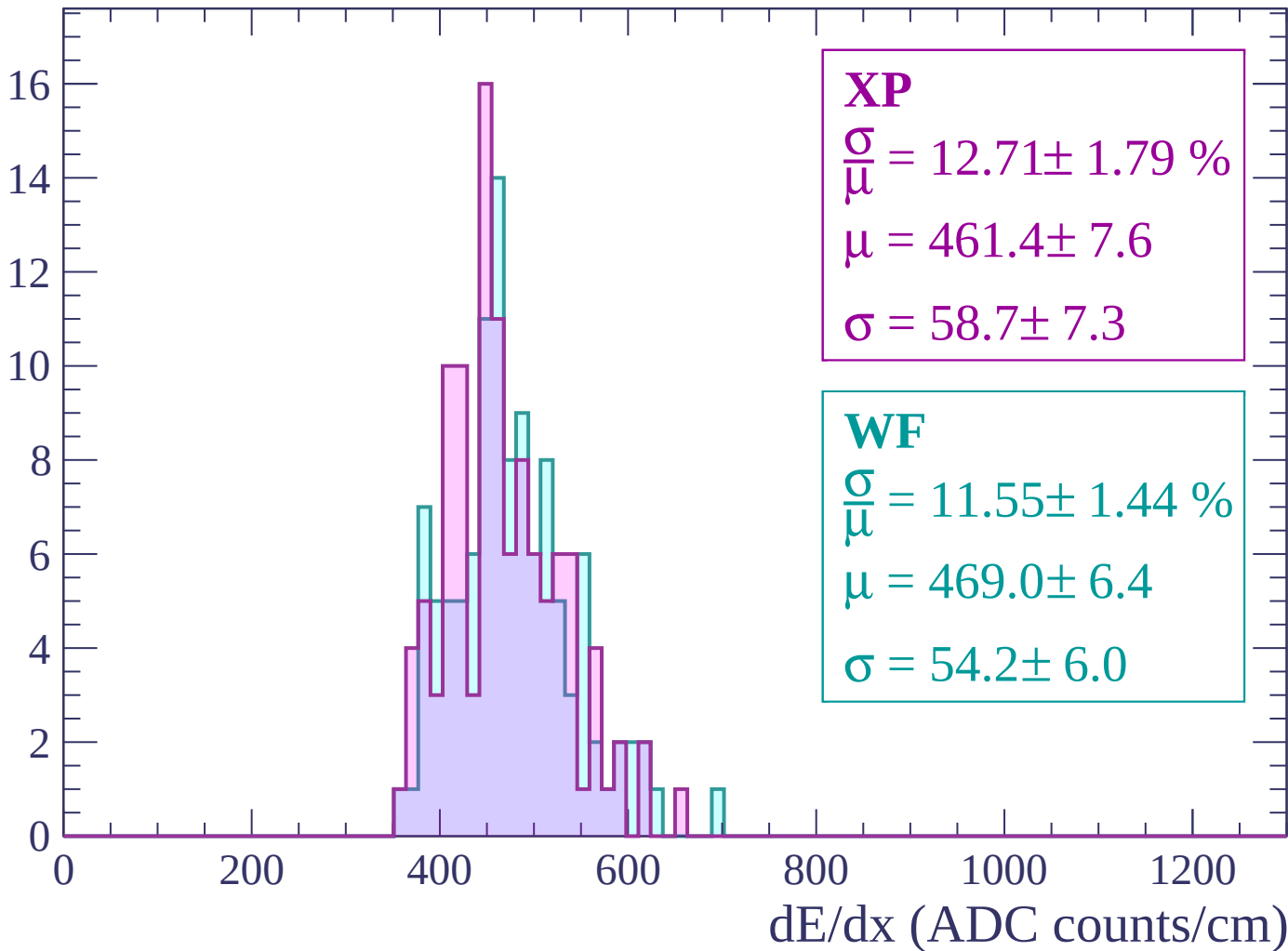
$$\mu = 476.2 \pm 6.6$$

$$\sigma = 48.8 \pm 6.1$$

dE/dx (ADC counts/cm)

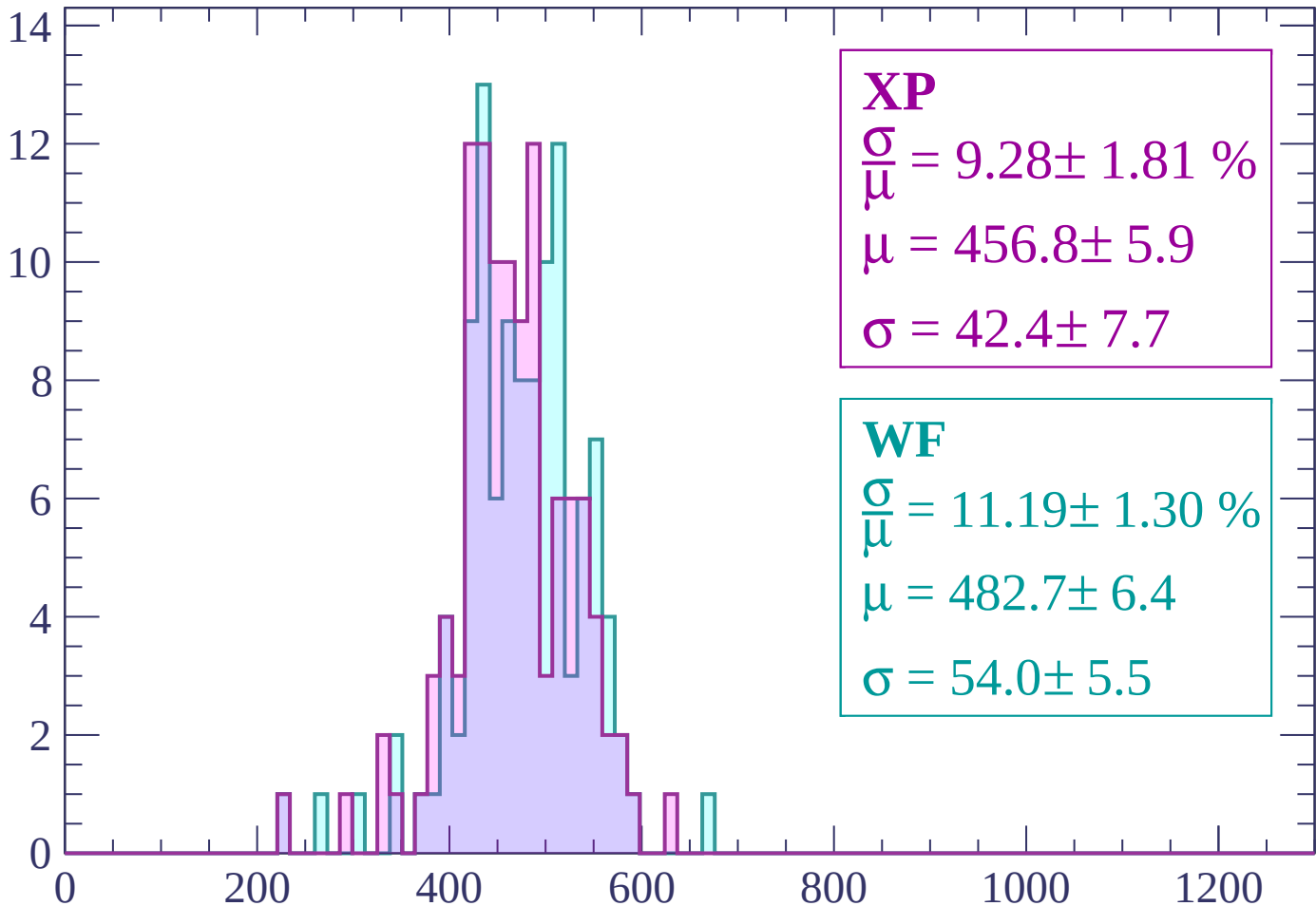
Energy loss | -1680 < p < -1620

Count



Energy loss | -1620 < p < -1560

Count



XP

$$\frac{\sigma}{\mu} = 9.28 \pm 1.81 \%$$

$$\mu = 456.8 \pm 5.9$$

$$\sigma = 42.4 \pm 7.7$$

WF

$$\frac{\sigma}{\mu} = 11.19 \pm 1.30 \%$$

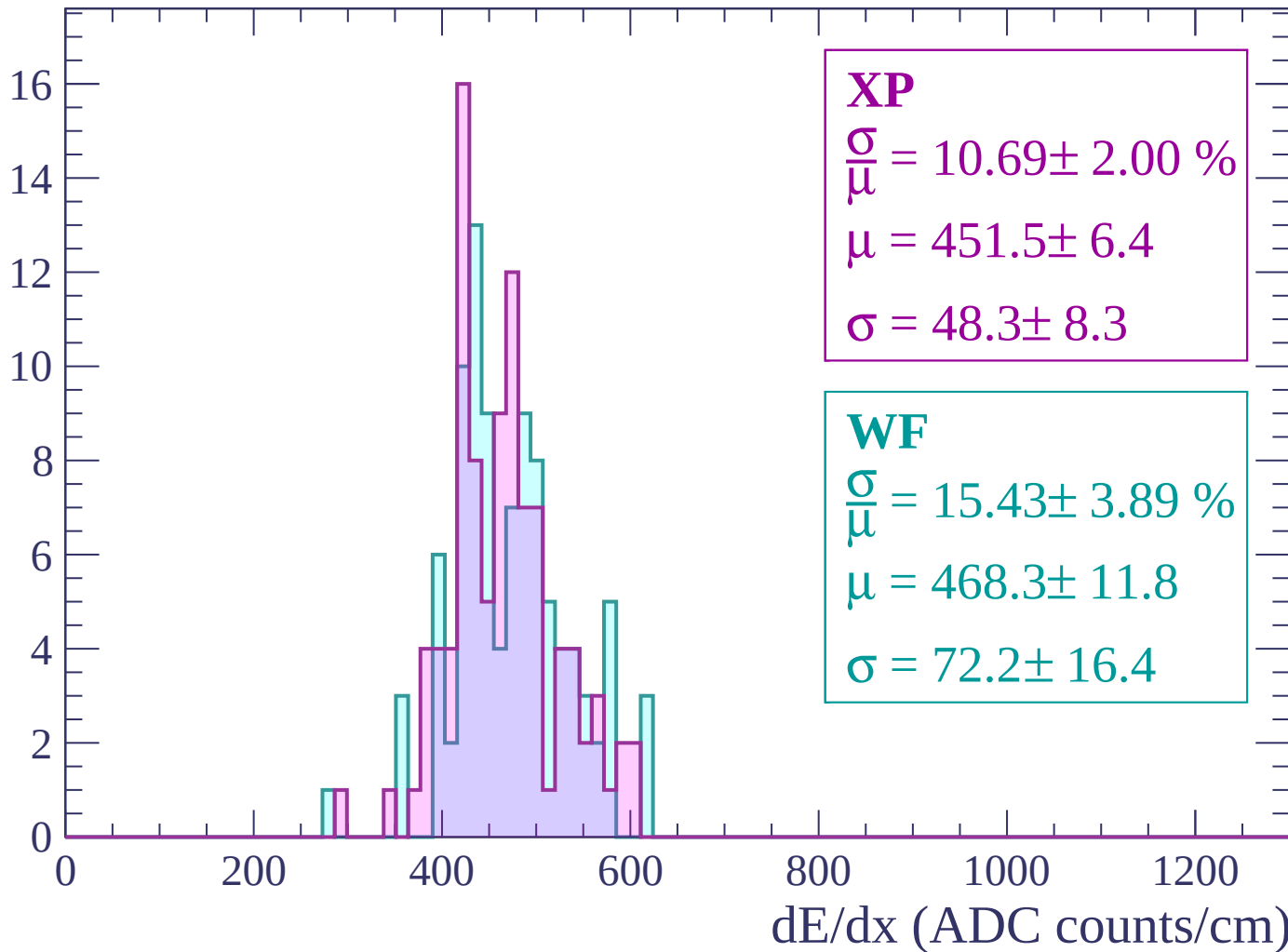
$$\mu = 482.7 \pm 6.4$$

$$\sigma = 54.0 \pm 5.5$$

dE/dx (ADC counts/cm)

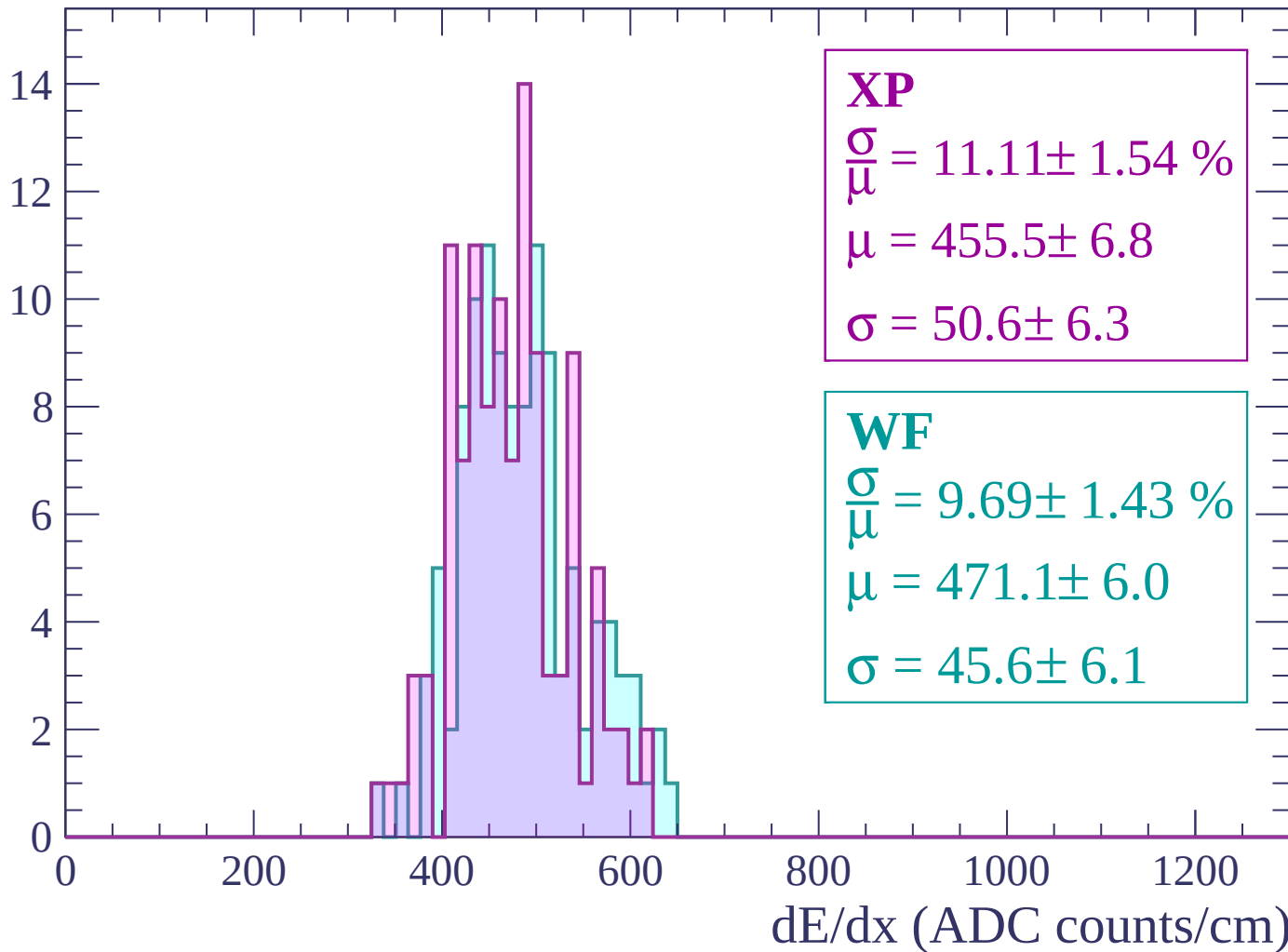
Energy loss | -1560 < p < -1500

Count



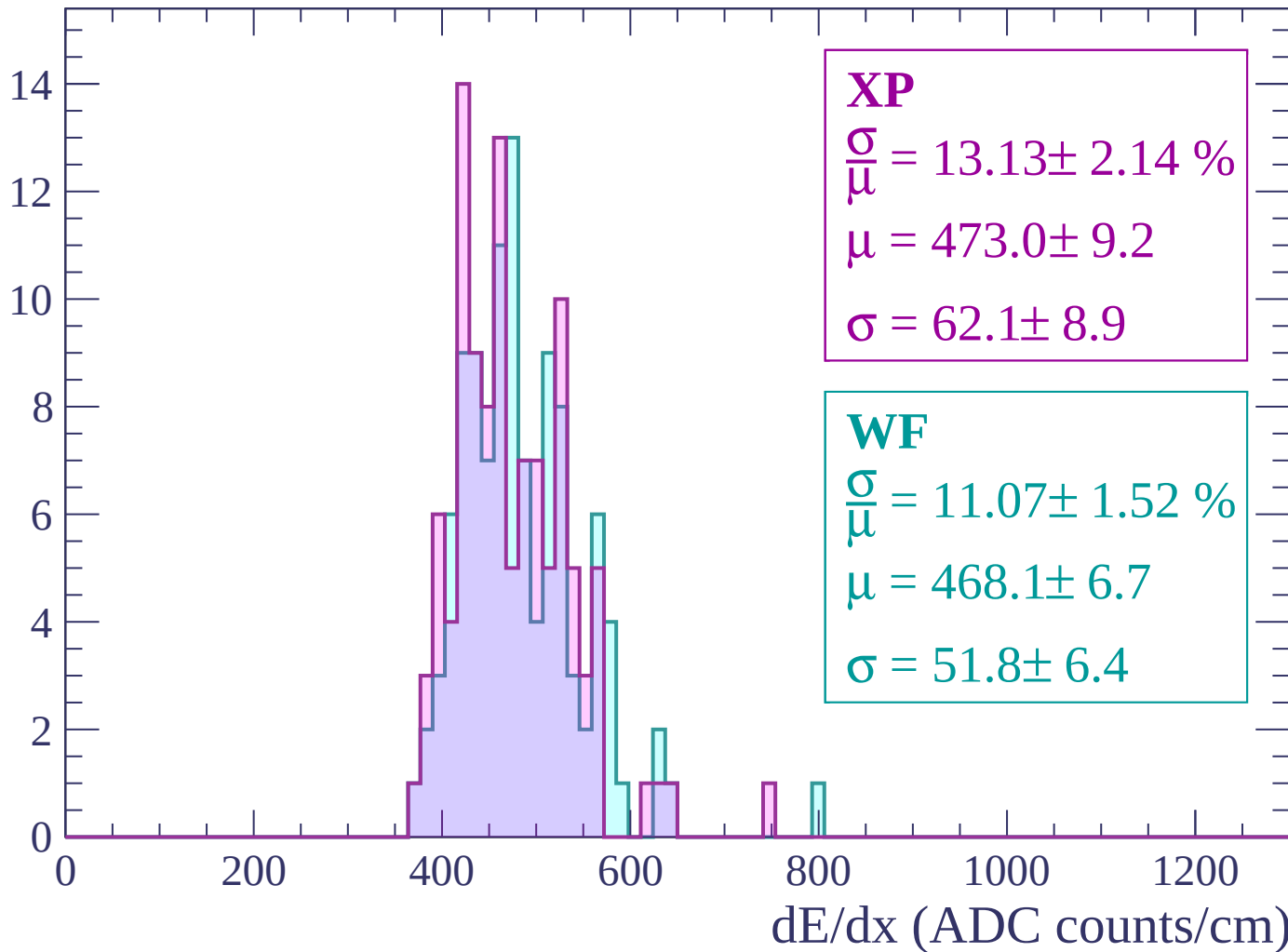
Energy loss | -1500 < p < -1440

Count



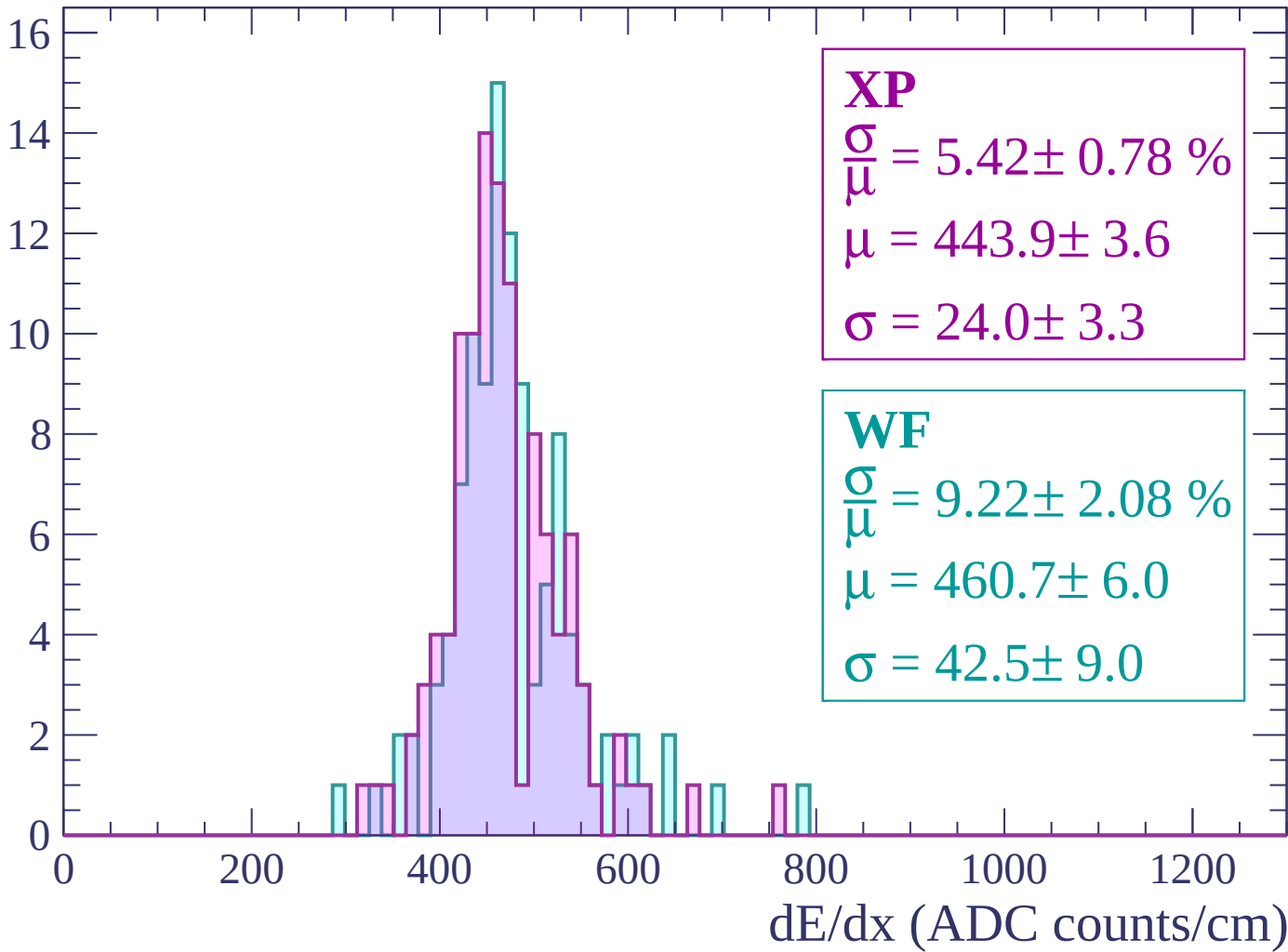
Energy loss | -1440 < p < -1380

Count



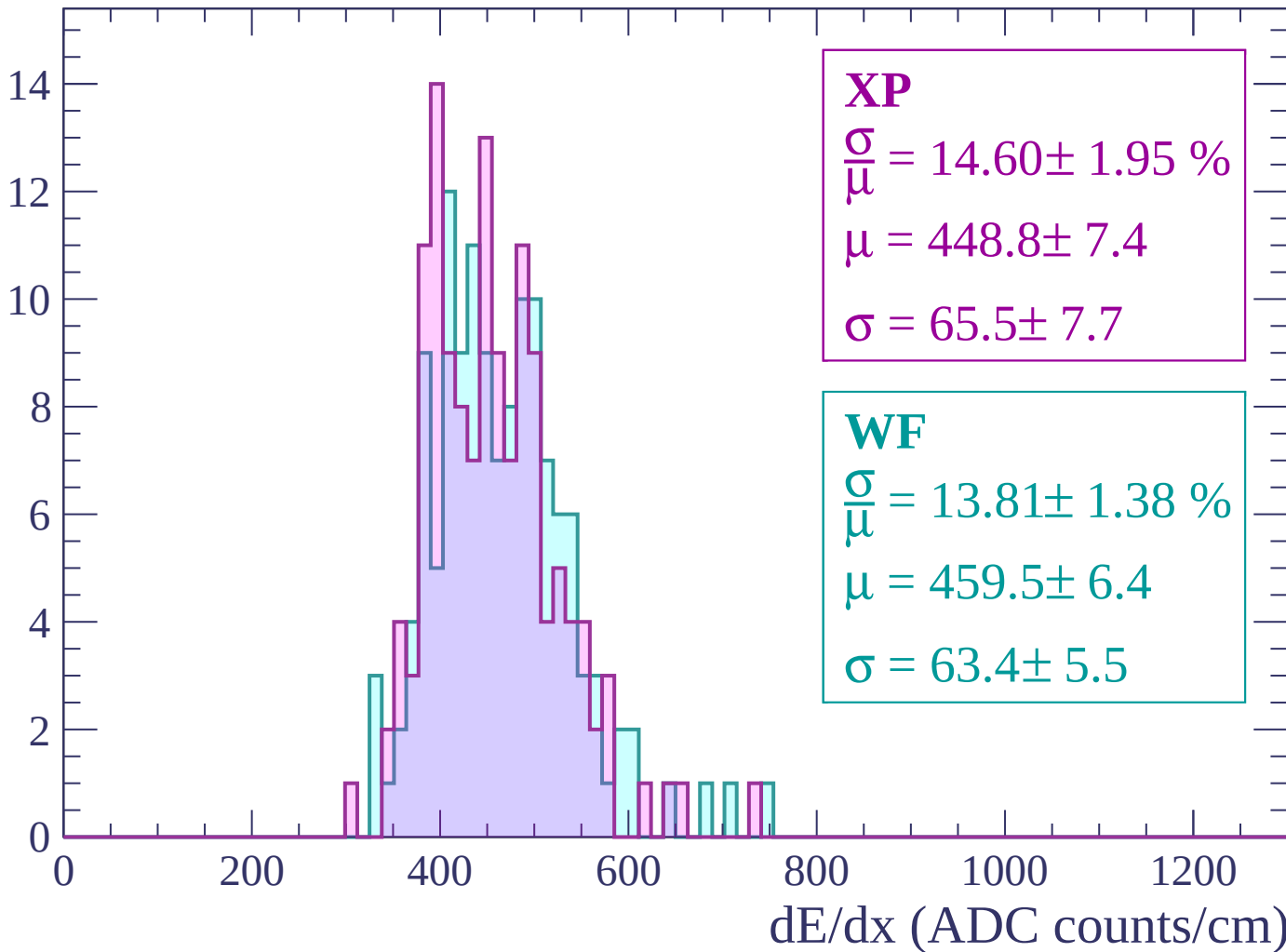
Energy loss | -1380 < p < -1320

Count



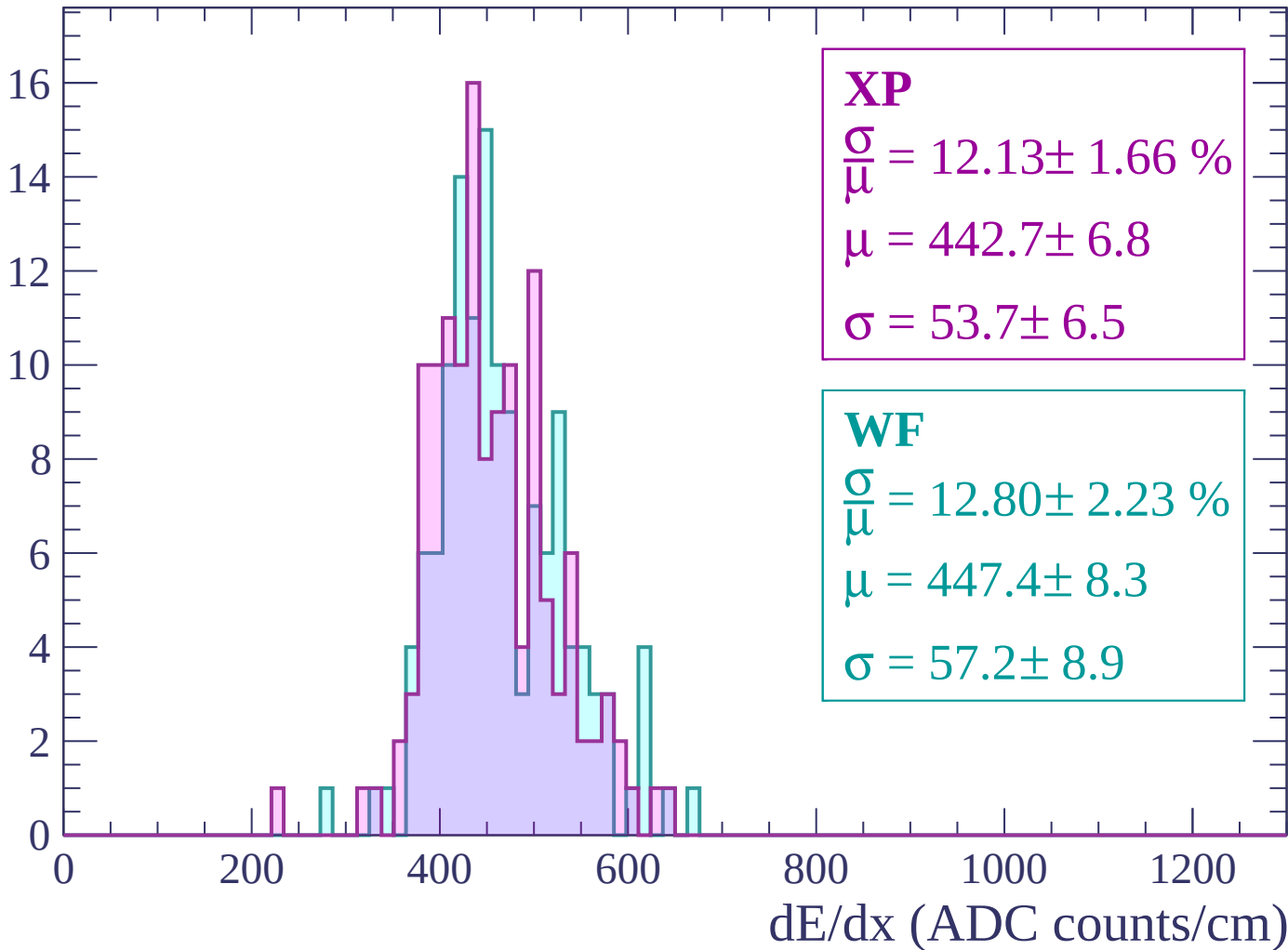
Energy loss | -1320 < p < -1260

Count



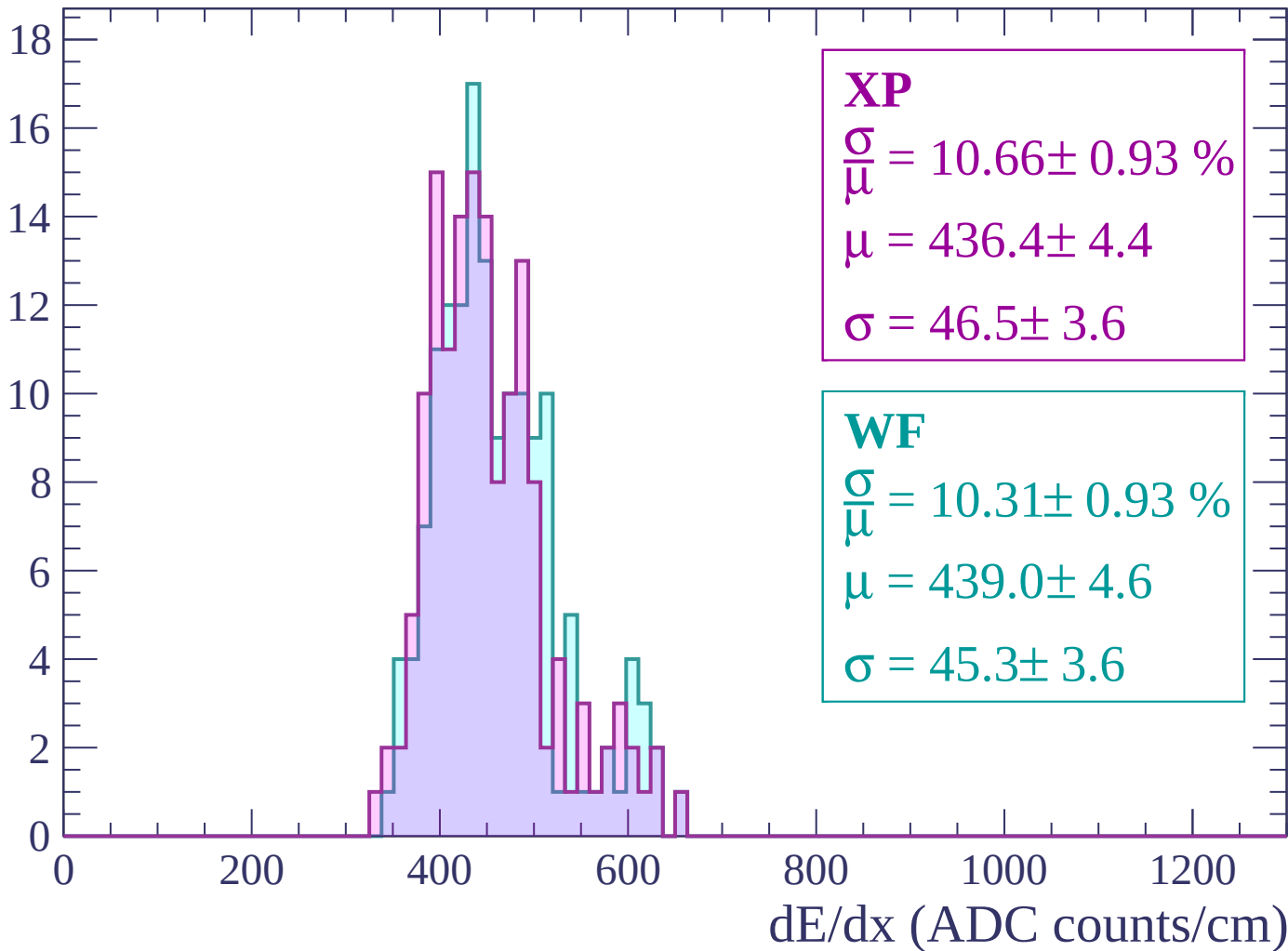
Energy loss | -1260 < p < -1200

Count



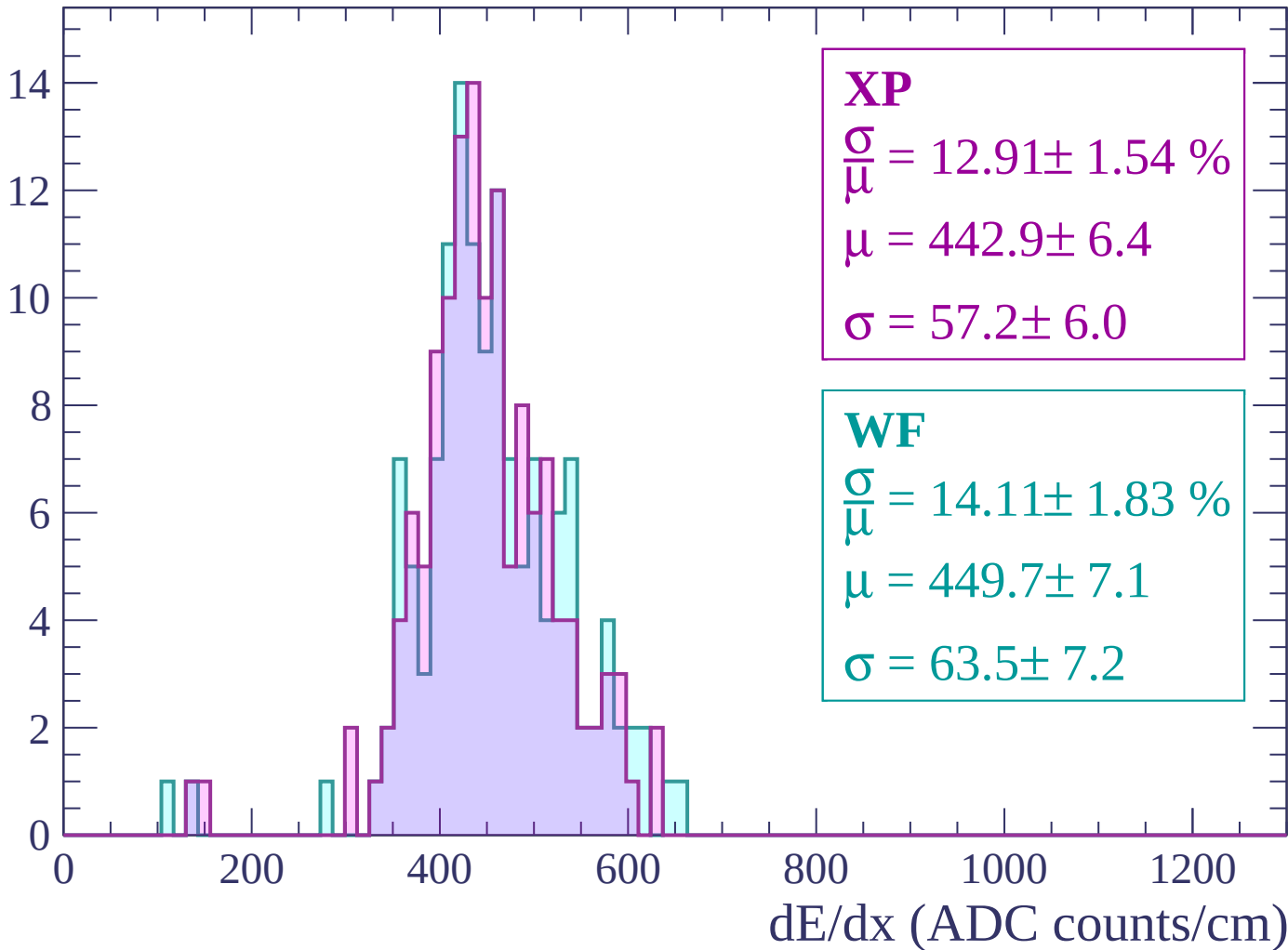
Energy loss | -1200 < p < -1140

Count



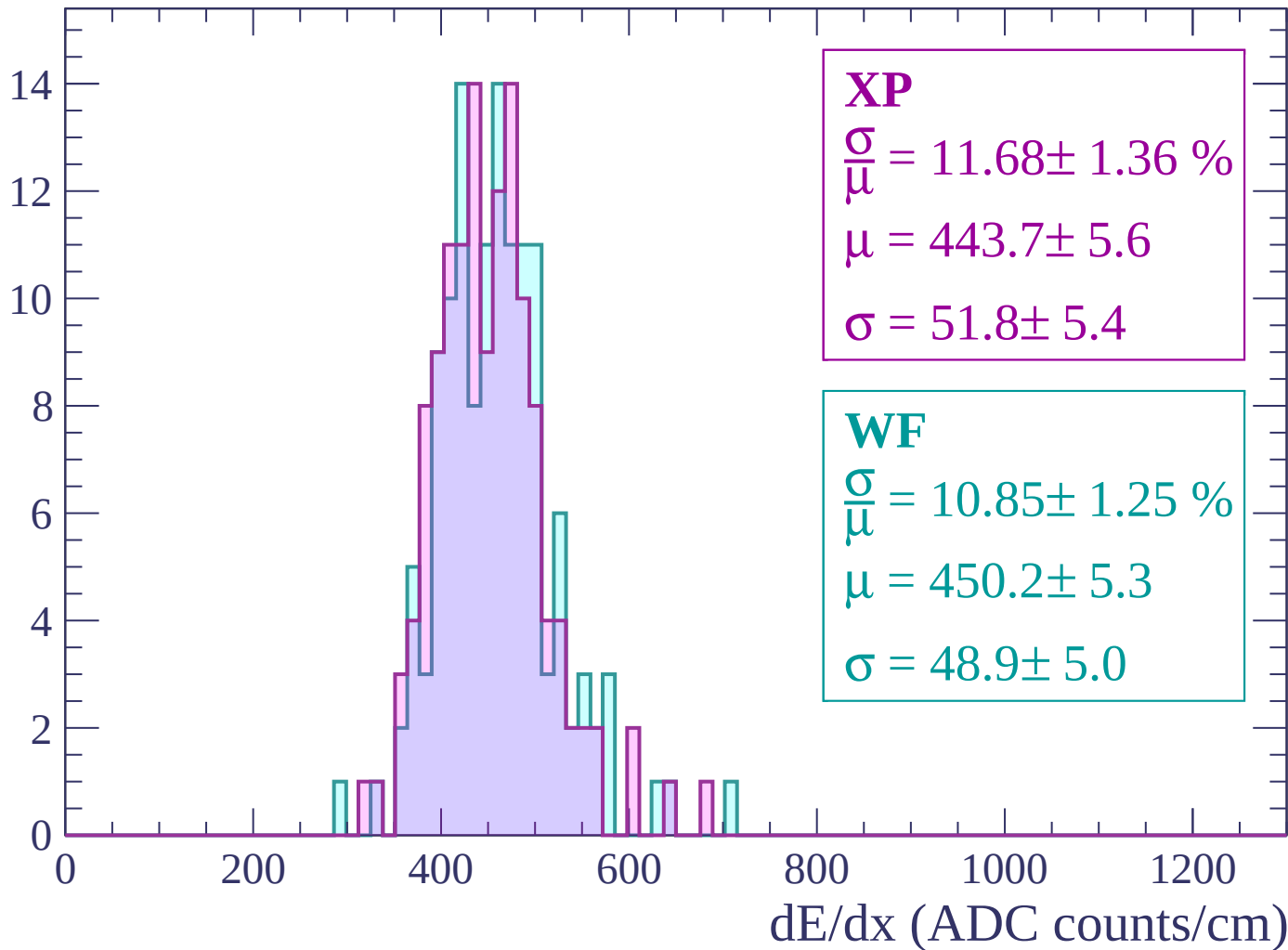
Energy loss | -1140 < p < -1080

Count



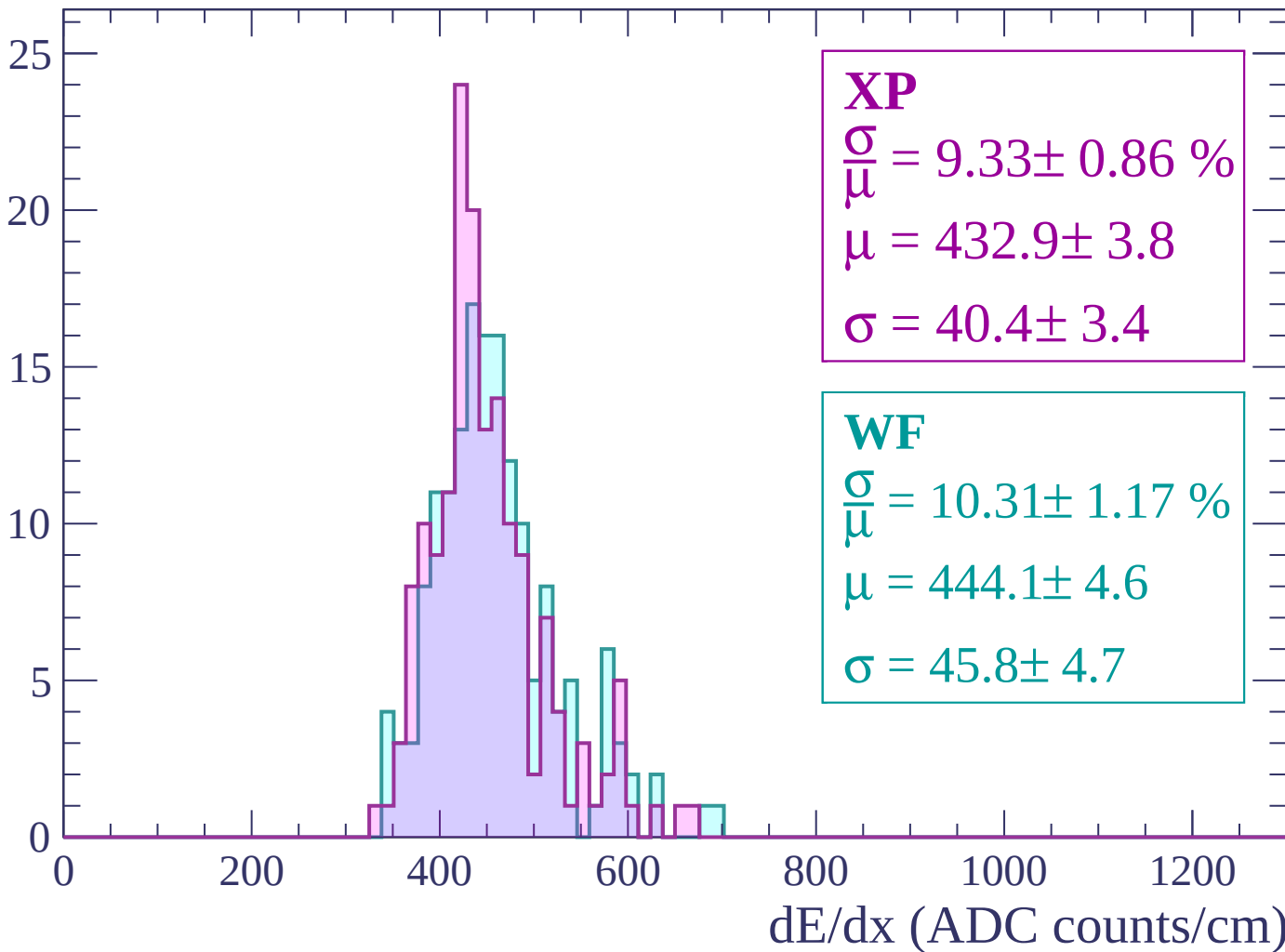
Energy loss | -1080 < p < -1020

Count



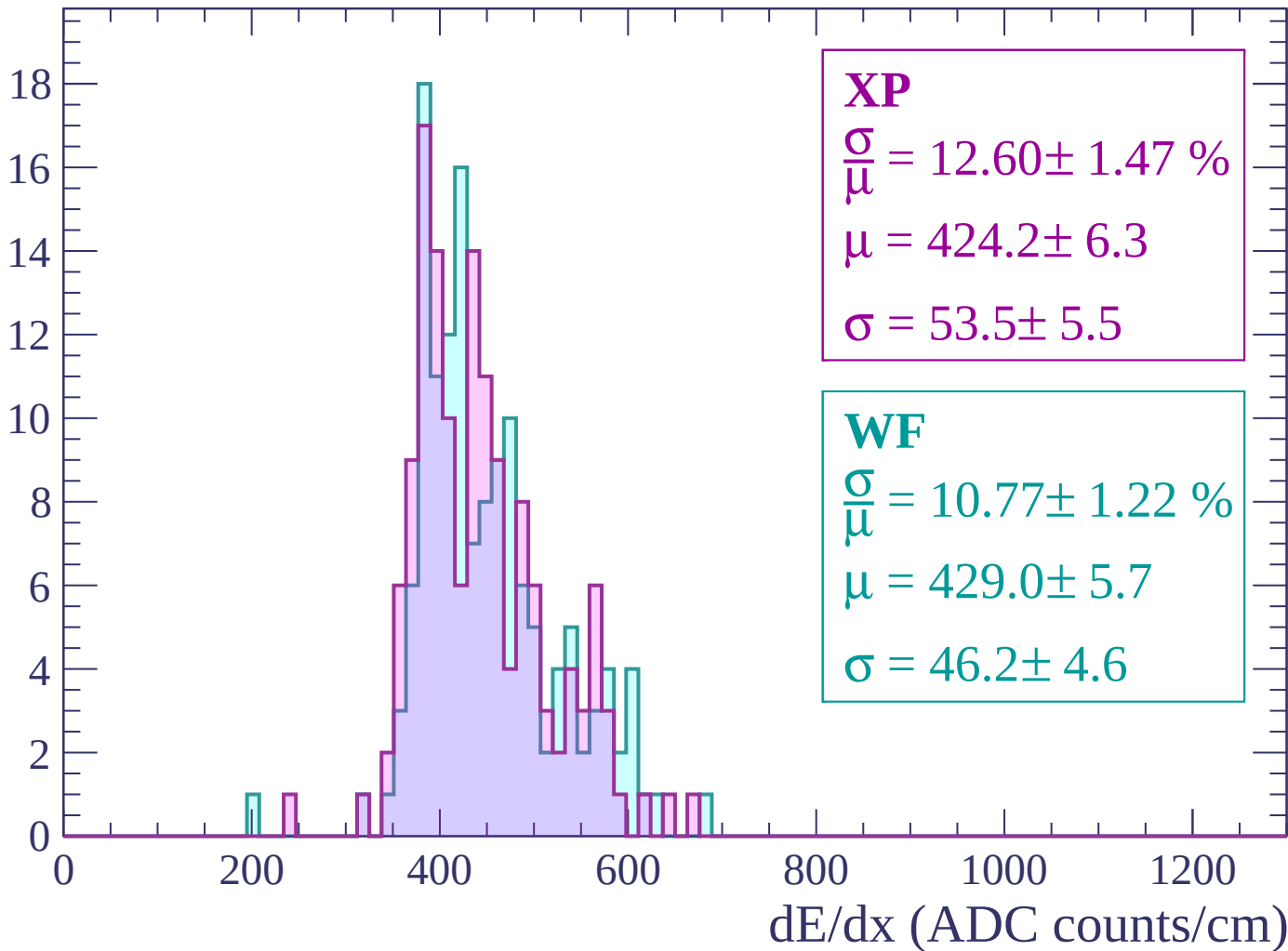
Energy loss | $-1020 < p < -960$

Count



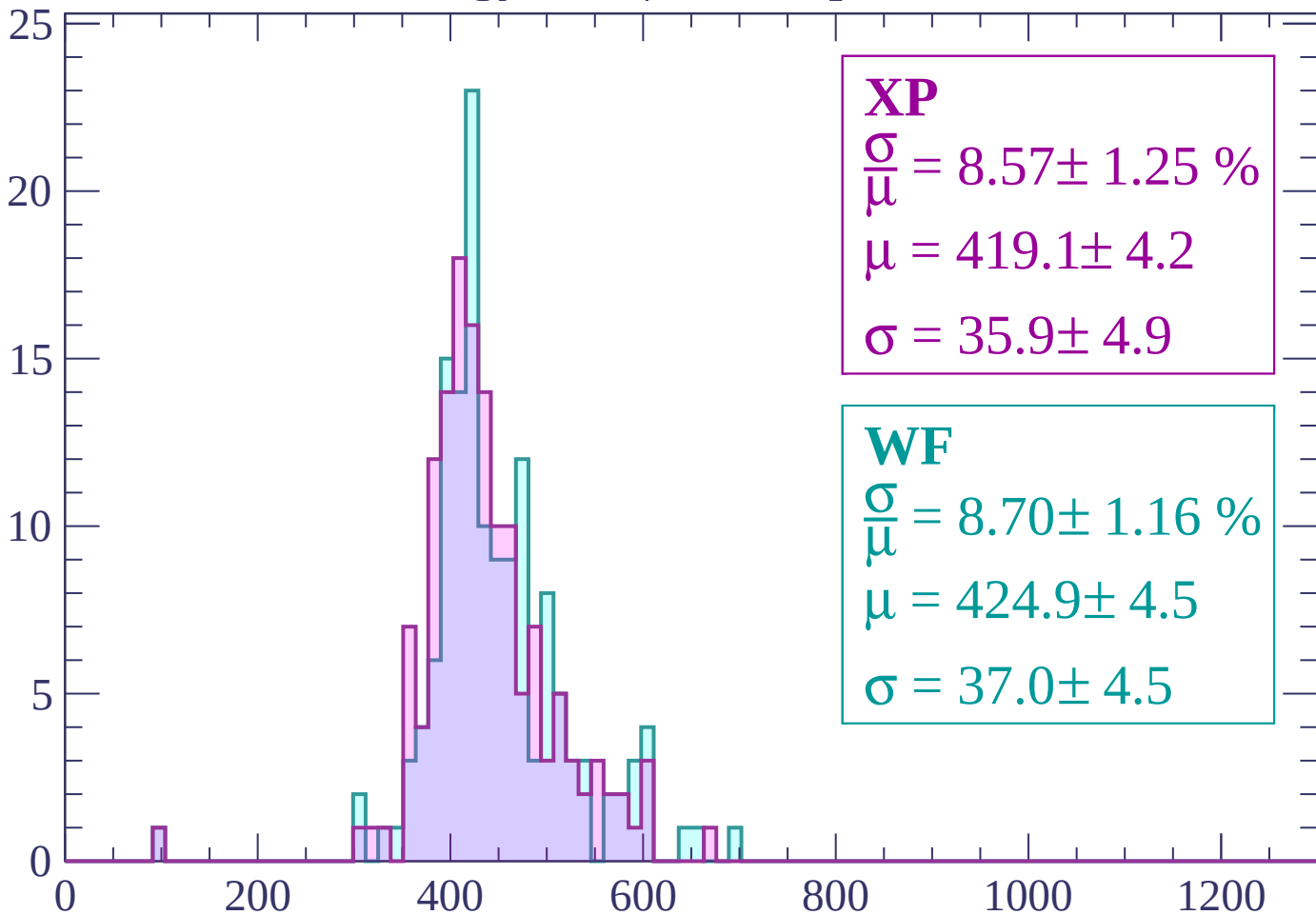
Energy loss | $-960 < p < -900$

Count



Energy loss | $-900 < p < -840$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.57 \pm 1.25 \%$$

$$\mu = 419.1 \pm 4.2$$

$$\sigma = 35.9 \pm 4.9$$

WF

$$\frac{\sigma}{\bar{\mu}} = 8.70 \pm 1.16 \%$$

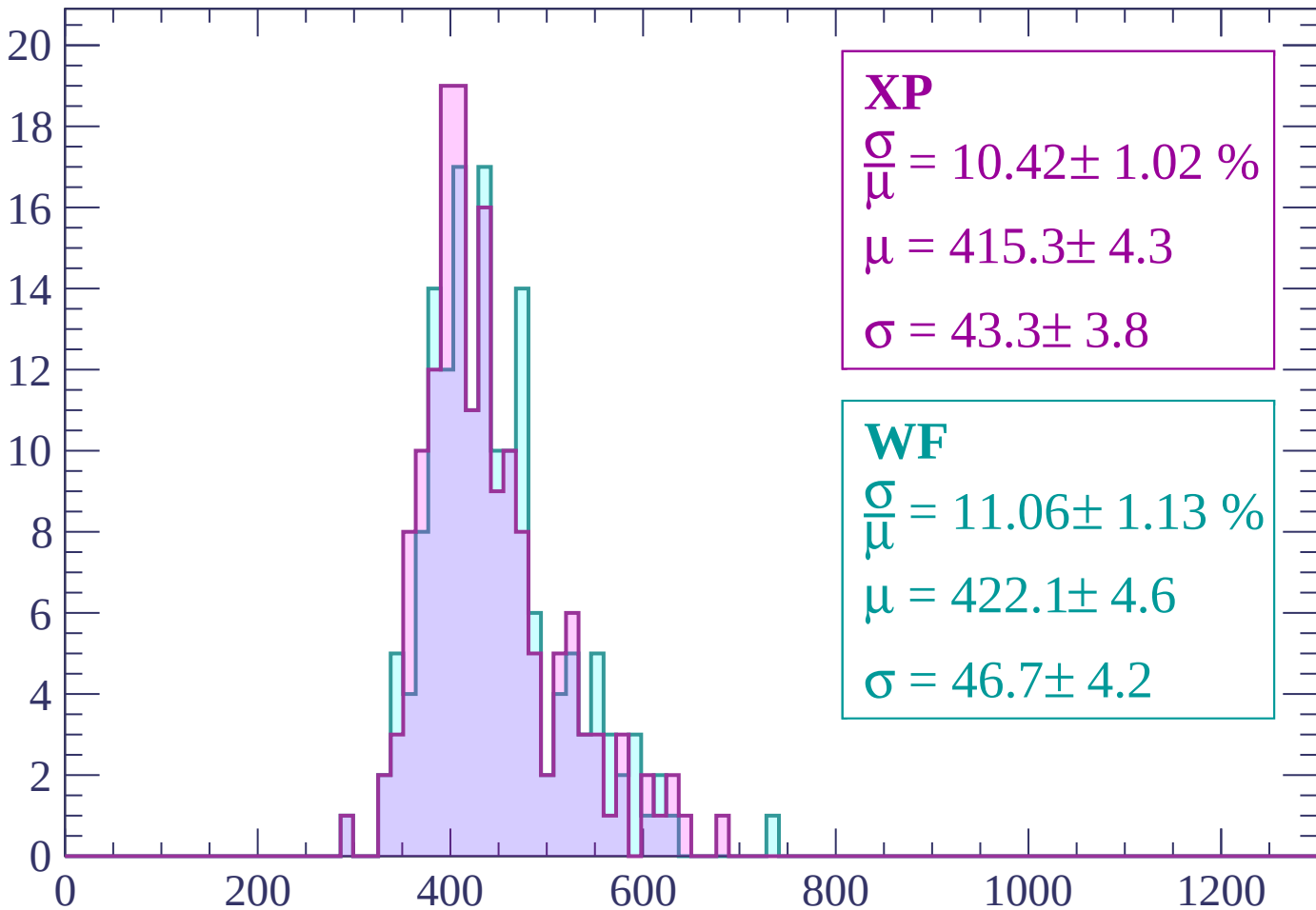
$$\mu = 424.9 \pm 4.5$$

$$\sigma = 37.0 \pm 4.5$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -780$

Count



XP

$$\frac{\sigma}{\mu} = 10.42 \pm 1.02 \%$$

$$\mu = 415.3 \pm 4.3$$

$$\sigma = 43.3 \pm 3.8$$

WF

$$\frac{\sigma}{\mu} = 11.06 \pm 1.13 \%$$

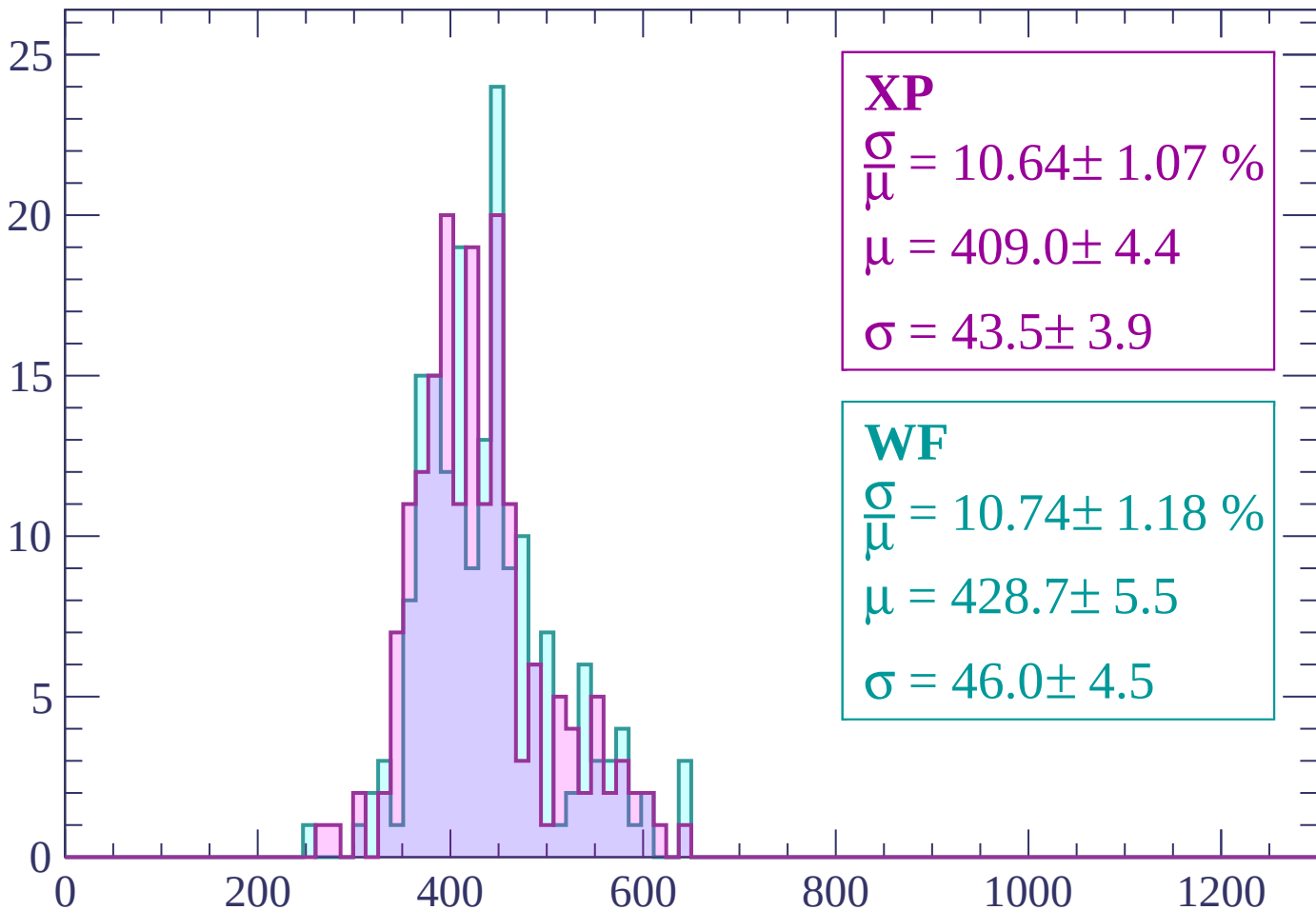
$$\mu = 422.1 \pm 4.6$$

$$\sigma = 46.7 \pm 4.2$$

dE/dx (ADC counts/cm)

Energy loss | $-780 < p < -720$

Count



XP

$$\frac{\sigma}{\mu} = 10.64 \pm 1.07 \%$$

$$\mu = 409.0 \pm 4.4$$

$$\sigma = 43.5 \pm 3.9$$

WF

$$\frac{\sigma}{\mu} = 10.74 \pm 1.18 \%$$

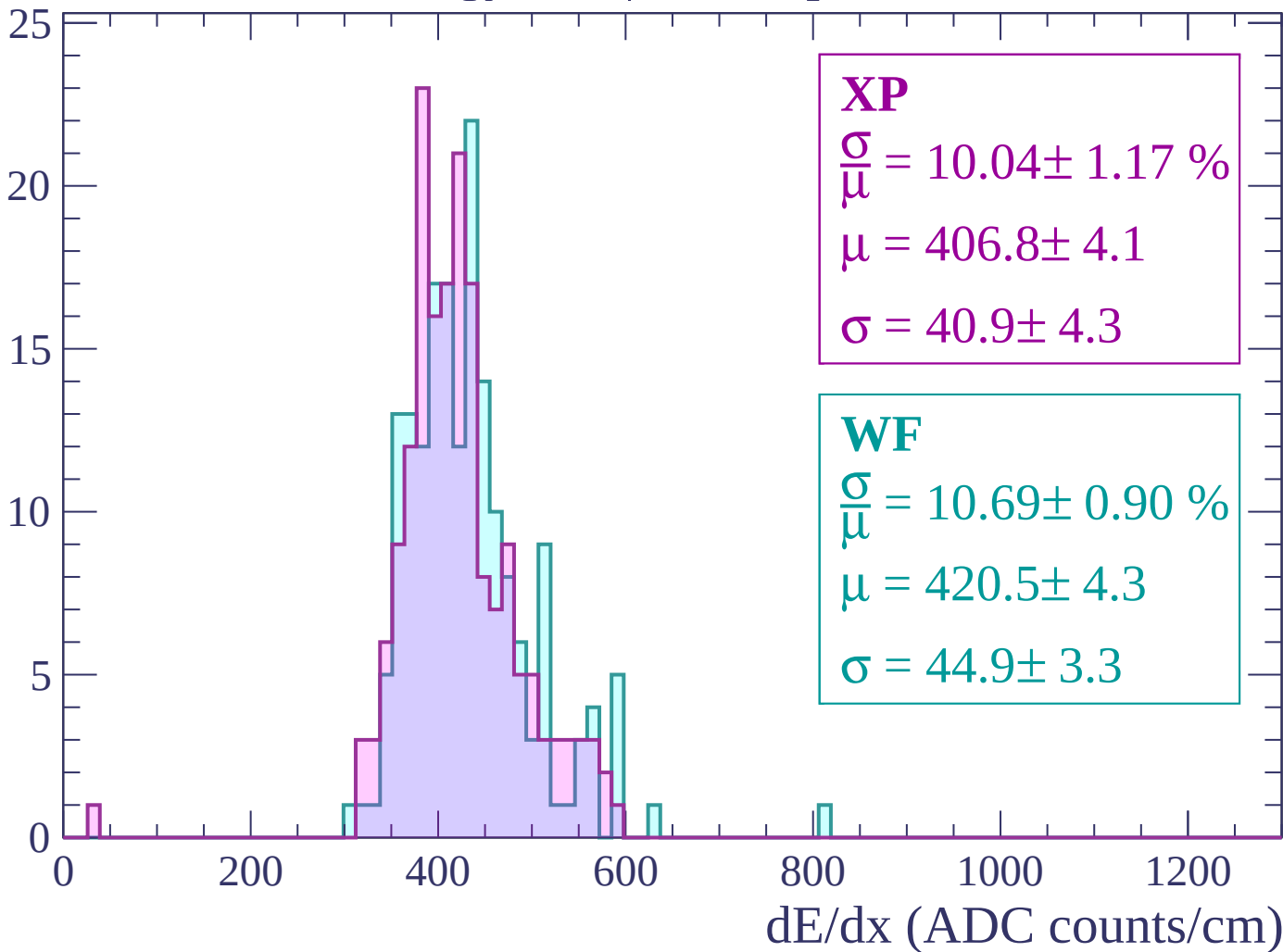
$$\mu = 428.7 \pm 5.5$$

$$\sigma = 46.0 \pm 4.5$$

dE/dx (ADC counts/cm)

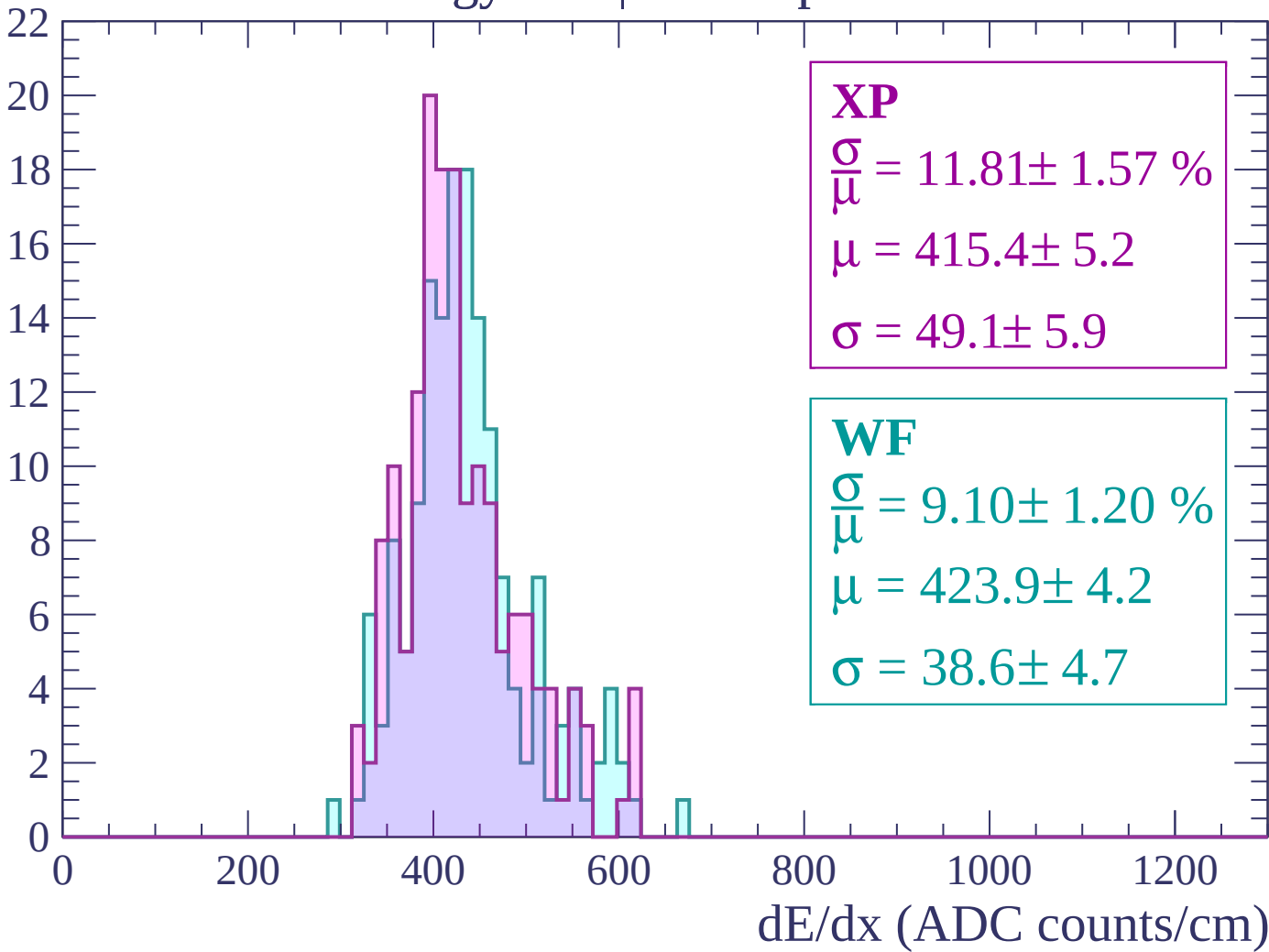
Energy loss | $-720 < p < -660$

Count



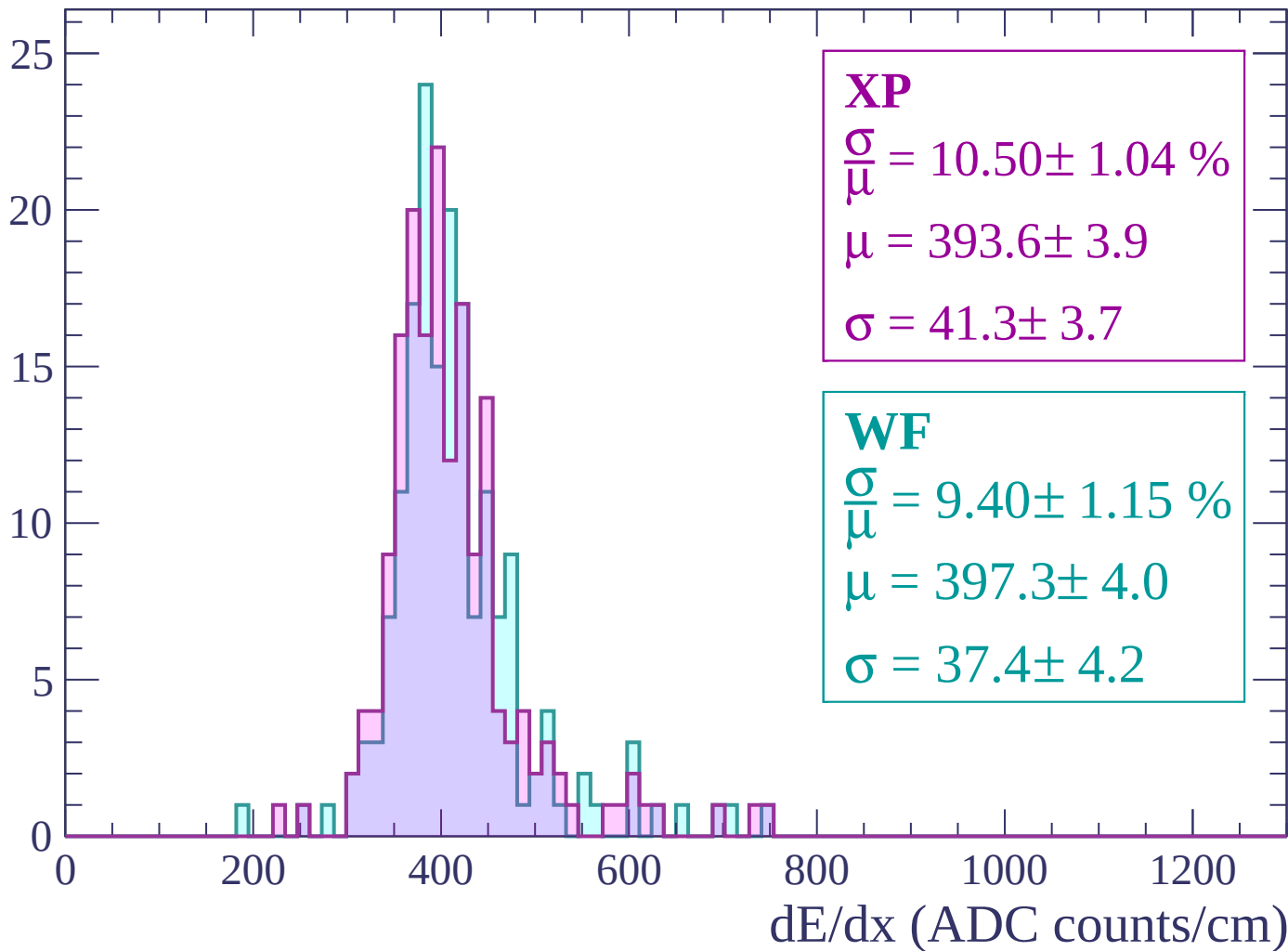
Energy loss | $-660 < p < -600$

Count



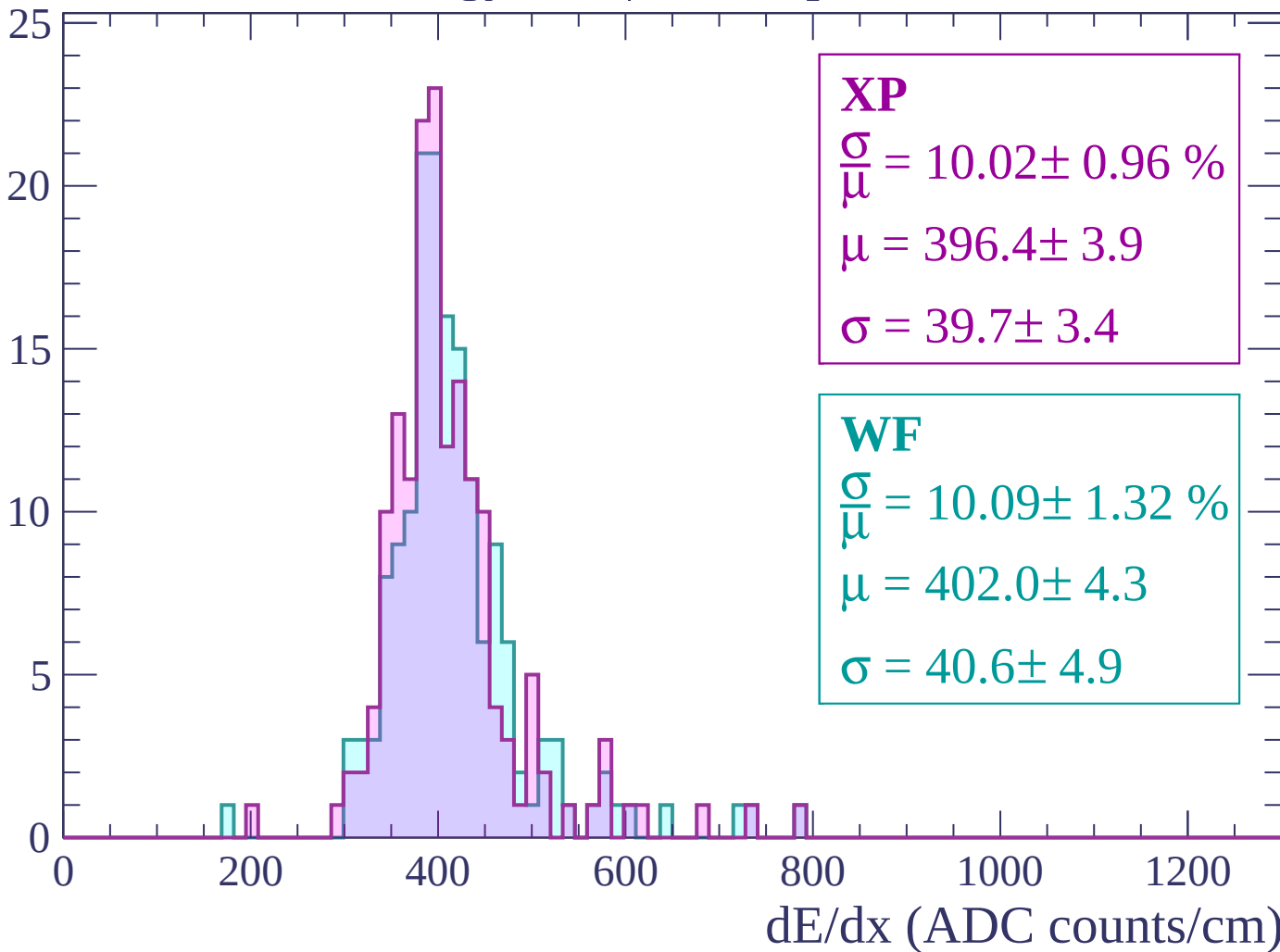
Energy loss | -600 < p < -540

Count



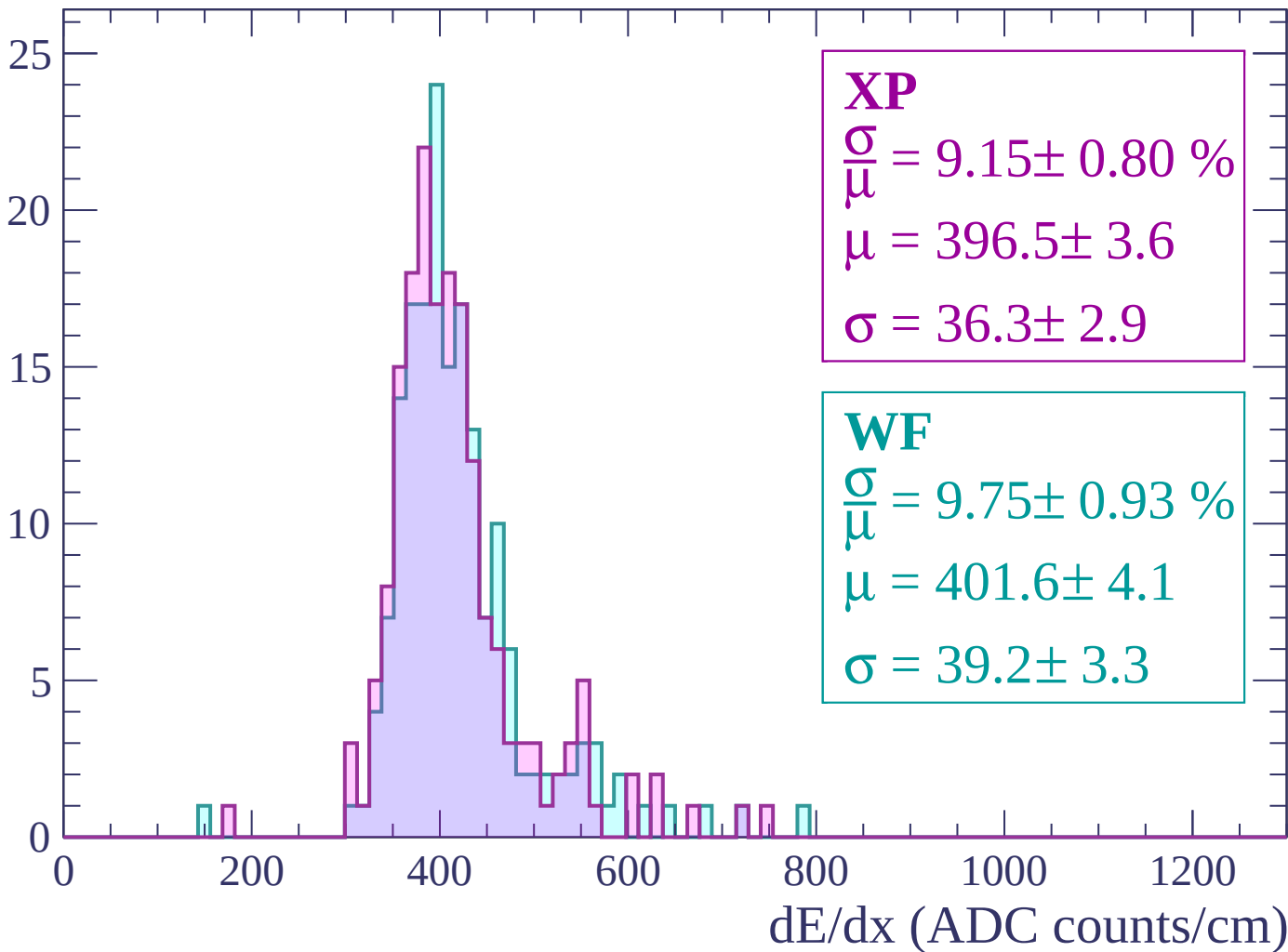
Energy loss | $-540 < p < -480$

Count



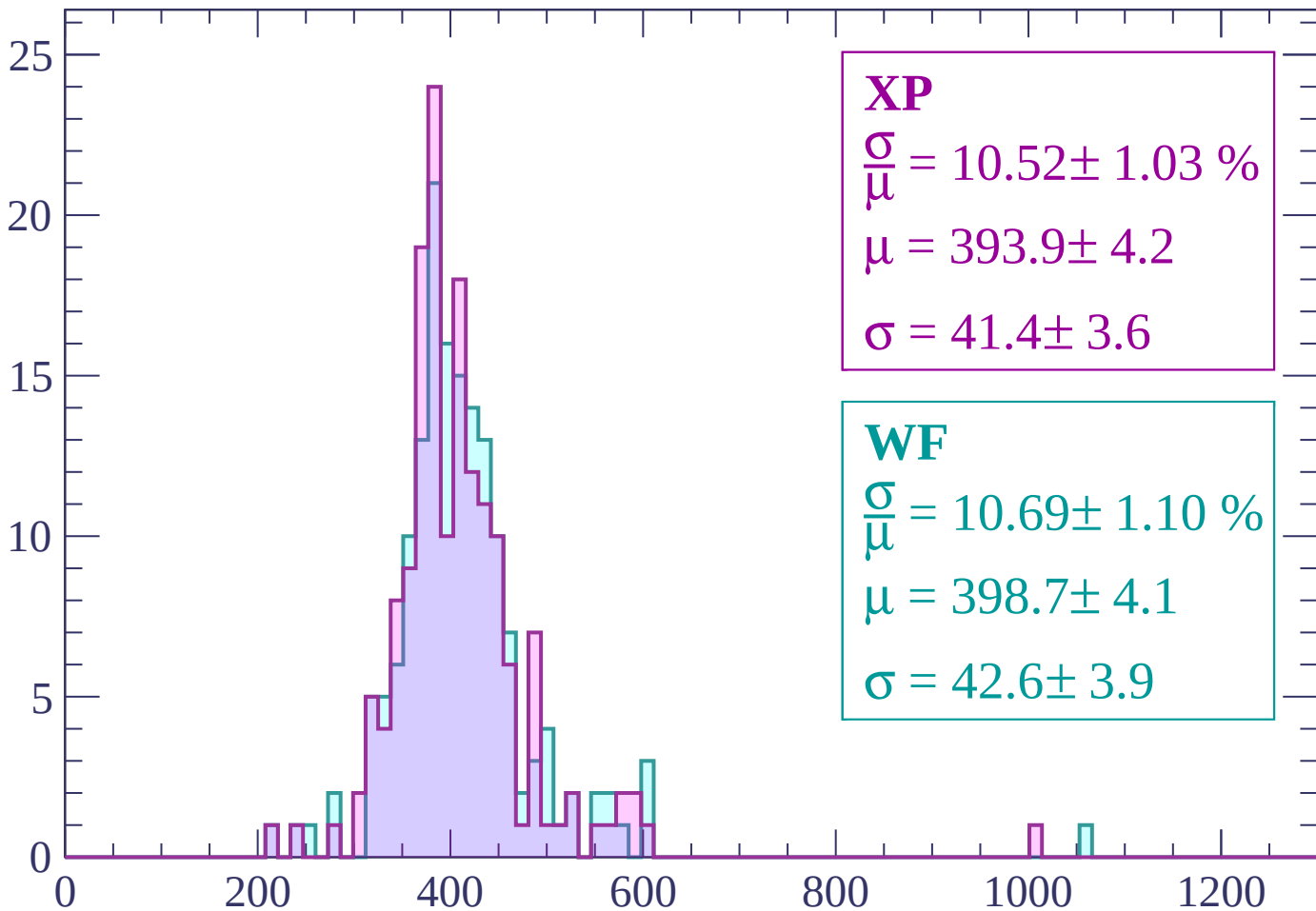
Energy loss | $-480 < p < -420$

Count



Energy loss | $-420 < p < -360$

Count



XP

$$\frac{\sigma}{\mu} = 10.52 \pm 1.03 \%$$

$$\mu = 393.9 \pm 4.2$$

$$\sigma = 41.4 \pm 3.6$$

WF

$$\frac{\sigma}{\mu} = 10.69 \pm 1.10 \%$$

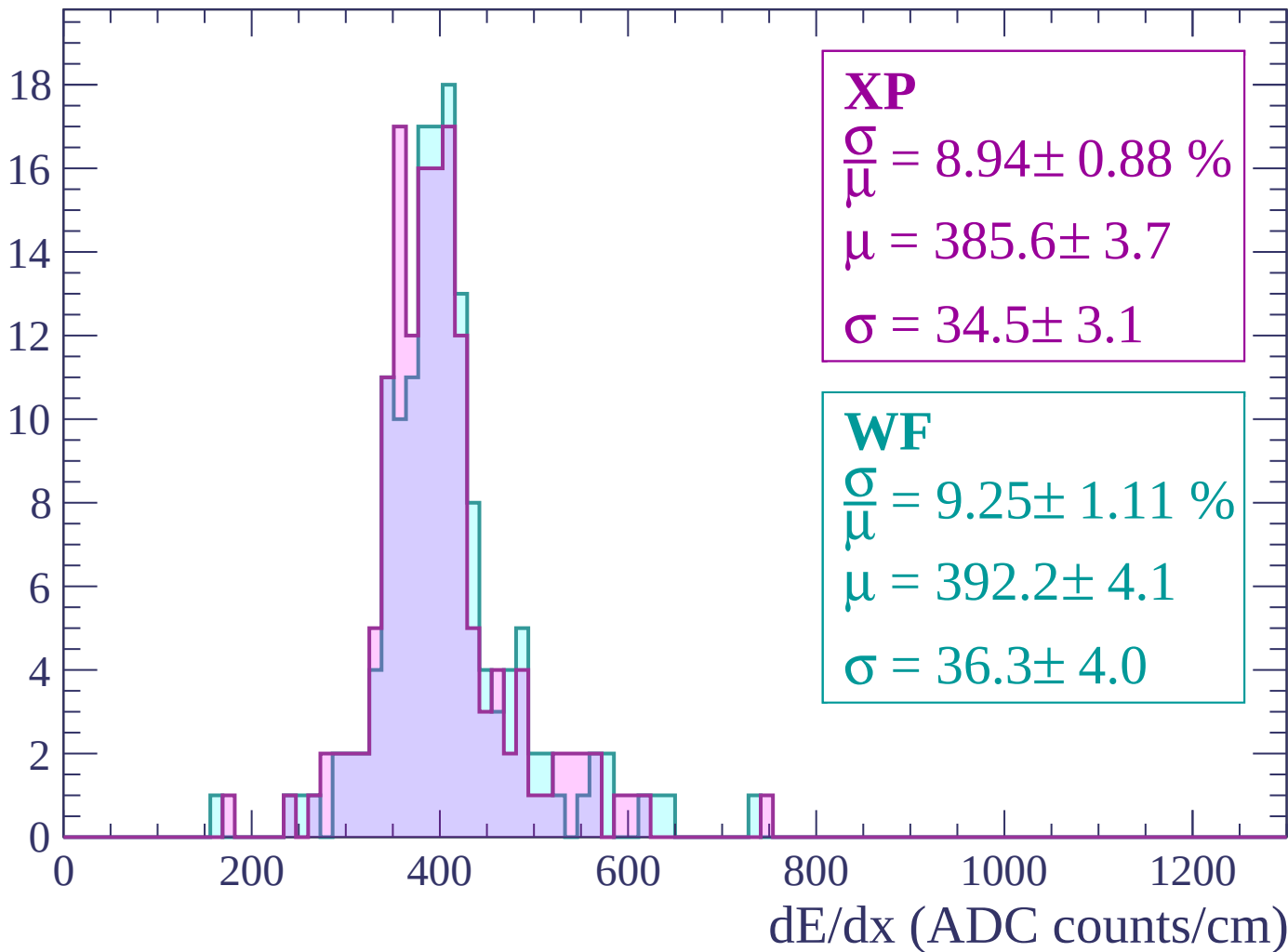
$$\mu = 398.7 \pm 4.1$$

$$\sigma = 42.6 \pm 3.9$$

dE/dx (ADC counts/cm)

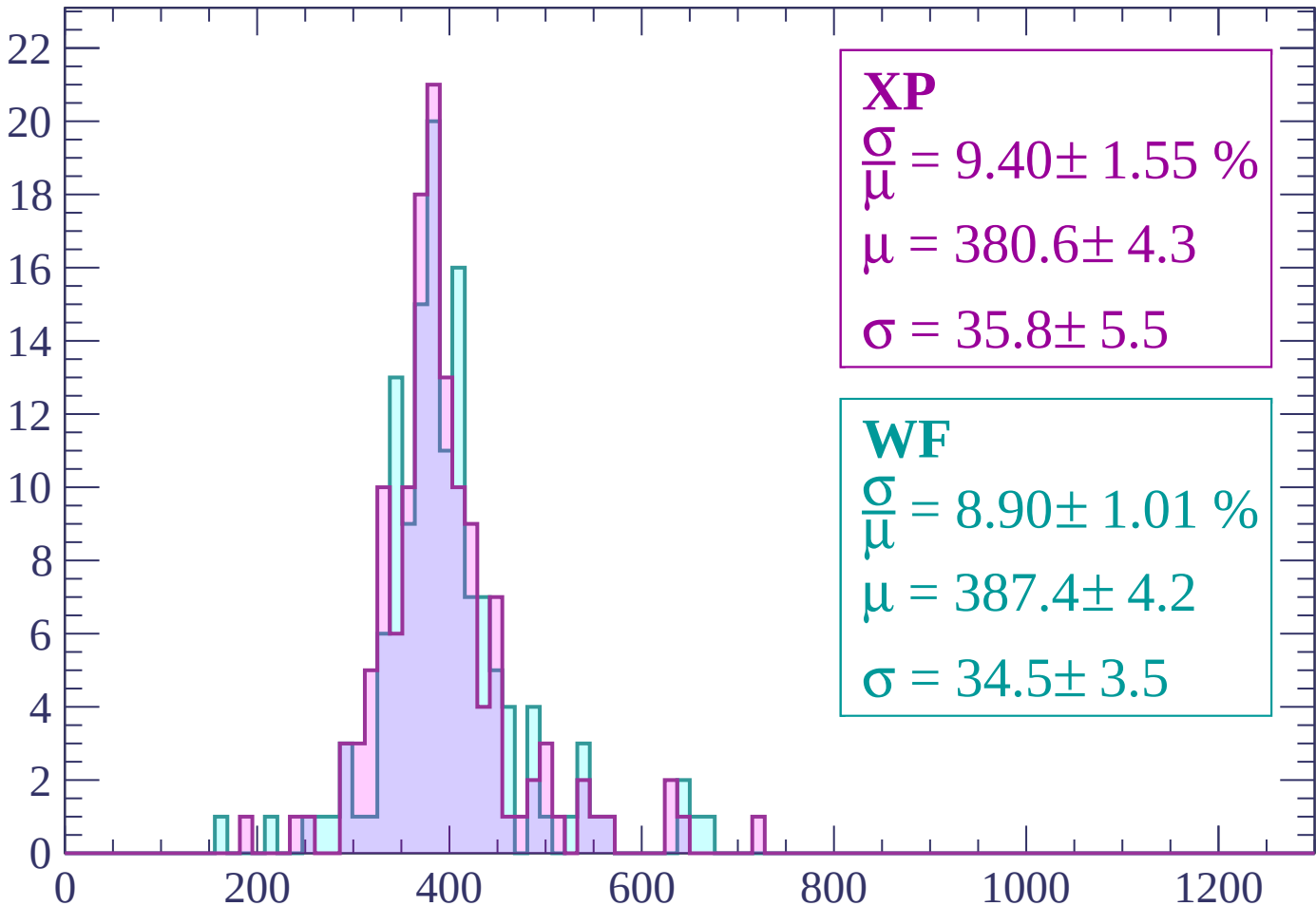
Energy loss | $-360 < p < -300$

Count



Energy loss | $-300 < p < -240$

Count



XP

$$\frac{\sigma}{\mu} = 9.40 \pm 1.55 \%$$

$$\mu = 380.6 \pm 4.3$$

$$\sigma = 35.8 \pm 5.5$$

WF

$$\frac{\sigma}{\mu} = 8.90 \pm 1.01 \%$$

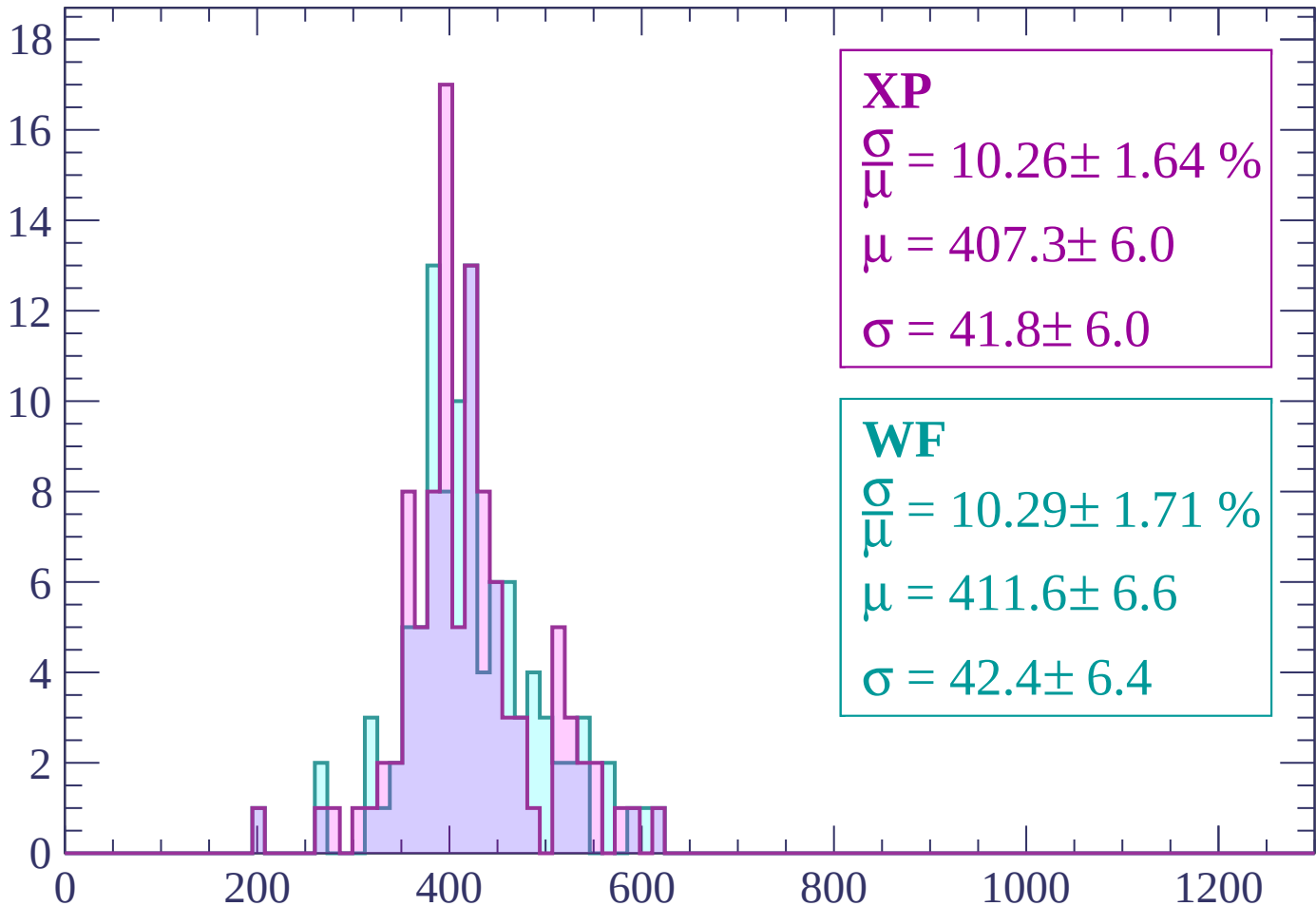
$$\mu = 387.4 \pm 4.2$$

$$\sigma = 34.5 \pm 3.5$$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -180$

Count



XP

$$\frac{\sigma}{\mu} = 10.26 \pm 1.64 \%$$

$$\mu = 407.3 \pm 6.0$$

$$\sigma = 41.8 \pm 6.0$$

WF

$$\frac{\sigma}{\mu} = 10.29 \pm 1.71 \%$$

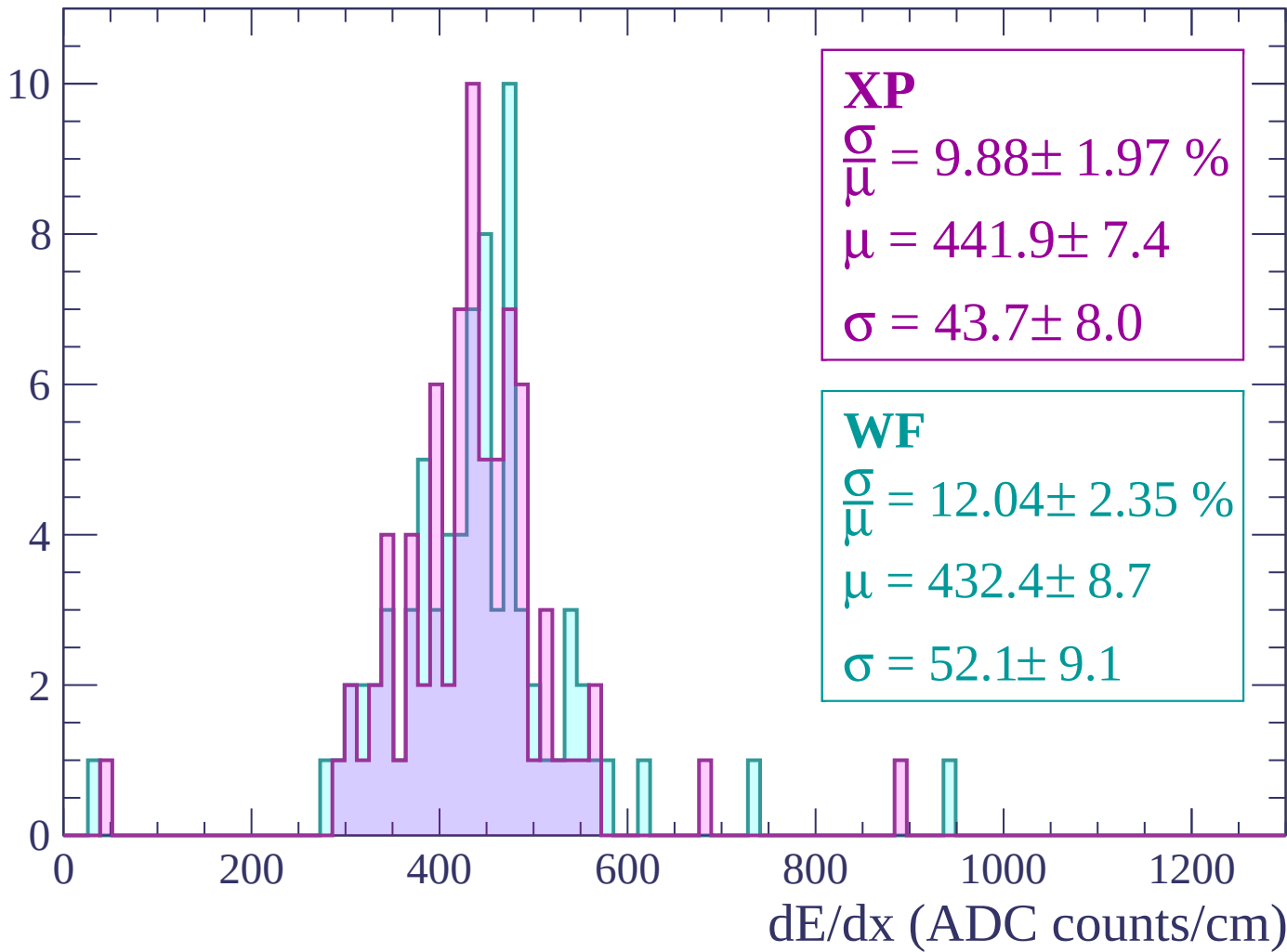
$$\mu = 411.6 \pm 6.6$$

$$\sigma = 42.4 \pm 6.4$$

dE/dx (ADC counts/cm)

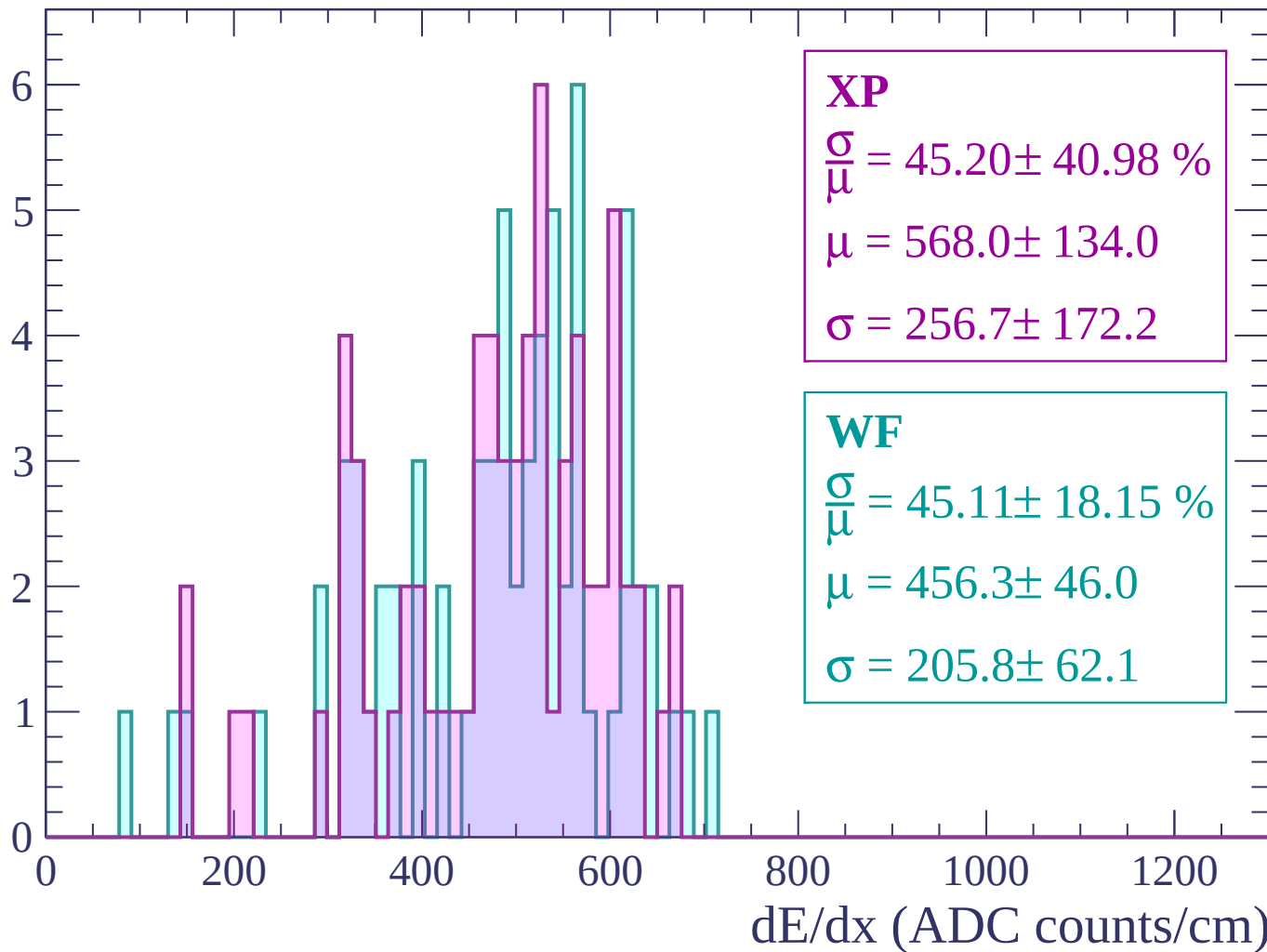
Energy loss | -180 < p < -120

Count



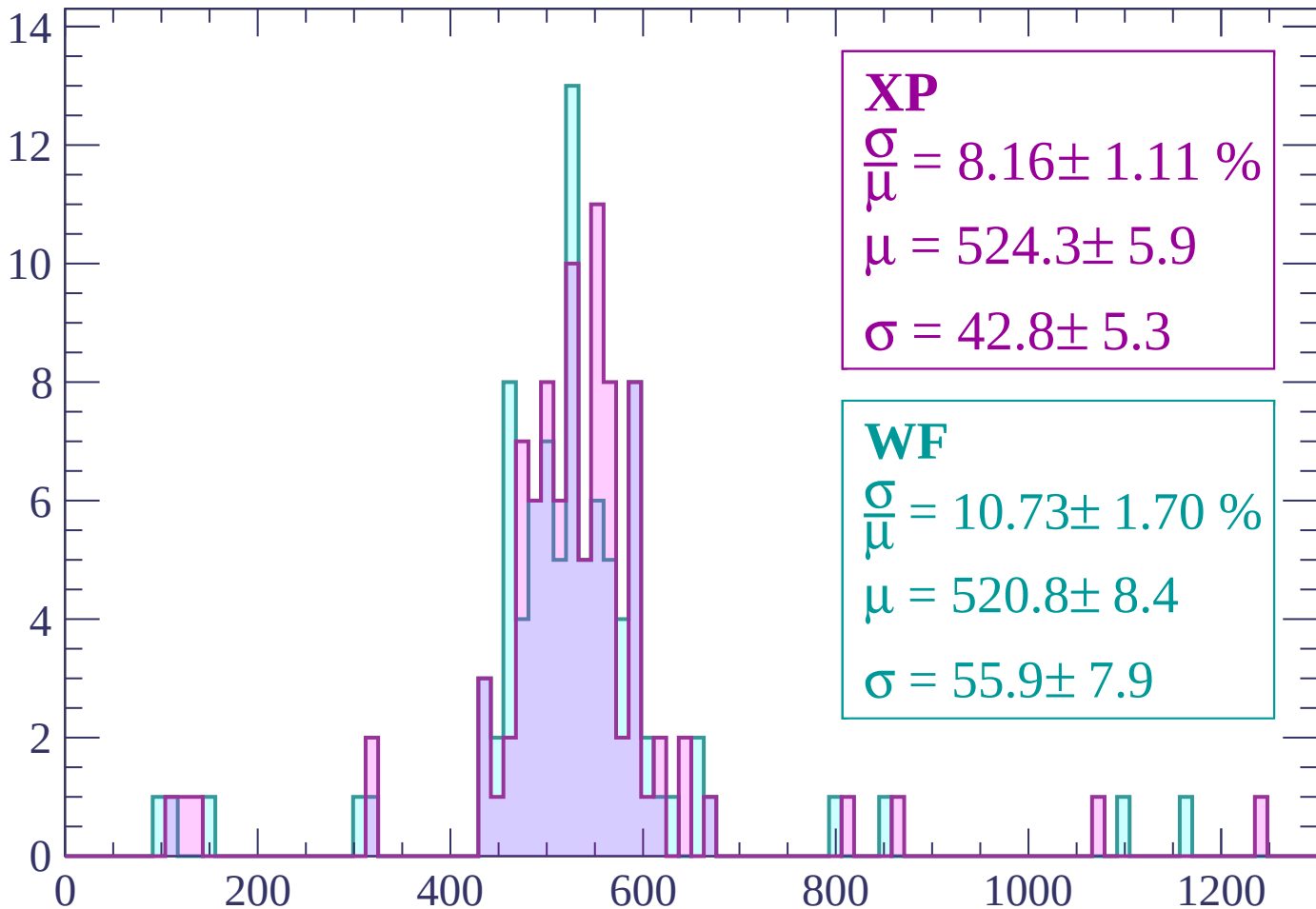
Energy loss | $-120 < p < -60$

Count



Energy loss | $-60 < p < 0$

Count



XP

$$\frac{\sigma}{\mu} = 8.16 \pm 1.11 \%$$

$$\mu = 524.3 \pm 5.9$$

$$\sigma = 42.8 \pm 5.3$$

WF

$$\frac{\sigma}{\mu} = 10.73 \pm 1.70 \%$$

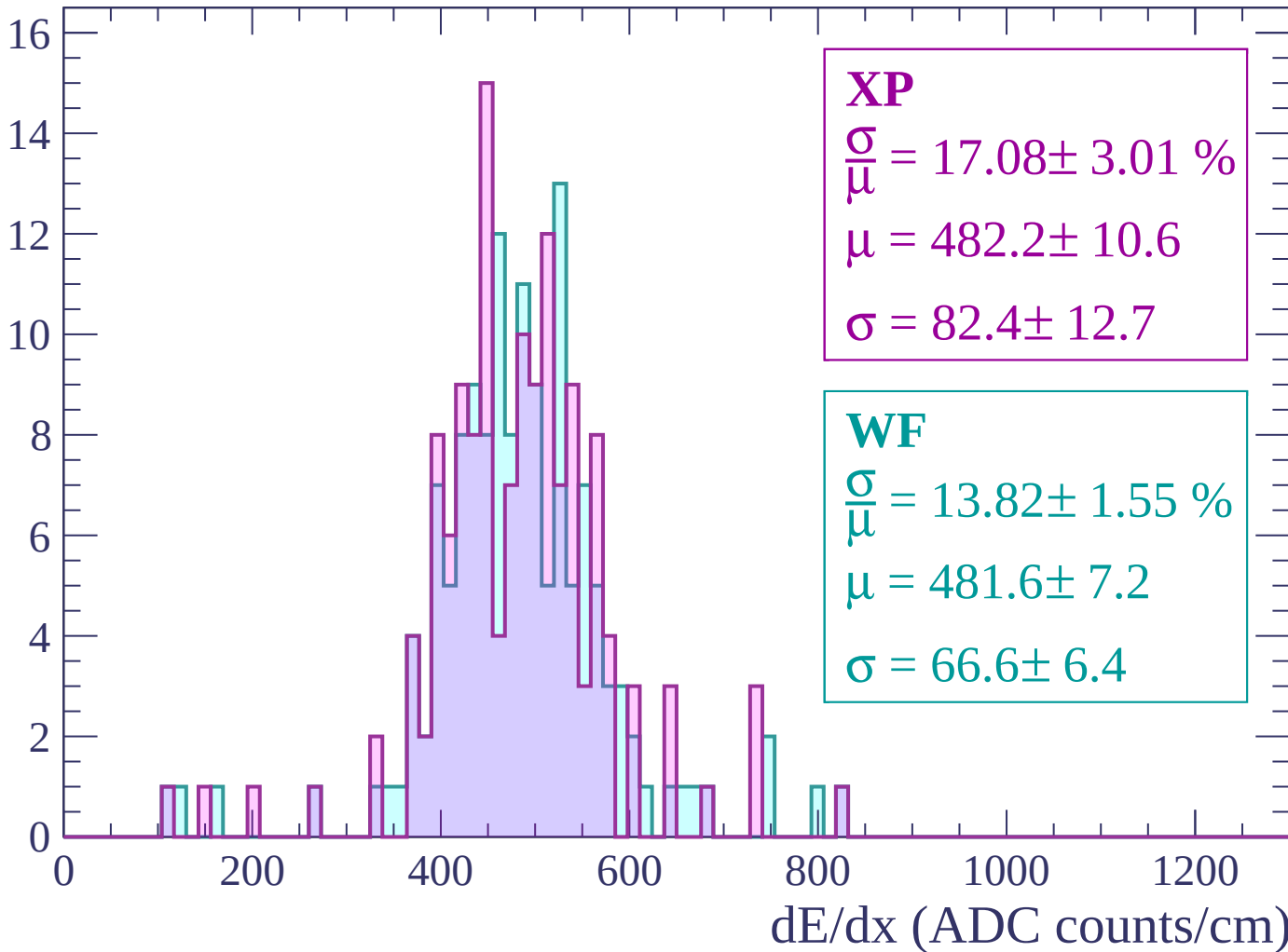
$$\mu = 520.8 \pm 8.4$$

$$\sigma = 55.9 \pm 7.9$$

dE/dx (ADC counts/cm)

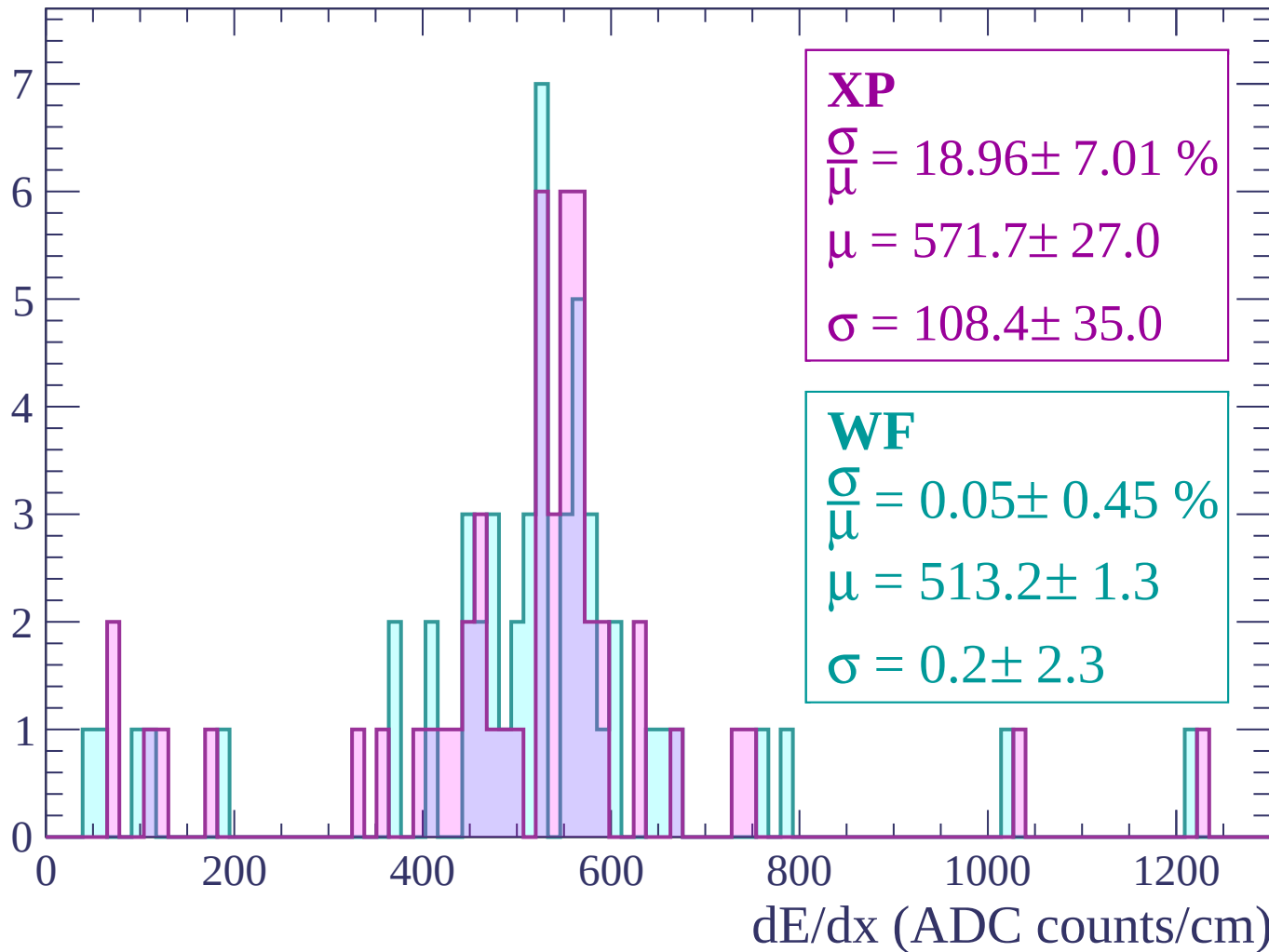
Energy loss | $0 < p < 60$

Count



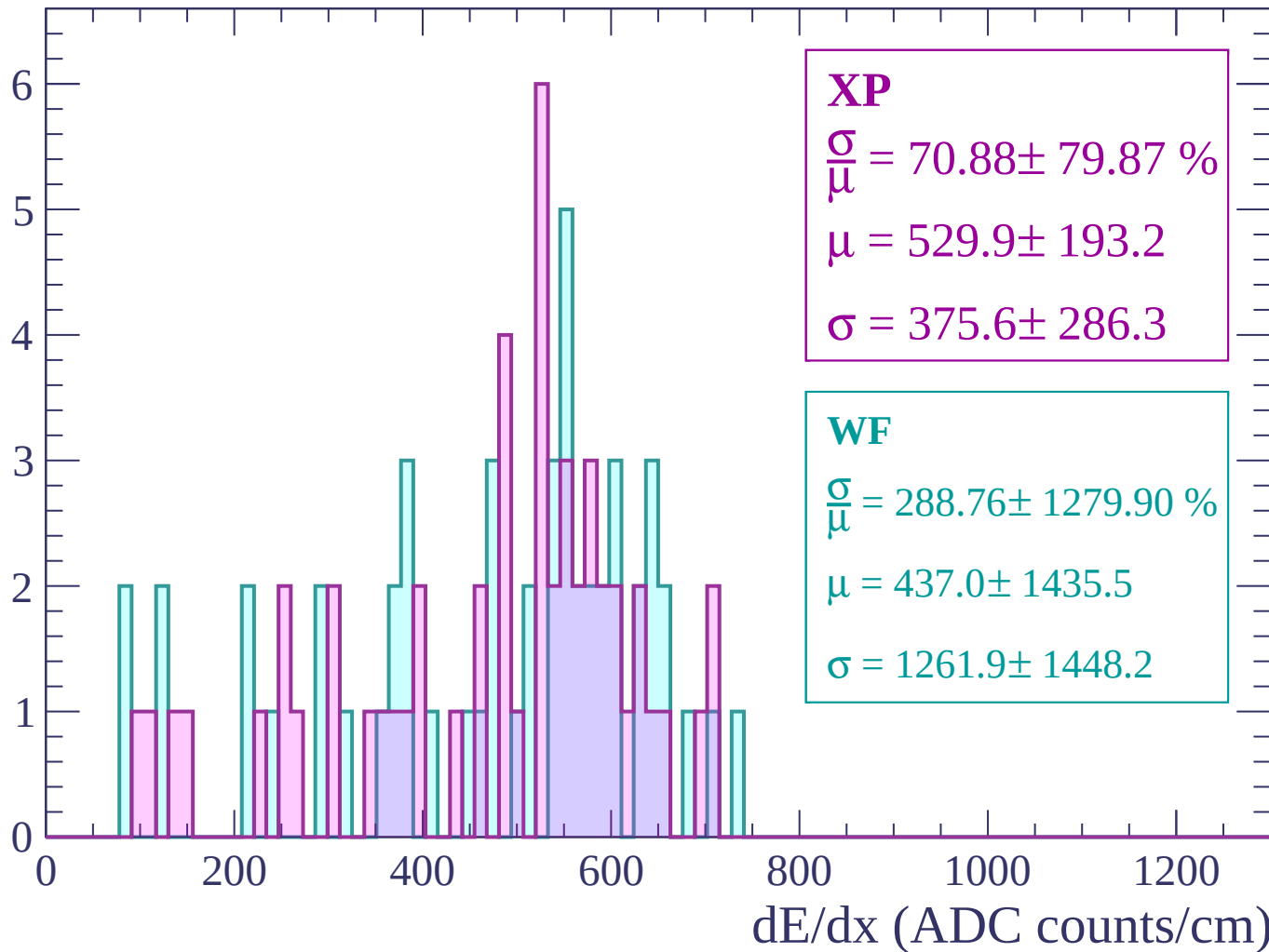
Energy loss | $60 < p < 120$

Count



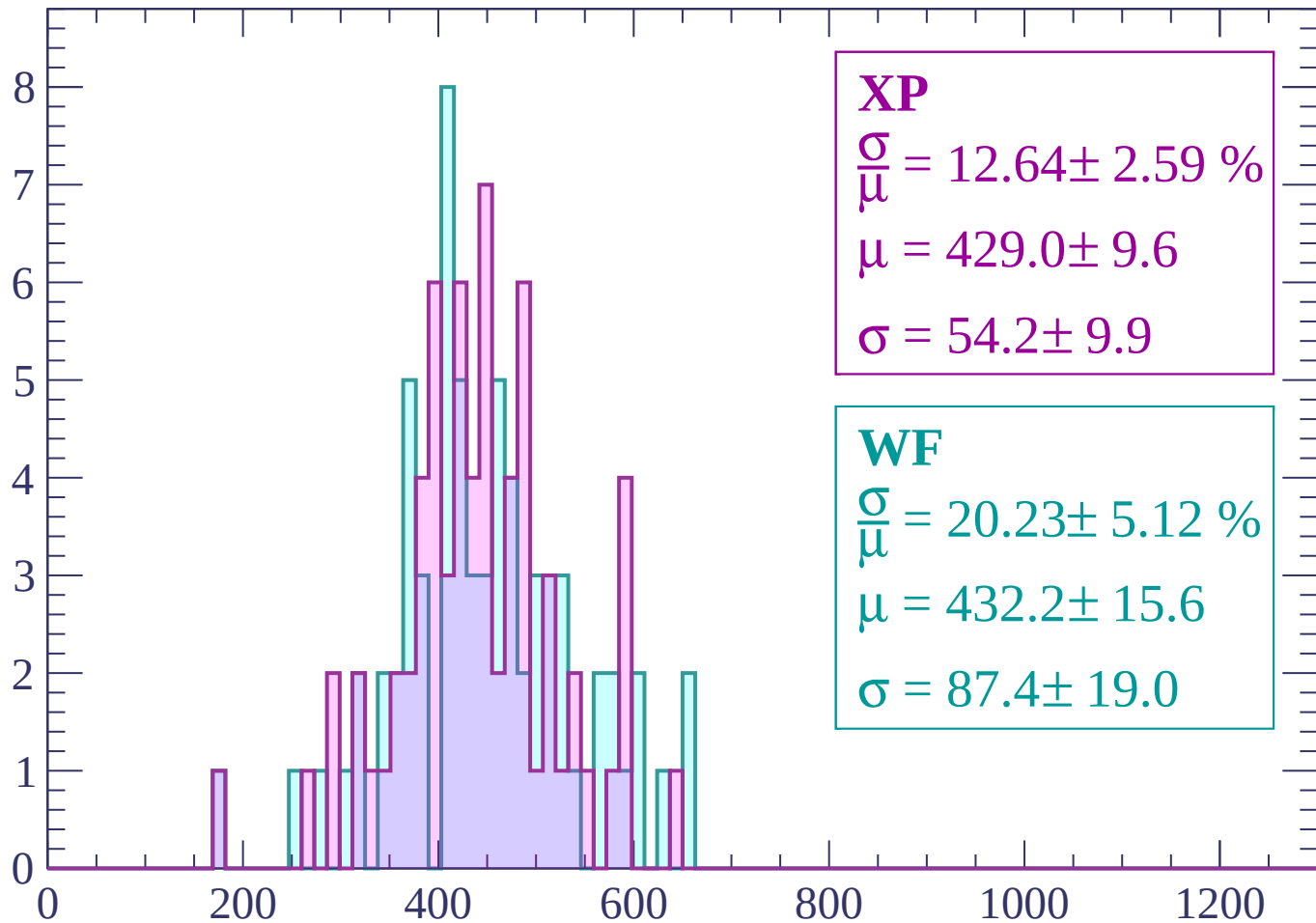
Energy loss | $120 < p < 180$

Count



Energy loss | $180 < p < 240$

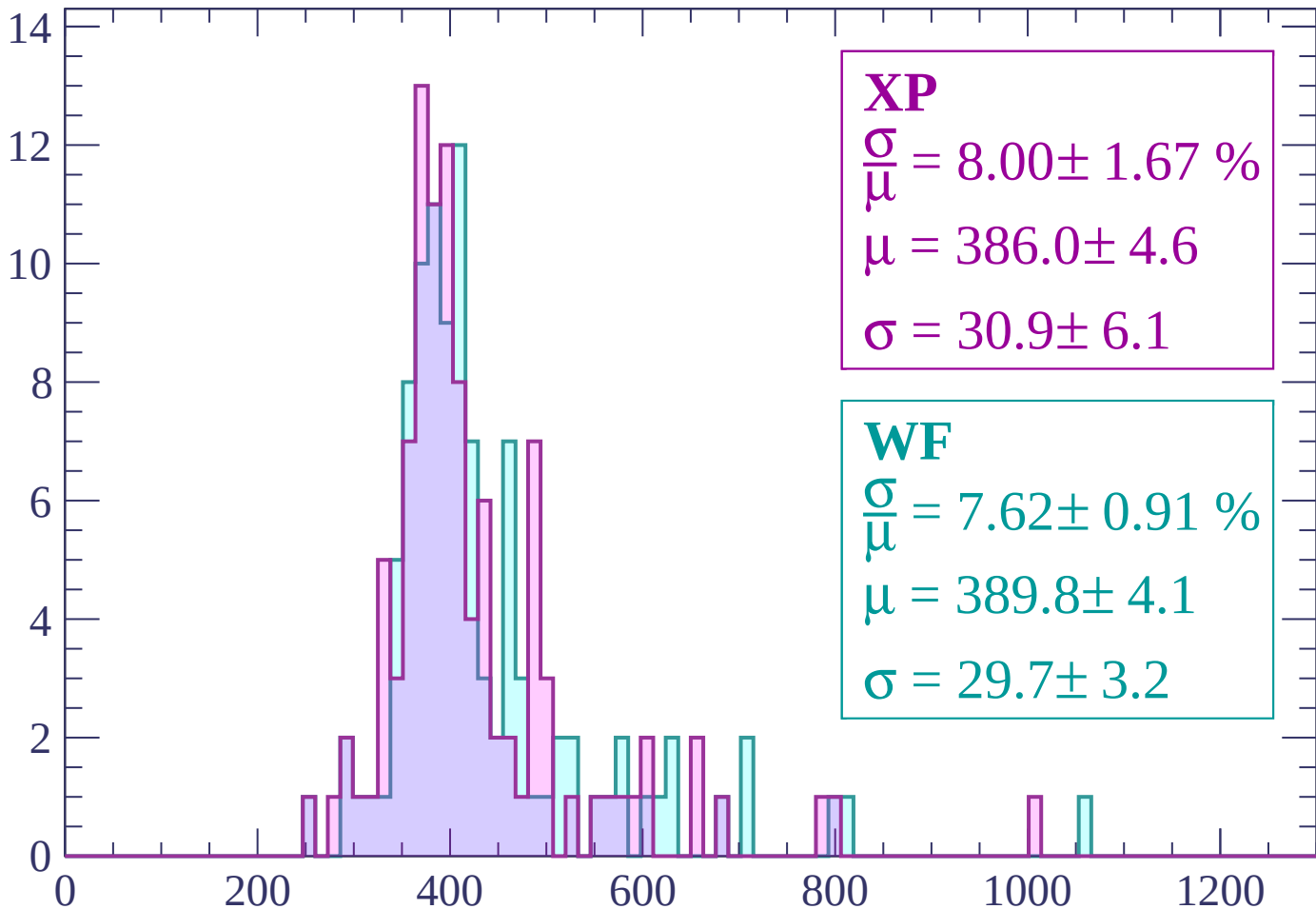
Count



dE/dx (ADC counts/cm)

Energy loss | $240 < p < 300$

Count



XP

$$\frac{\sigma}{\mu} = 8.00 \pm 1.67 \%$$

$$\mu = 386.0 \pm 4.6$$

$$\sigma = 30.9 \pm 6.1$$

WF

$$\frac{\sigma}{\mu} = 7.62 \pm 0.91 \%$$

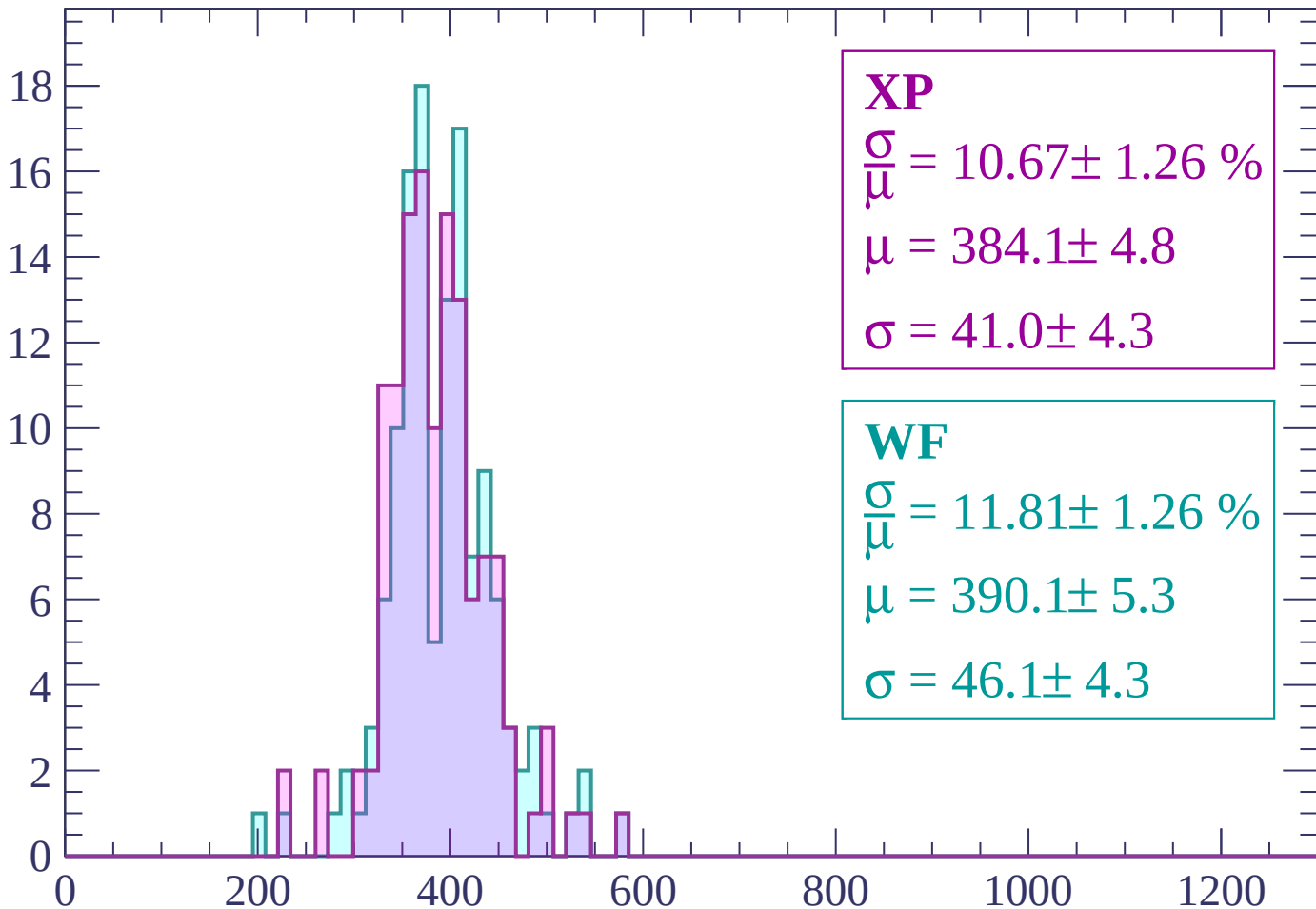
$$\mu = 389.8 \pm 4.1$$

$$\sigma = 29.7 \pm 3.2$$

dE/dx (ADC counts/cm)

Energy loss | $300 < p < 360$

Count



XP

$$\frac{\sigma}{\mu} = 10.67 \pm 1.26 \%$$

$$\mu = 384.1 \pm 4.8$$

$$\sigma = 41.0 \pm 4.3$$

WF

$$\frac{\sigma}{\mu} = 11.81 \pm 1.26 \%$$

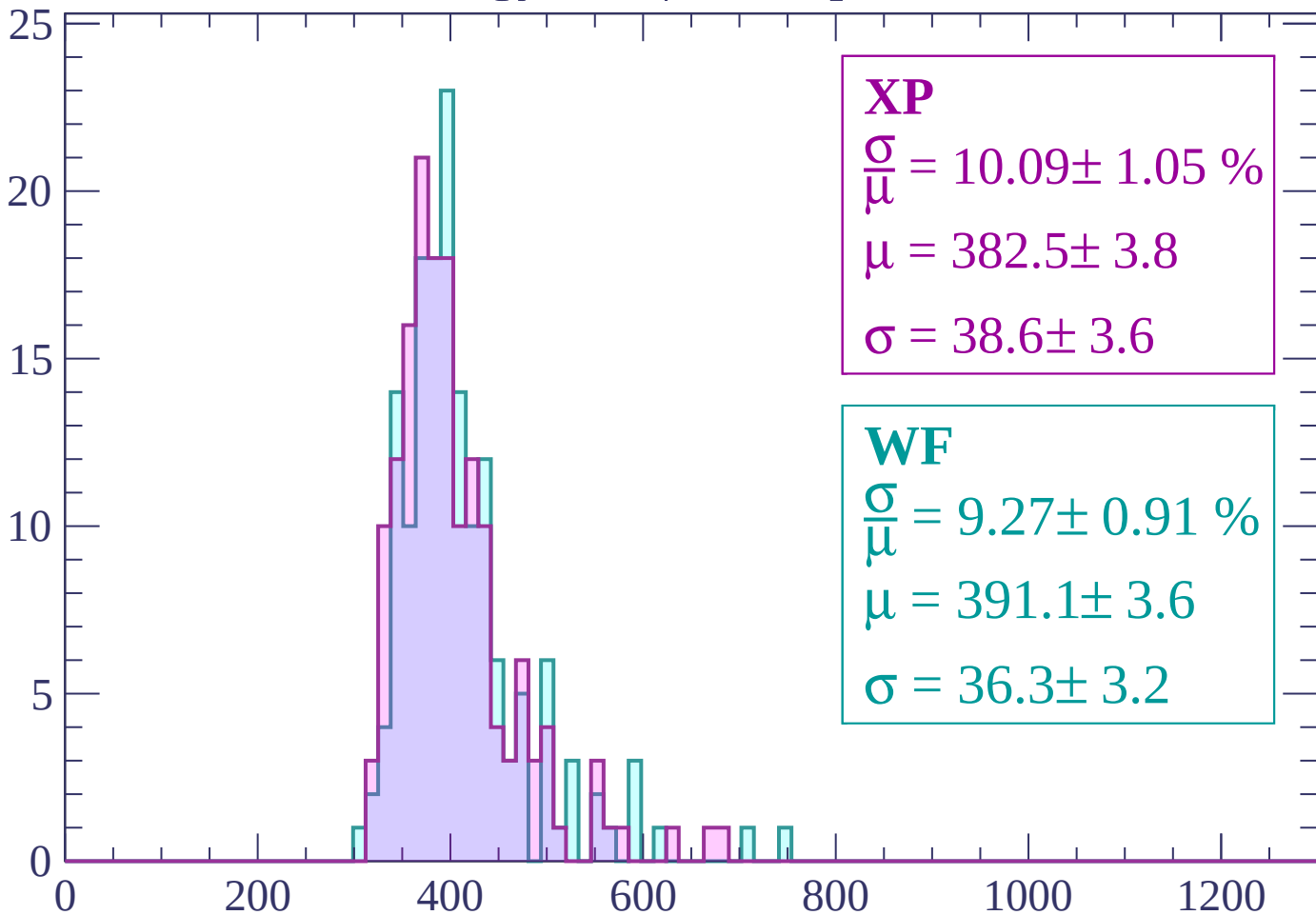
$$\mu = 390.1 \pm 5.3$$

$$\sigma = 46.1 \pm 4.3$$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 420$

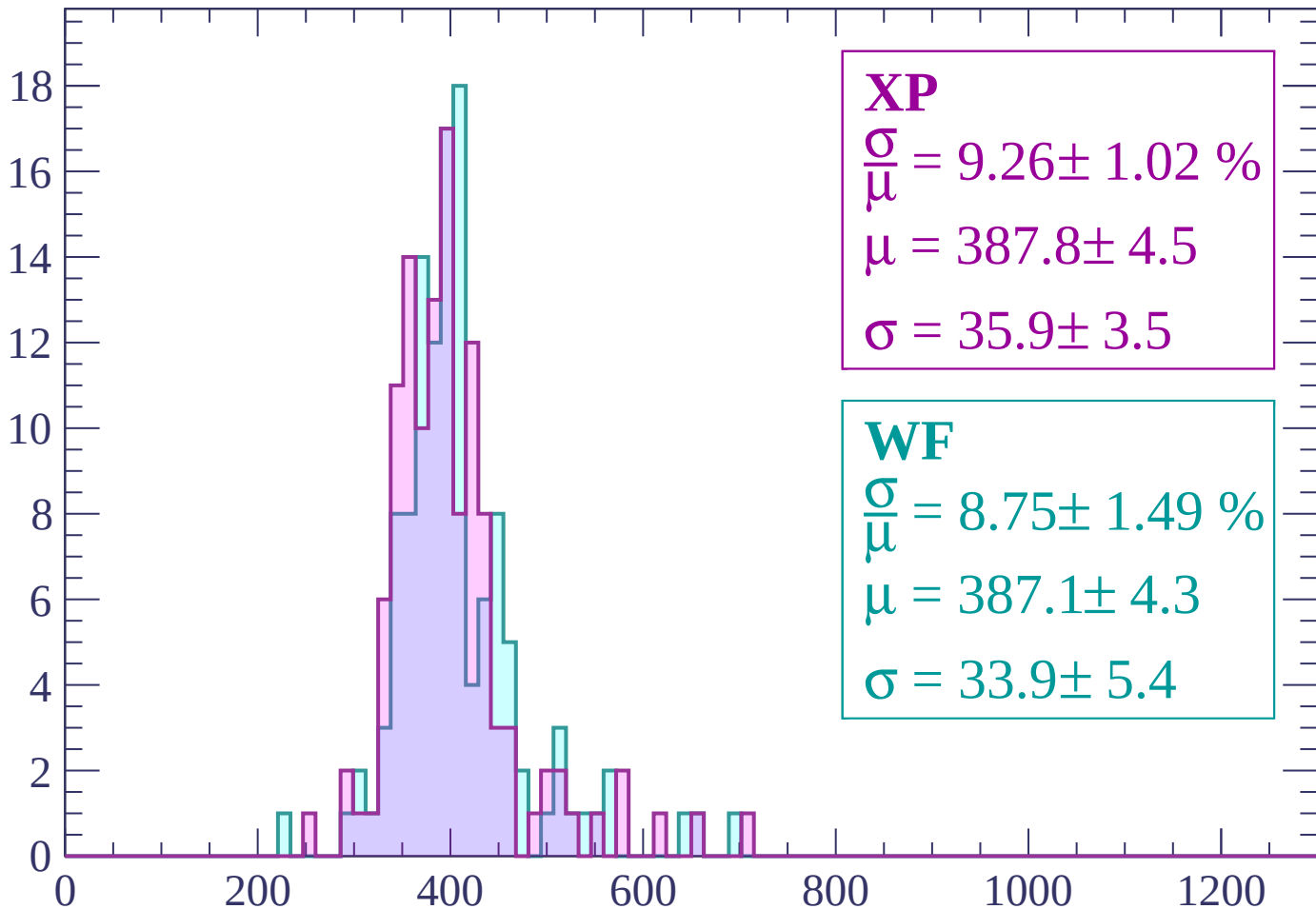
Count



dE/dx (ADC counts/cm)

Energy loss | $420 < p < 480$

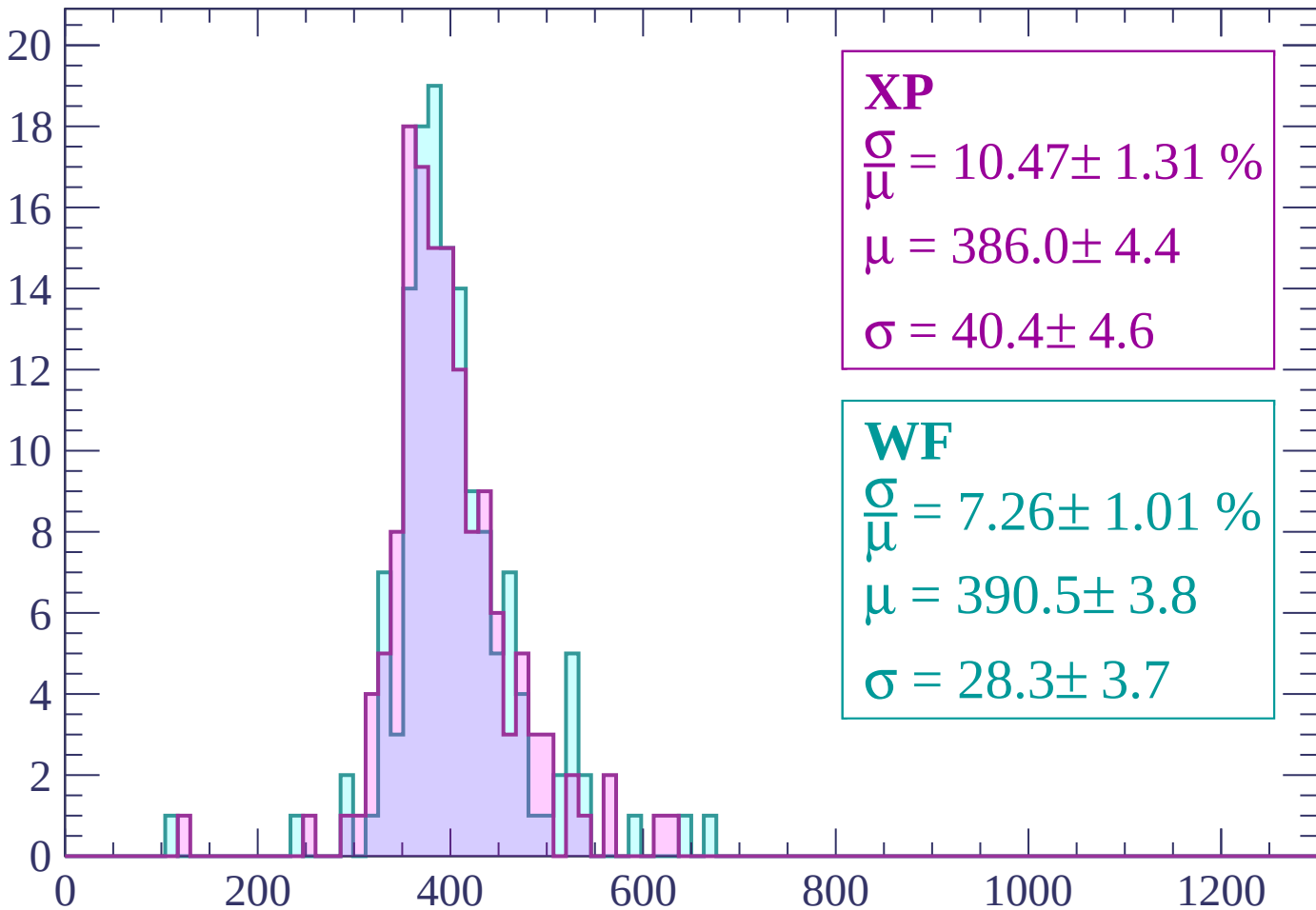
Count



dE/dx (ADC counts/cm)

Energy loss | $480 < p < 540$

Count



XP

$$\frac{\sigma}{\mu} = 10.47 \pm 1.31 \%$$

$$\mu = 386.0 \pm 4.4$$

$$\sigma = 40.4 \pm 4.6$$

WF

$$\frac{\sigma}{\mu} = 7.26 \pm 1.01 \%$$

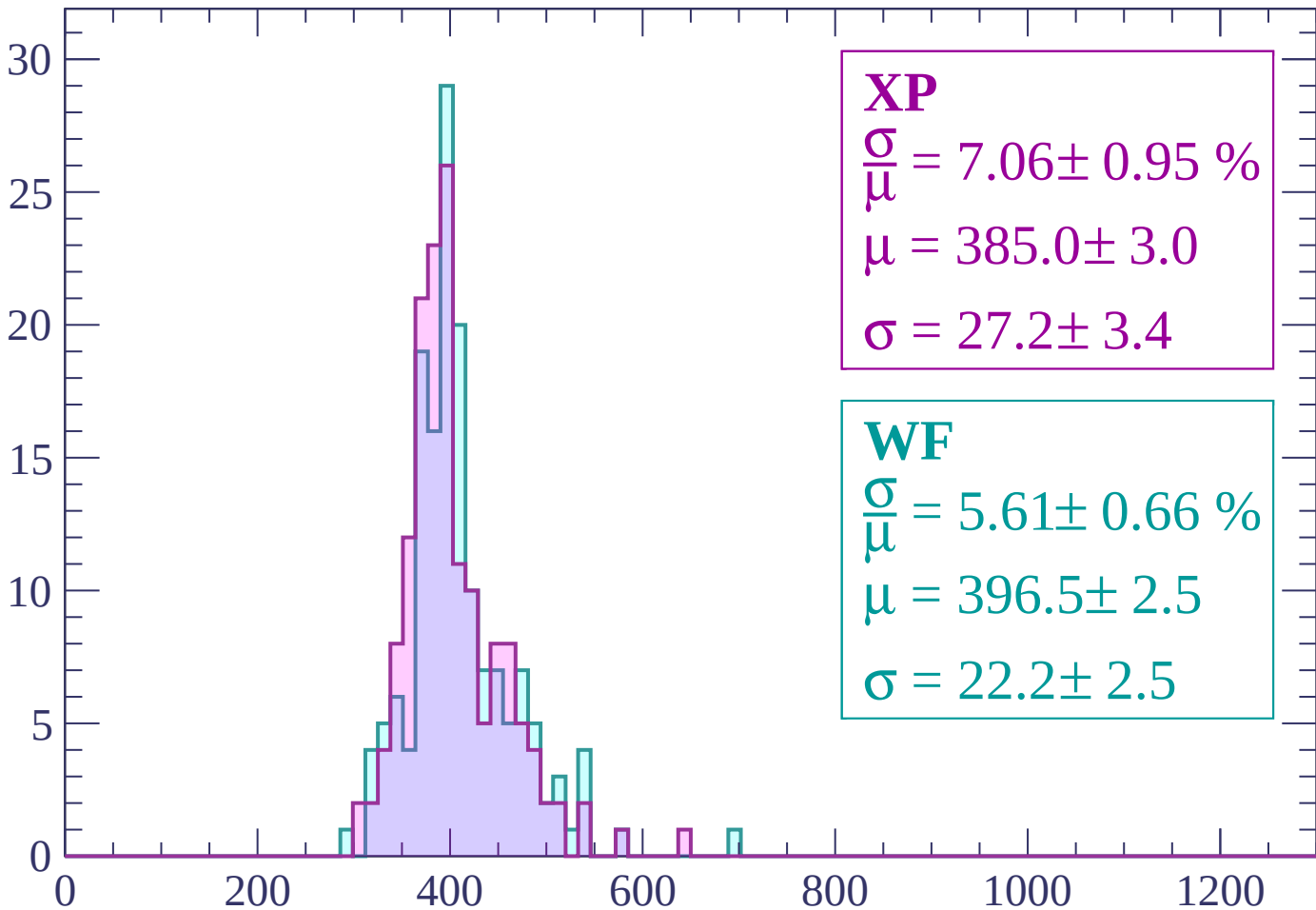
$$\mu = 390.5 \pm 3.8$$

$$\sigma = 28.3 \pm 3.7$$

dE/dx (ADC counts/cm)

Energy loss | $540 < p < 600$

Count



XP

$$\frac{\sigma}{\mu} = 7.06 \pm 0.95 \%$$

$$\mu = 385.0 \pm 3.0$$

$$\sigma = 27.2 \pm 3.4$$

WF

$$\frac{\sigma}{\mu} = 5.61 \pm 0.66 \%$$

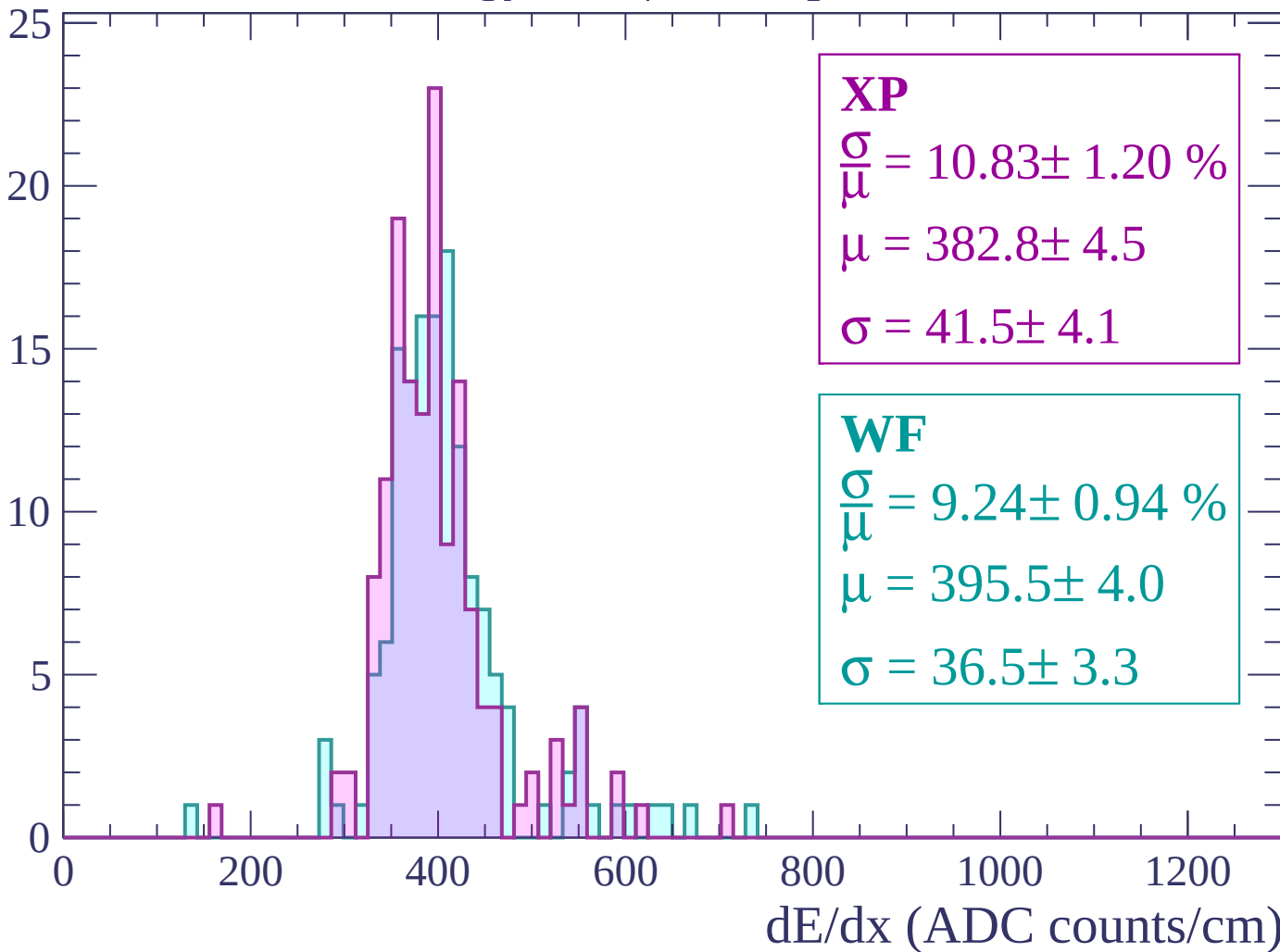
$$\mu = 396.5 \pm 2.5$$

$$\sigma = 22.2 \pm 2.5$$

dE/dx (ADC counts/cm)

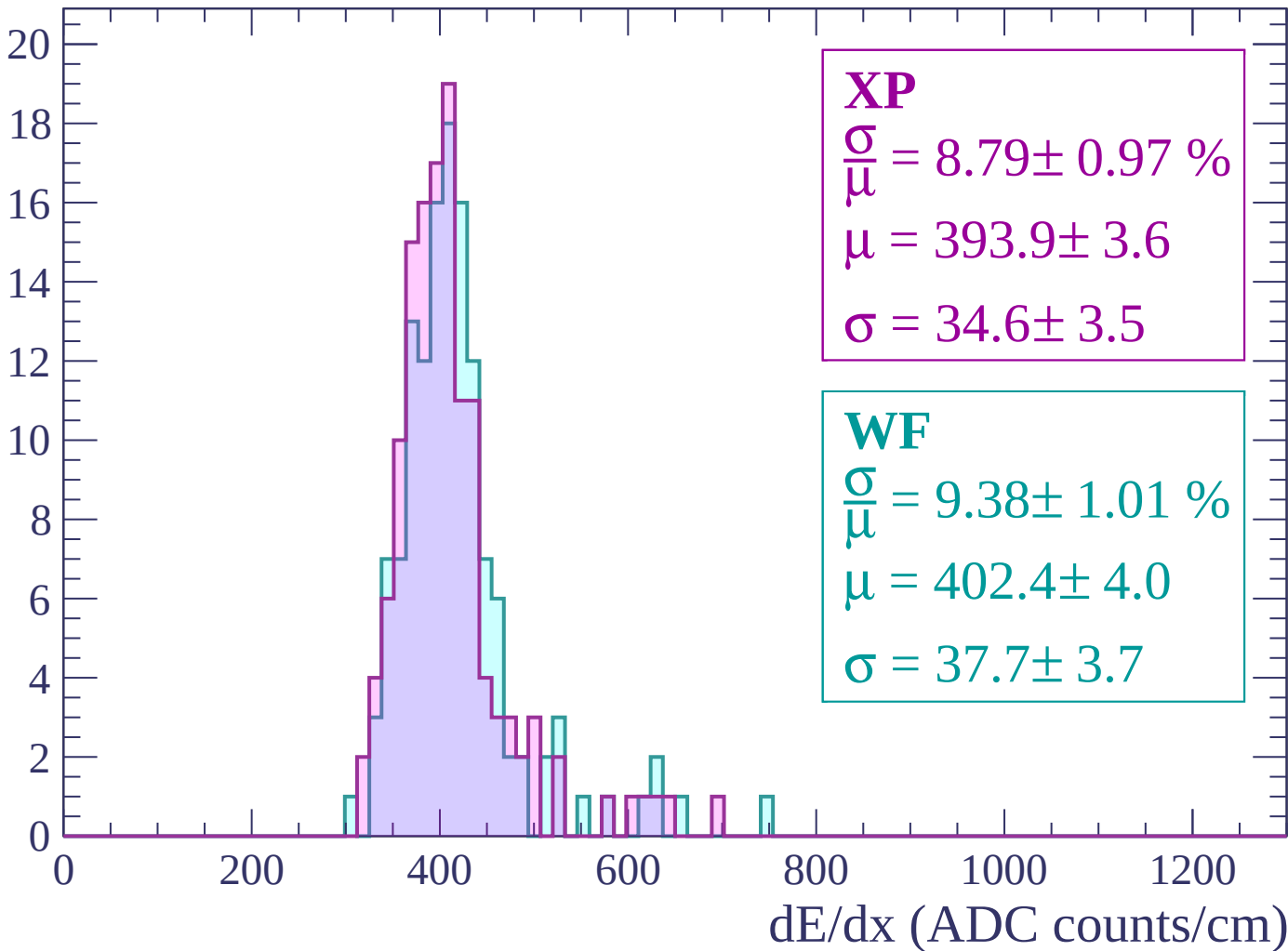
Energy loss | $600 < p < 660$

Count



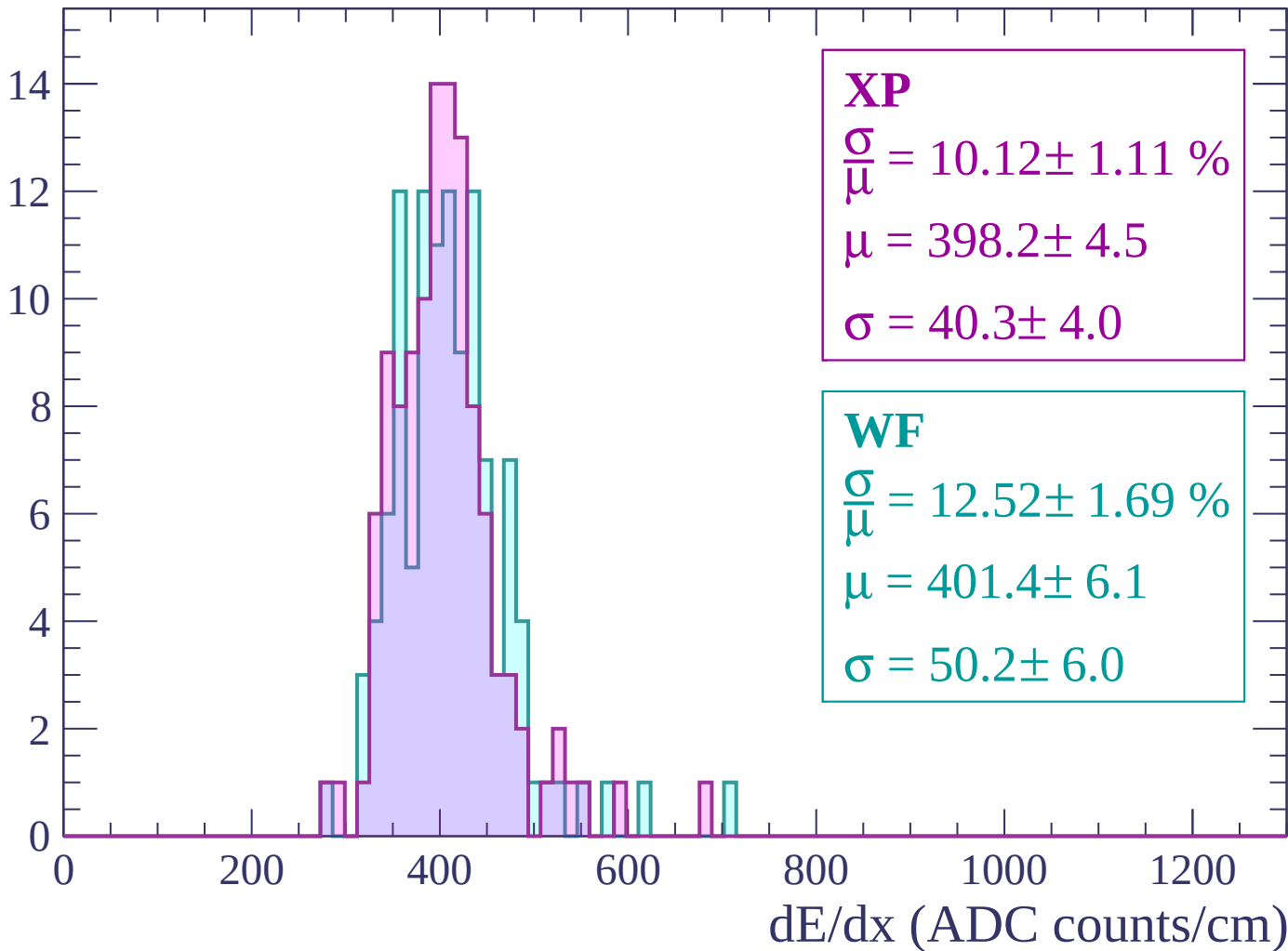
Energy loss | $660 < p < 720$

Count



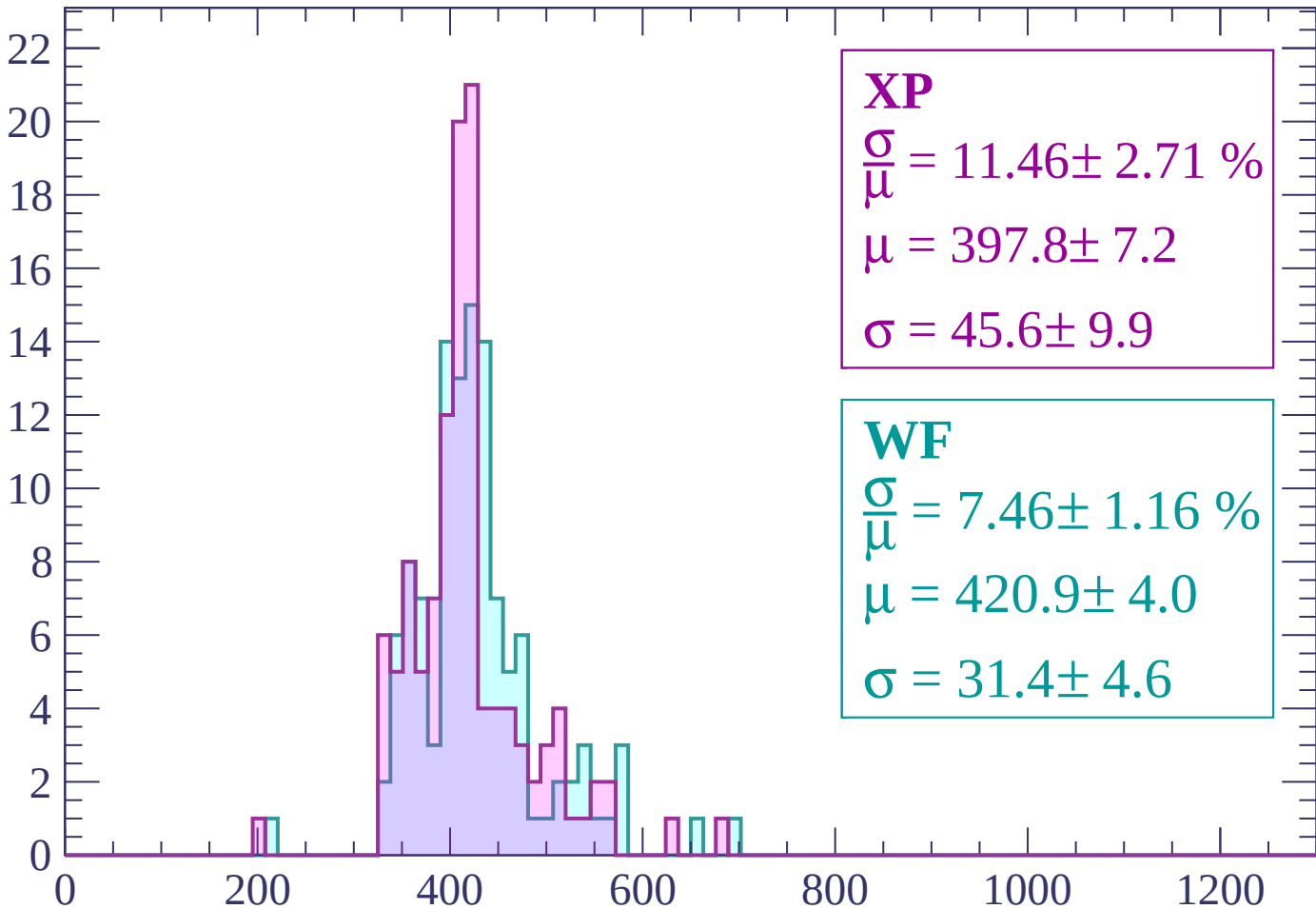
Energy loss | $720 < p < 780$

Count



Energy loss | $780 < p < 840$

Count



XP

$$\frac{\sigma}{\mu} = 11.46 \pm 2.71 \%$$

$$\mu = 397.8 \pm 7.2$$

$$\sigma = 45.6 \pm 9.9$$

WF

$$\frac{\sigma}{\mu} = 7.46 \pm 1.16 \%$$

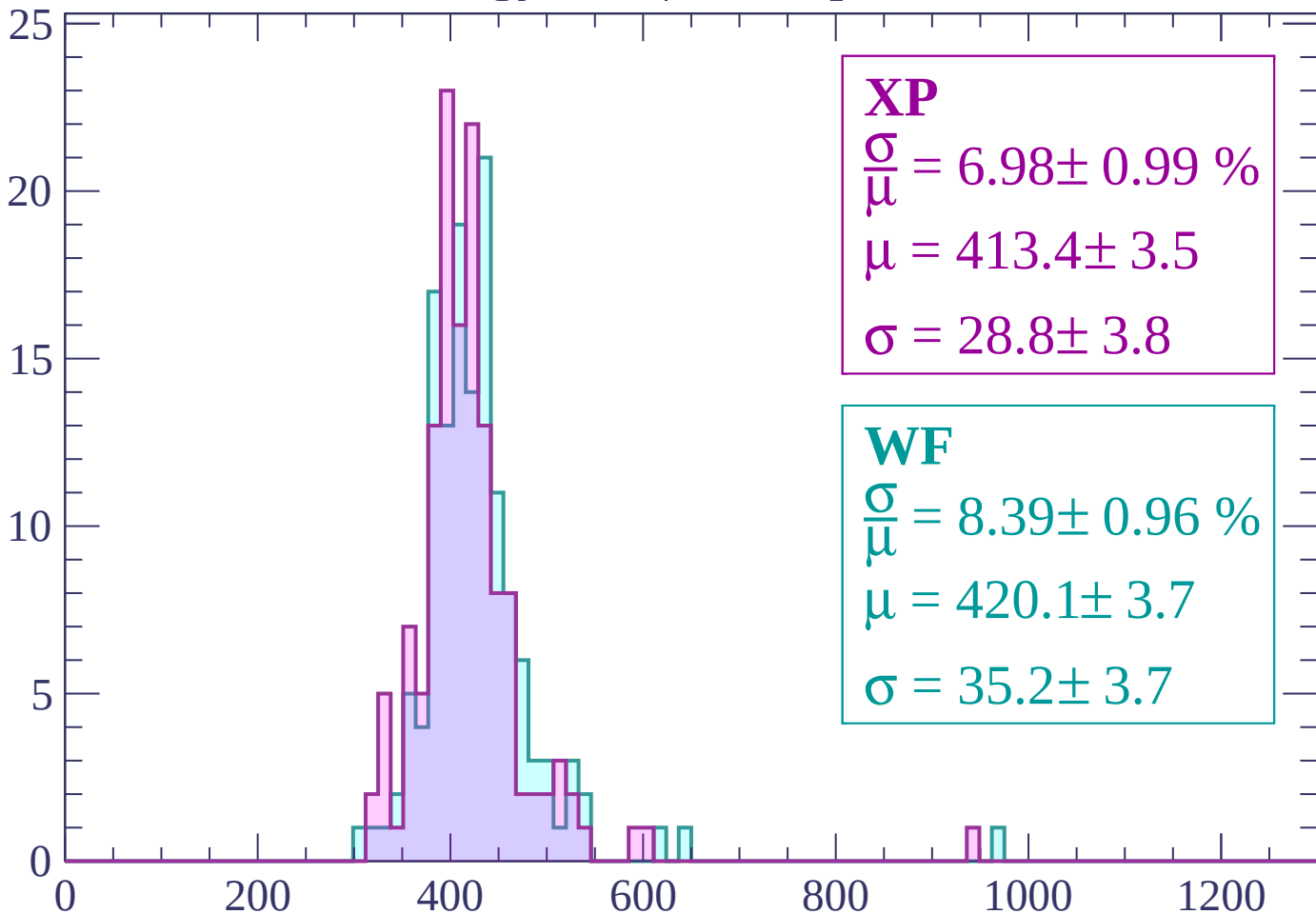
$$\mu = 420.9 \pm 4.0$$

$$\sigma = 31.4 \pm 4.6$$

dE/dx (ADC counts/cm)

Energy loss | $840 < p < 900$

Count



XP

$$\frac{\sigma}{\mu} = 6.98 \pm 0.99 \%$$

$$\mu = 413.4 \pm 3.5$$

$$\sigma = 28.8 \pm 3.8$$

WF

$$\frac{\sigma}{\mu} = 8.39 \pm 0.96 \%$$

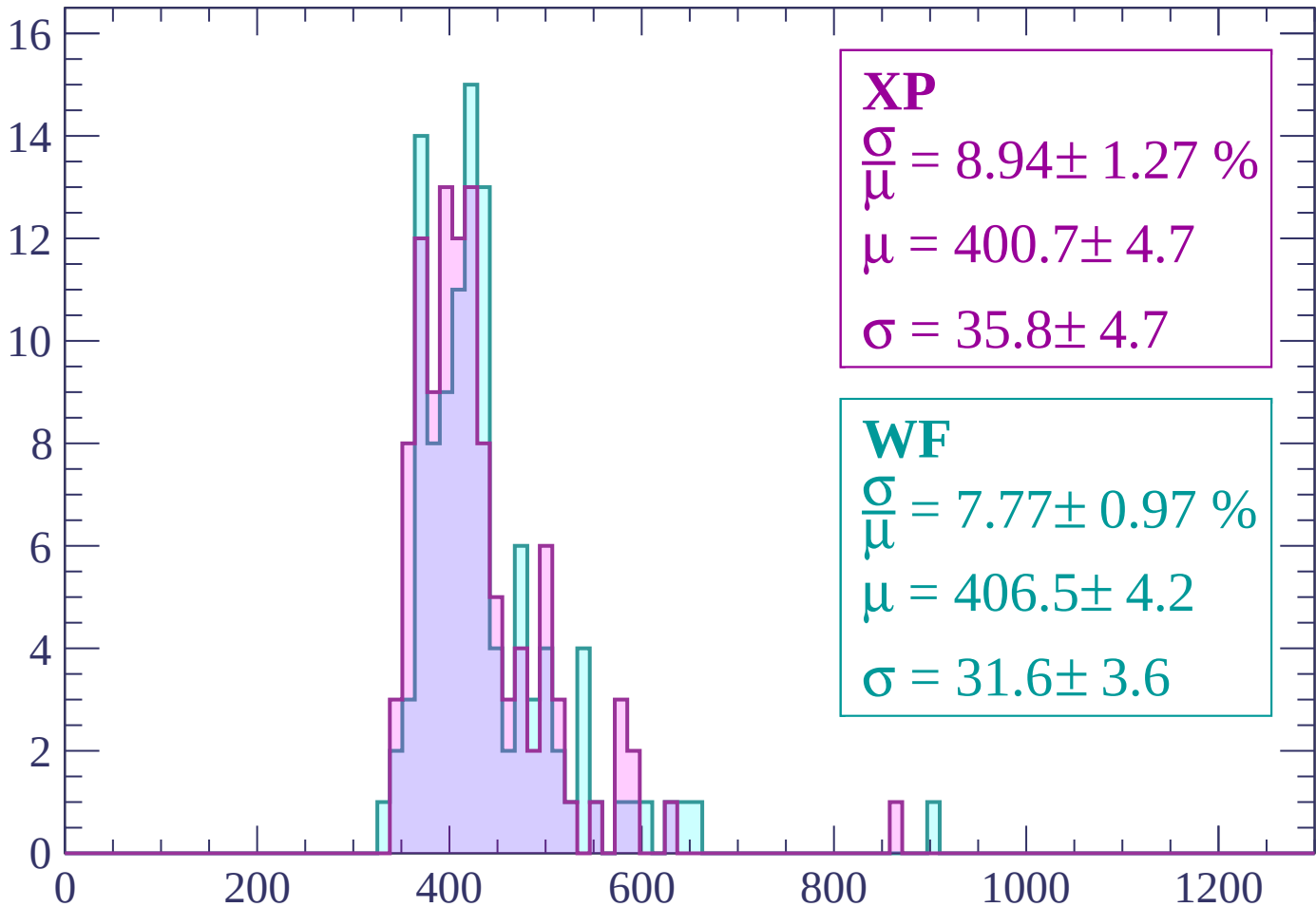
$$\mu = 420.1 \pm 3.7$$

$$\sigma = 35.2 \pm 3.7$$

dE/dx (ADC counts/cm)

Energy loss | $900 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 8.94 \pm 1.27 \%$$

$$\mu = 400.7 \pm 4.7$$

$$\sigma = 35.8 \pm 4.7$$

WF

$$\frac{\sigma}{\mu} = 7.77 \pm 0.97 \%$$

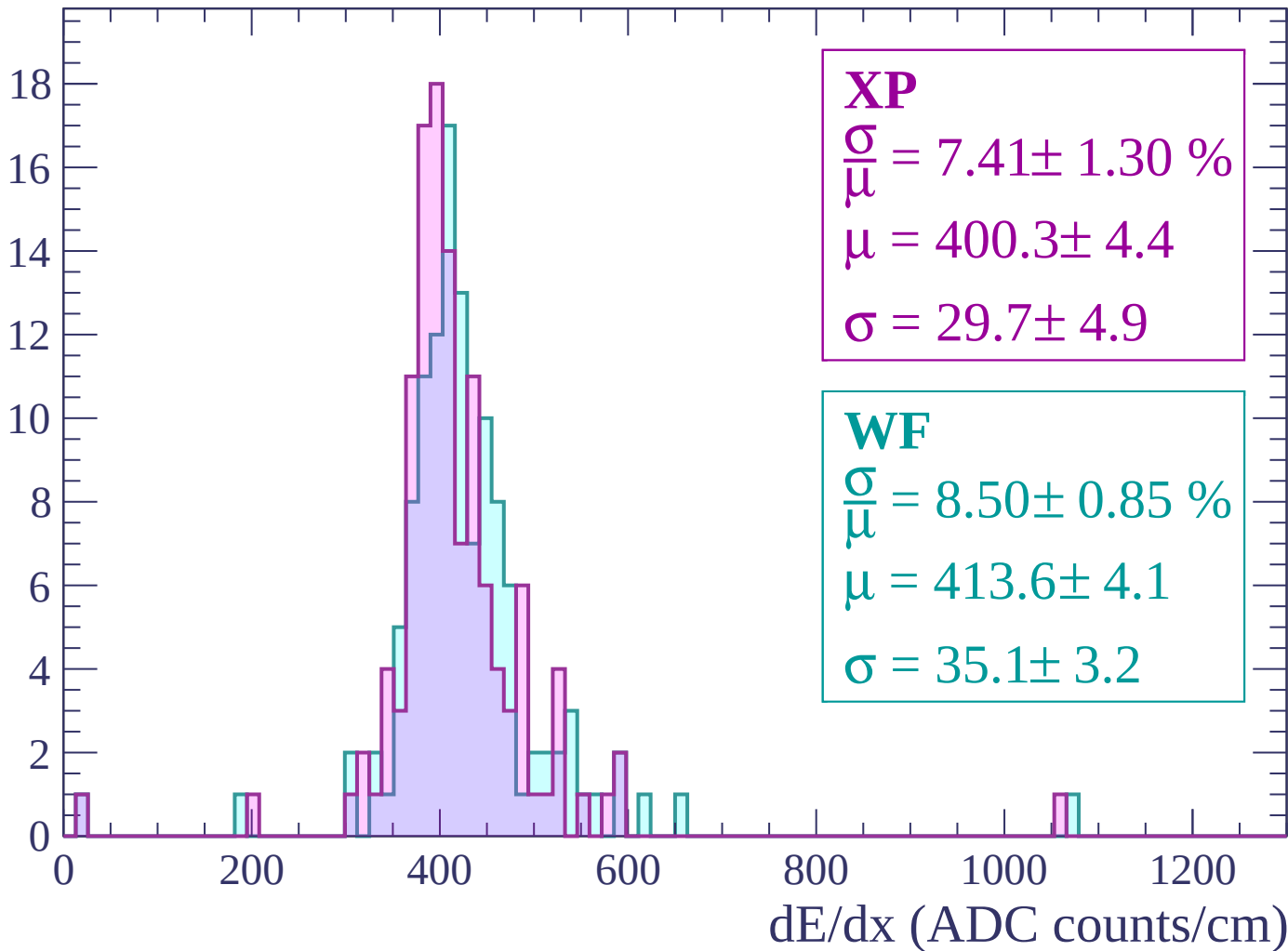
$$\mu = 406.5 \pm 4.2$$

$$\sigma = 31.6 \pm 3.6$$

dE/dx (ADC counts/cm)

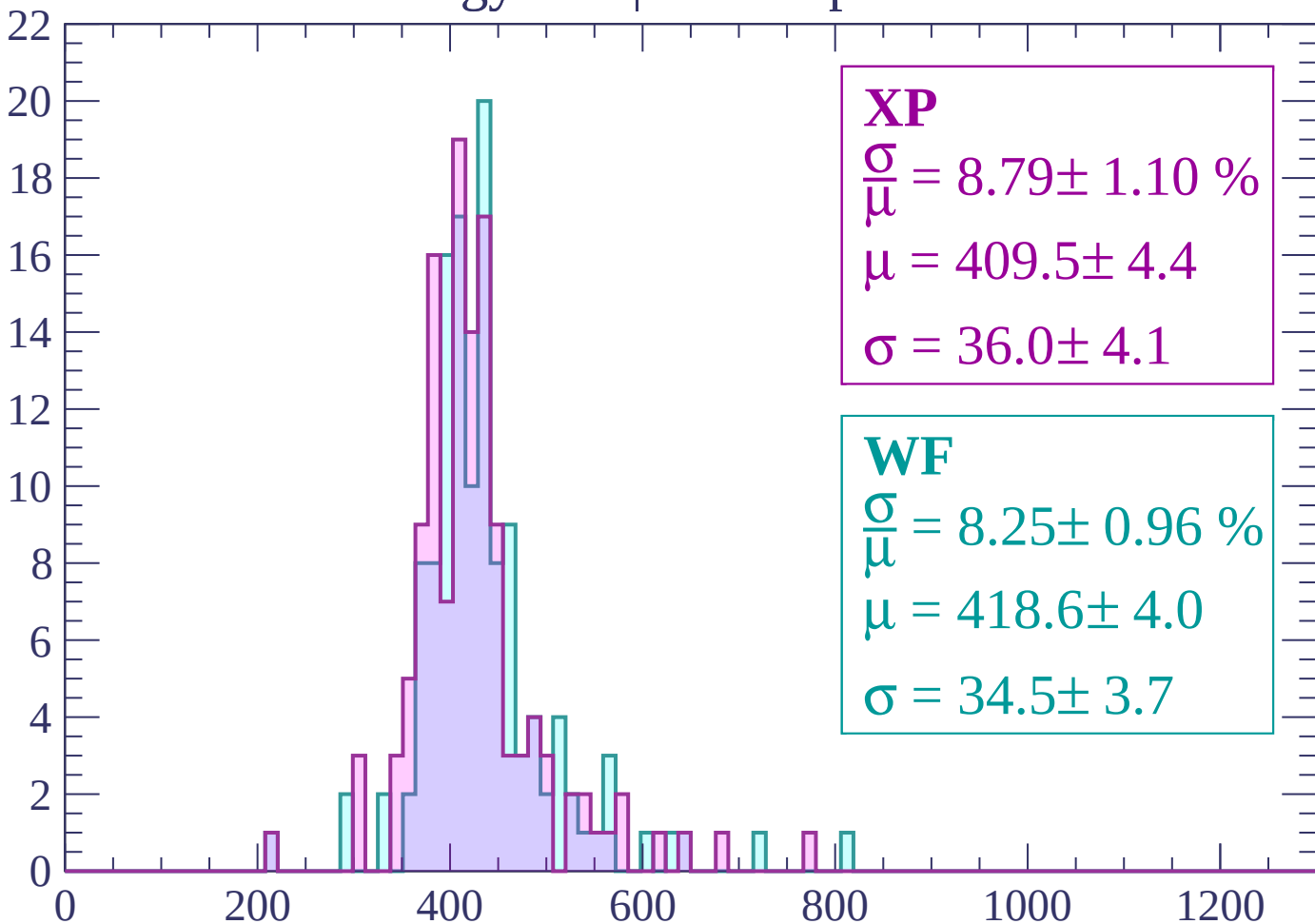
Energy loss | $960 < p < 1020$

Count



Energy loss | $1020 < p < 1080$

Count



XP

$$\frac{\sigma}{\mu} = 8.79 \pm 1.10 \%$$

$$\mu = 409.5 \pm 4.4$$

$$\sigma = 36.0 \pm 4.1$$

WF

$$\frac{\sigma}{\mu} = 8.25 \pm 0.96 \%$$

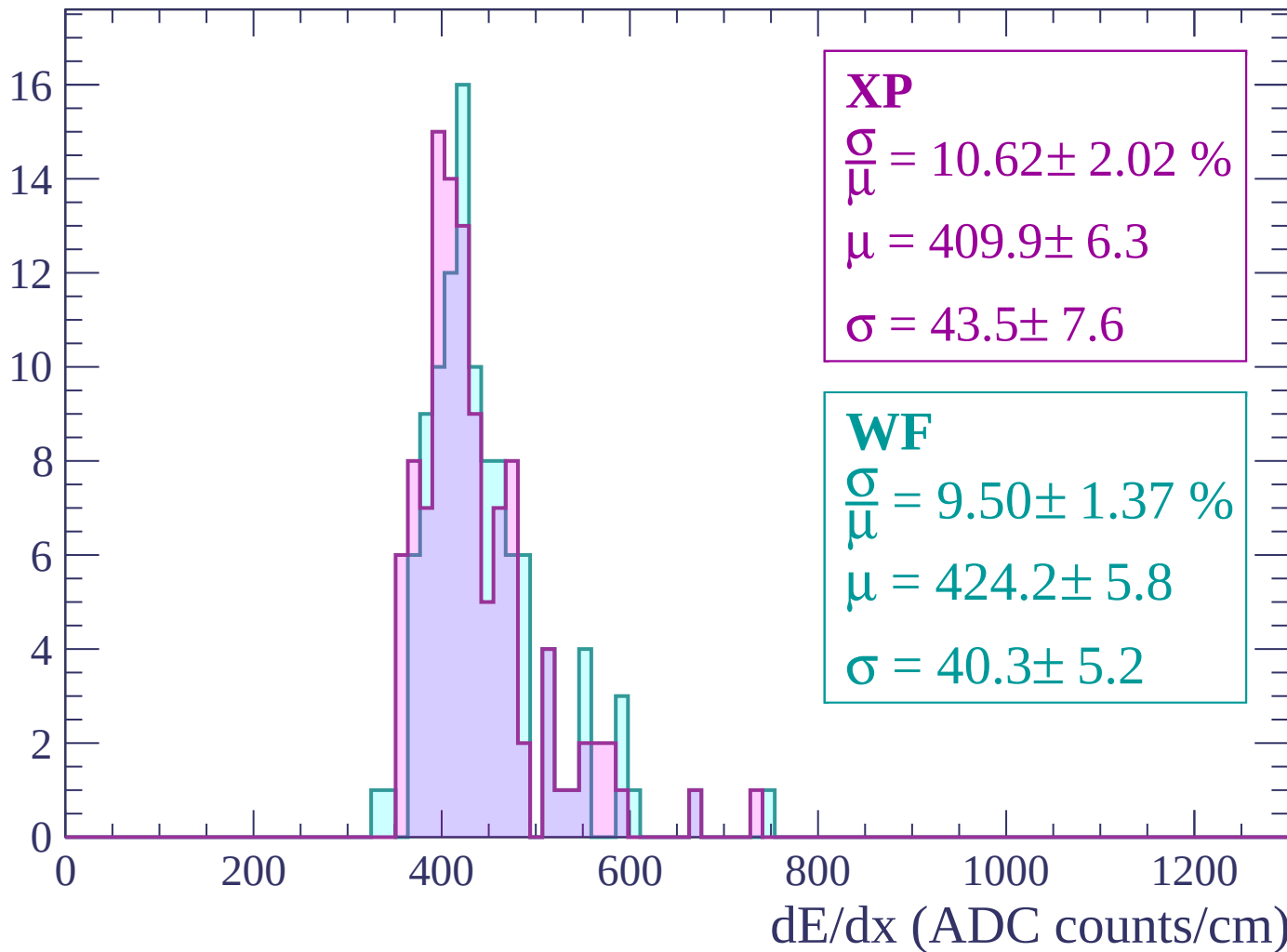
$$\mu = 418.6 \pm 4.0$$

$$\sigma = 34.5 \pm 3.7$$

dE/dx (ADC counts/cm)

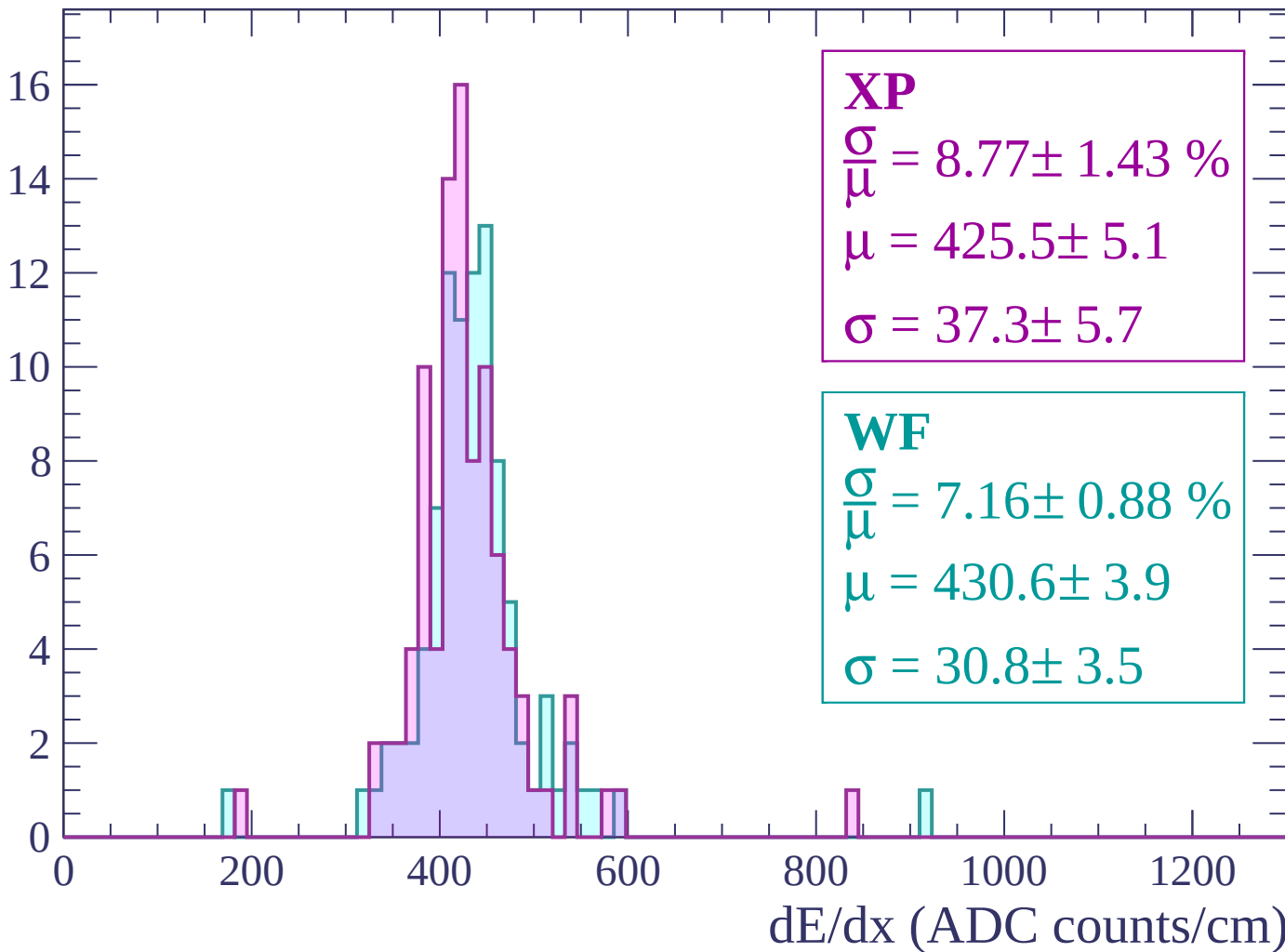
Energy loss | $1080 < p < 1140$

Count



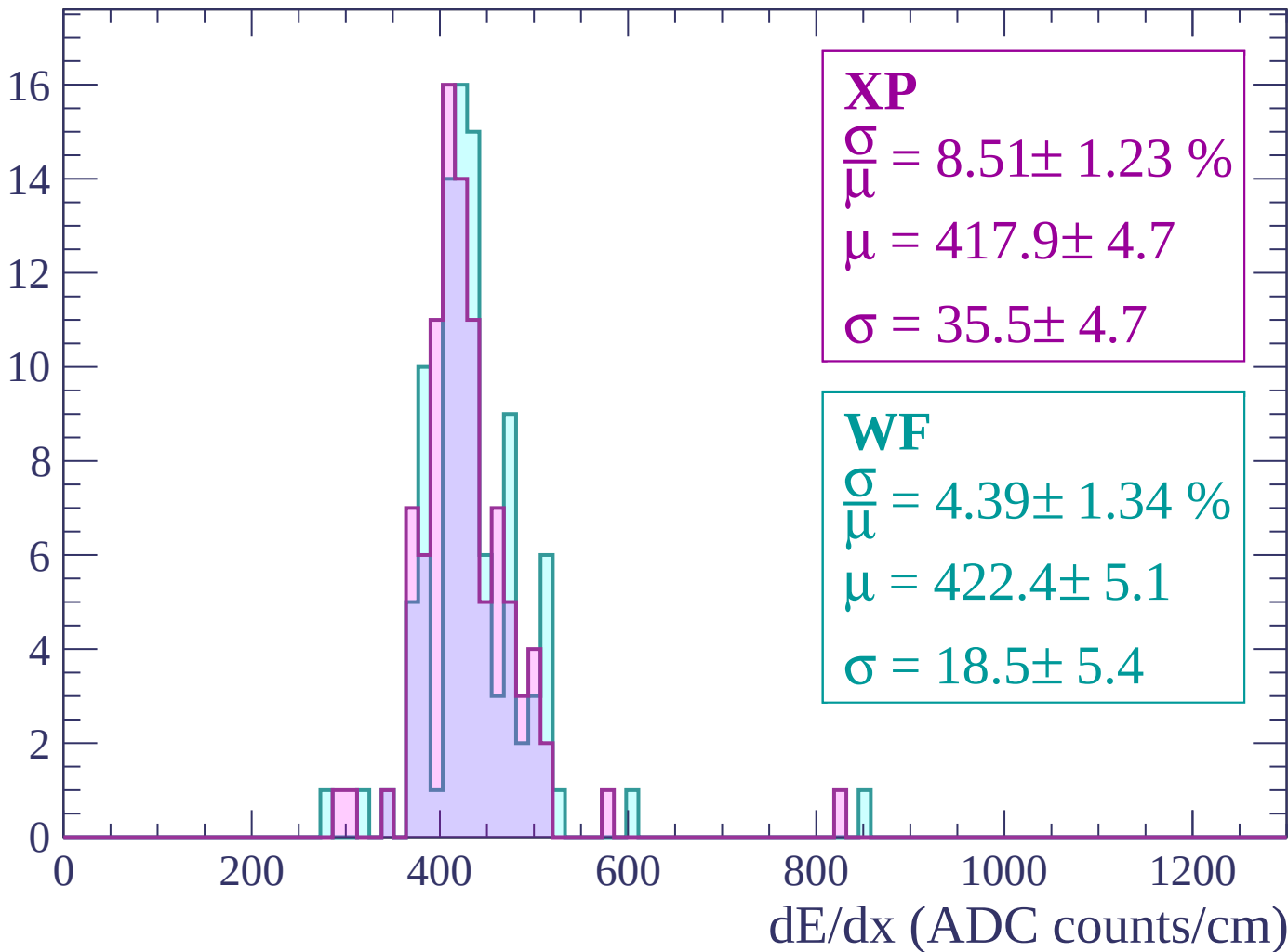
Energy loss | $1140 < p < 1200$

Count



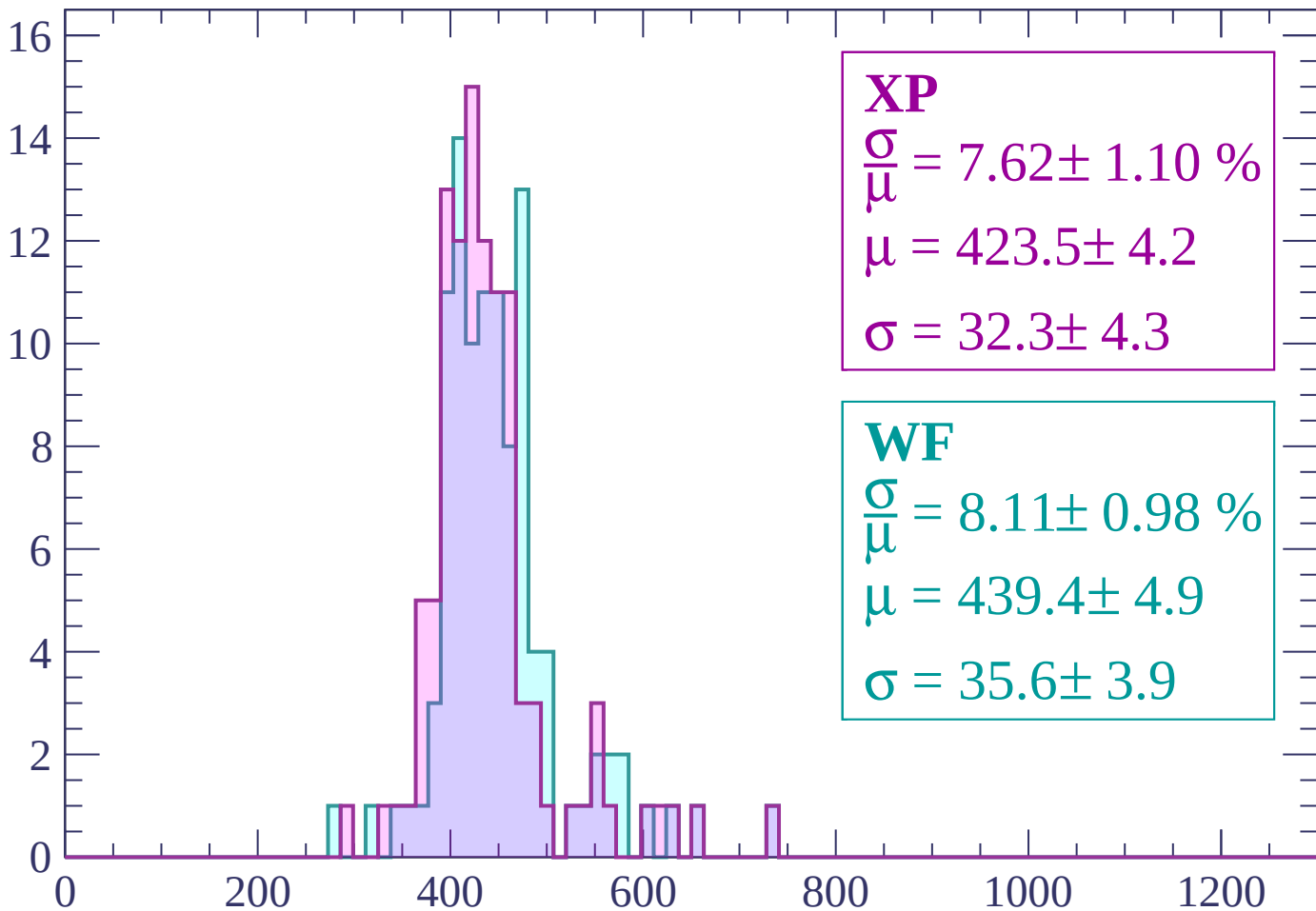
Energy loss | $1200 < p < 1260$

Count



Energy loss | $1260 < p < 1320$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 7.62 \pm 1.10 \%$$

$$\mu = 423.5 \pm 4.2$$

$$\sigma = 32.3 \pm 4.3$$

WF

$$\frac{\sigma}{\bar{\mu}} = 8.11 \pm 0.98 \%$$

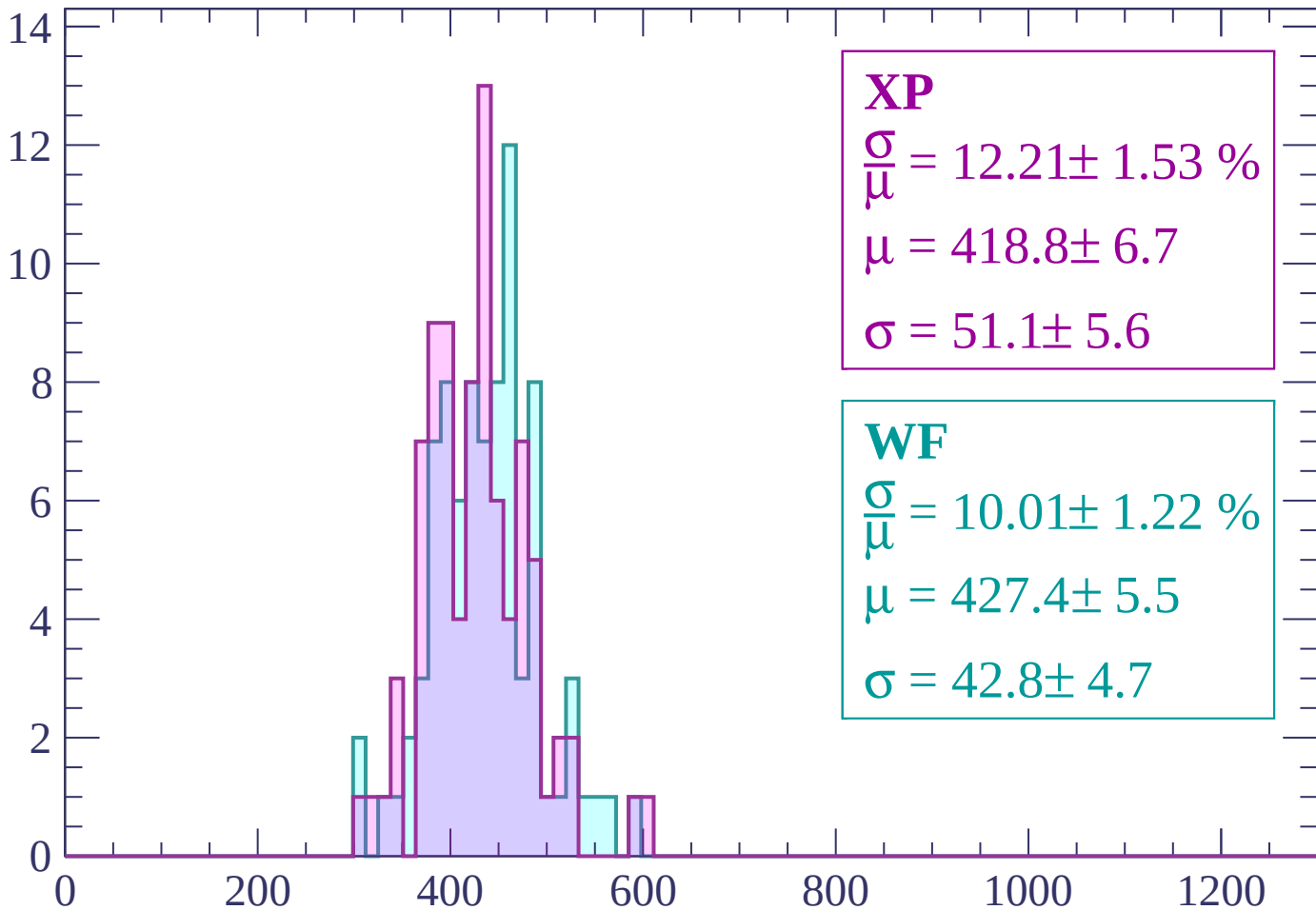
$$\mu = 439.4 \pm 4.9$$

$$\sigma = 35.6 \pm 3.9$$

dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1380$

Count



XP

$$\frac{\sigma}{\mu} = 12.21 \pm 1.53 \%$$

$$\mu = 418.8 \pm 6.7$$

$$\sigma = 51.1 \pm 5.6$$

WF

$$\frac{\sigma}{\mu} = 10.01 \pm 1.22 \%$$

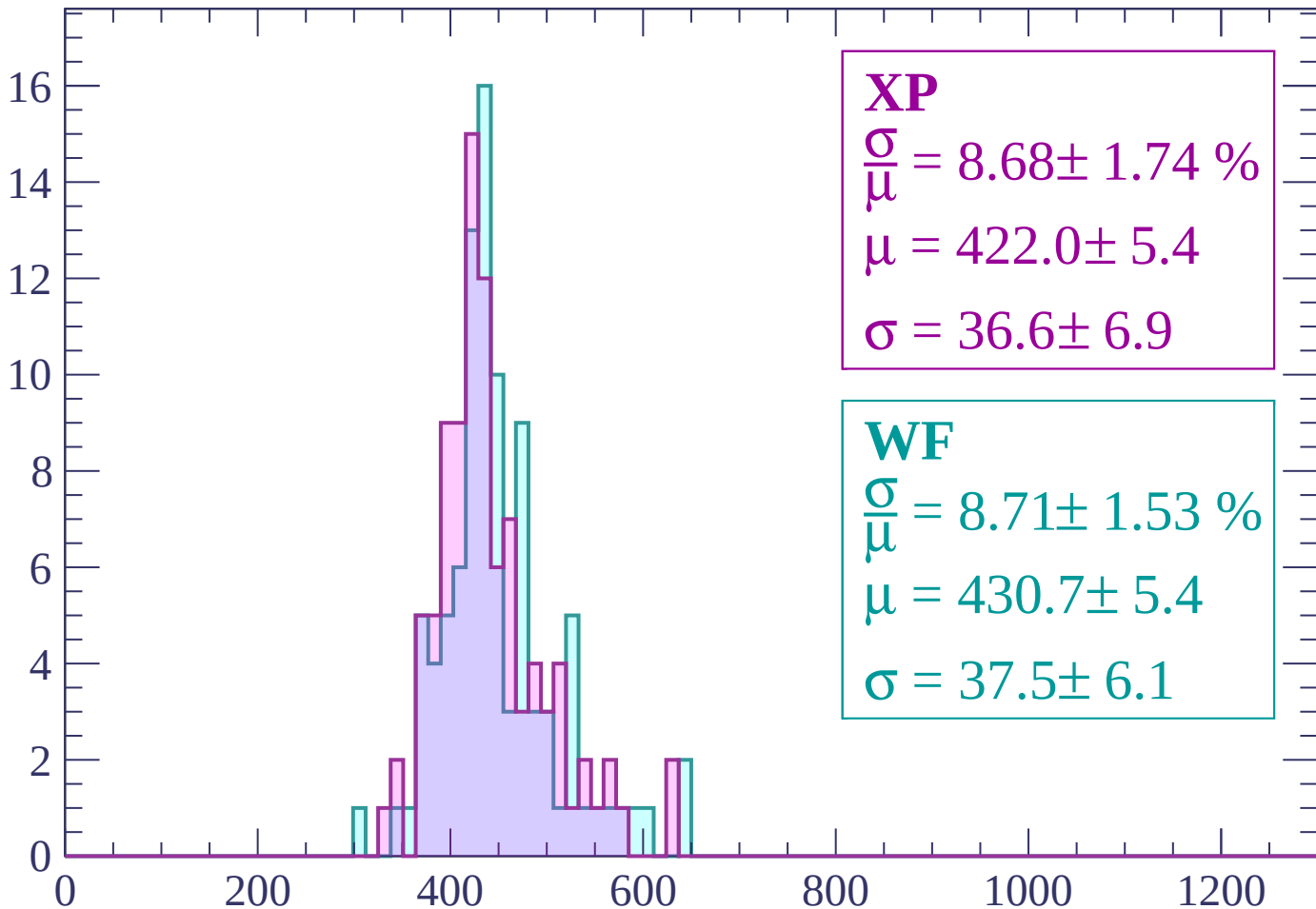
$$\mu = 427.4 \pm 5.5$$

$$\sigma = 42.8 \pm 4.7$$

dE/dx (ADC counts/cm)

Energy loss | $1380 < p < 1440$

Count



XP

$$\frac{\sigma}{\mu} = 8.68 \pm 1.74 \%$$

$$\mu = 422.0 \pm 5.4$$

$$\sigma = 36.6 \pm 6.9$$

WF

$$\frac{\sigma}{\mu} = 8.71 \pm 1.53 \%$$

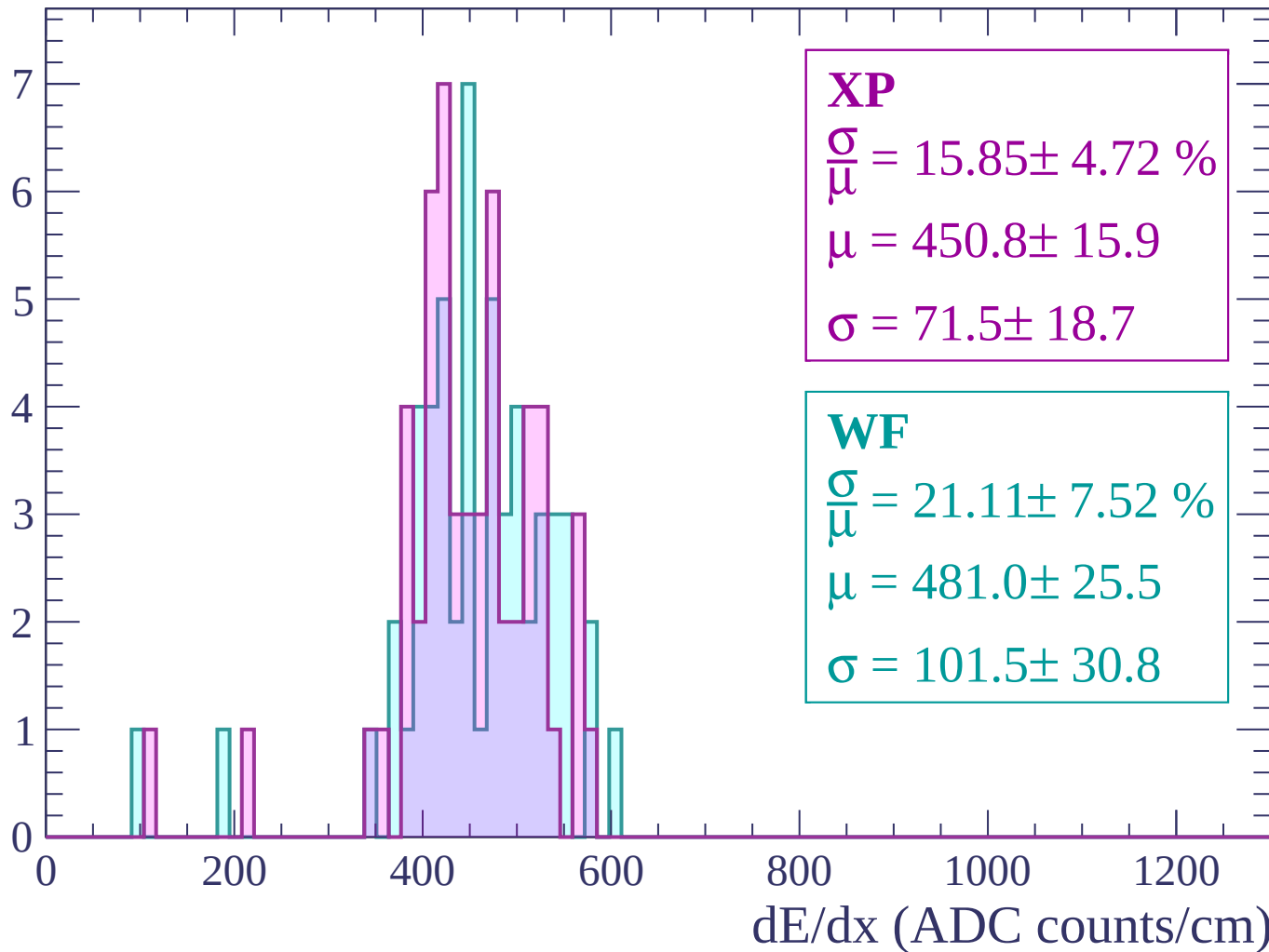
$$\mu = 430.7 \pm 5.4$$

$$\sigma = 37.5 \pm 6.1$$

dE/dx (ADC counts/cm)

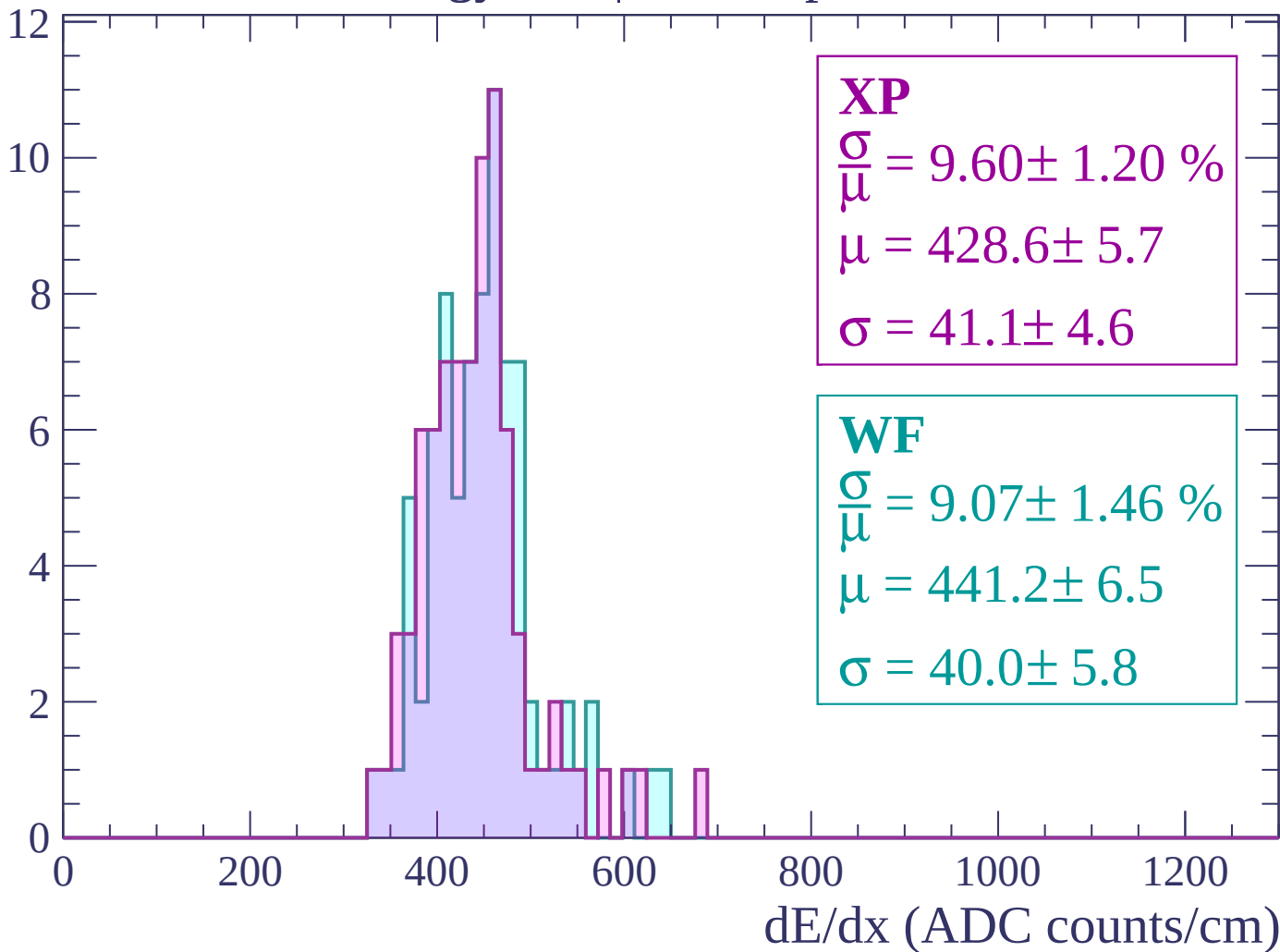
Energy loss | $1440 < p < 1500$

Count



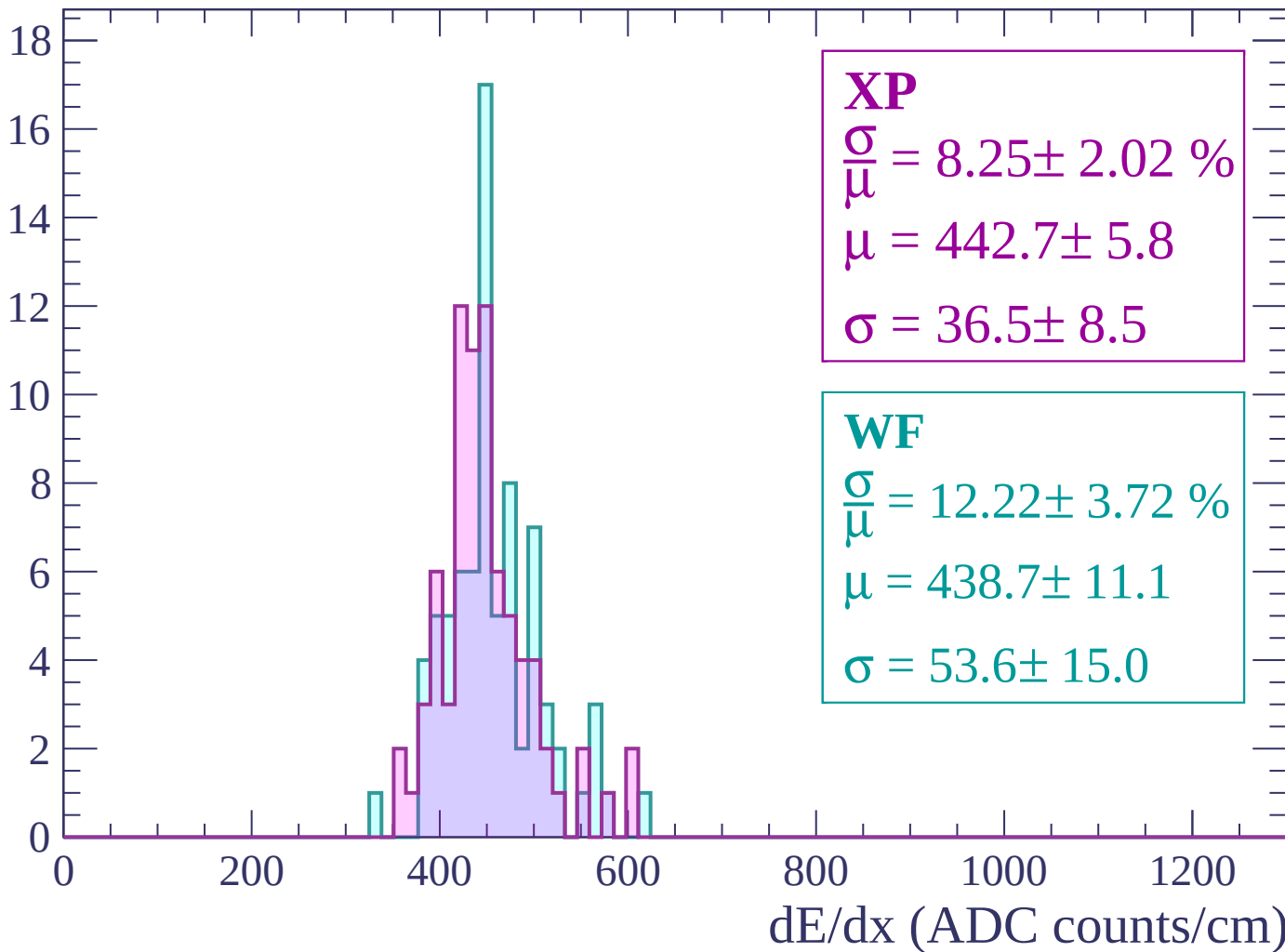
Energy loss | $1500 < p < 1560$

Count



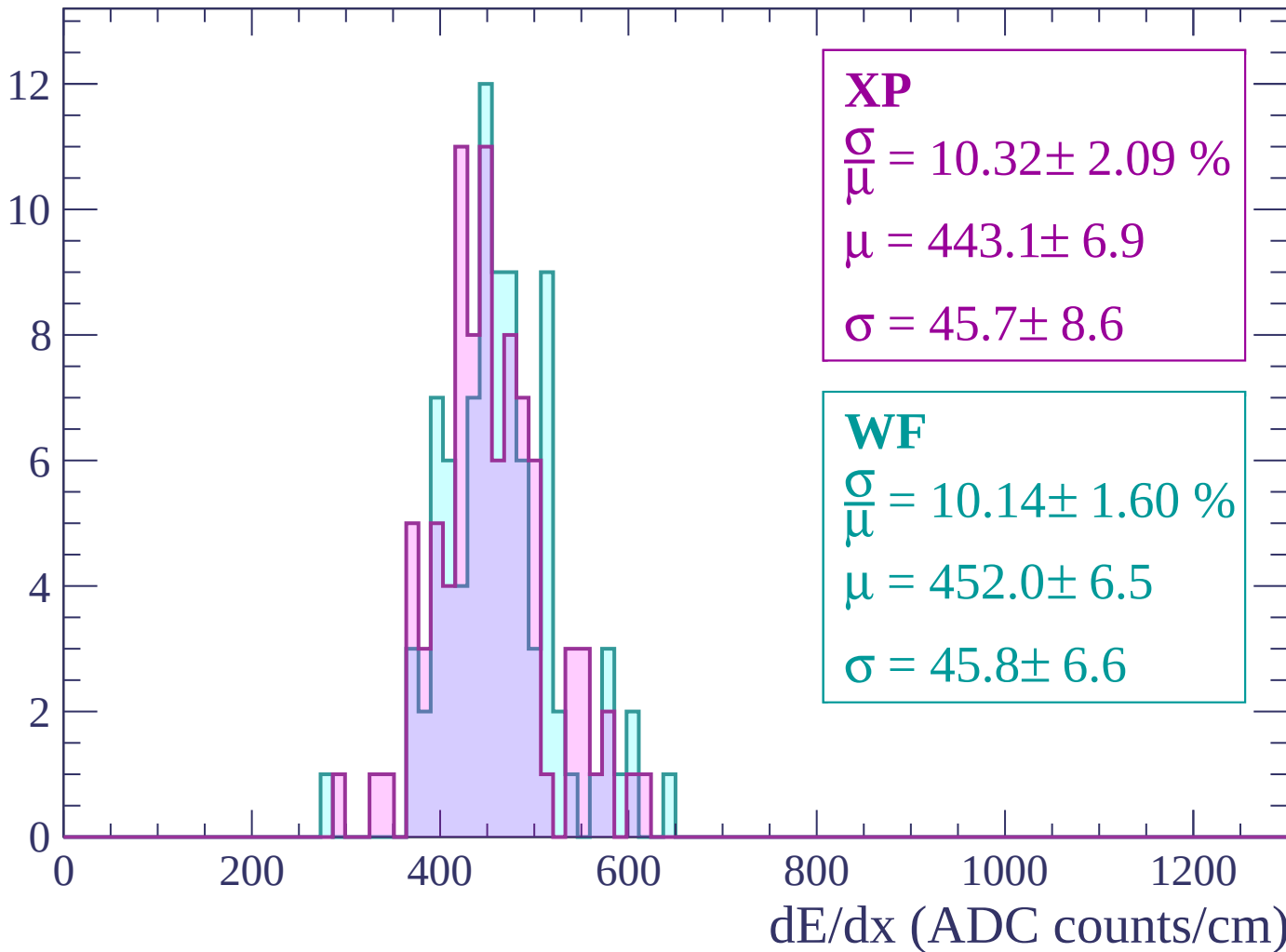
Energy loss | $1560 < p < 1620$

Count



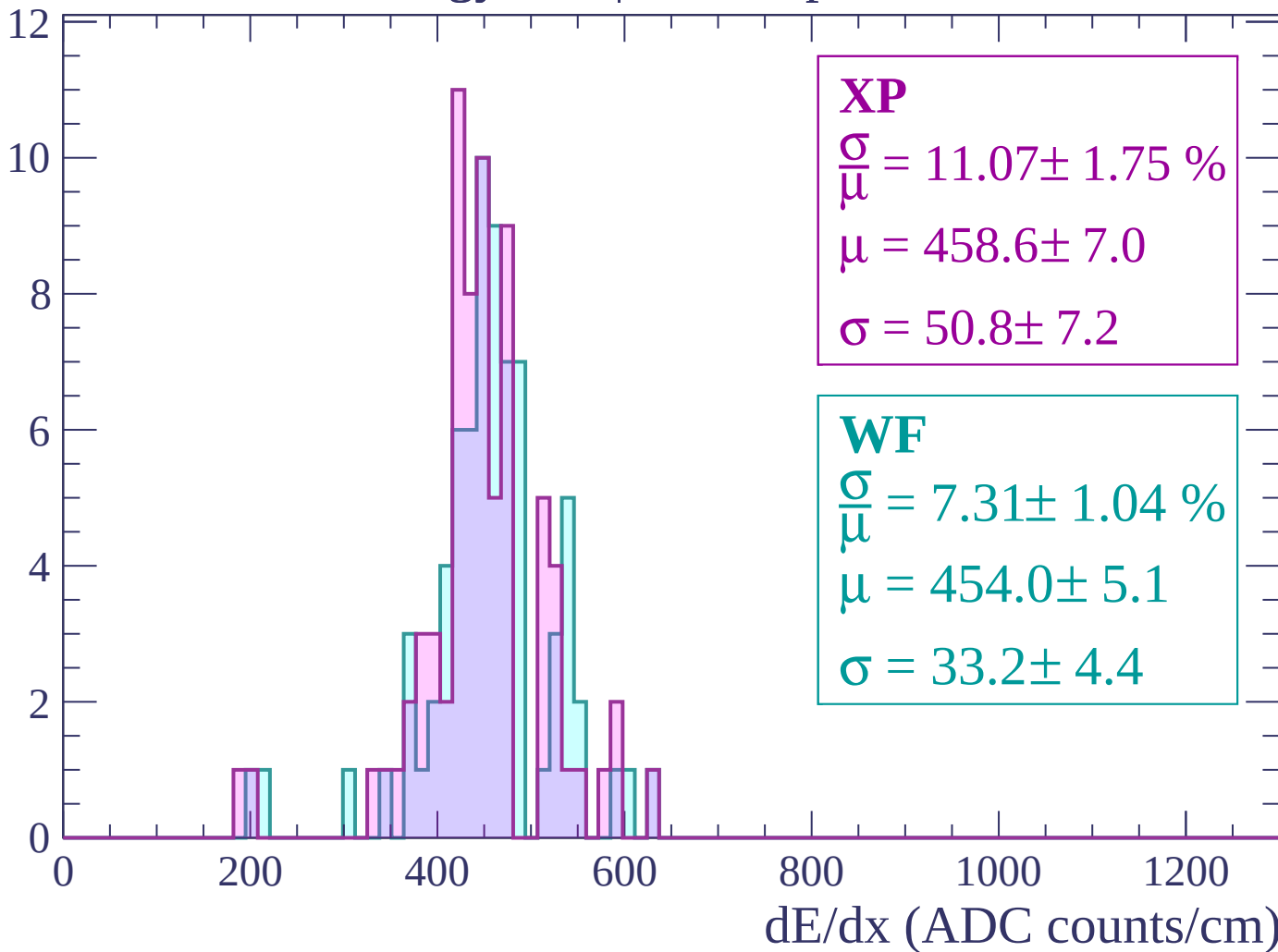
Energy loss | $1620 < p < 1680$

Count



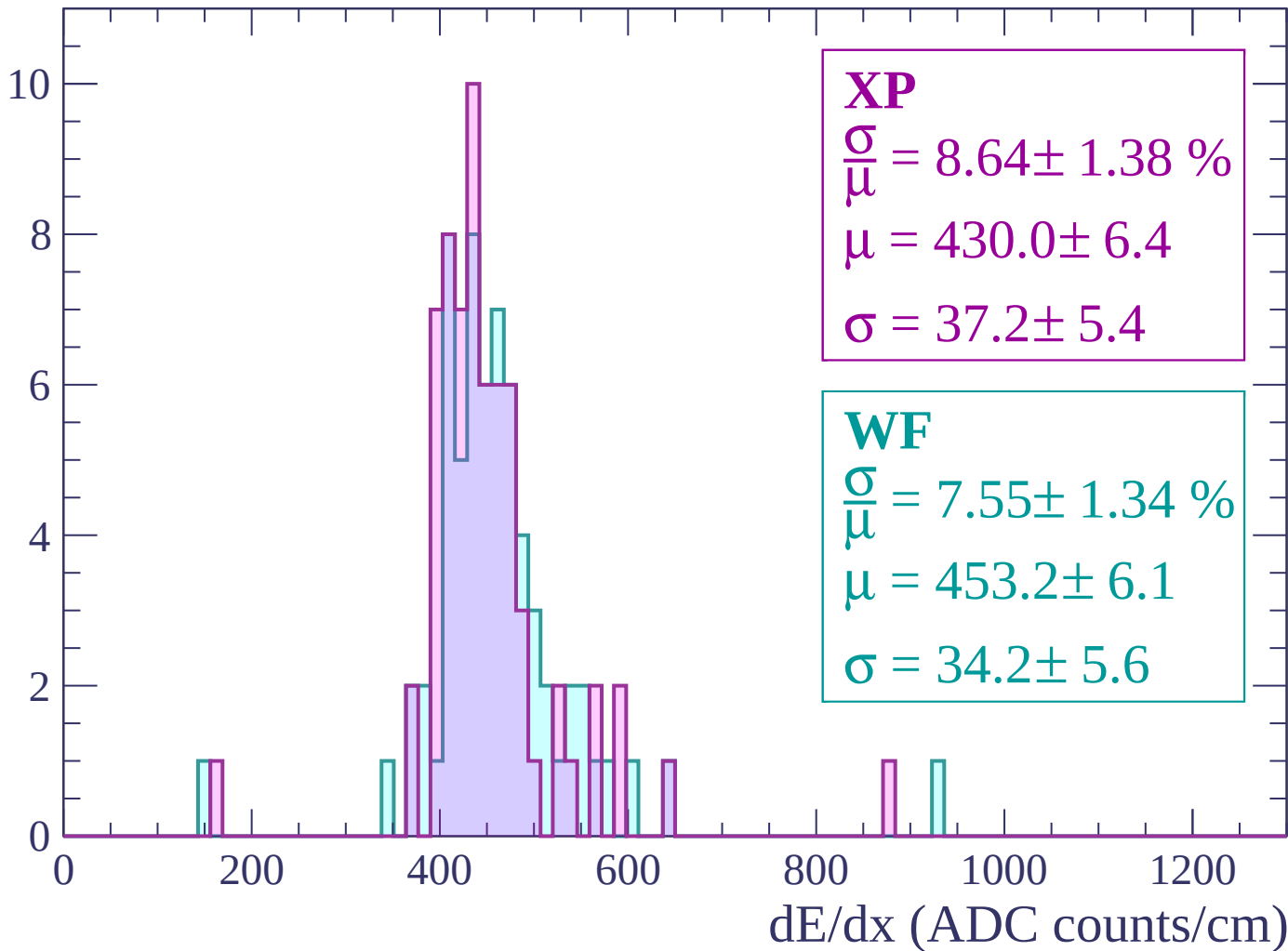
Energy loss | $1680 < p < 1740$

Count



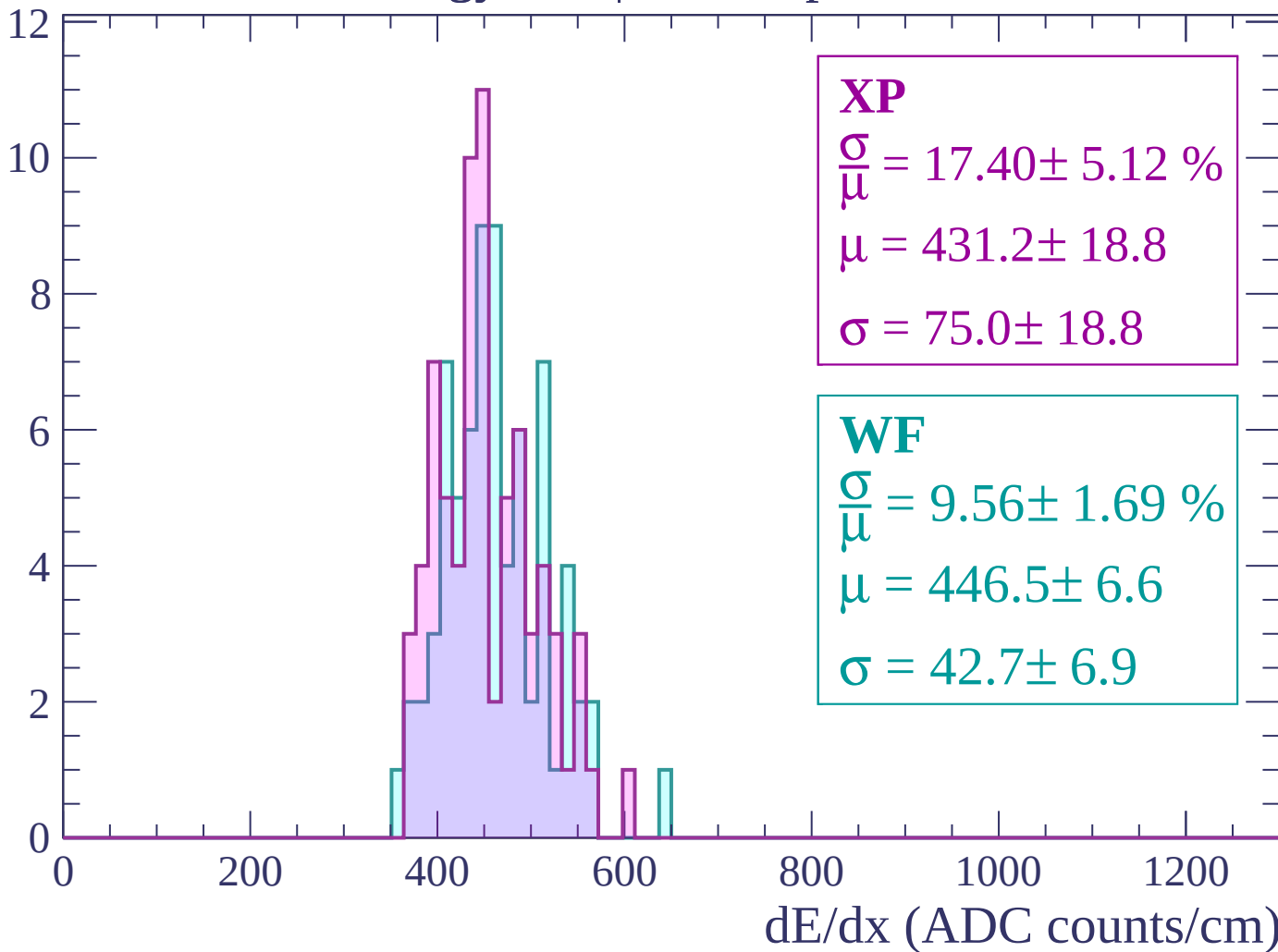
Energy loss | $1740 < p < 1800$

Count



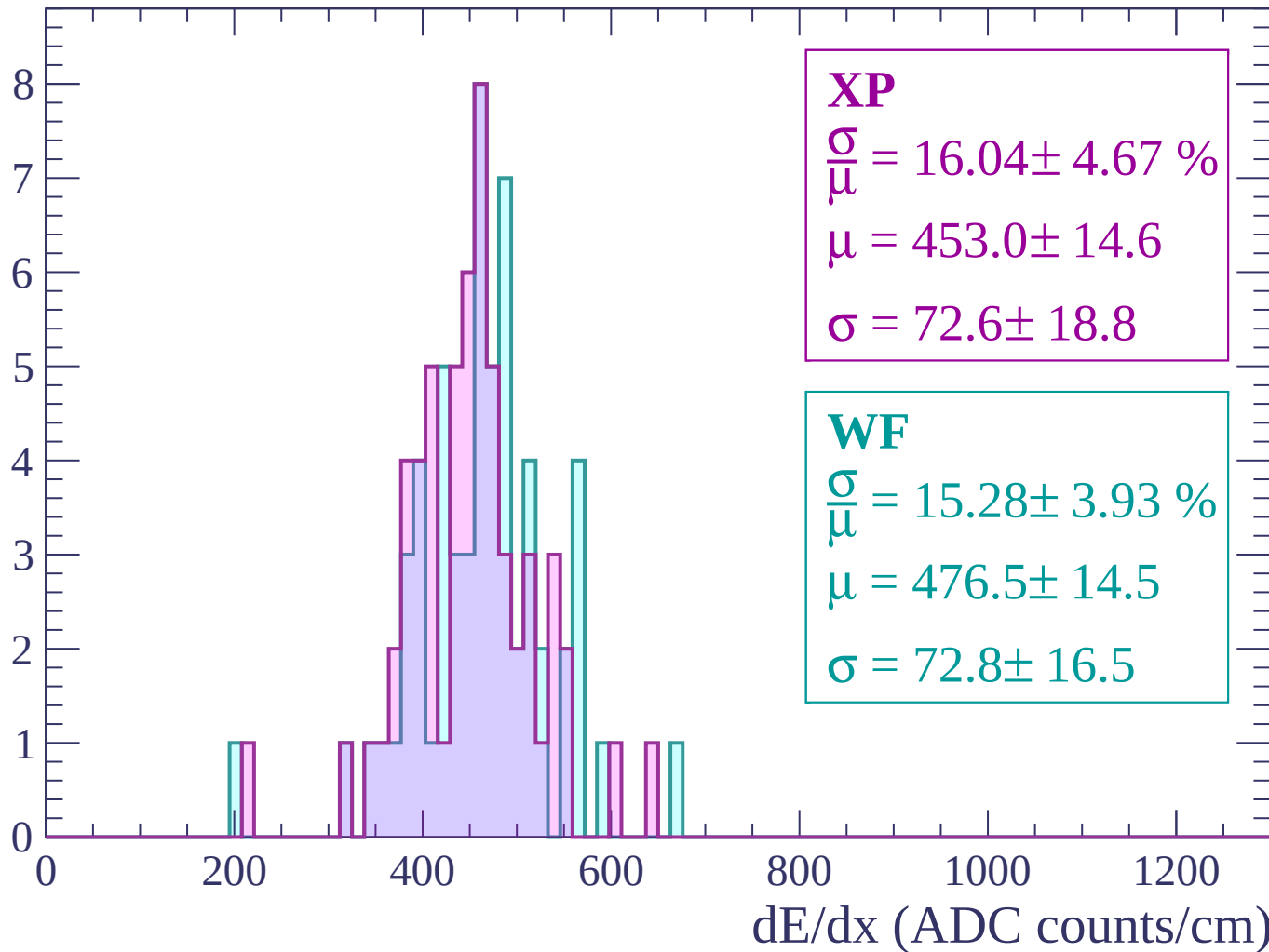
Energy loss | $1800 < p < 1860$

Count



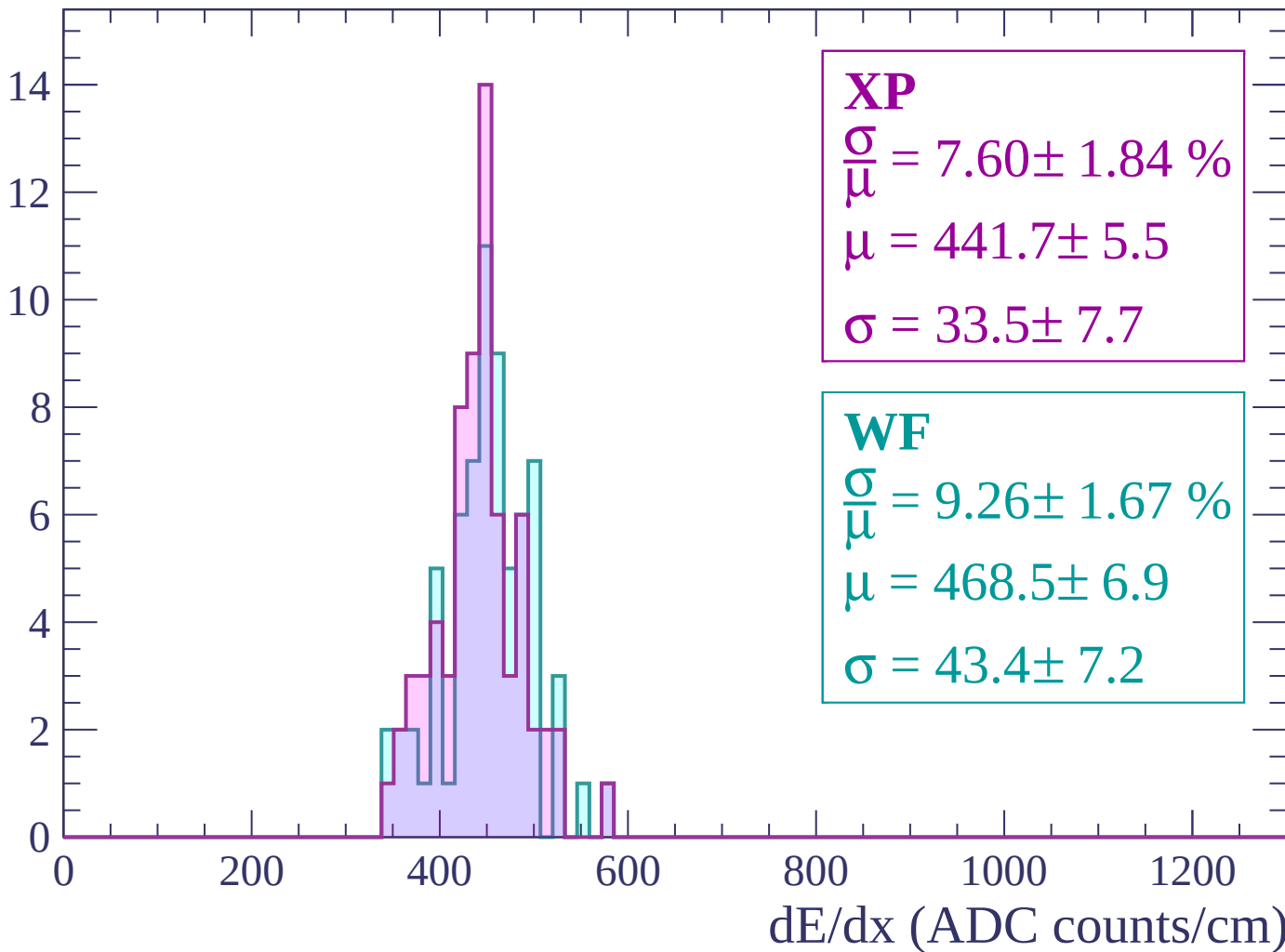
Energy loss | $1860 < p < 1920$

Count



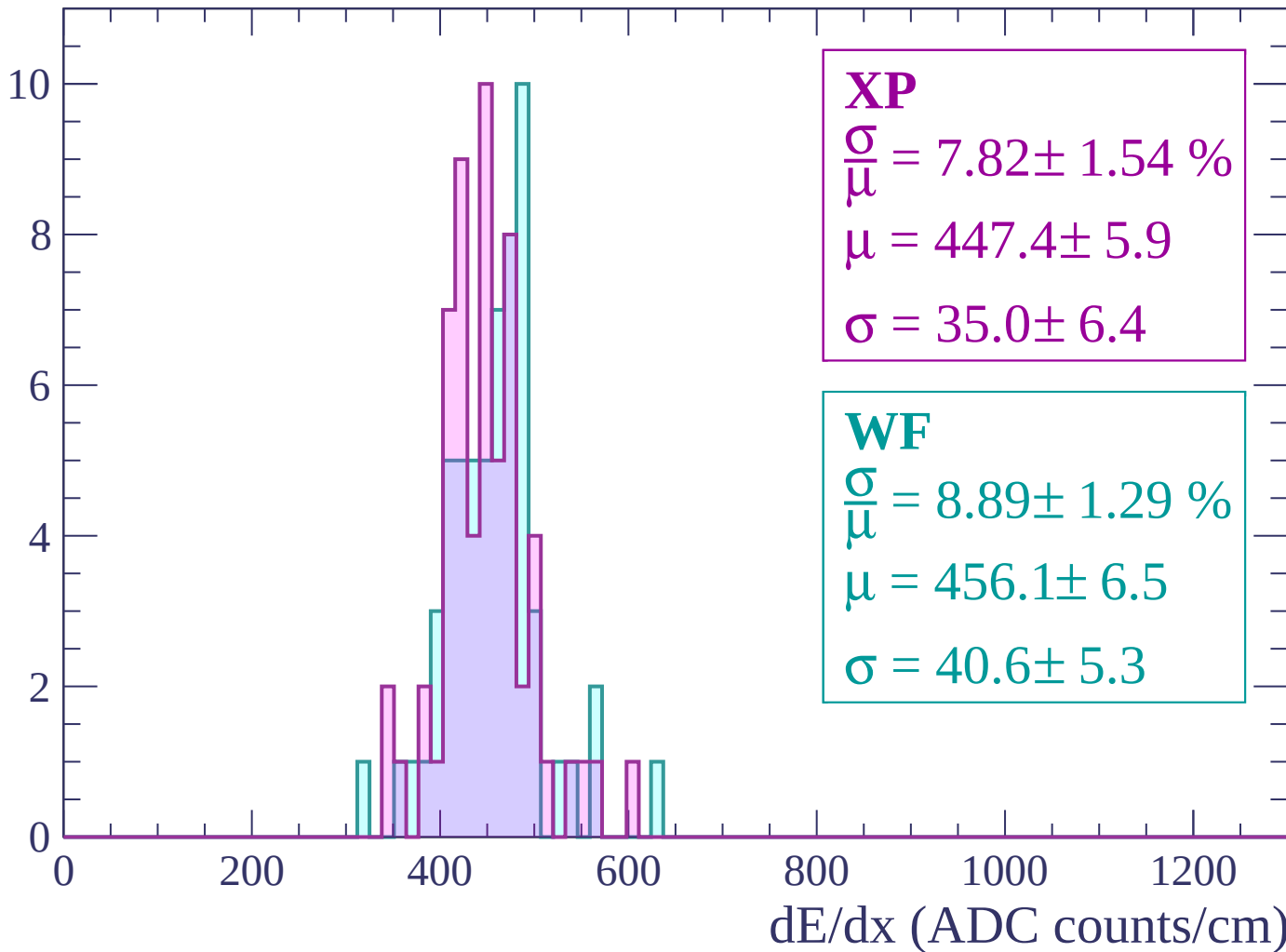
Energy loss | $1920 < p < 1980$

Count



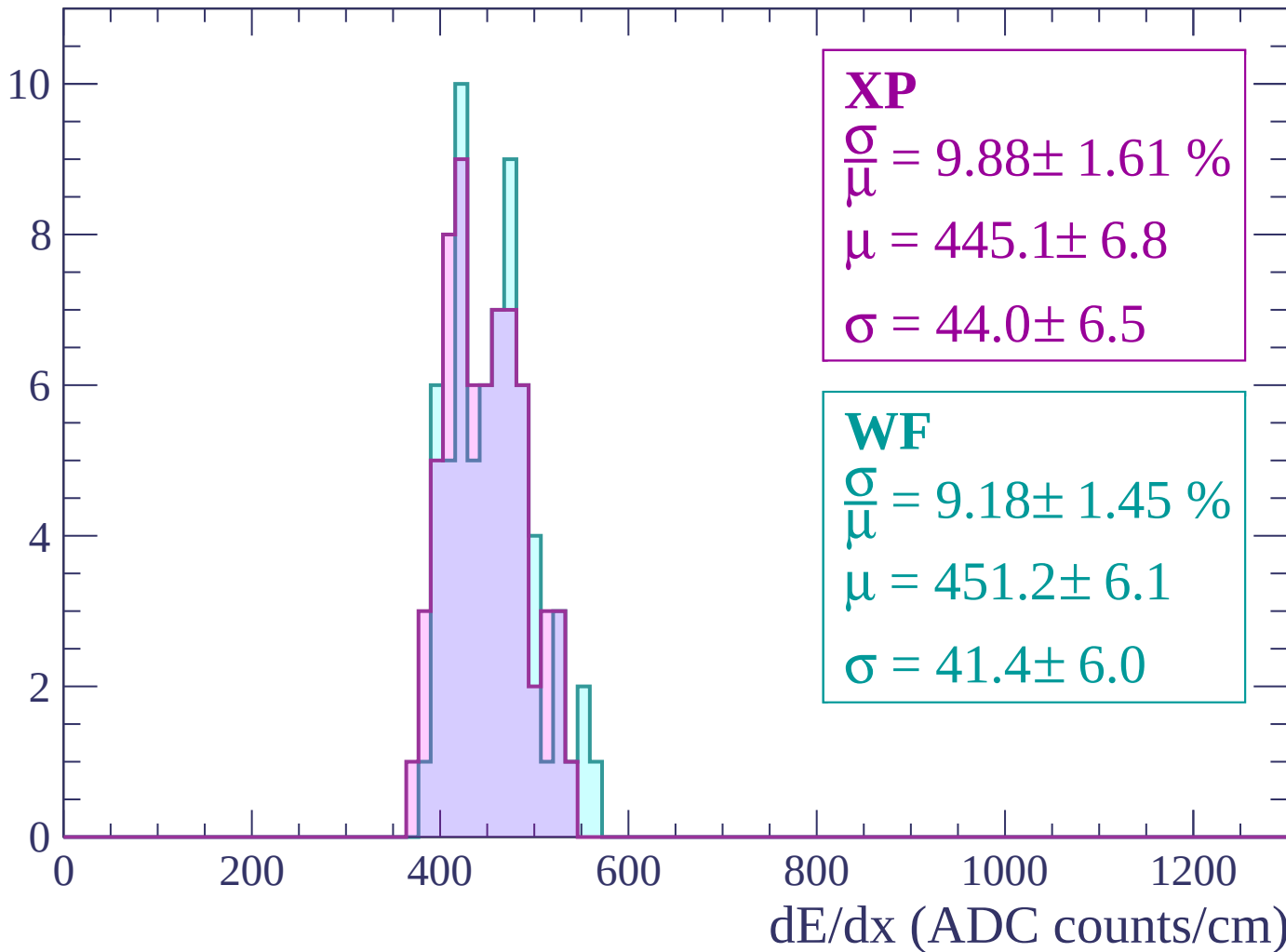
Energy loss | $1980 < p < 2040$

Count



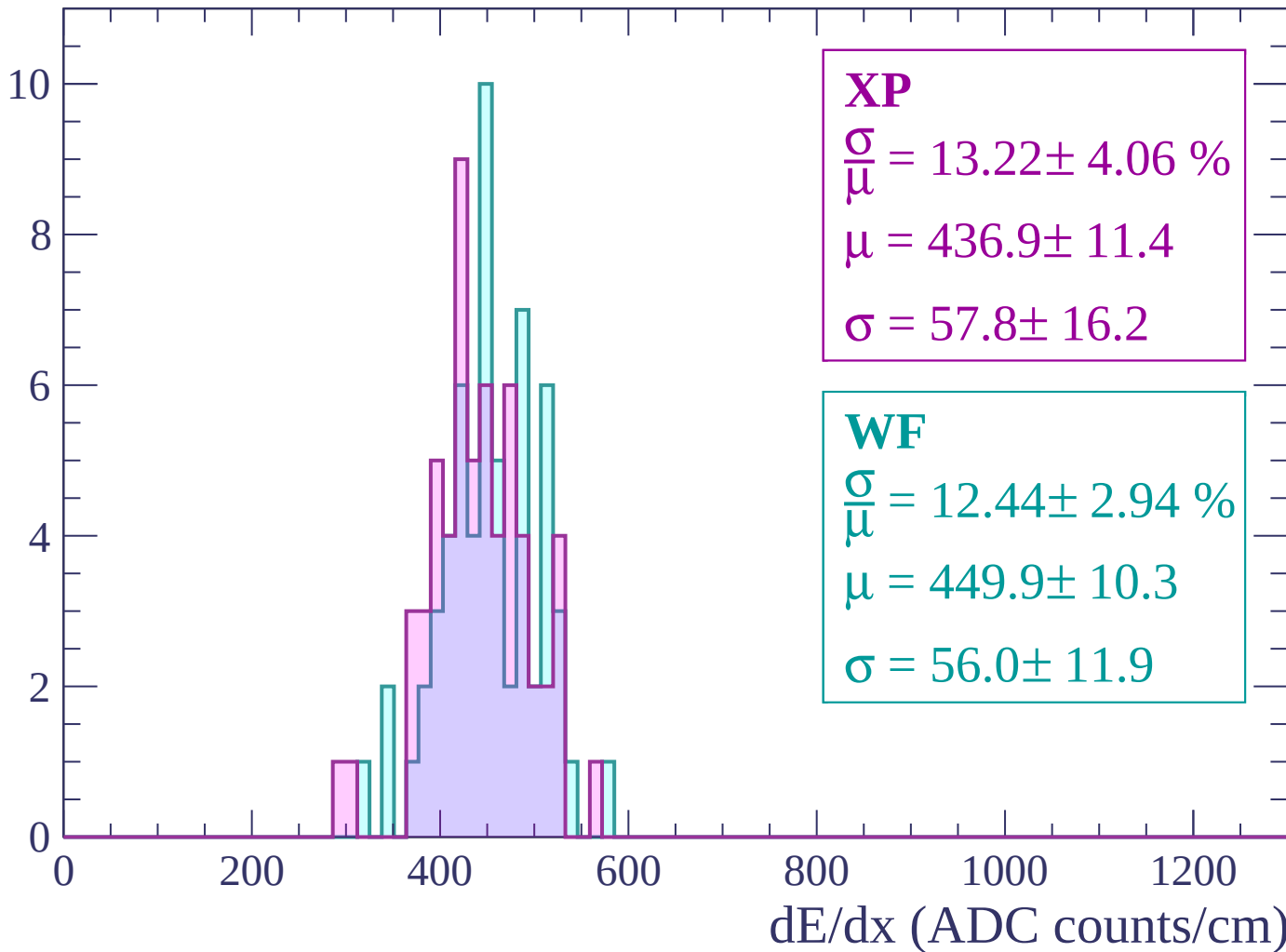
Energy loss | $2040 < p < 2100$

Count



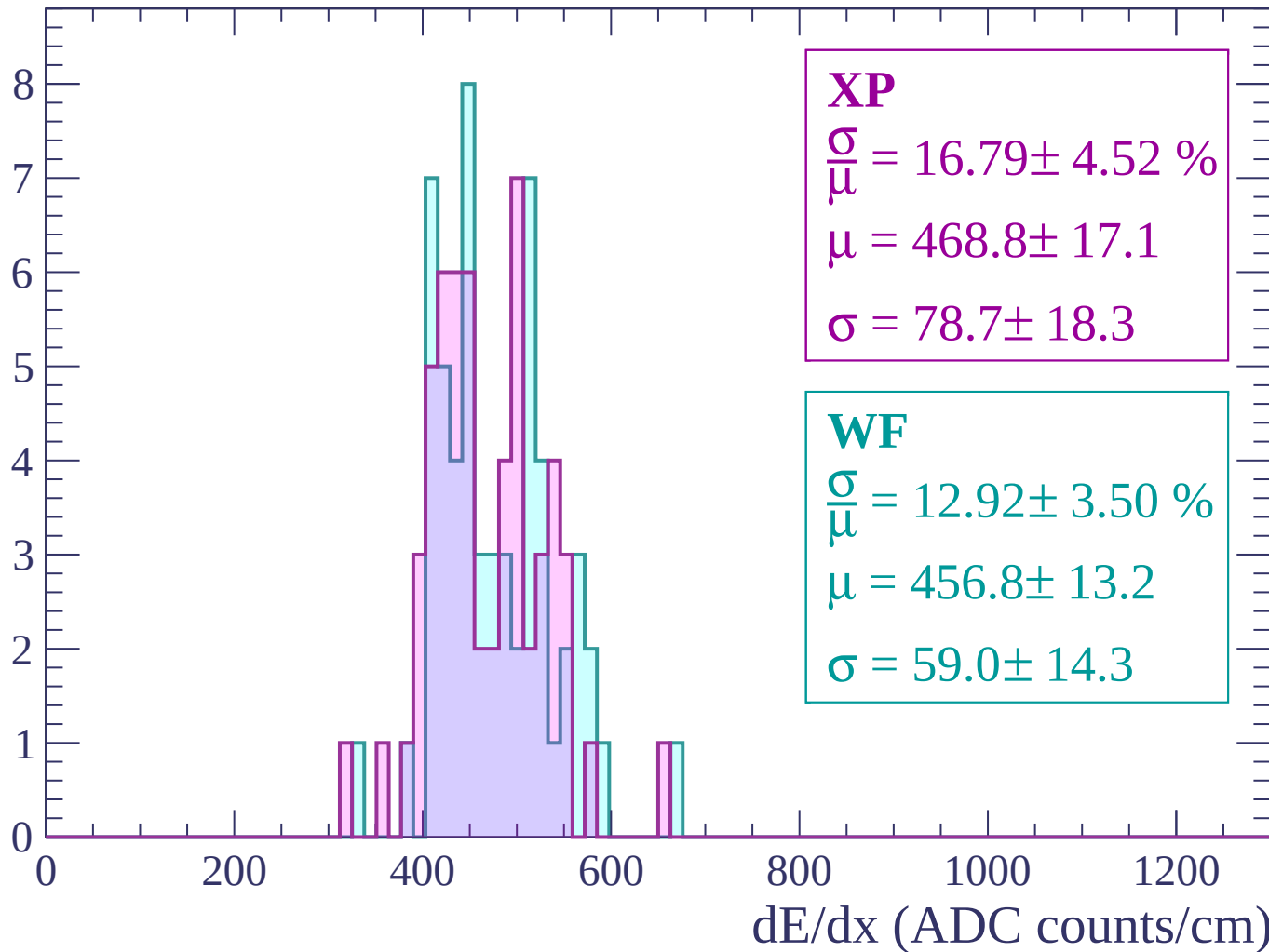
Energy loss | $2100 < p < 2160$

Count



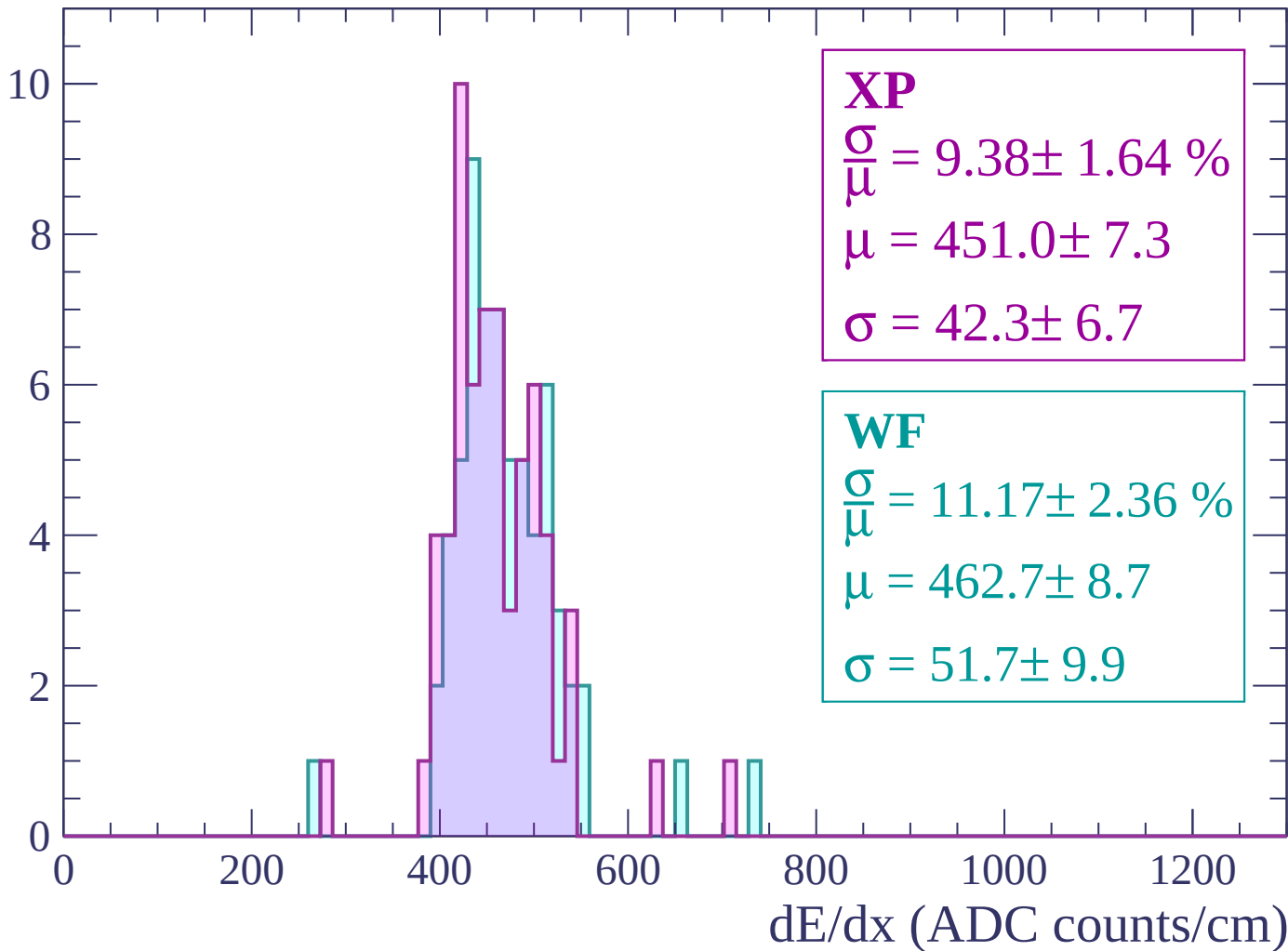
Energy loss | $2160 < p < 2220$

Count



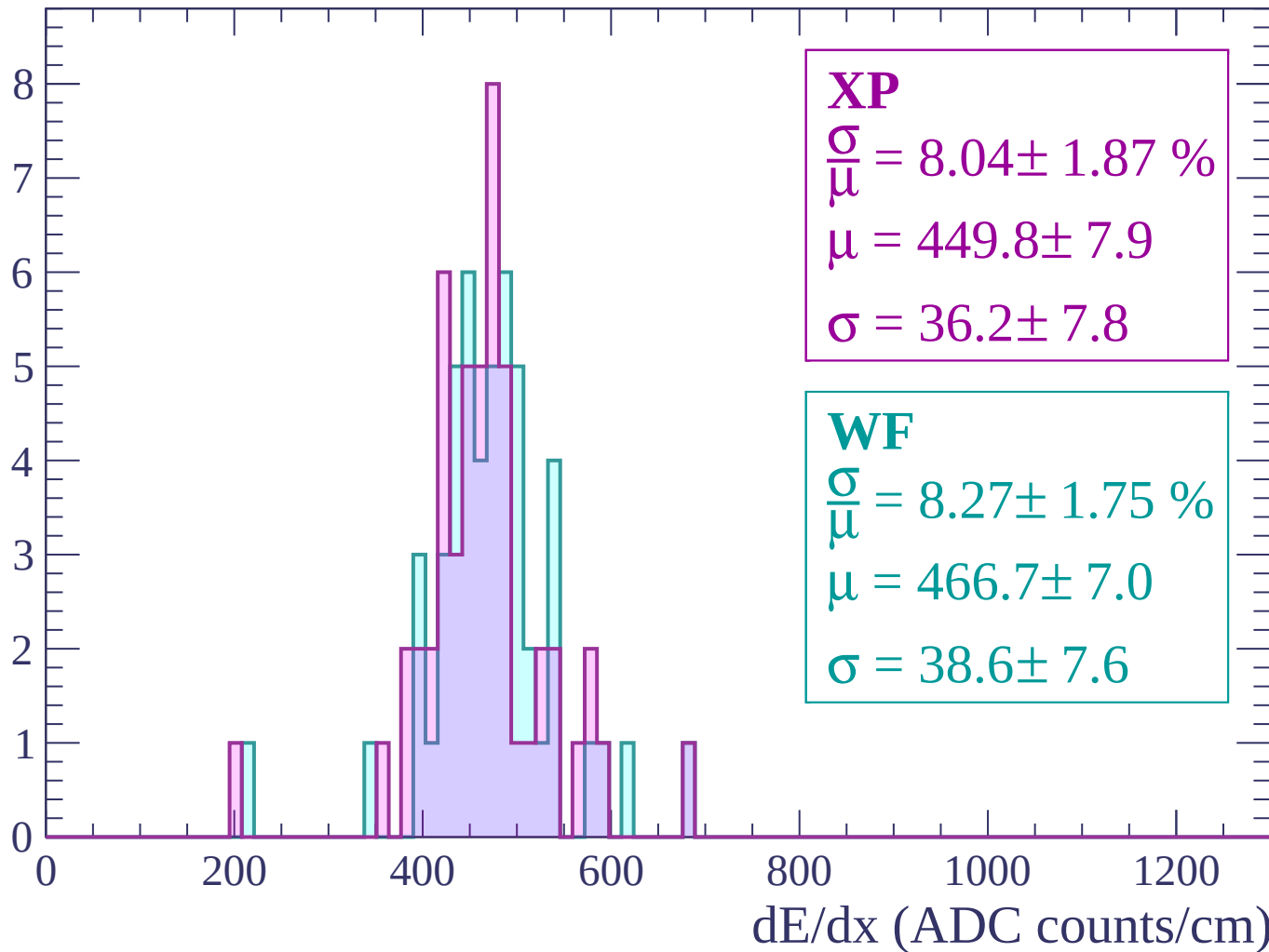
Energy loss | $2220 < p < 2280$

Count



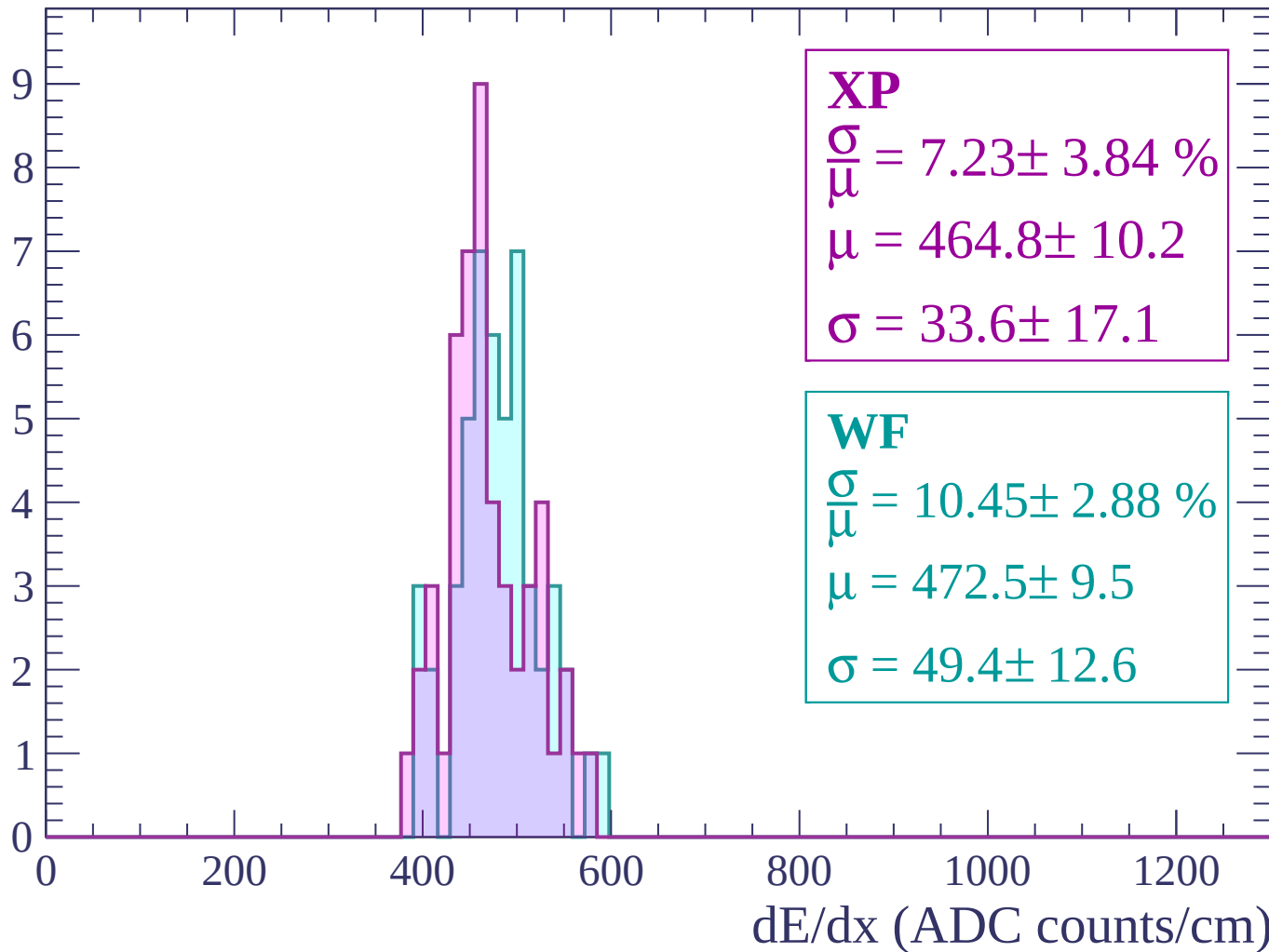
Energy loss | $2280 < p < 2340$

Count



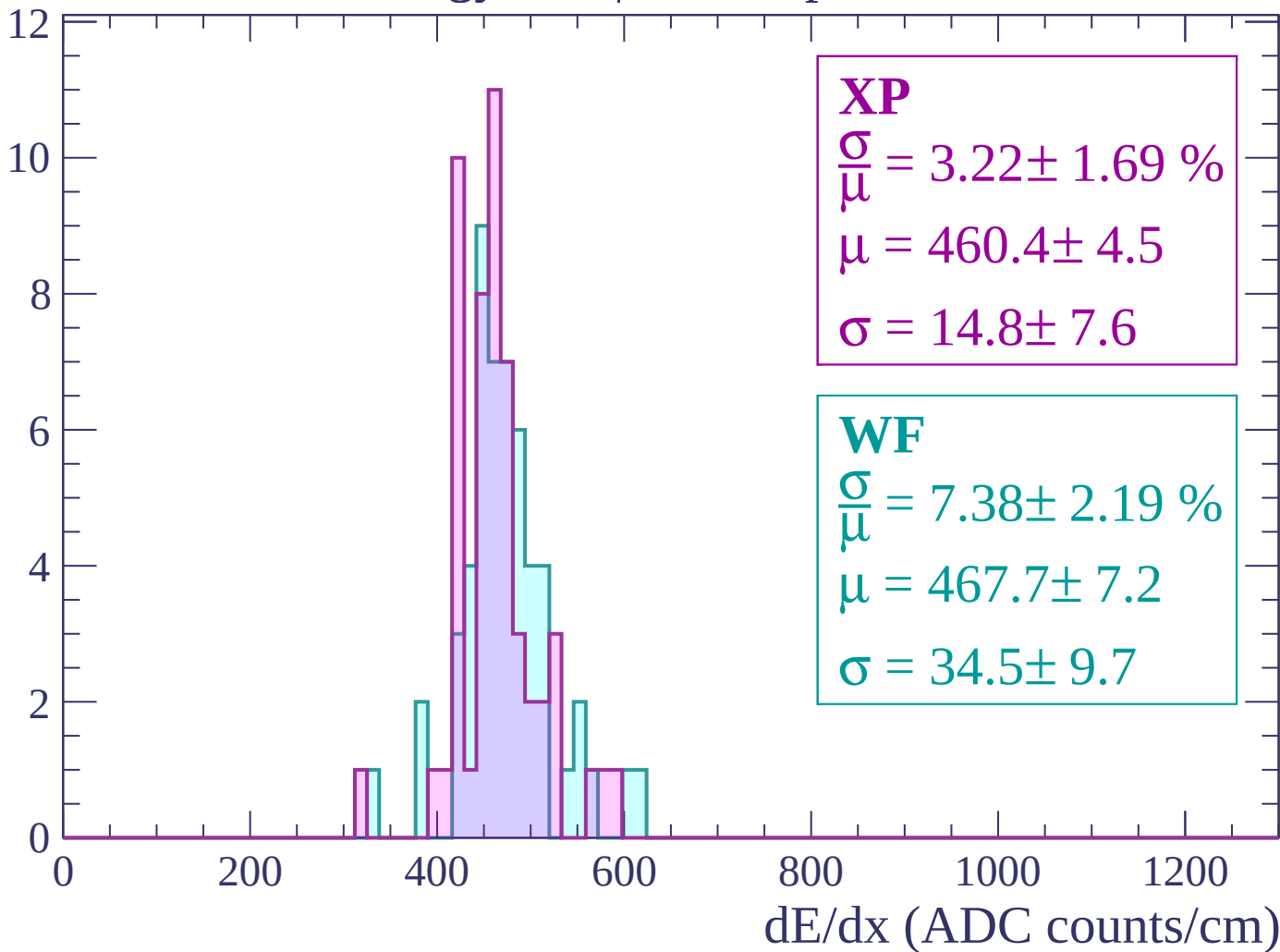
Energy loss | $2340 < p < 2400$

Count



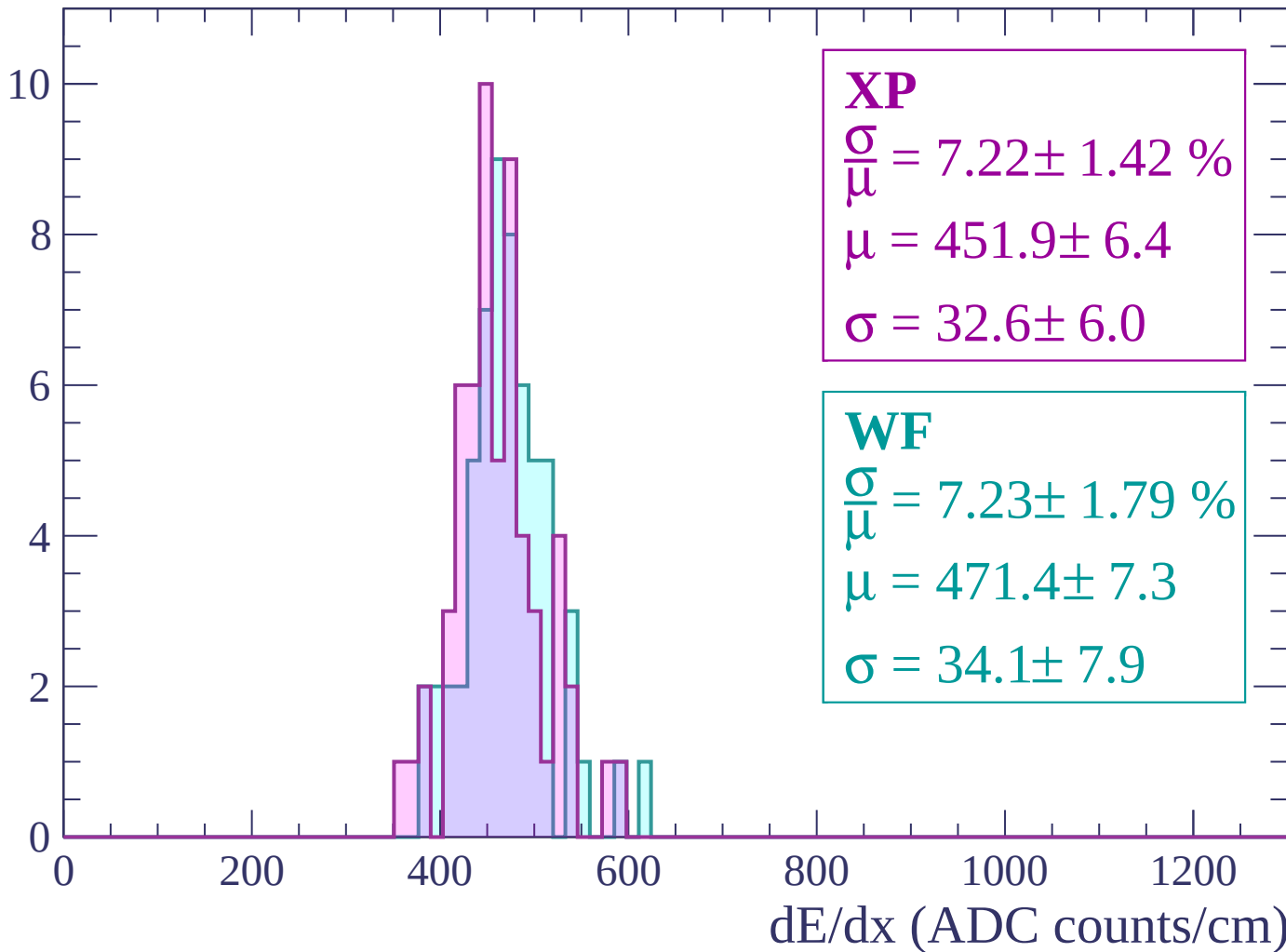
Energy loss | $2400 < p < 2460$

Count



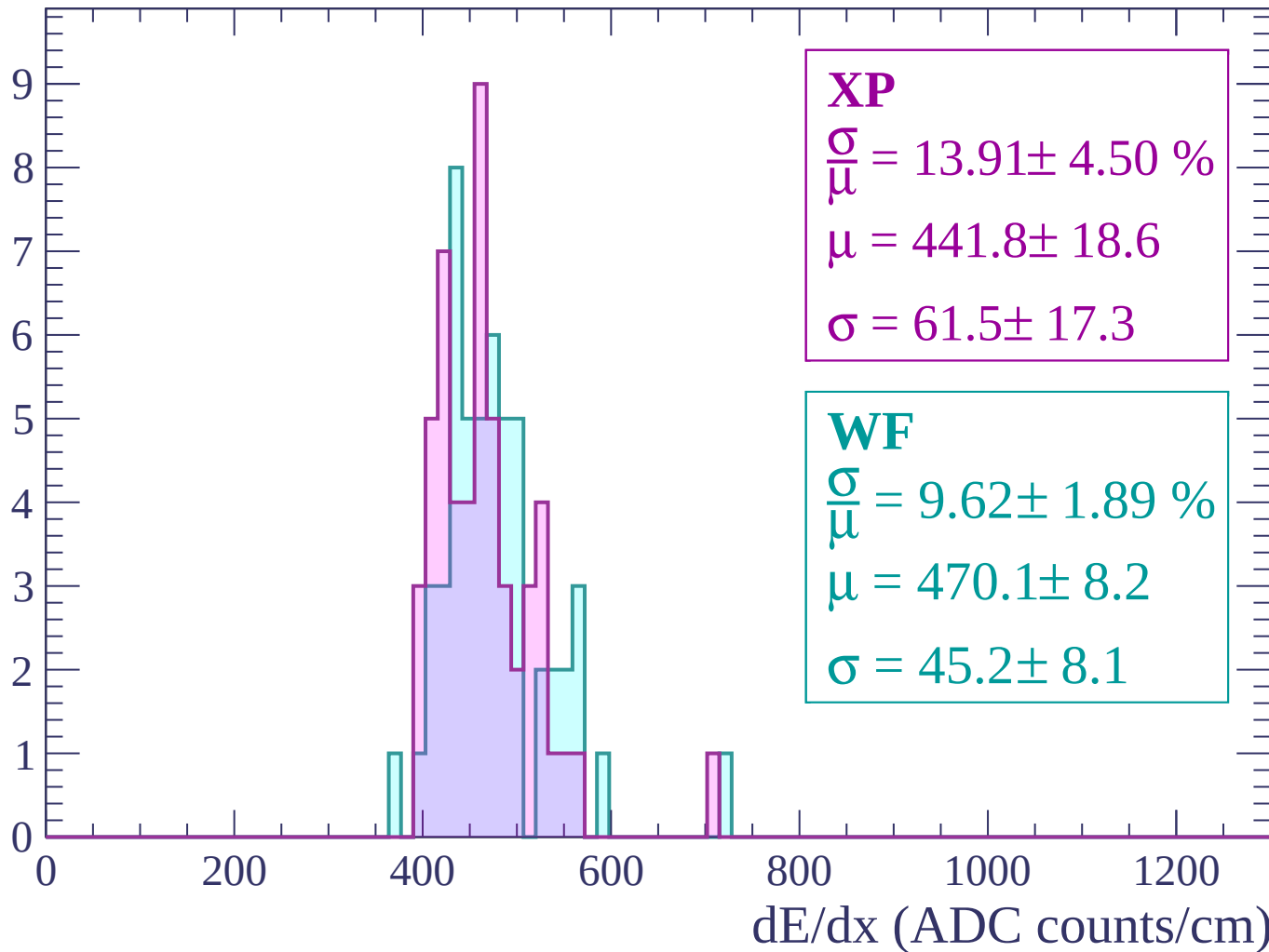
Energy loss | $2460 < p < 2520$

Count



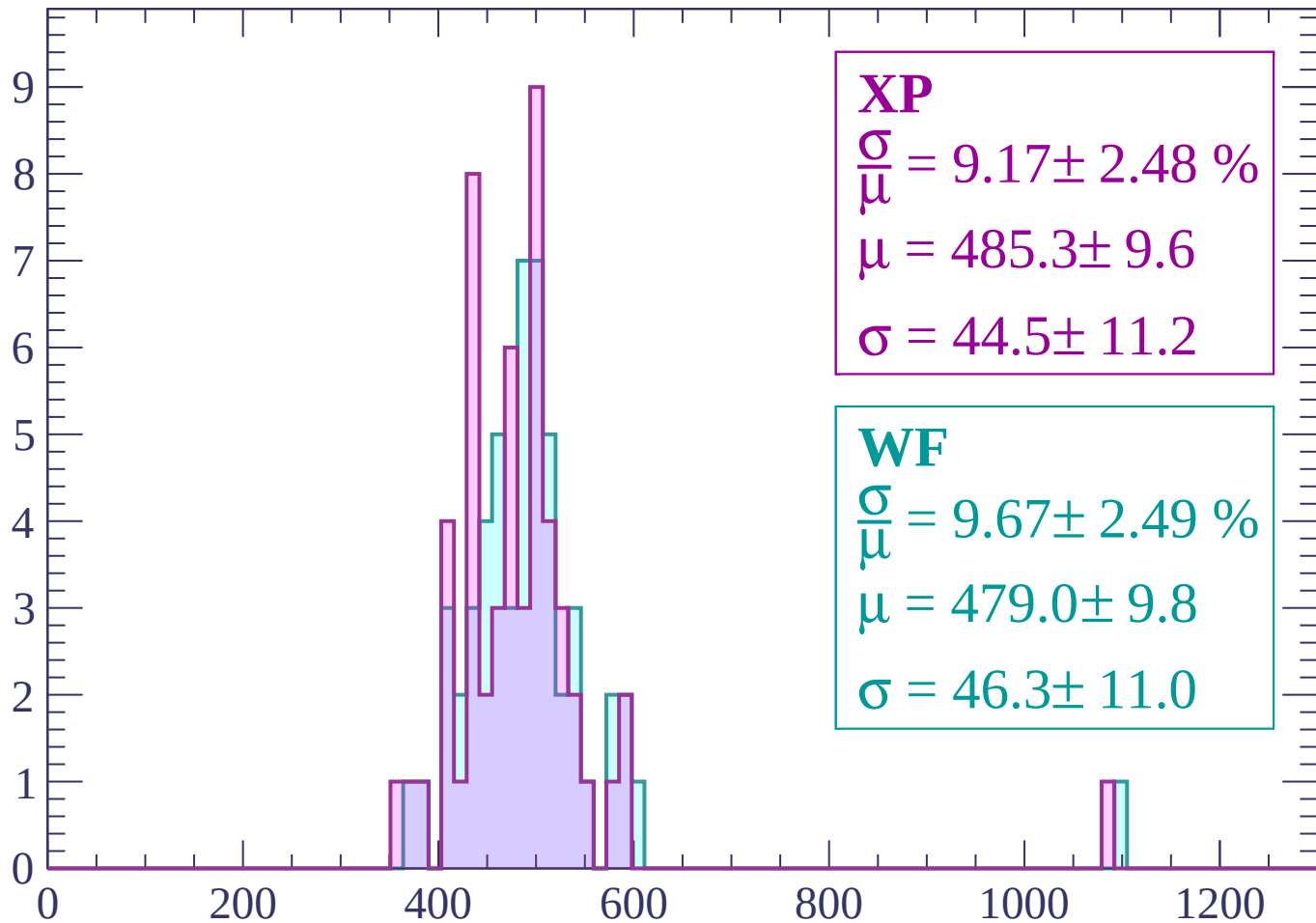
Energy loss | $2520 < p < 2580$

Count



Energy loss | $2580 < p < 2640$

Count



XP

$$\frac{\sigma}{\mu} = 9.17 \pm 2.48 \%$$

$$\mu = 485.3 \pm 9.6$$

$$\sigma = 44.5 \pm 11.2$$

WF

$$\frac{\sigma}{\mu} = 9.67 \pm 2.49 \%$$

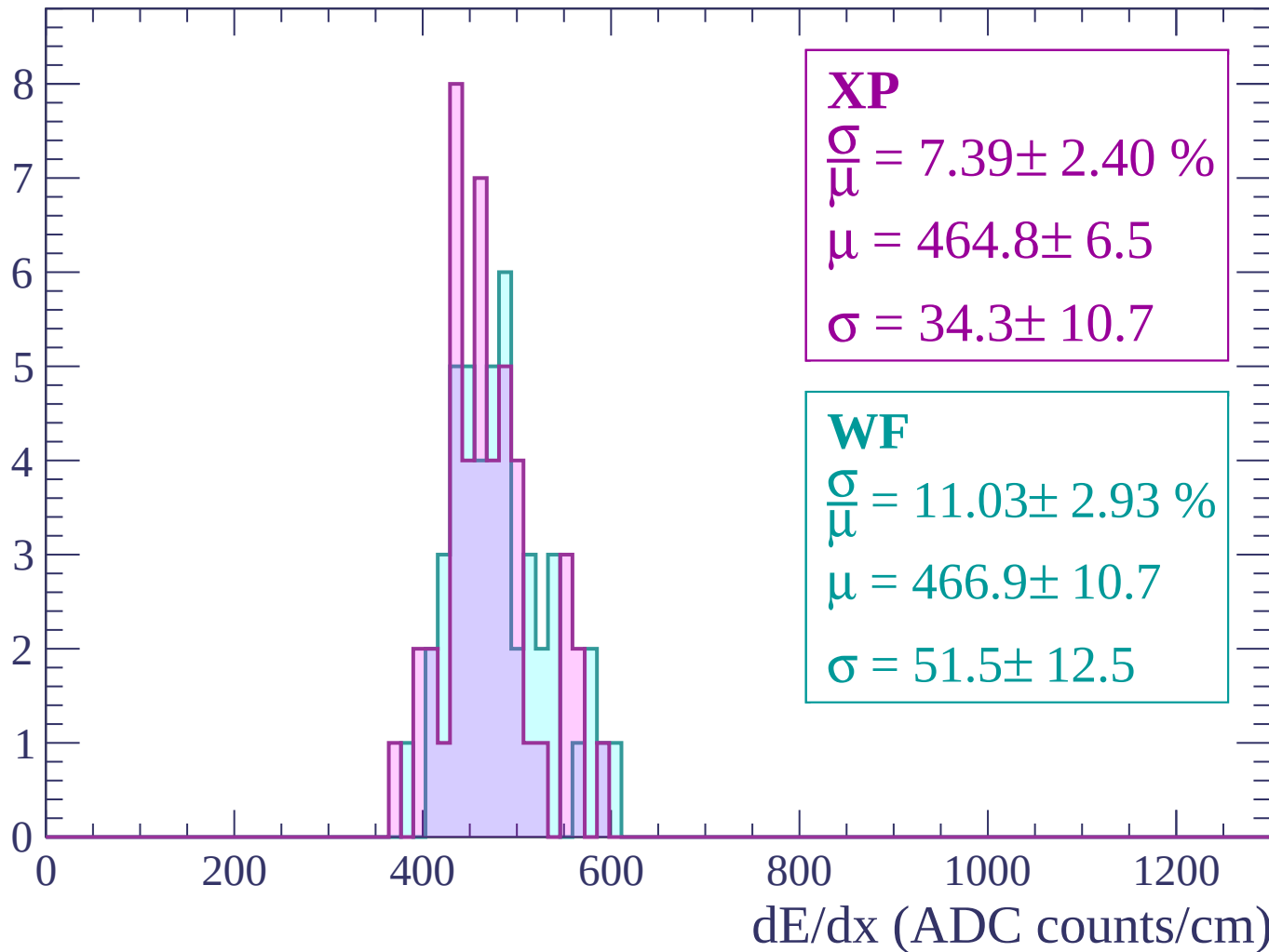
$$\mu = 479.0 \pm 9.8$$

$$\sigma = 46.3 \pm 11.0$$

dE/dx (ADC counts/cm)

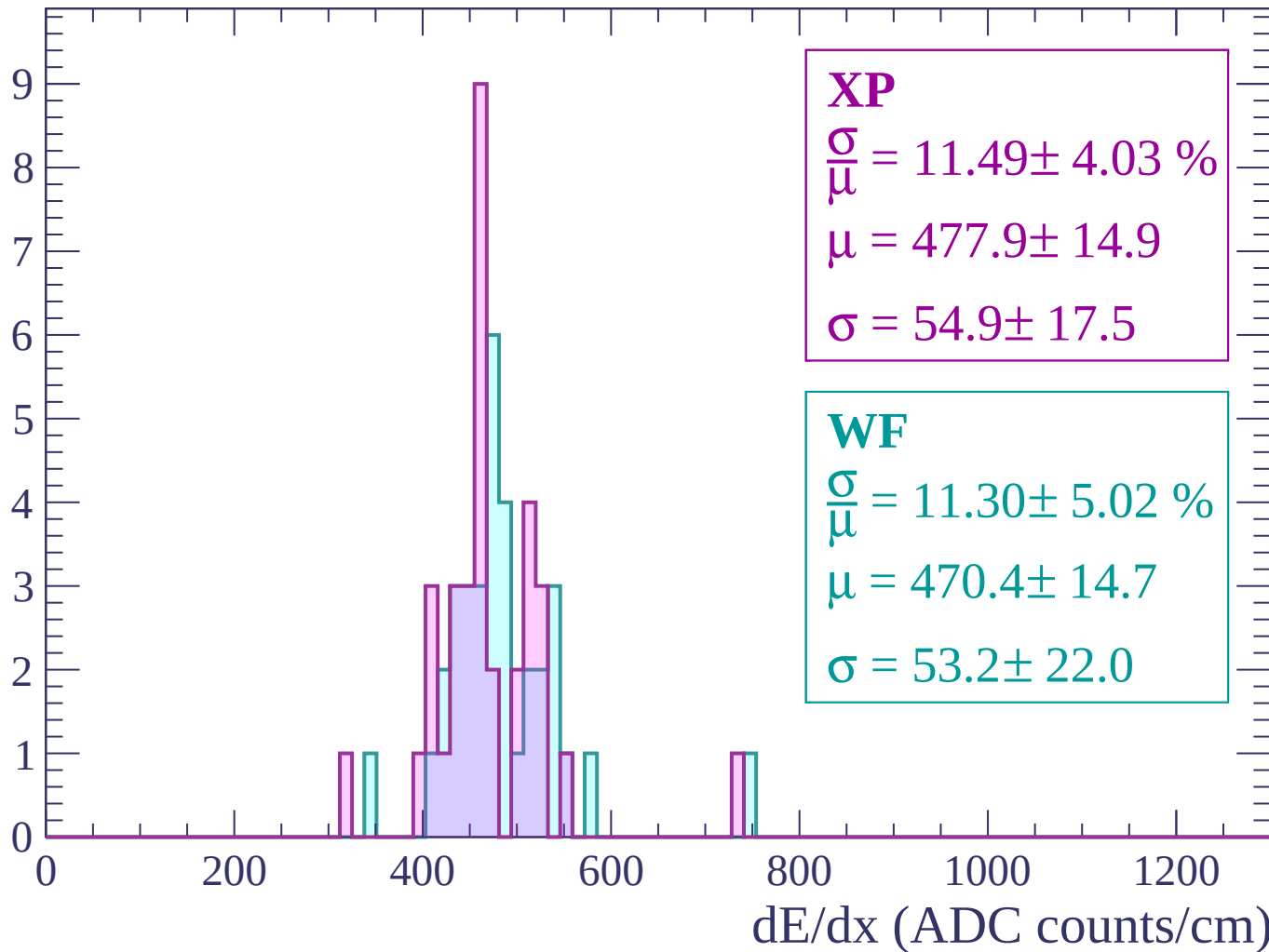
Energy loss | $2640 < p < 2700$

Count



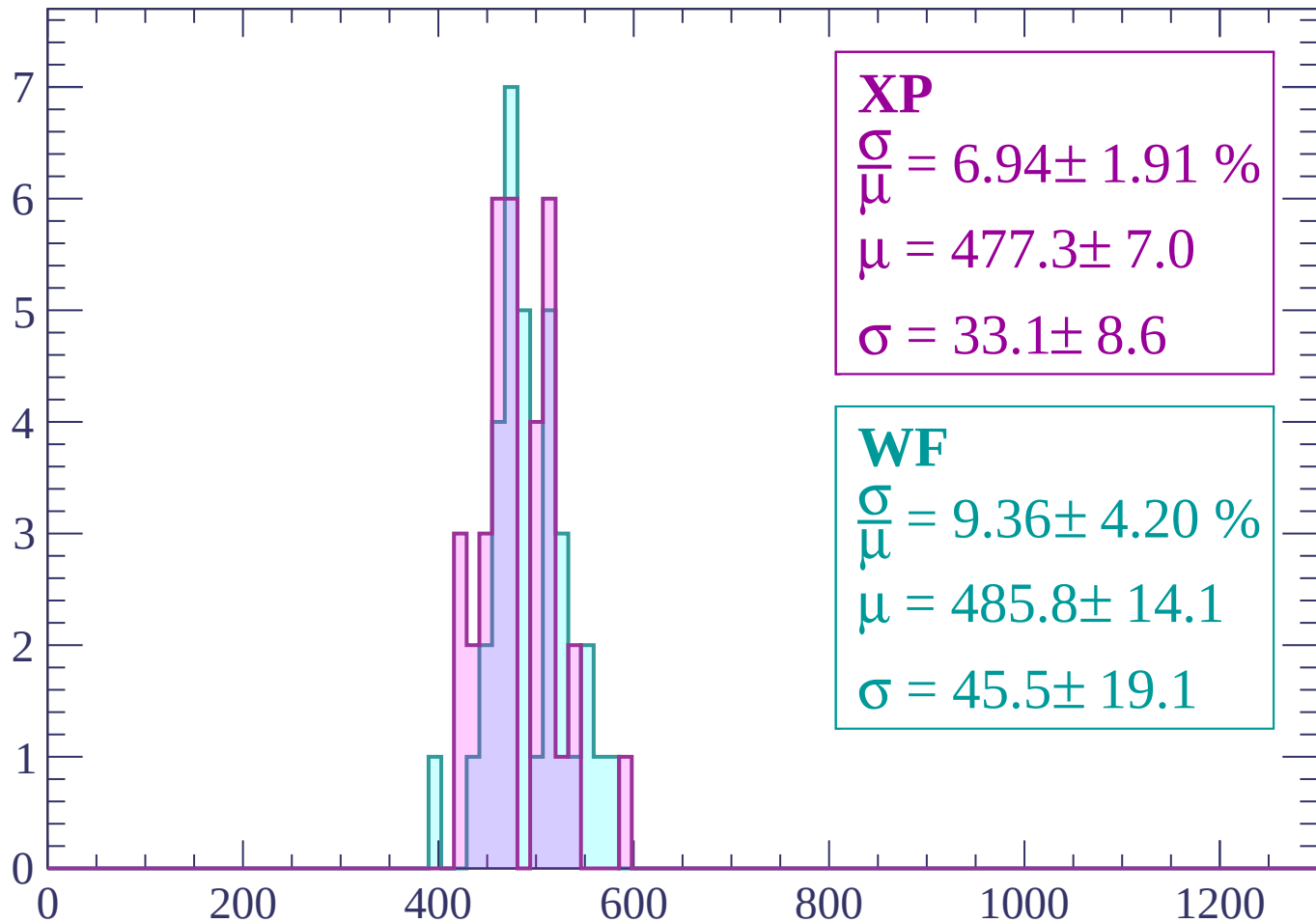
Energy loss | $2700 < p < 2760$

Count



Energy loss | $2760 < p < 2820$

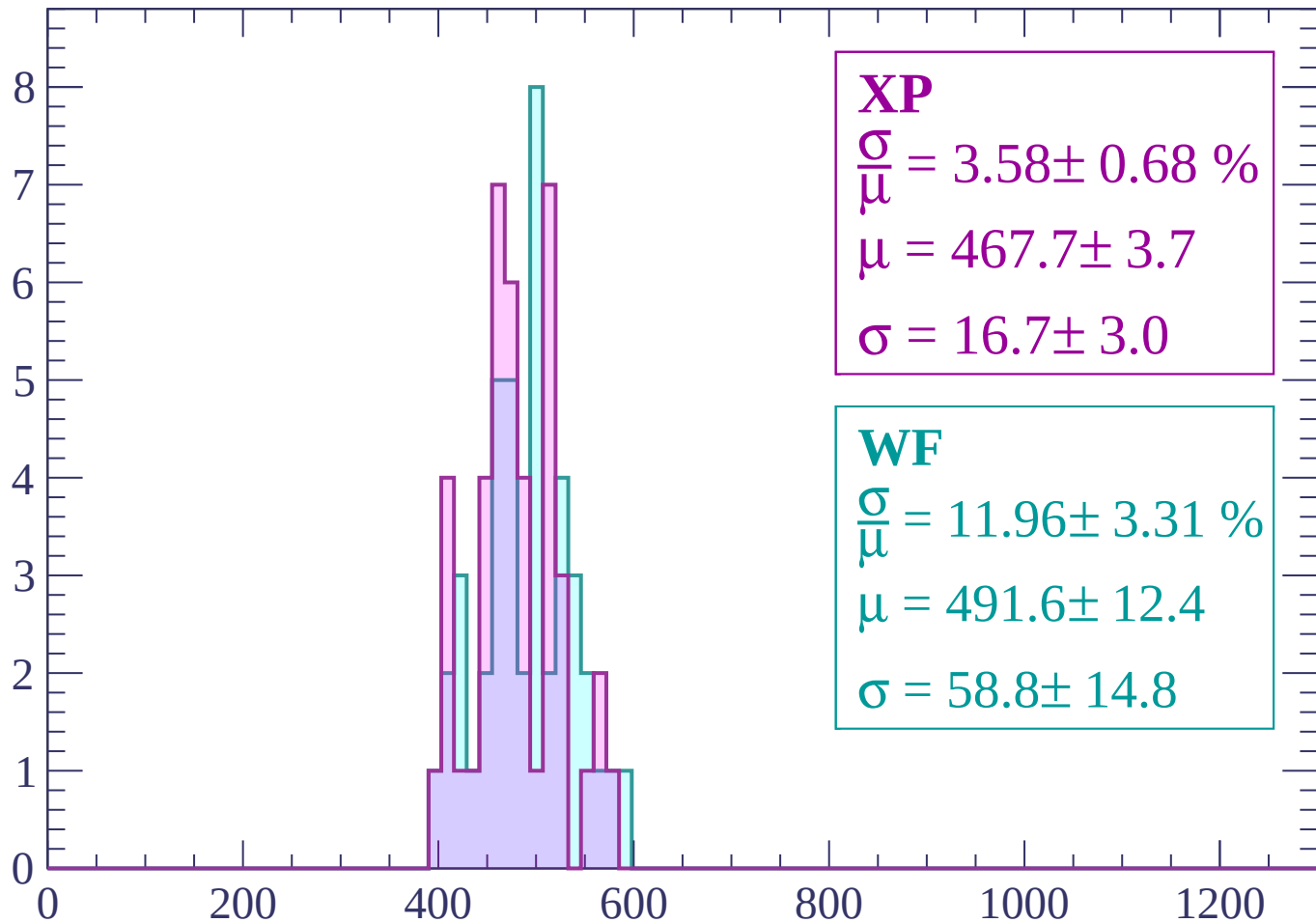
Count



dE/dx (ADC counts/cm)

Energy loss | $2820 < p < 2880$

Count



XP

$$\frac{\sigma}{\mu} = 3.58 \pm 0.68 \%$$

$$\mu = 467.7 \pm 3.7$$

$$\sigma = 16.7 \pm 3.0$$

WF

$$\frac{\sigma}{\mu} = 11.96 \pm 3.31 \%$$

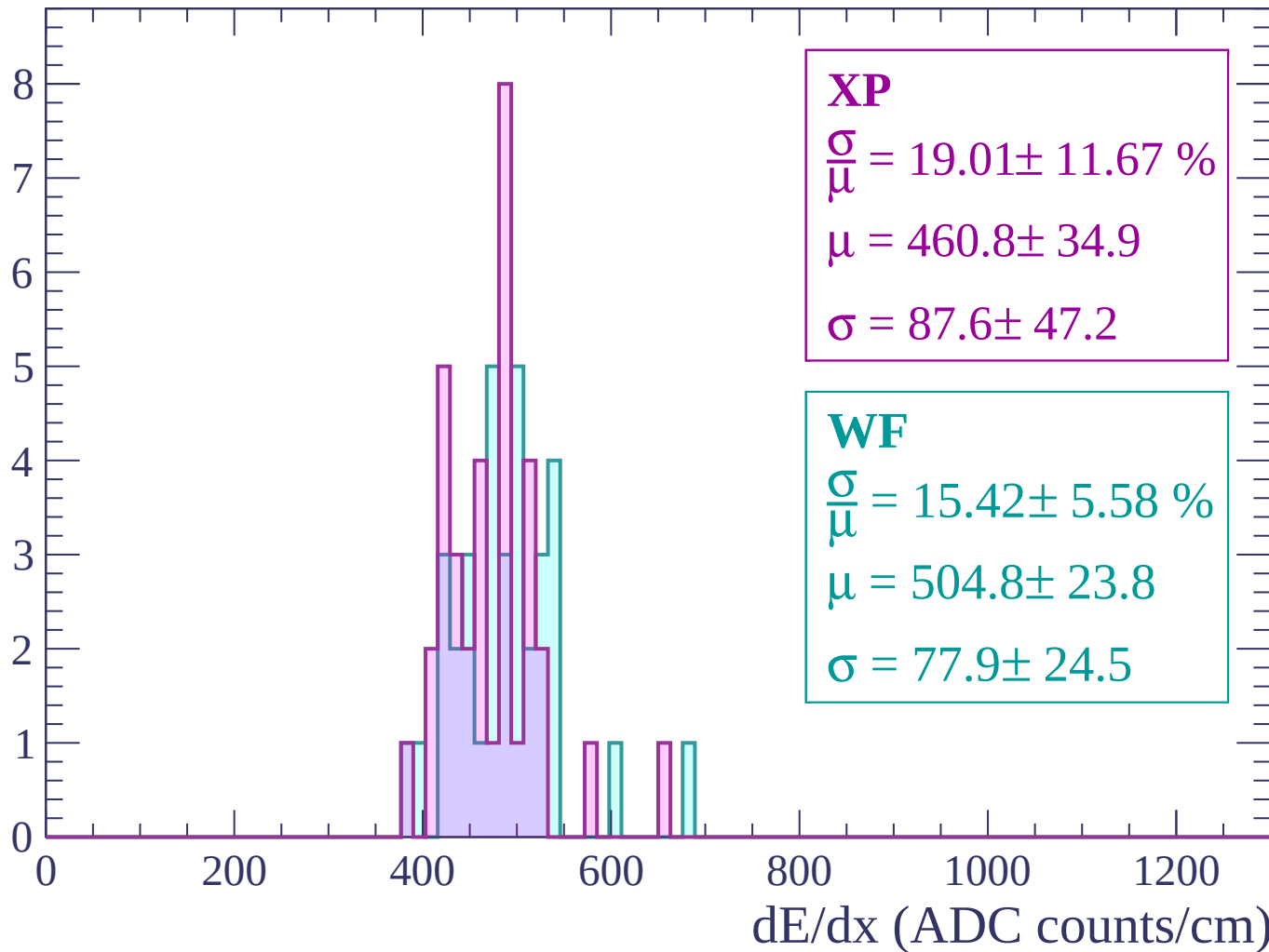
$$\mu = 491.6 \pm 12.4$$

$$\sigma = 58.8 \pm 14.8$$

dE/dx (ADC counts/cm)

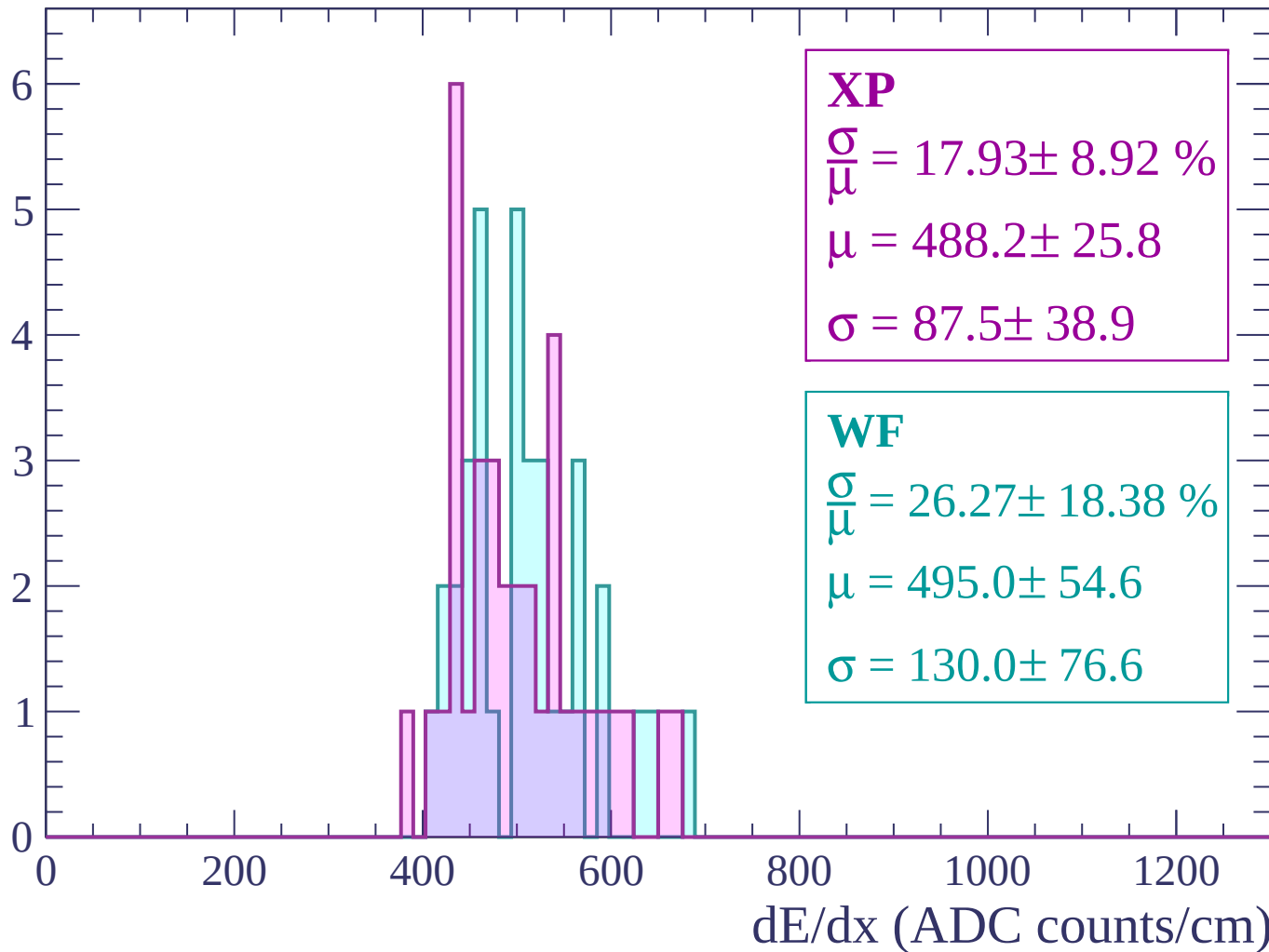
Energy loss | $2880 < p < 2940$

Count



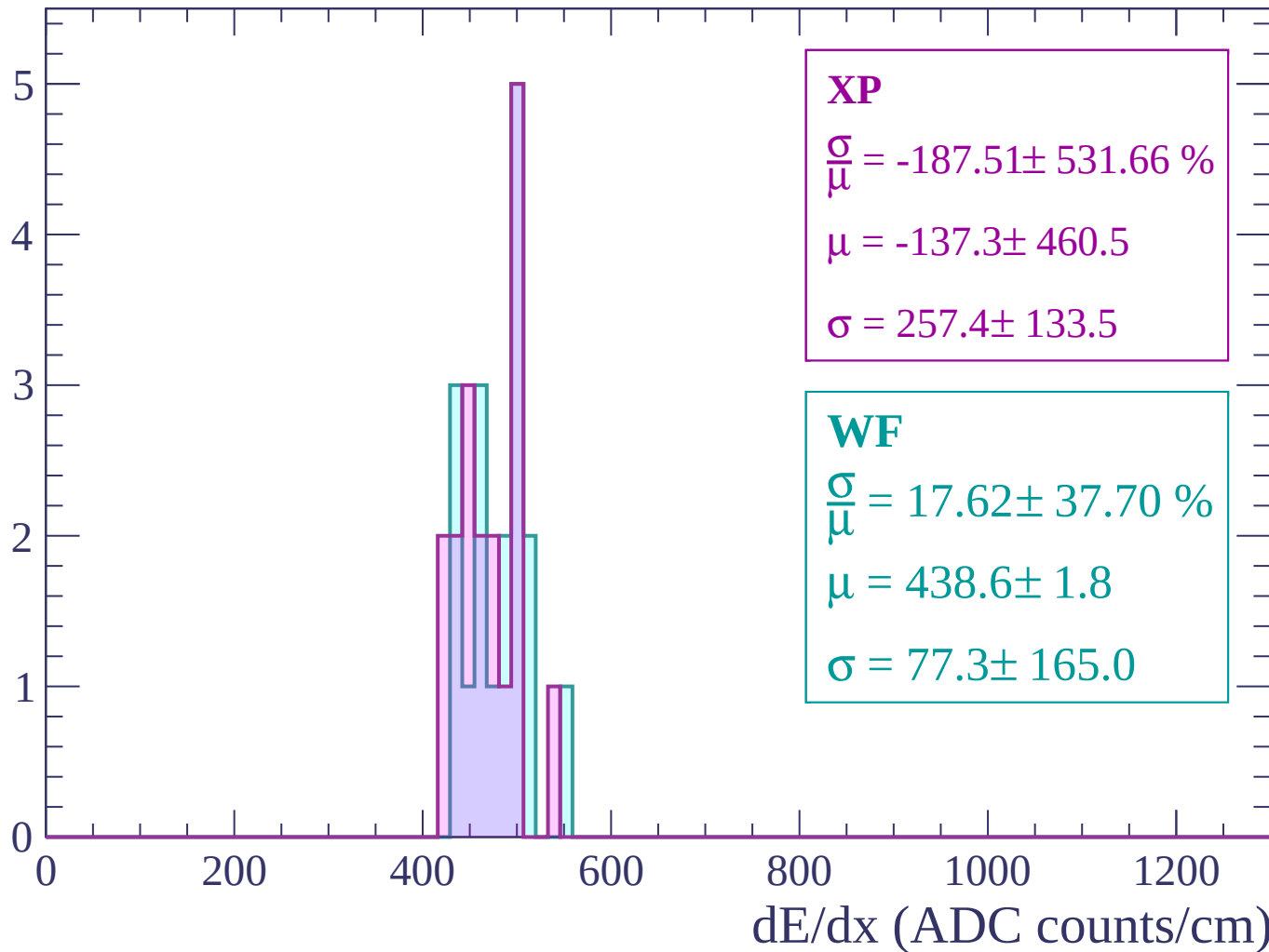
Energy loss | $2940 < p < 3000$

Count



Energy loss | $3000 < p < 3060$

Count



Energy loss | $-90 < \theta < -88$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-88 < \theta < -86$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-86 < \theta < -84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-84 < \theta < -82$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

Energy loss | $-82 < \theta < -80$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-80 < \theta < -78$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-78 < \theta < -76$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-76 < \theta < -74$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-74 < \theta < -72$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

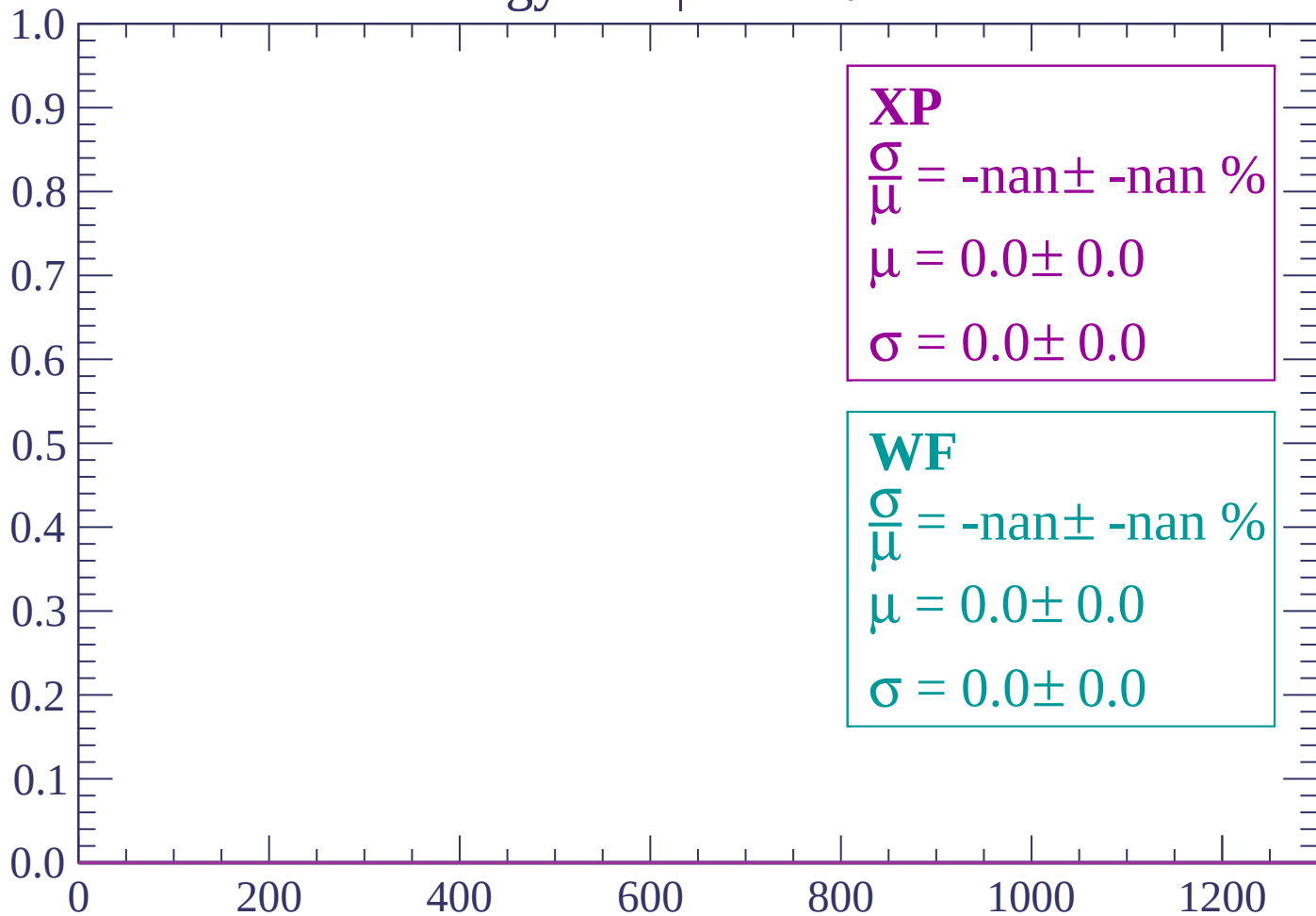
$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-72 < \theta < -70$

Count



dE/dx (ADC counts/cm)

Energy loss | $-70 < \theta < -68$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-68 < \theta < -66$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

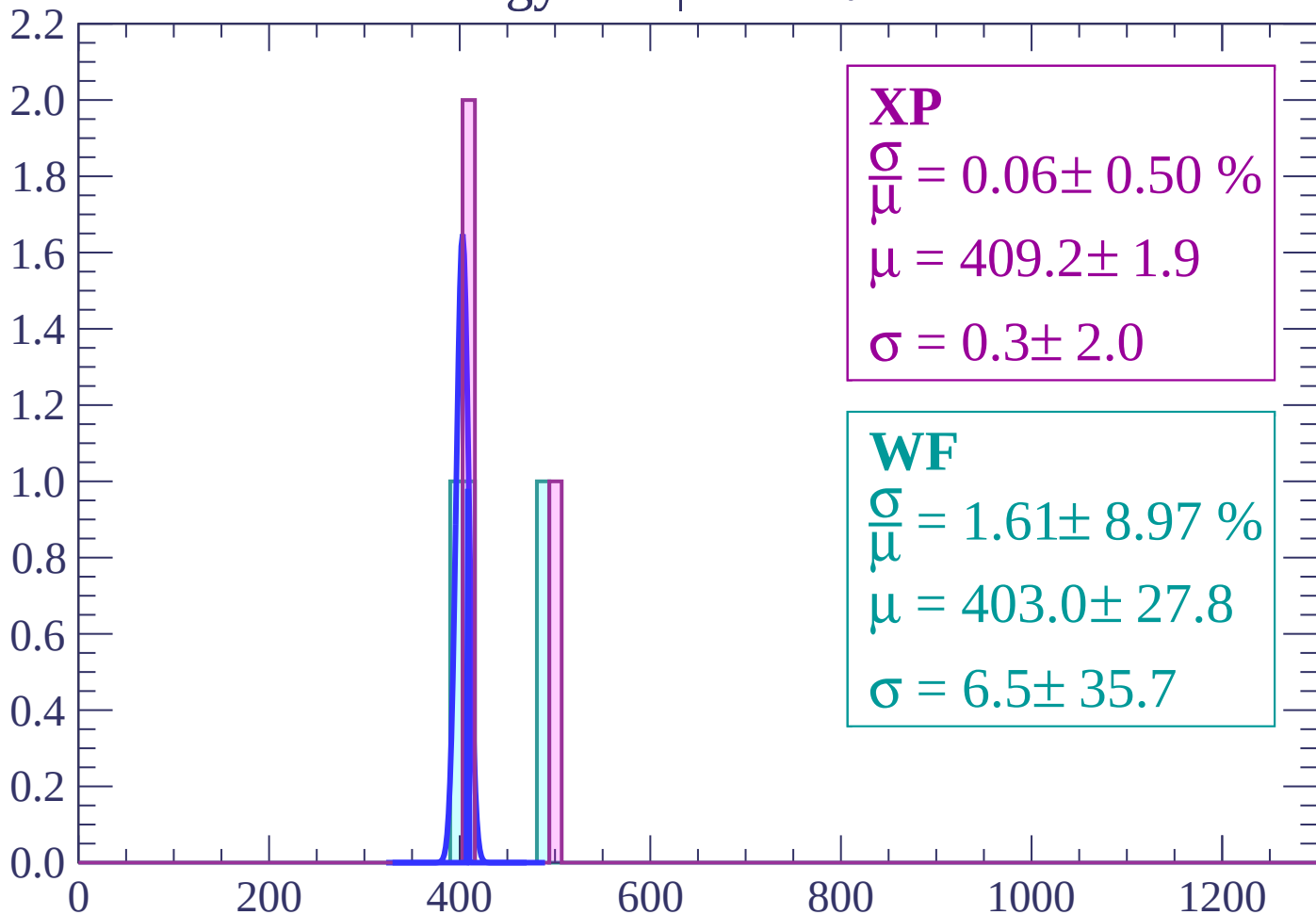
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-66 < \theta < -64$

Count



XP

$$\frac{\sigma}{\mu} = 0.06 \pm 0.50 \%$$

$$\mu = 409.2 \pm 1.9$$

$$\sigma = 0.3 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 1.61 \pm 8.97 \%$$

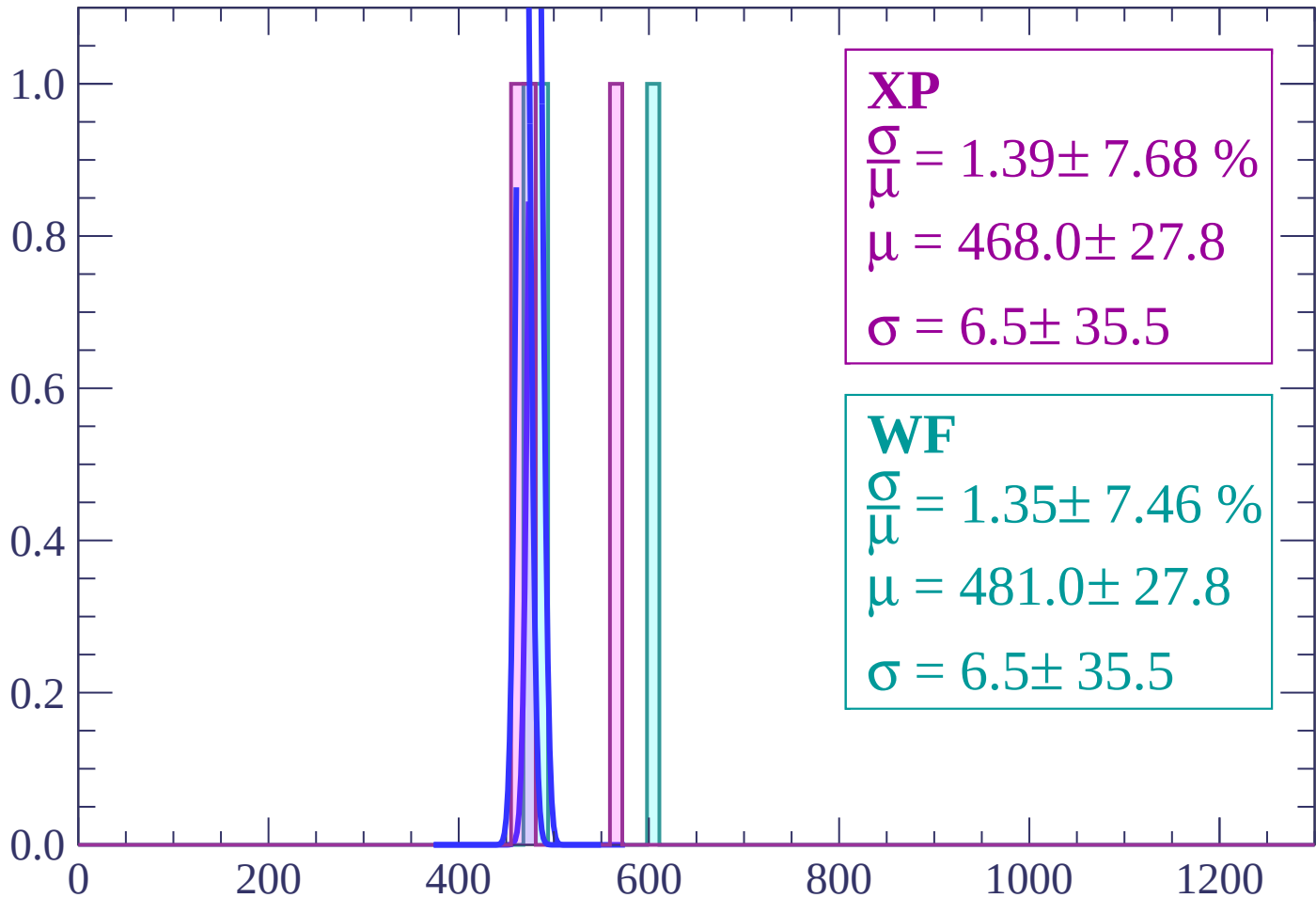
$$\mu = 403.0 \pm 27.8$$

$$\sigma = 6.5 \pm 35.7$$

dE/dx (ADC counts/cm)

Energy loss | $-64 < \theta < -62$

Count



XP

$$\frac{\sigma}{\mu} = 1.39 \pm 7.68 \%$$

$$\mu = 468.0 \pm 27.8$$

$$\sigma = 6.5 \pm 35.5$$

WF

$$\frac{\sigma}{\mu} = 1.35 \pm 7.46 \%$$

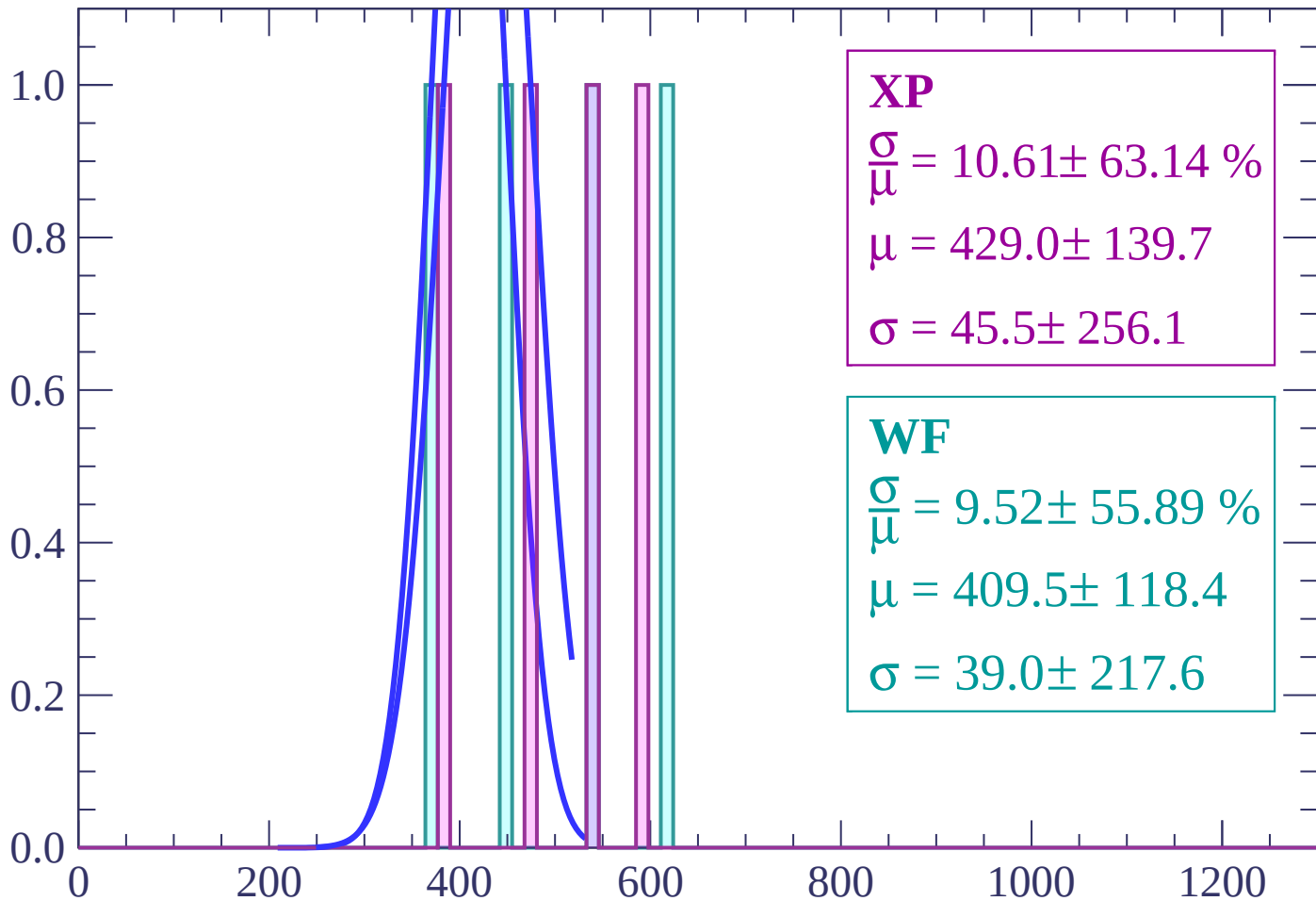
$$\mu = 481.0 \pm 27.8$$

$$\sigma = 6.5 \pm 35.5$$

dE/dx (ADC counts/cm)

Energy loss | $-62 < \theta < -60$

Count



XP

$$\frac{\sigma}{\mu} = 10.61 \pm 63.14 \%$$

$$\mu = 429.0 \pm 139.7$$

$$\sigma = 45.5 \pm 256.1$$

WF

$$\frac{\sigma}{\mu} = 9.52 \pm 55.89 \%$$

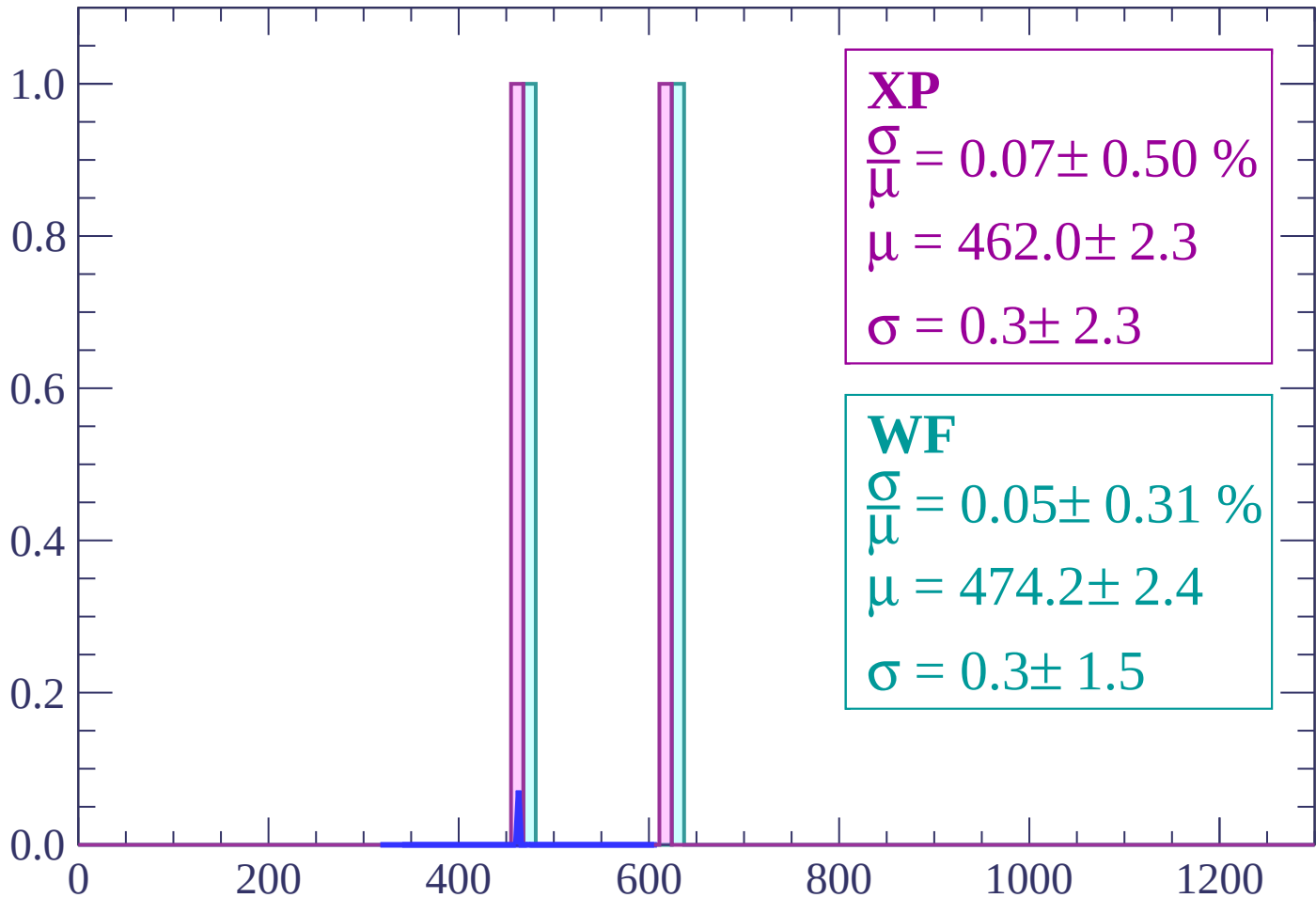
$$\mu = 409.5 \pm 118.4$$

$$\sigma = 39.0 \pm 217.6$$

dE/dx (ADC counts/cm)

Energy loss | $-60 < \theta < -58$

Count



XP

$$\frac{\sigma}{\mu} = 0.07 \pm 0.50 \%$$

$$\mu = 462.0 \pm 2.3$$

$$\sigma = 0.3 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 0.05 \pm 0.31 \%$$

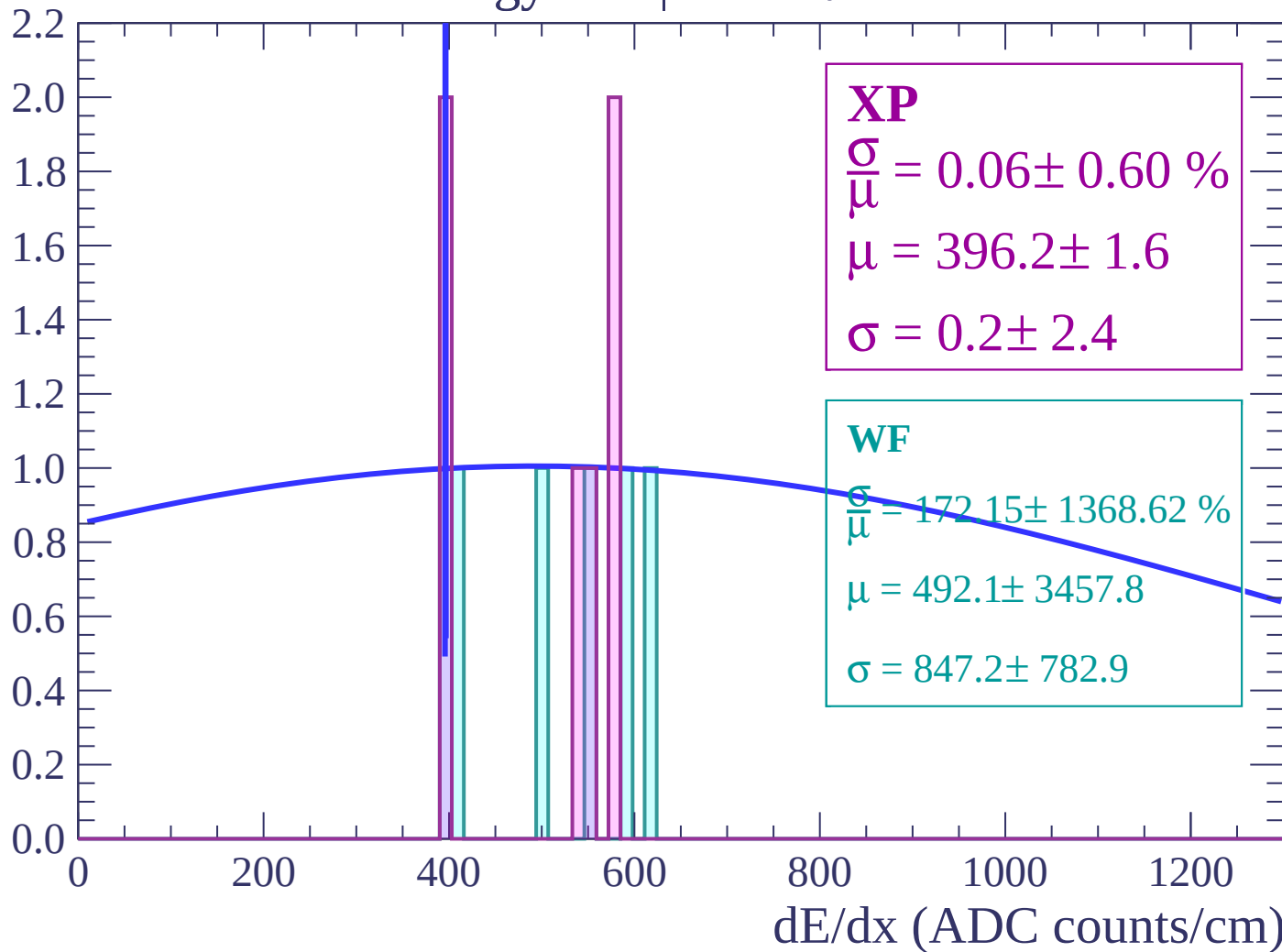
$$\mu = 474.2 \pm 2.4$$

$$\sigma = 0.3 \pm 1.5$$

dE/dx (ADC counts/cm)

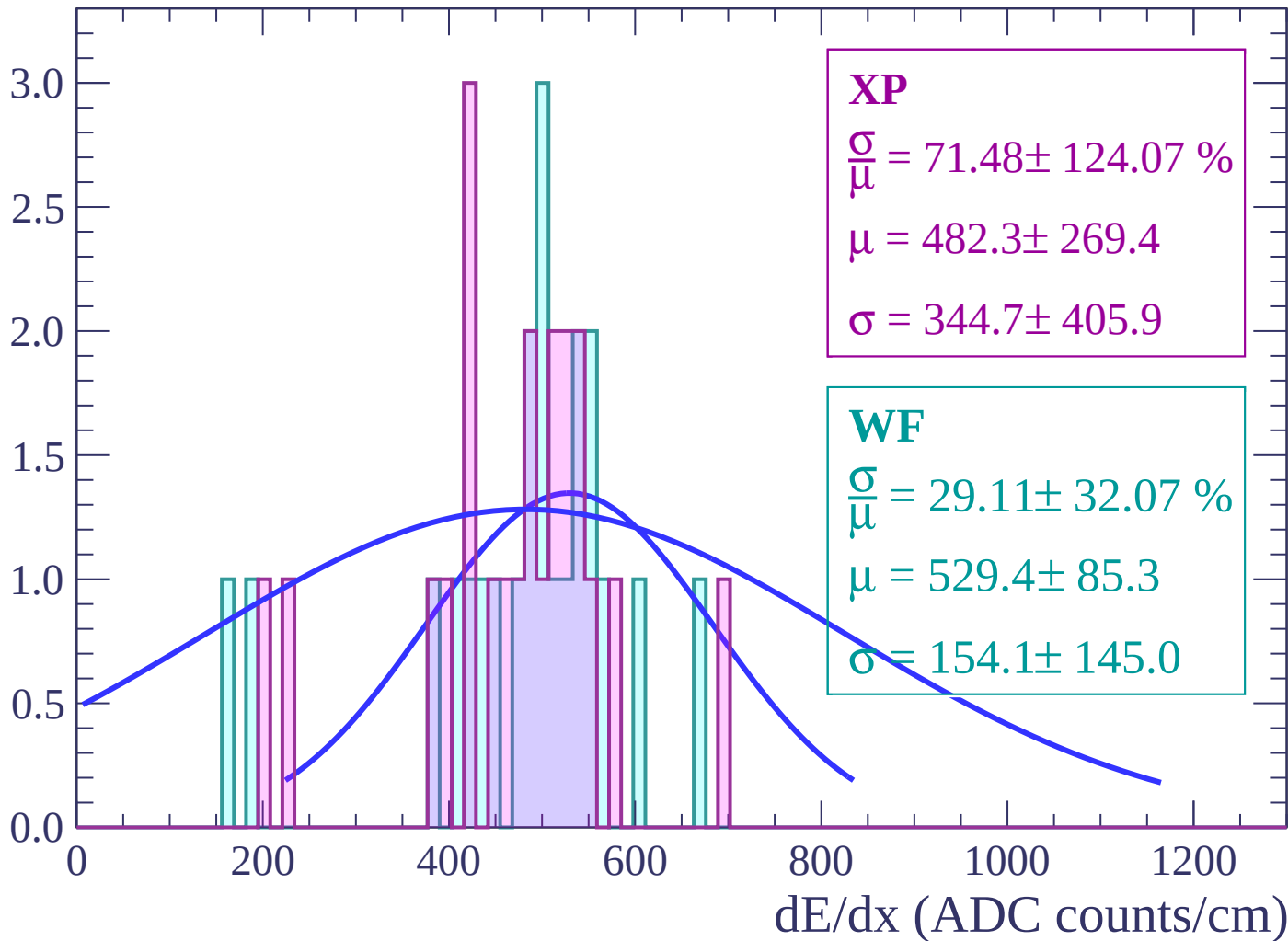
Energy loss | $-58 < \theta < -56$

Count



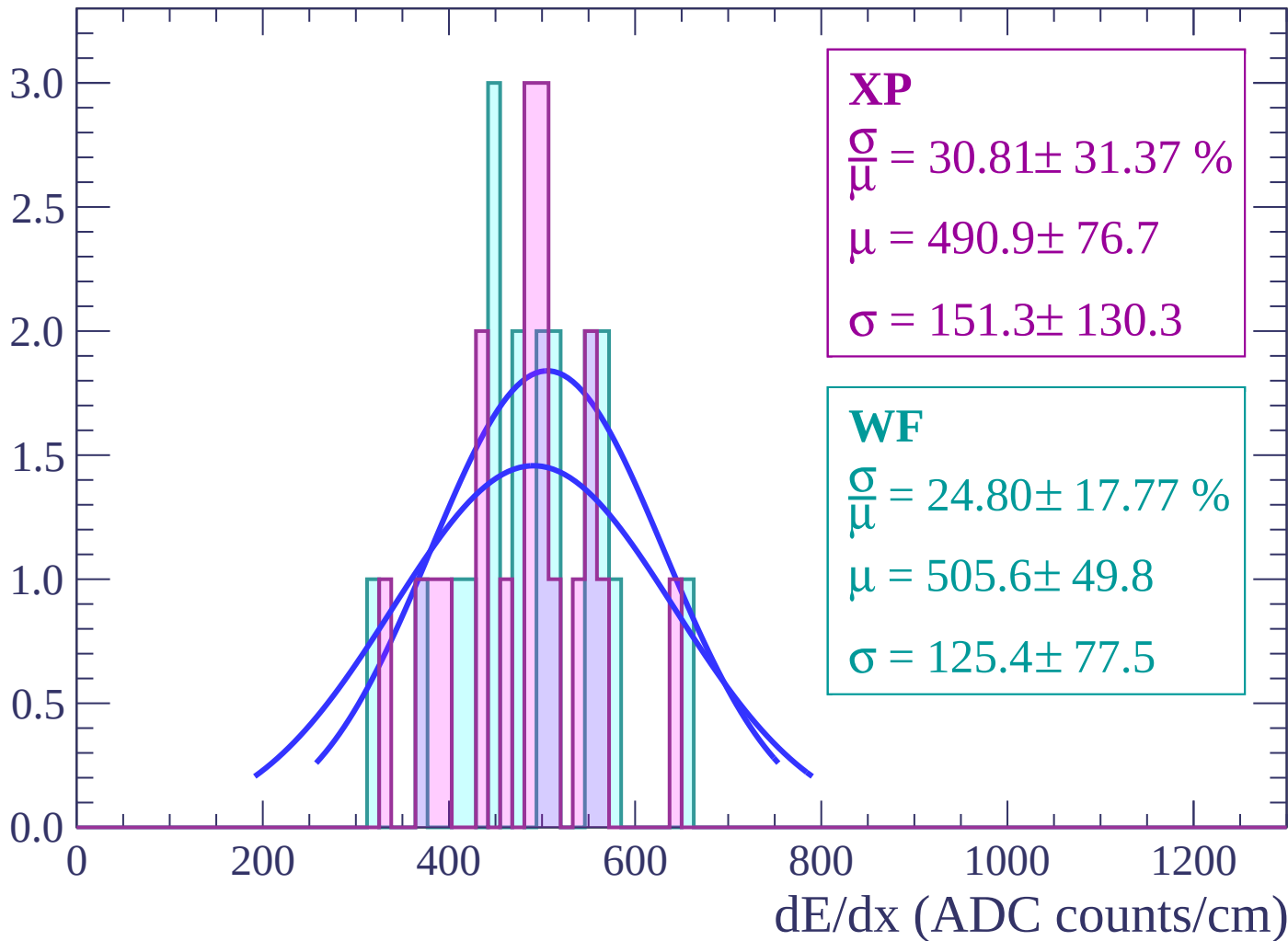
Energy loss | $-56 < \theta < -54$

Count



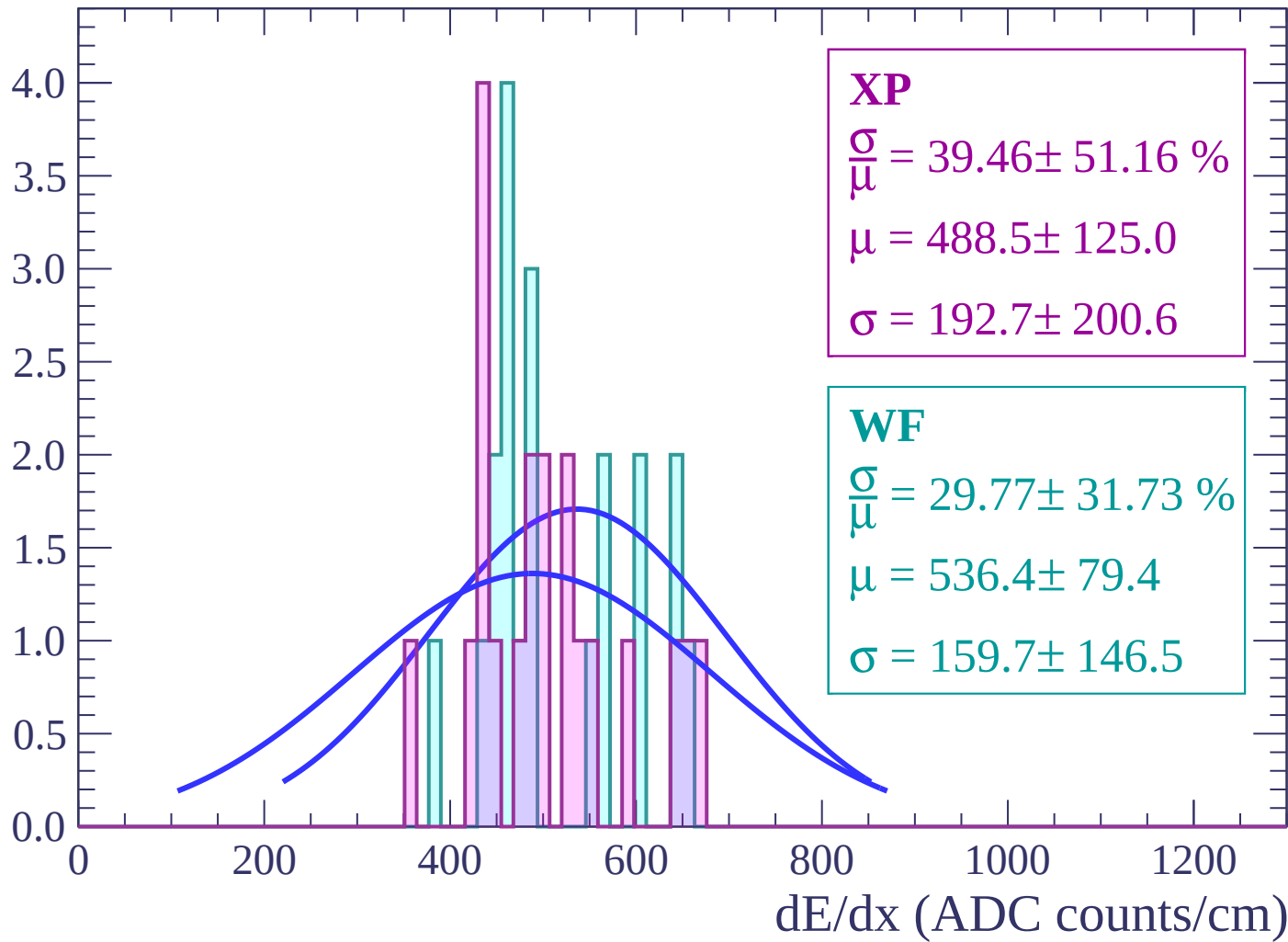
Energy loss | $-54 < \theta < -52$

Count



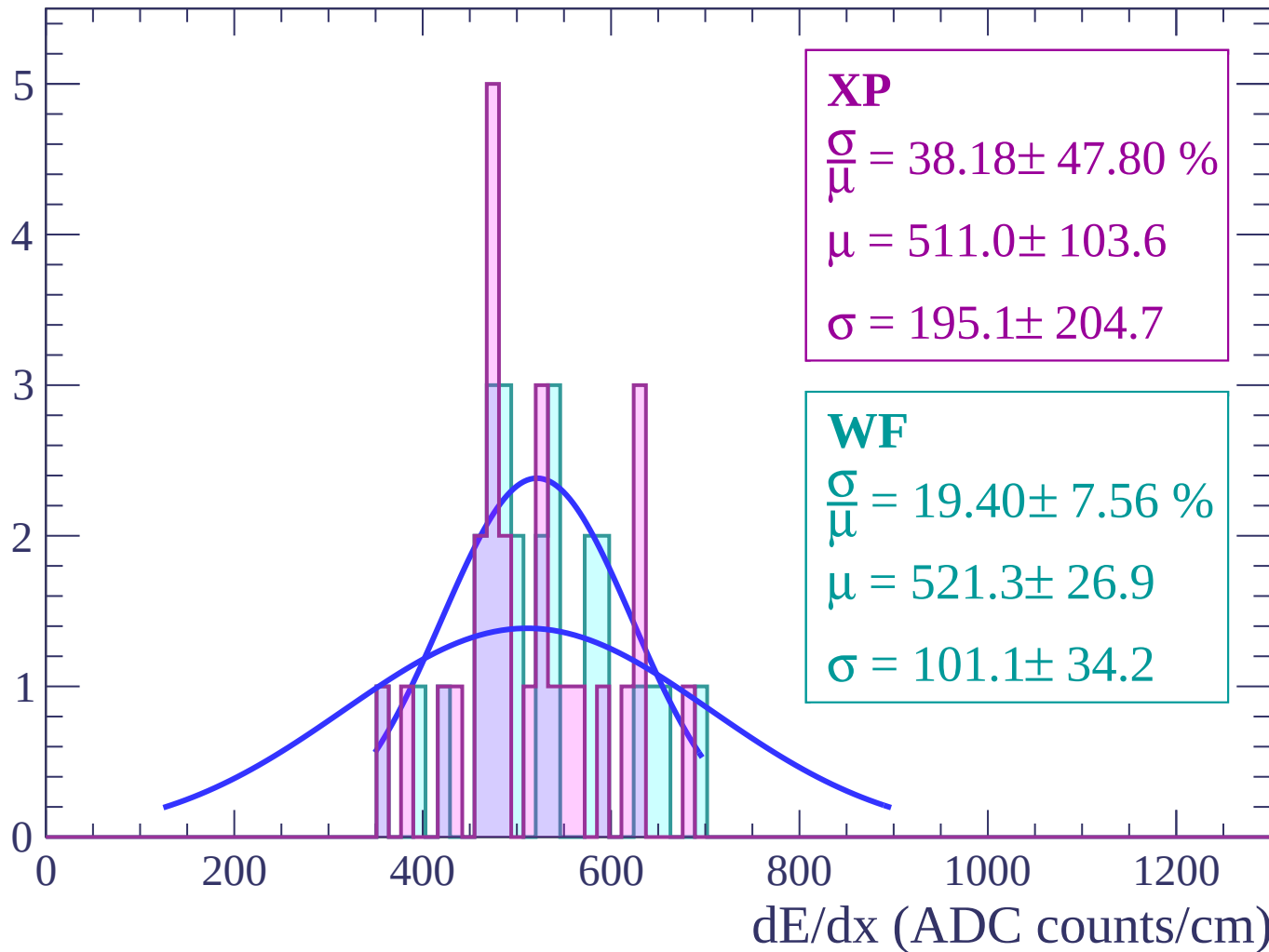
Energy loss | $-52 < \theta < -50$

Count



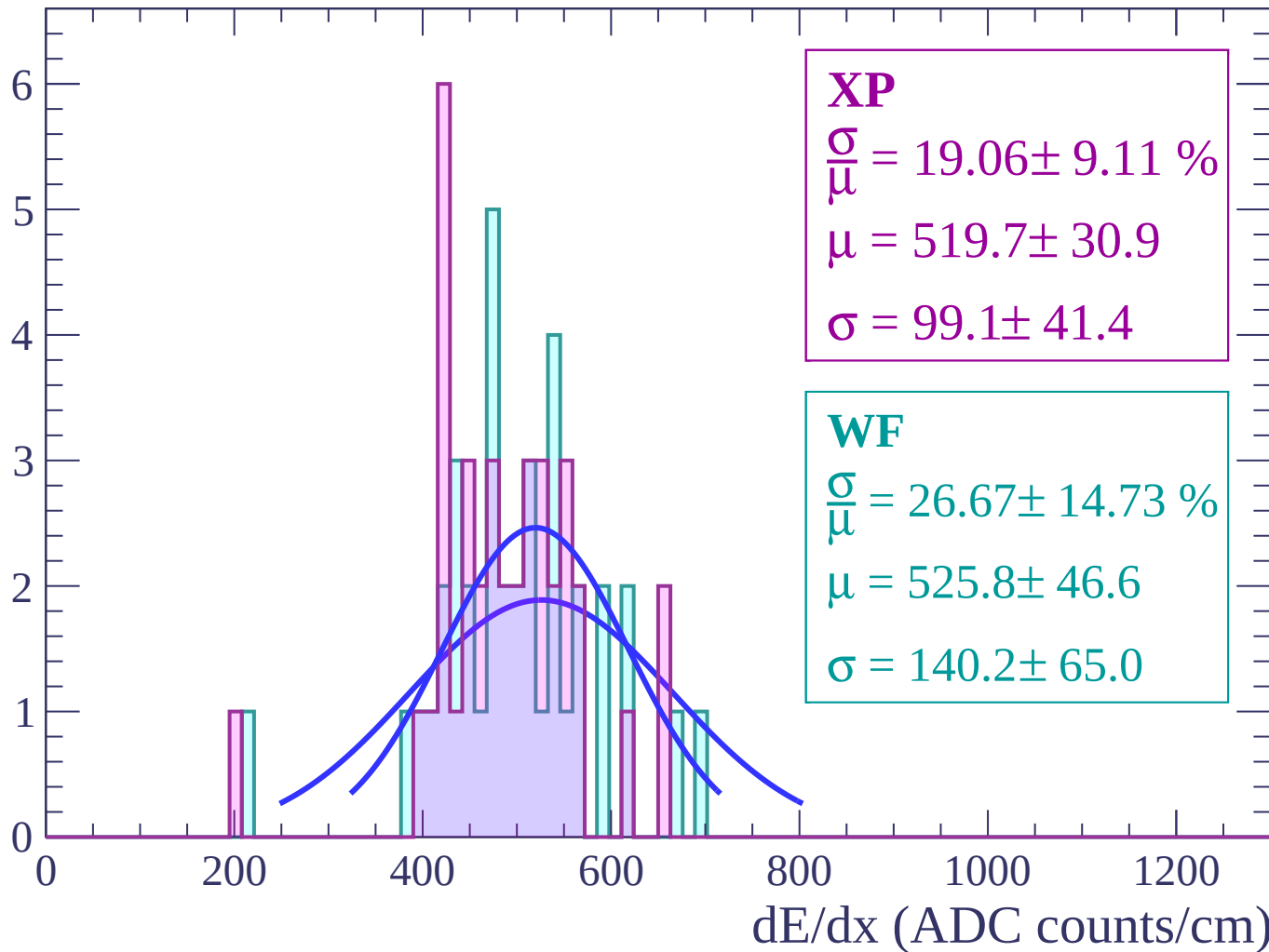
Energy loss | $-50 < \theta < -48$

Count



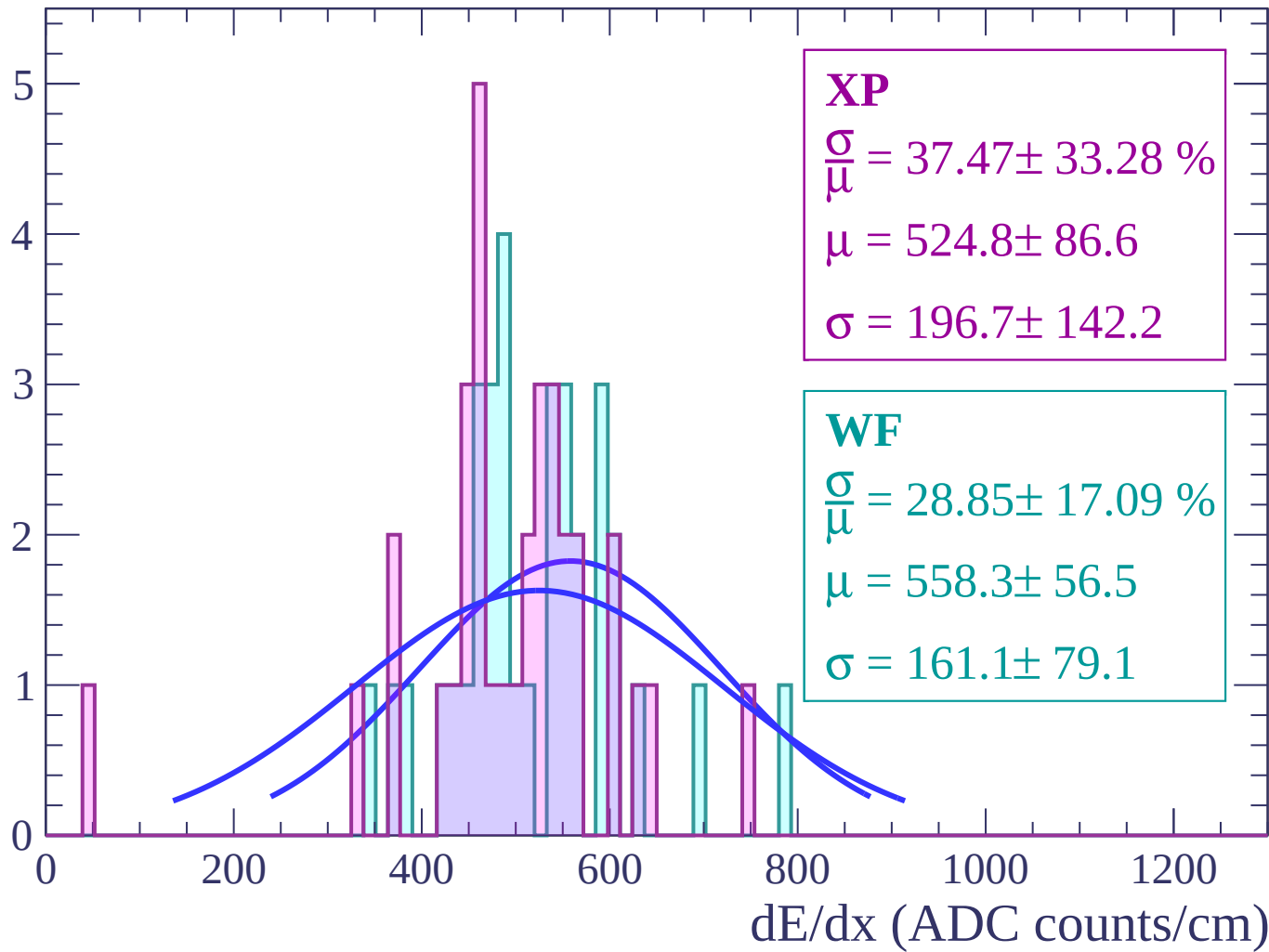
Energy loss | $-48 < \theta < -46$

Count



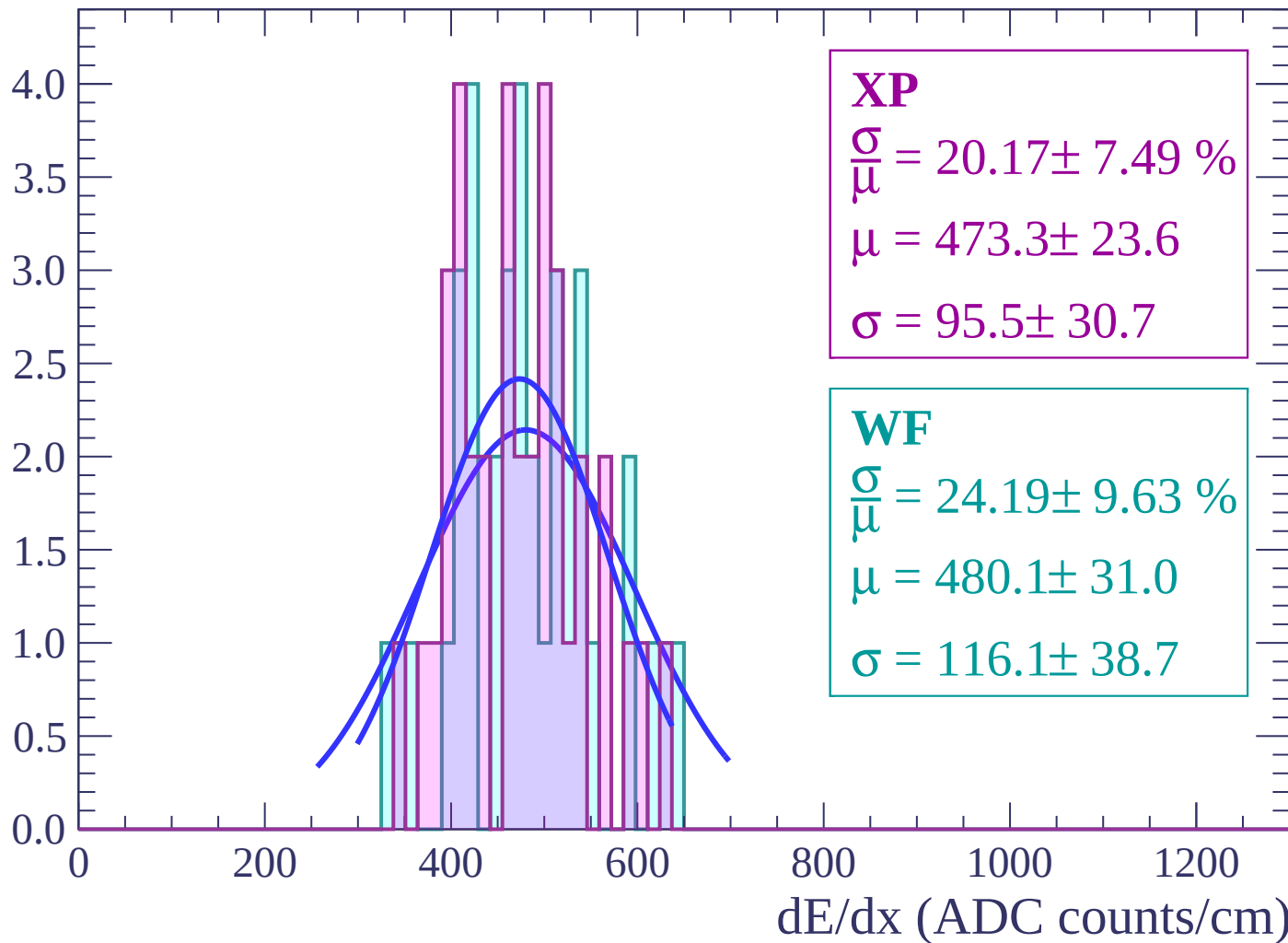
Energy loss | $-46 < \theta < -44$

Count



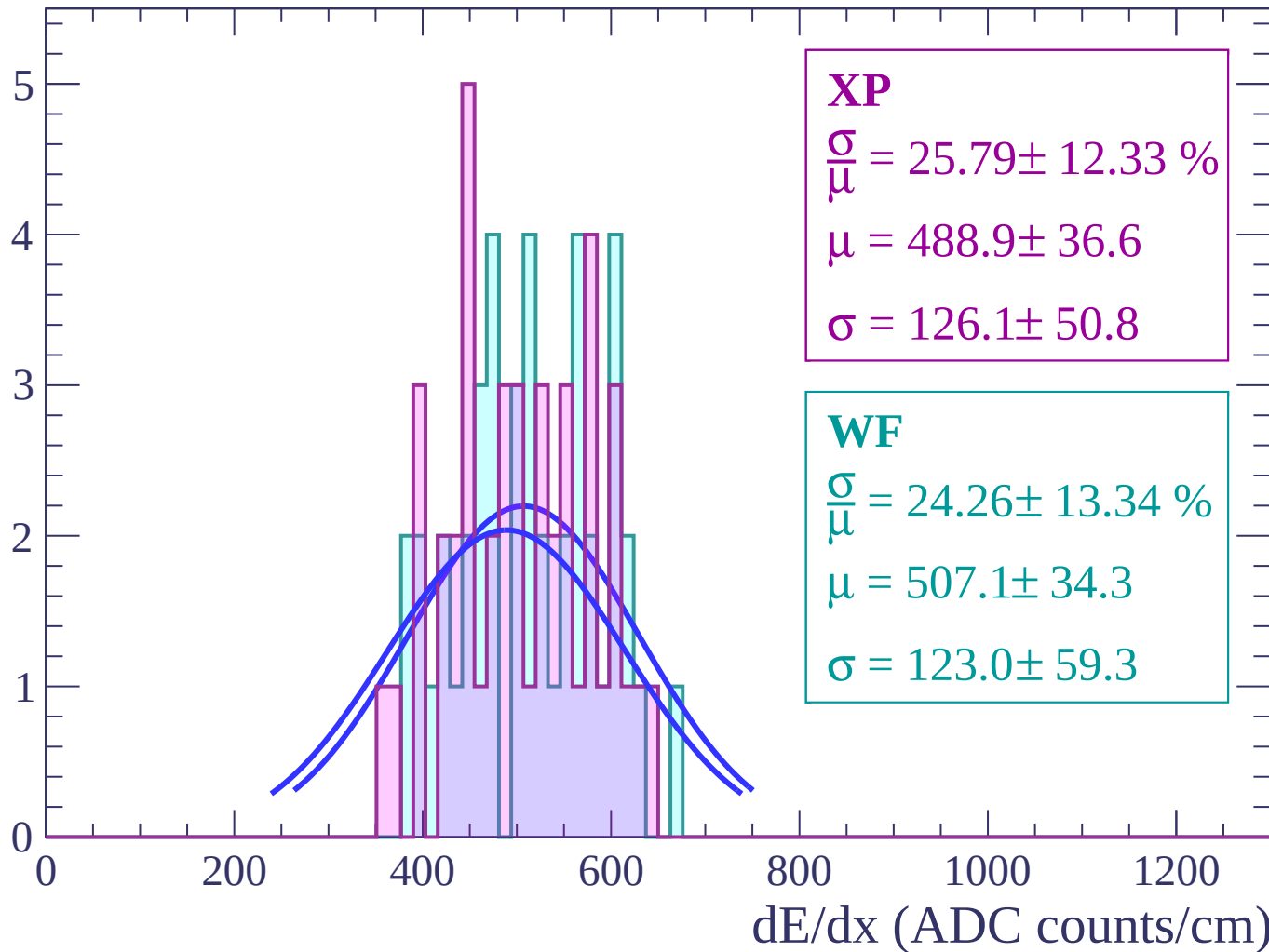
Energy loss | $-44 < \theta < -42$

Count



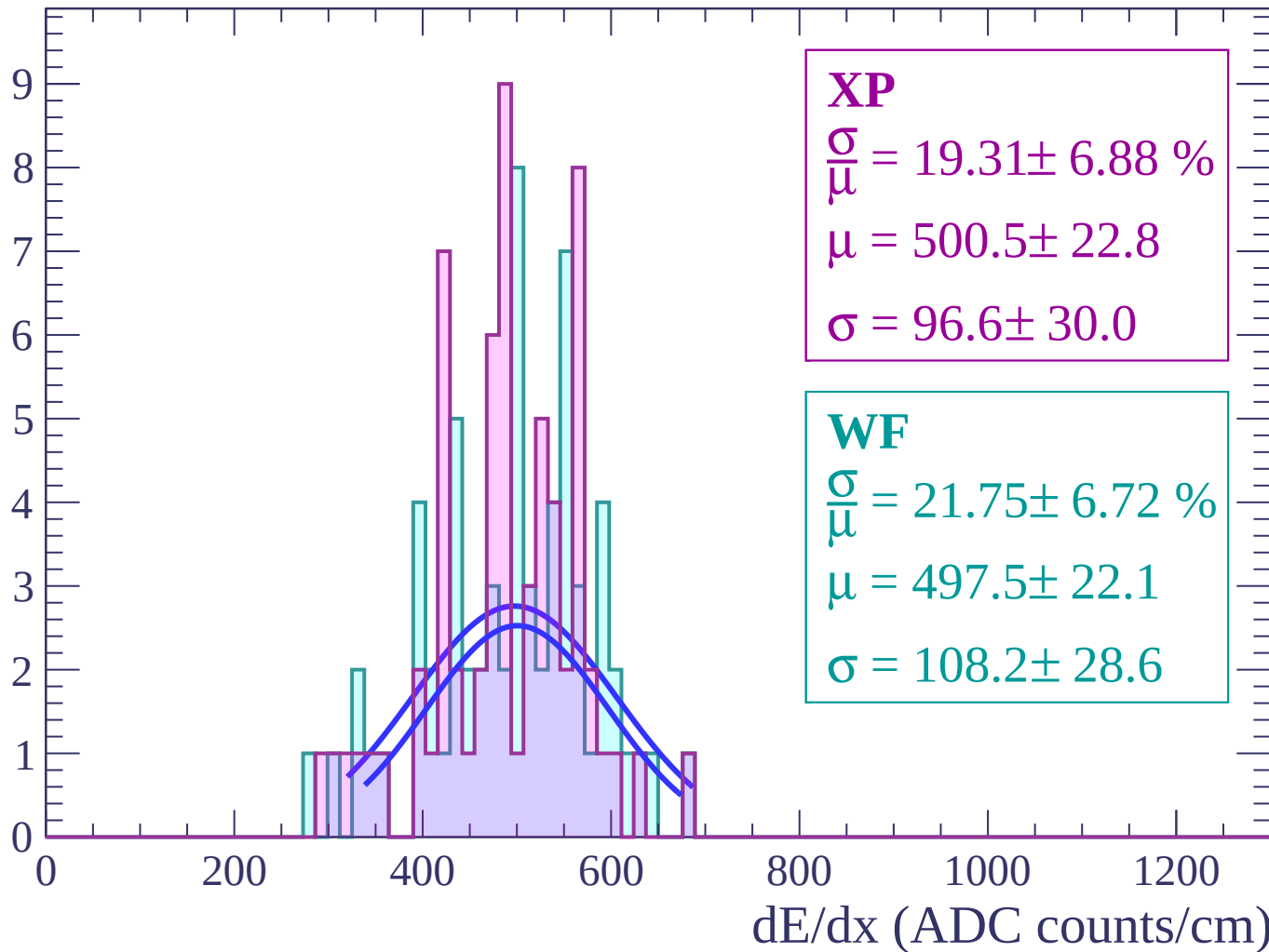
Energy loss | $-42 < \theta < -40$

Count



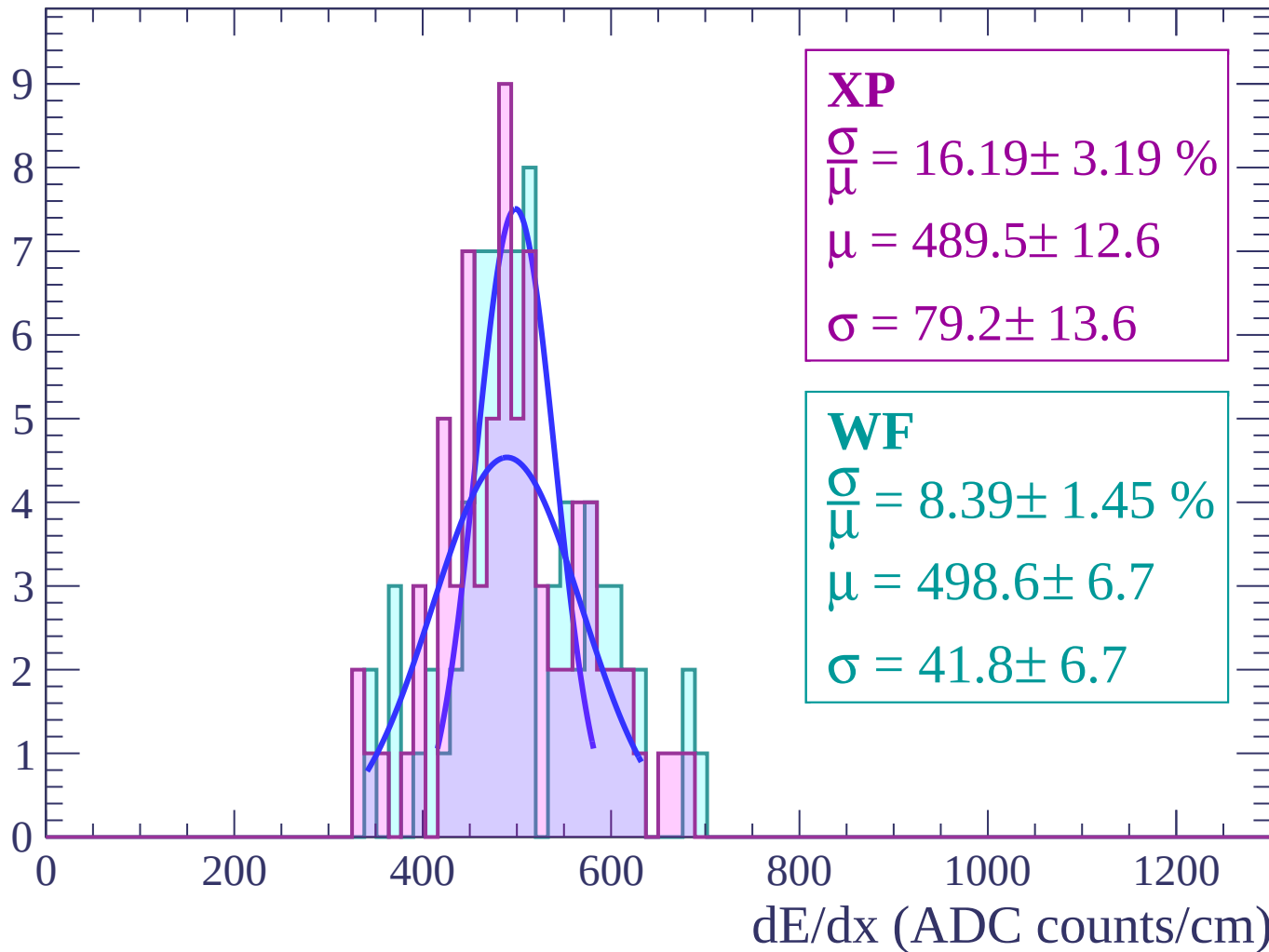
Energy loss | $-40 < \theta < -38$

Count



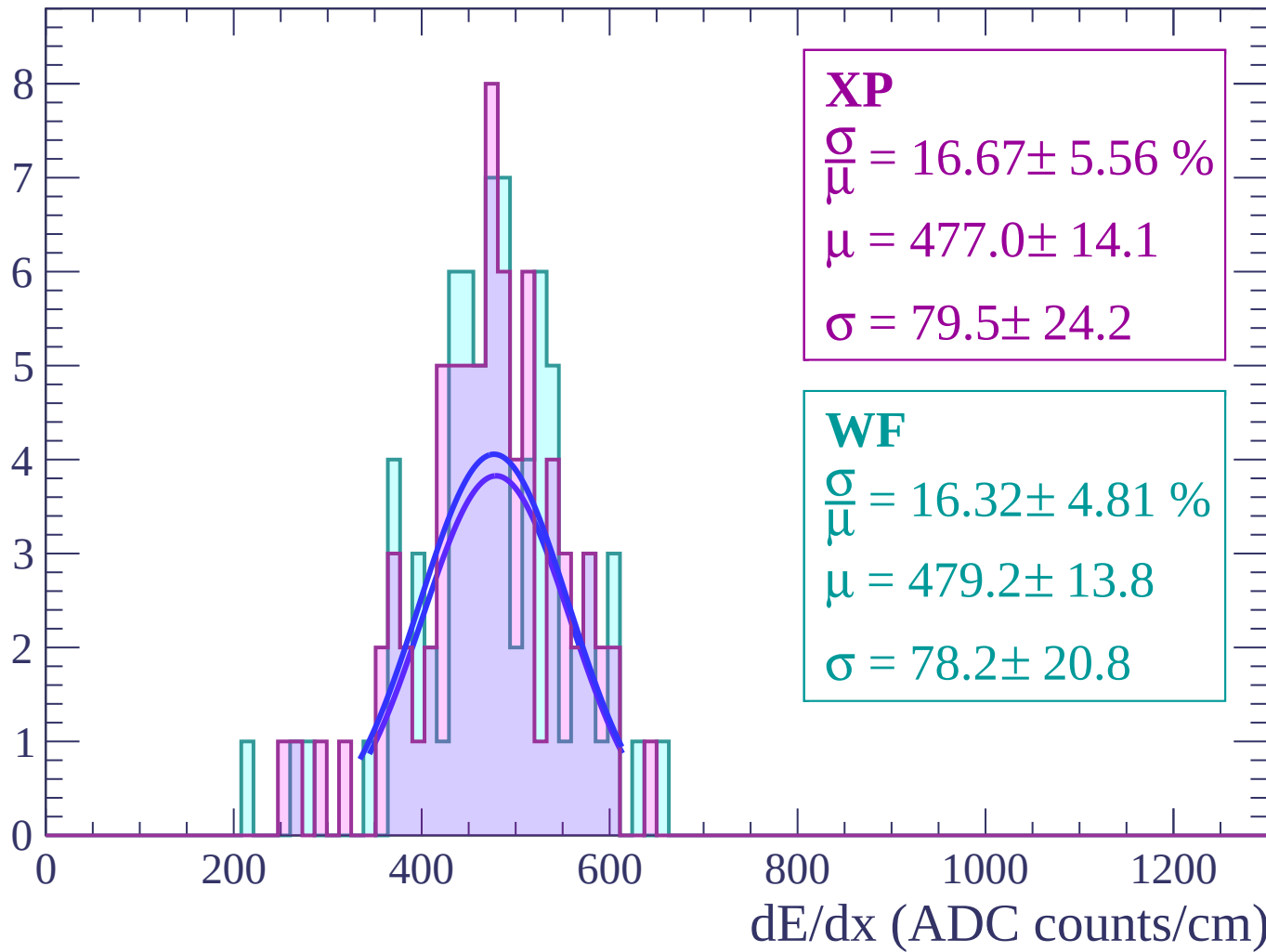
Energy loss | $-38 < \theta < -36$

Count



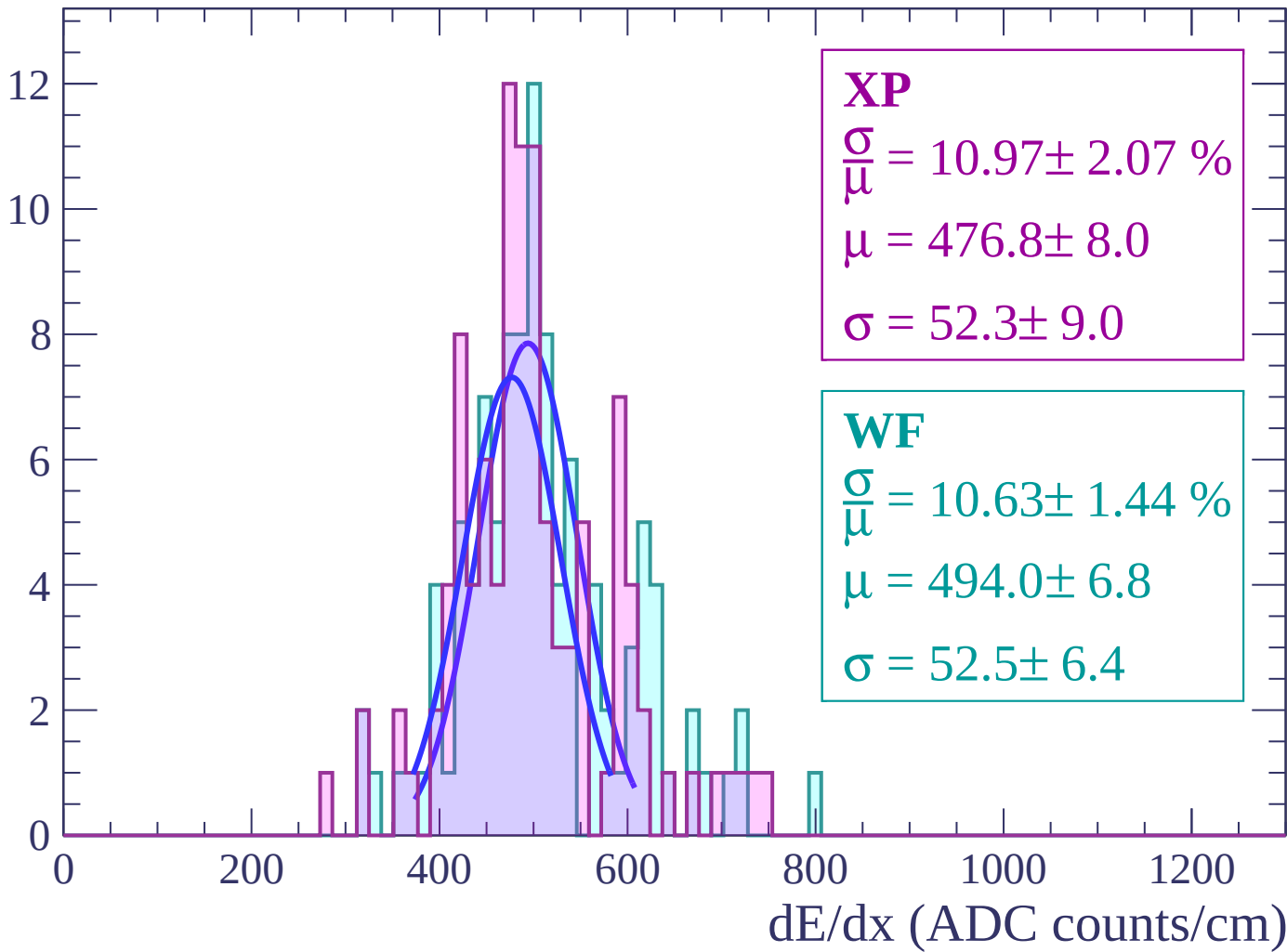
Energy loss | $-36 < \theta < -34$

Count



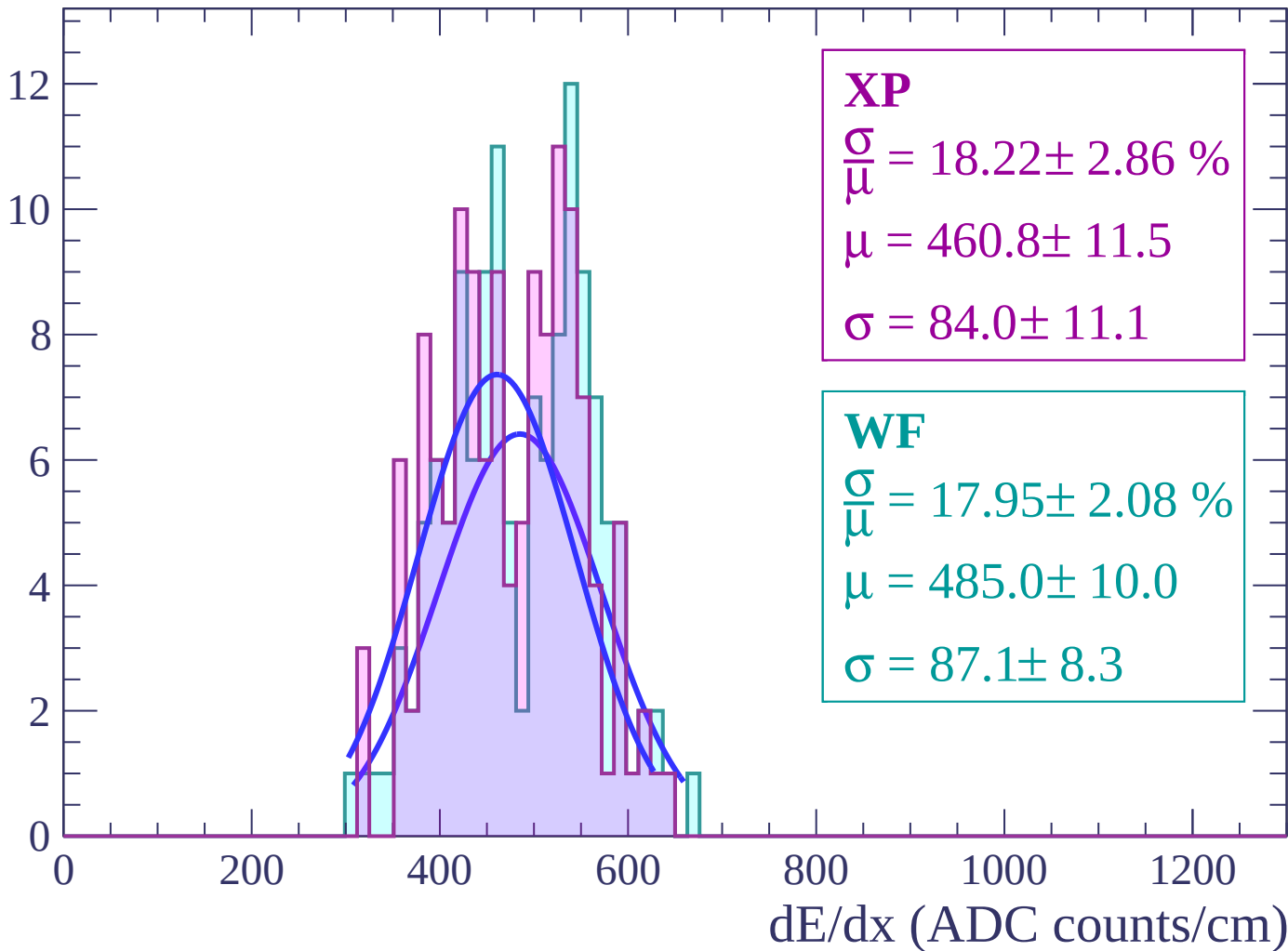
Energy loss | $-34 < \theta < -32$

Count



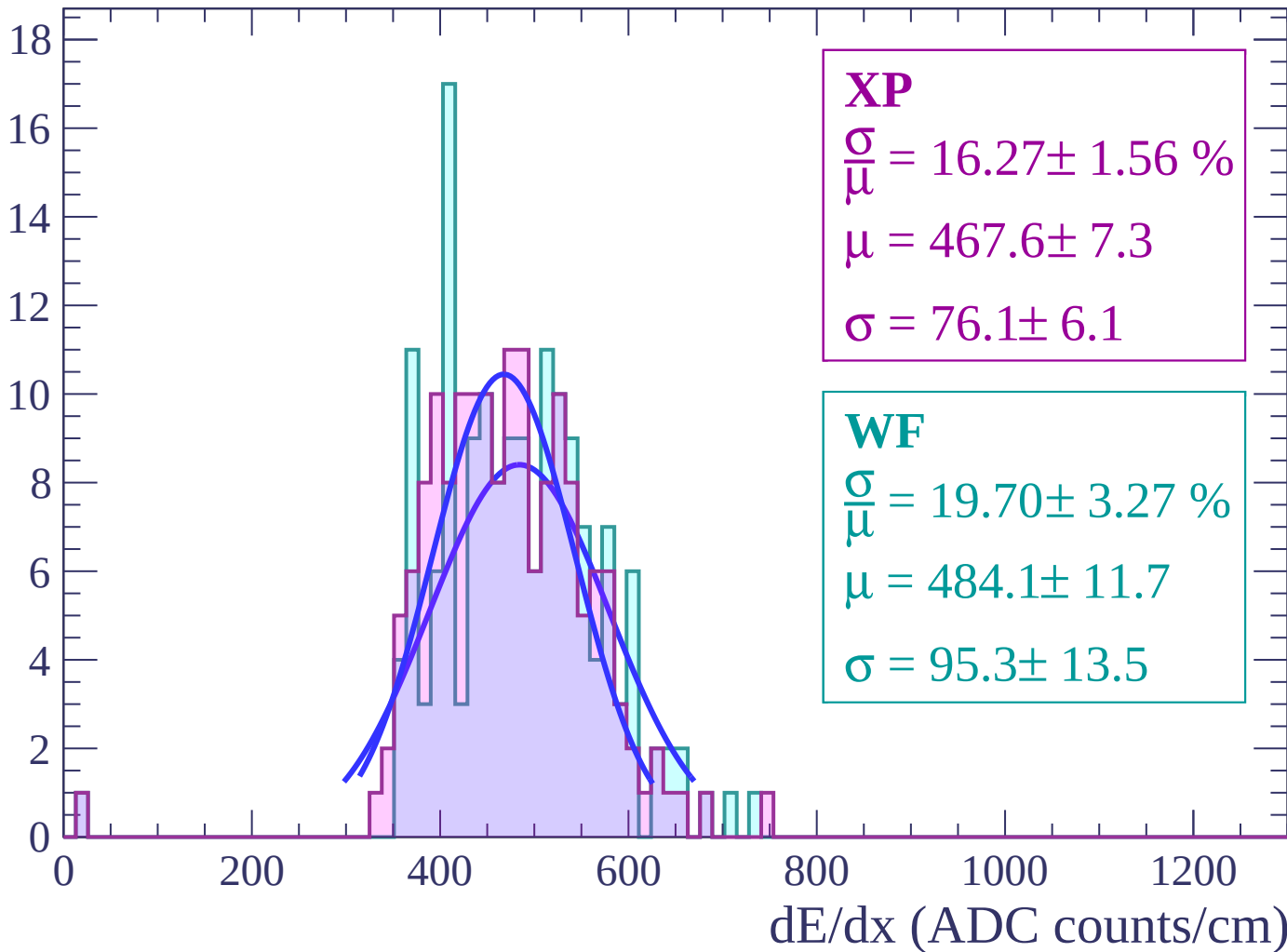
Energy loss | $-32 < \theta < -30$

Count



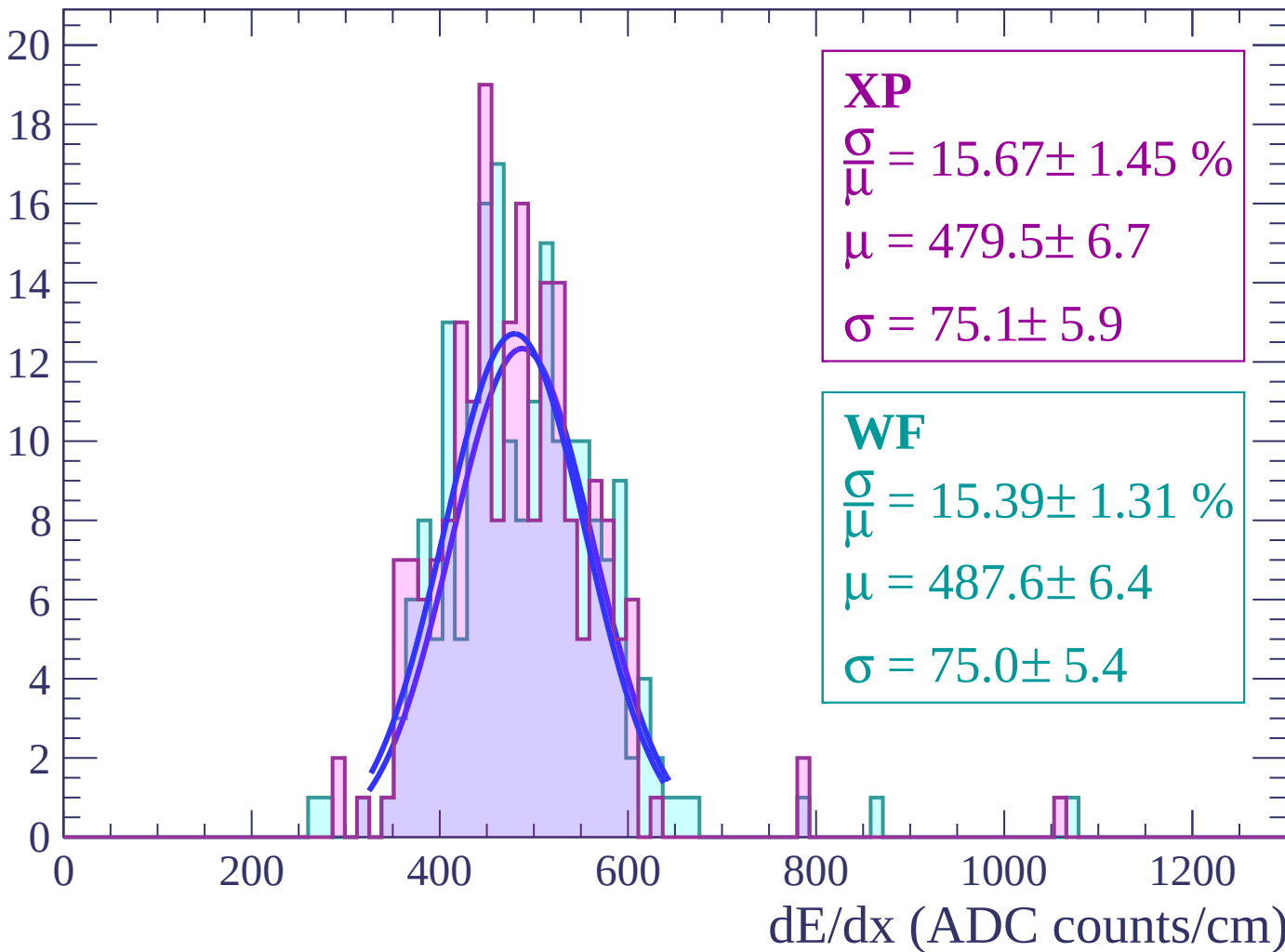
Energy loss | $-30 < \theta < -28$

Count



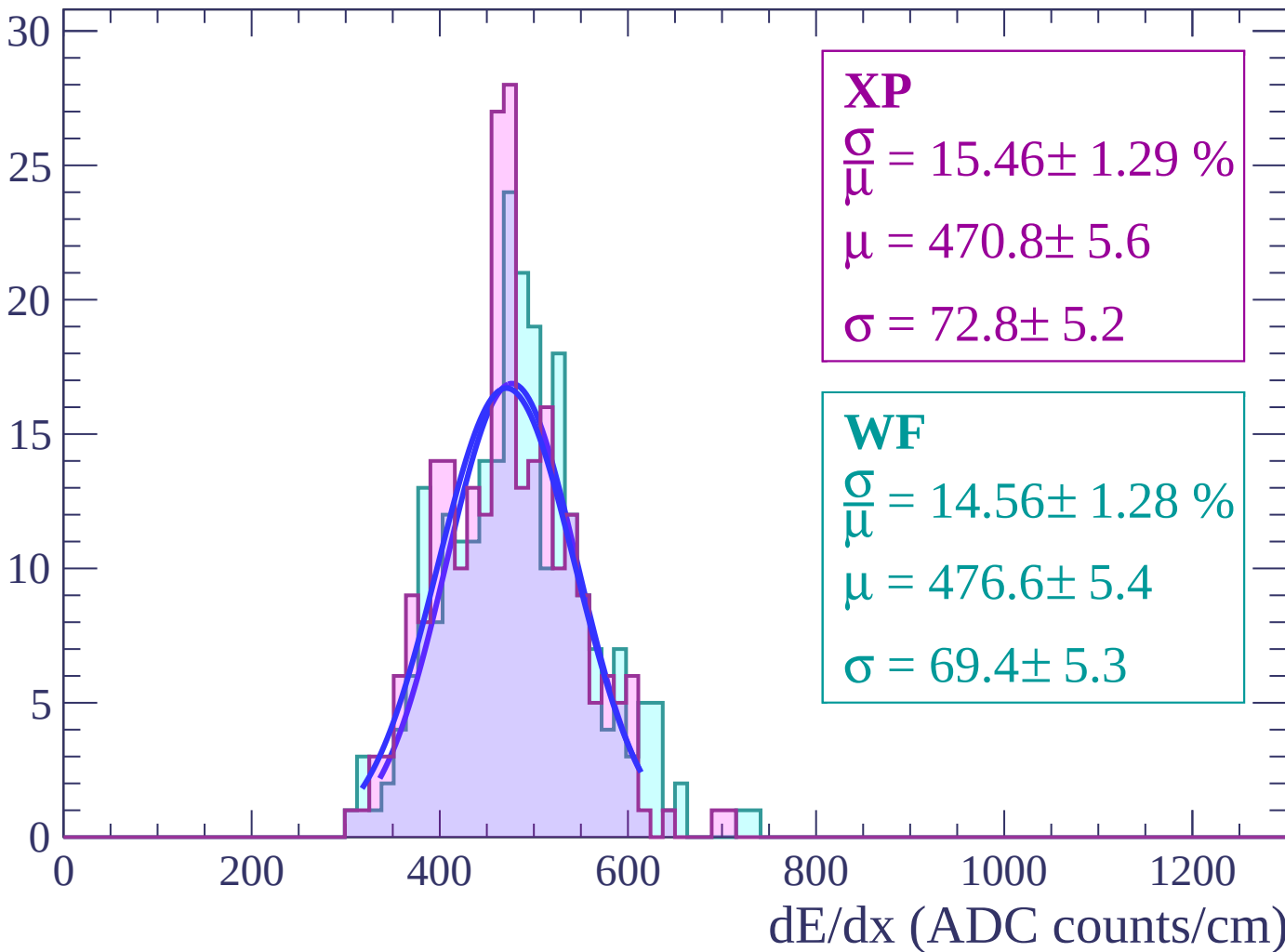
Energy loss | $-28 < \theta < -26$

Count



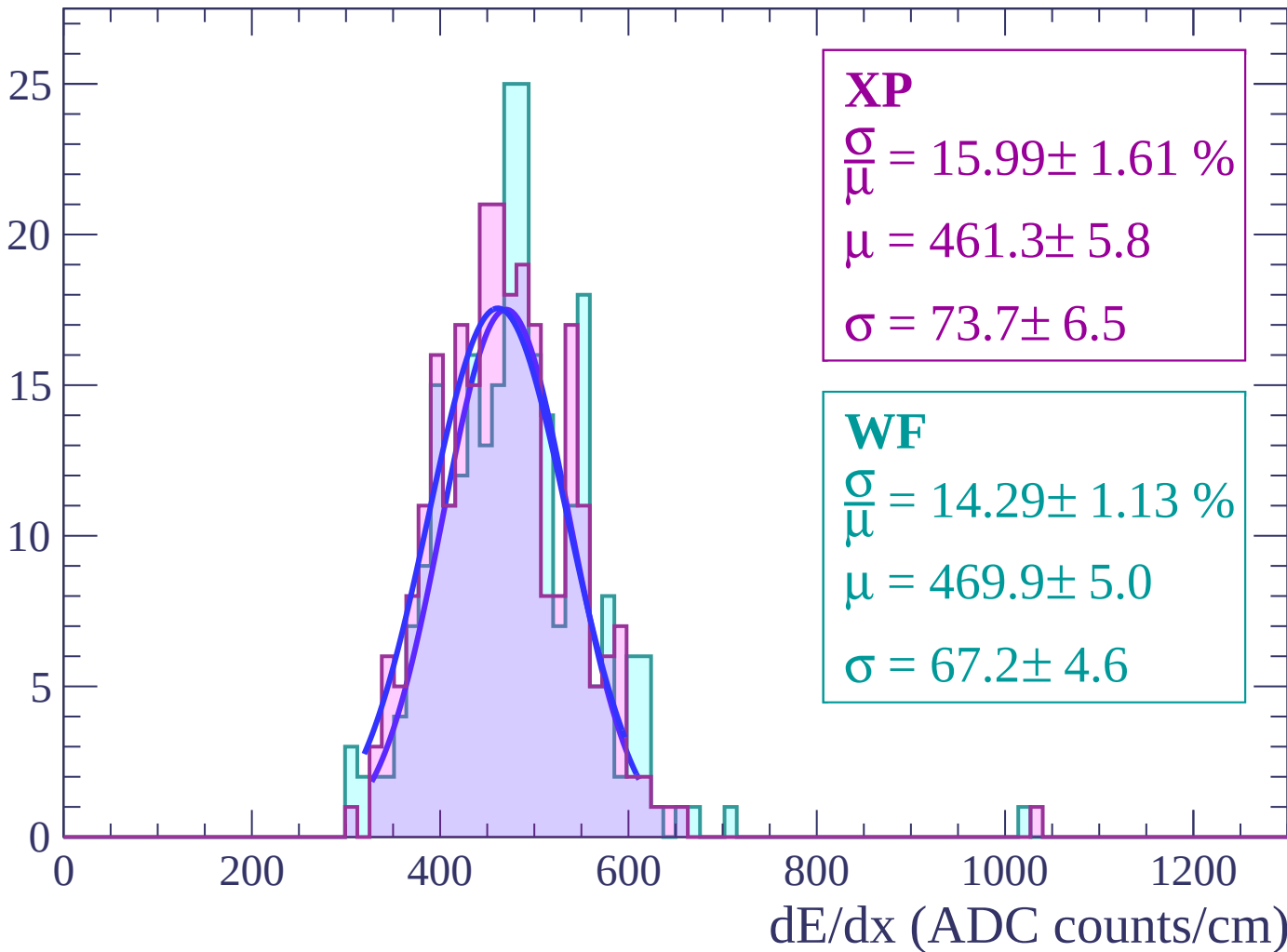
Energy loss | $-26 < \theta < -24$

Count



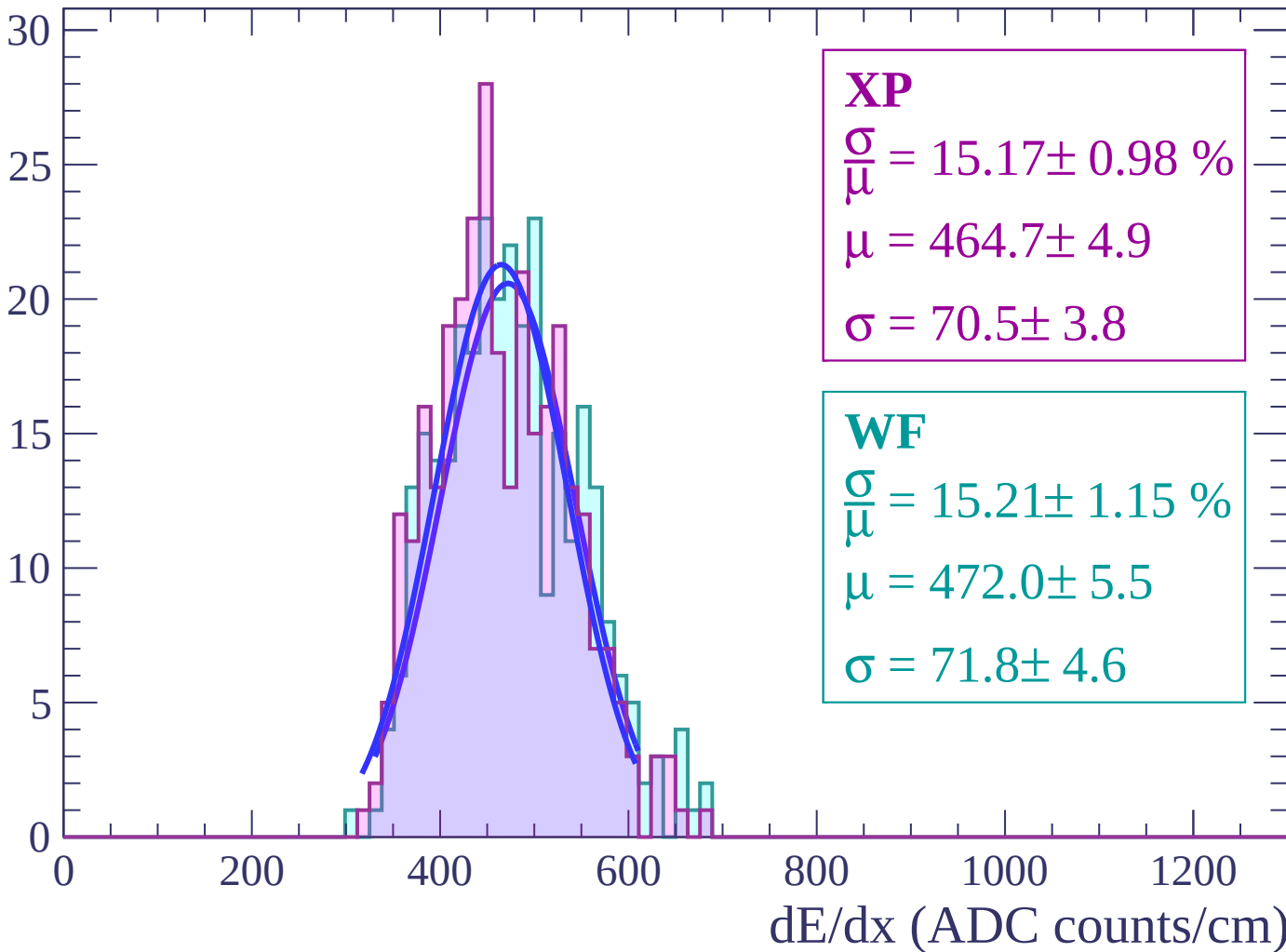
Energy loss | $-24 < \theta < -22$

Count



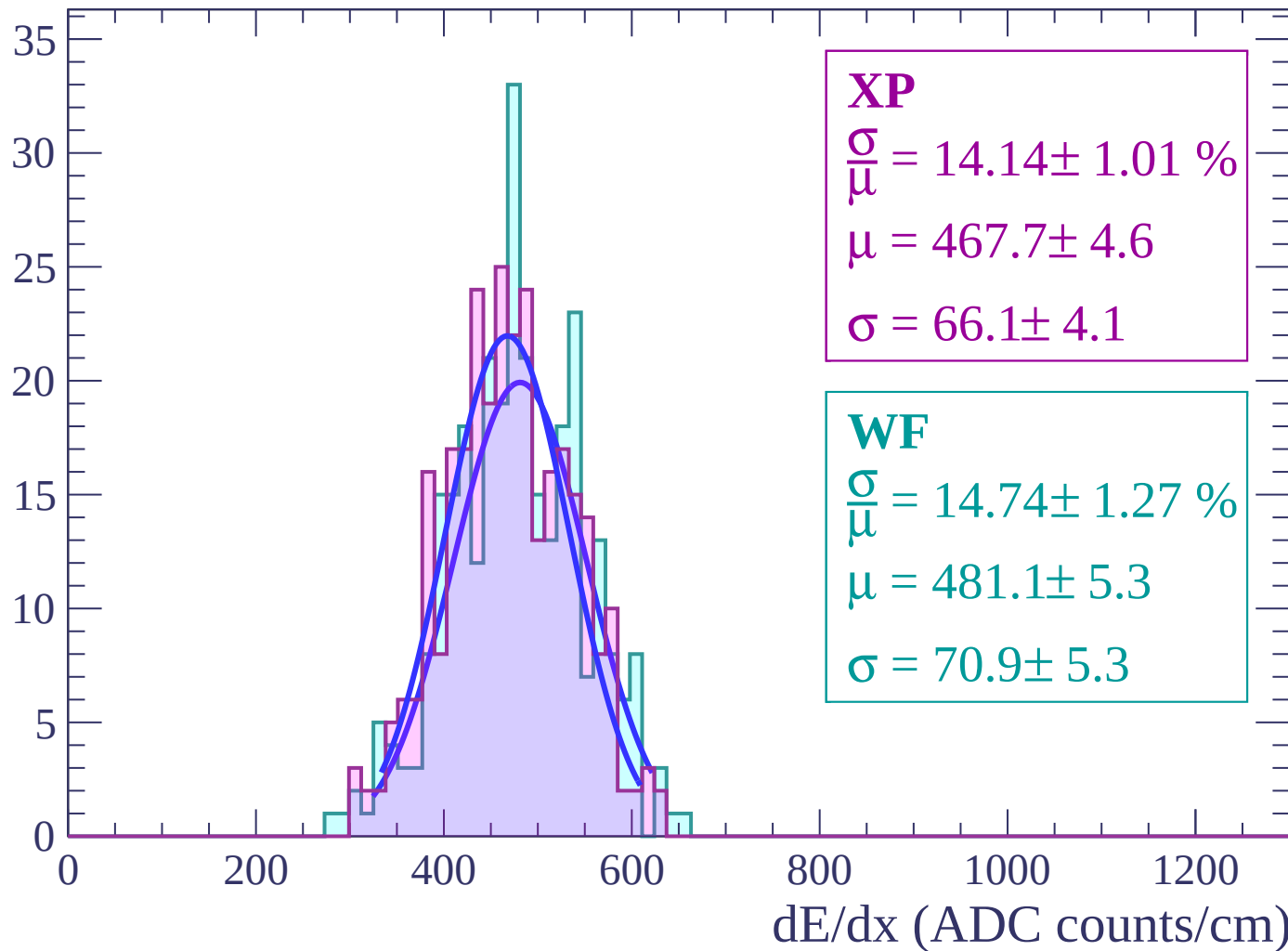
Energy loss | $-22 < \theta < -20$

Count



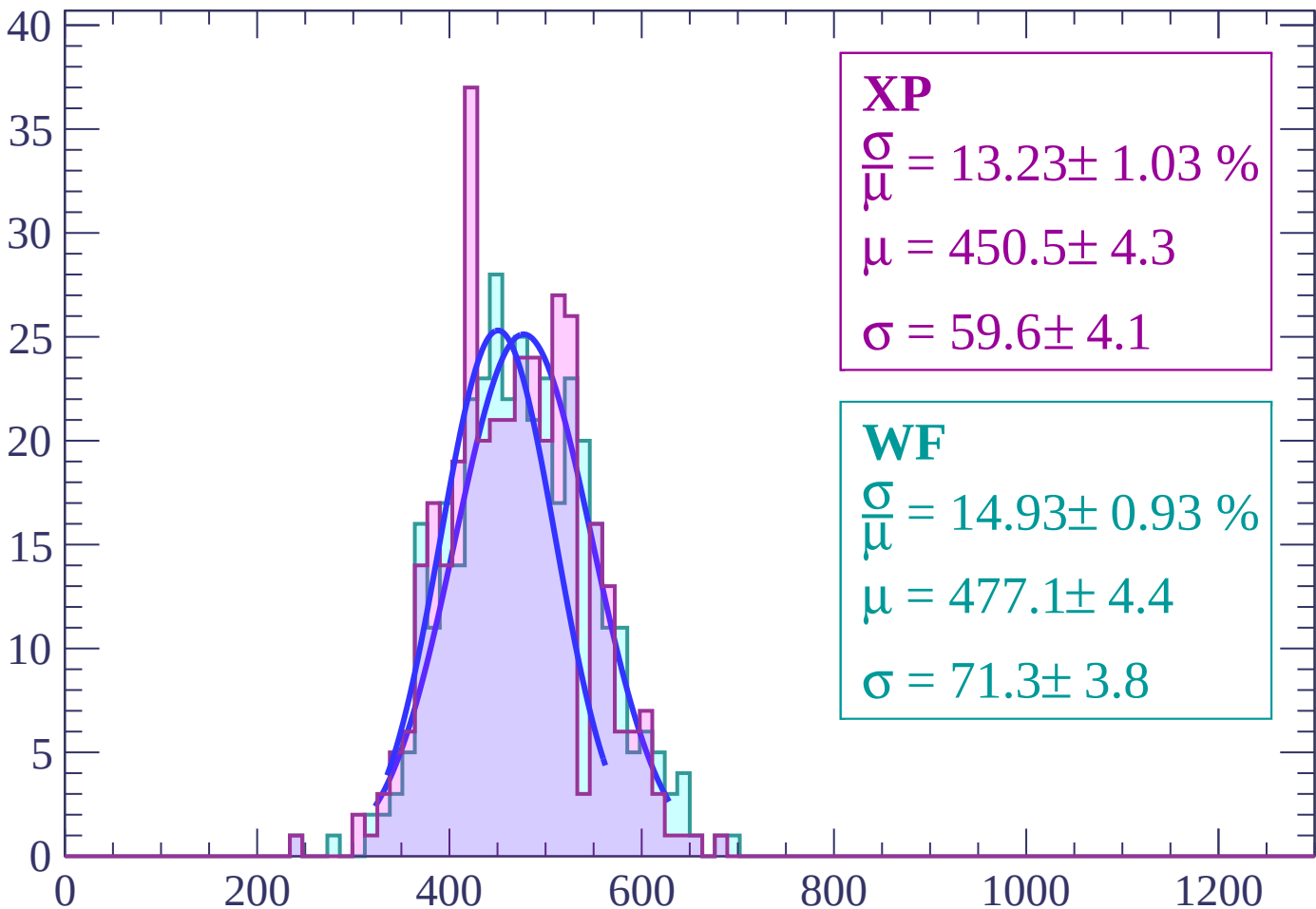
Energy loss | $-20 < \theta < -18$

Count



Energy loss | $-18 < \theta < -16$

Count



XP

$$\frac{\sigma}{\mu} = 13.23 \pm 1.03 \%$$

$$\mu = 450.5 \pm 4.3$$

$$\sigma = 59.6 \pm 4.1$$

WF

$$\frac{\sigma}{\mu} = 14.93 \pm 0.93 \%$$

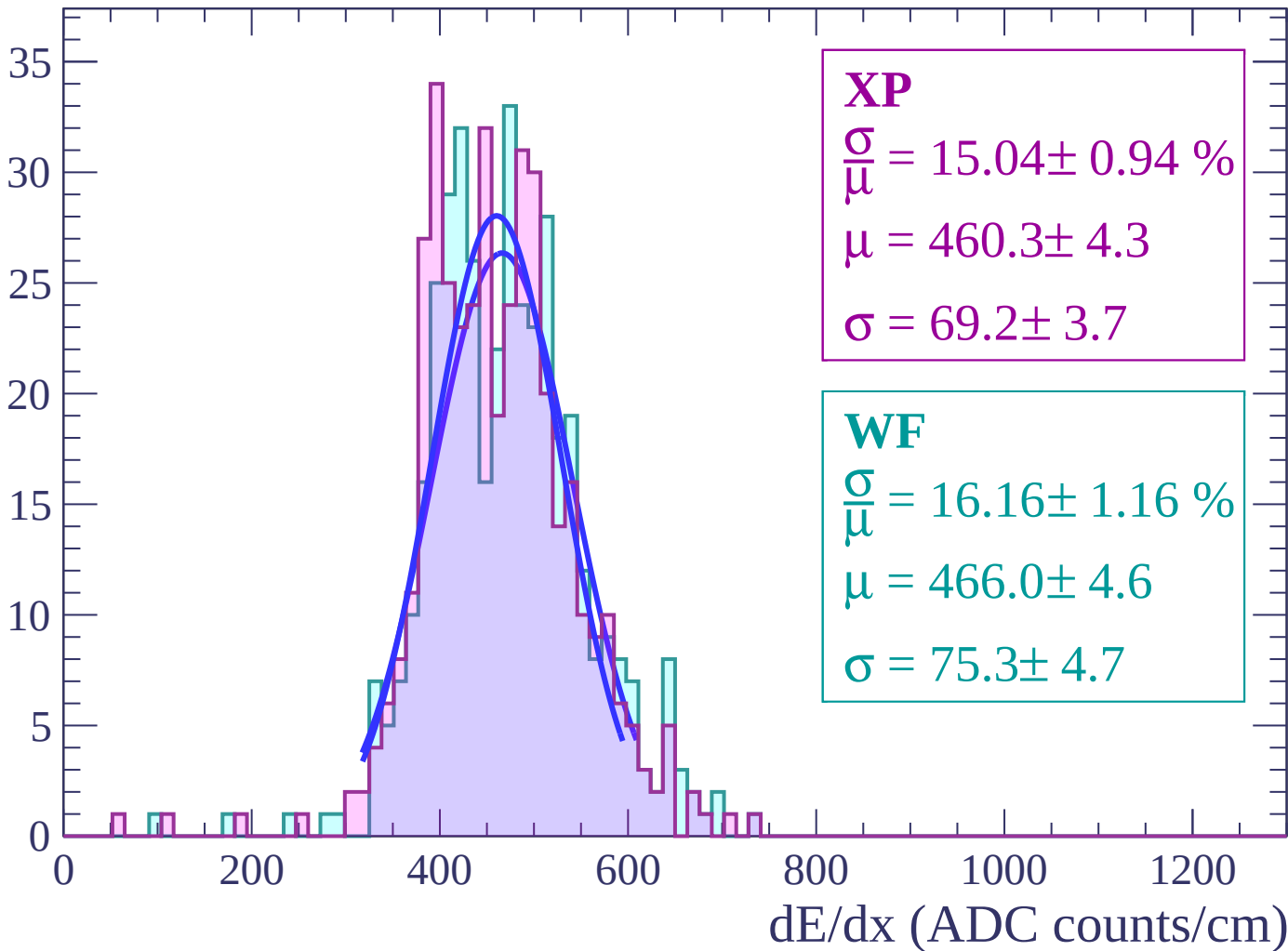
$$\mu = 477.1 \pm 4.4$$

$$\sigma = 71.3 \pm 3.8$$

dE/dx (ADC counts/cm)

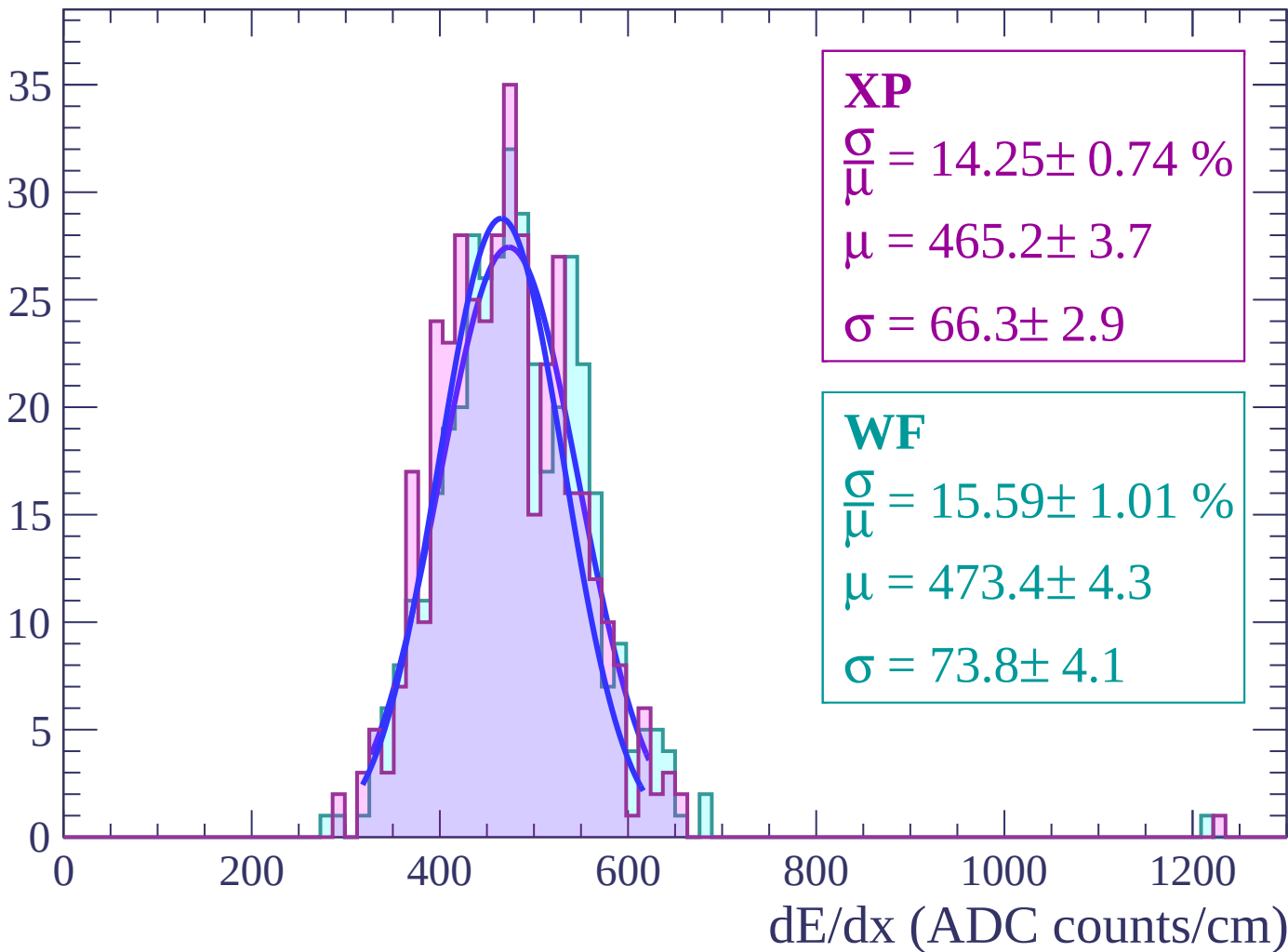
Energy loss | $-16 < \theta < -14$

Count



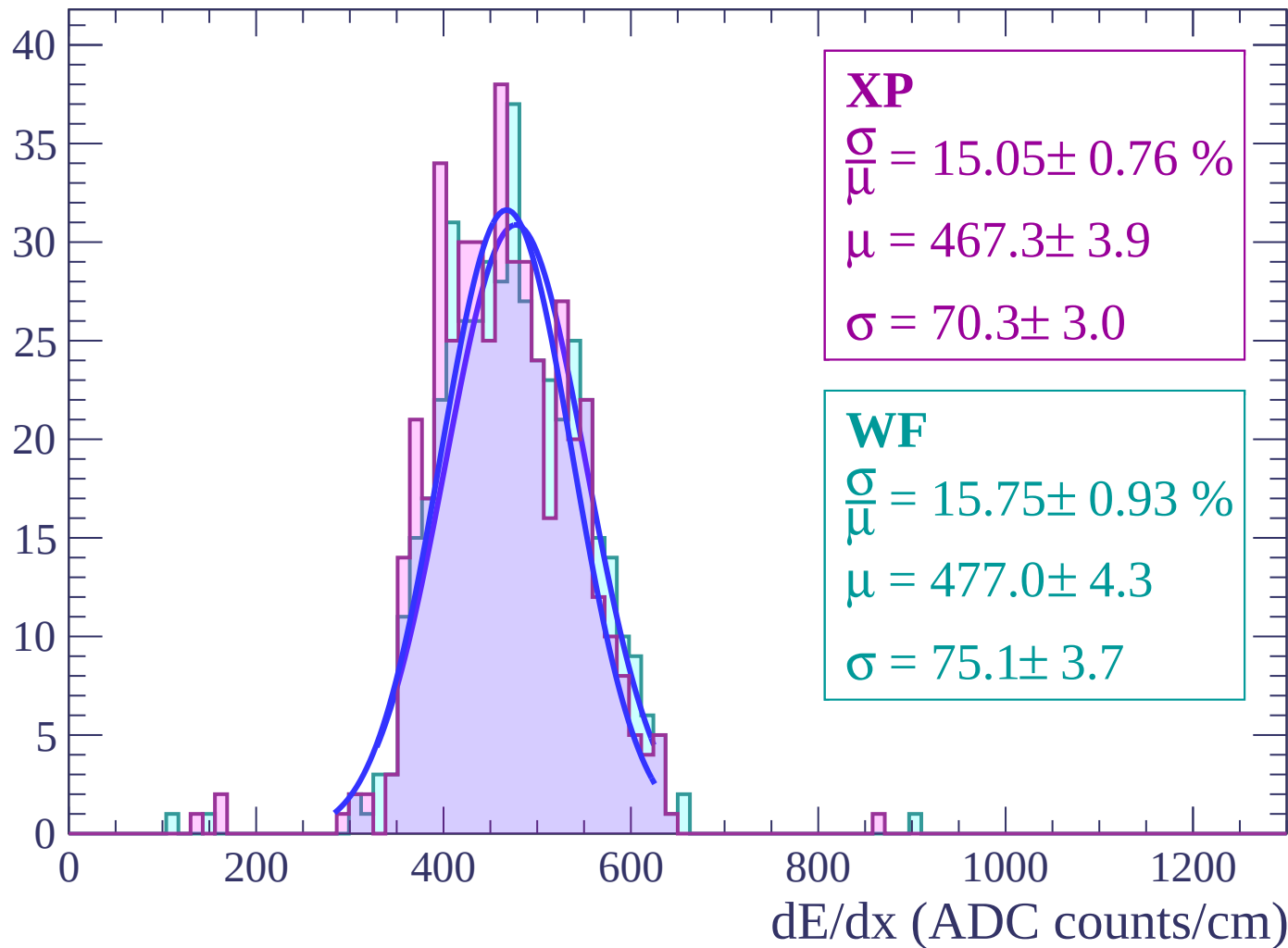
Energy loss | $-14 < \theta < -12$

Count



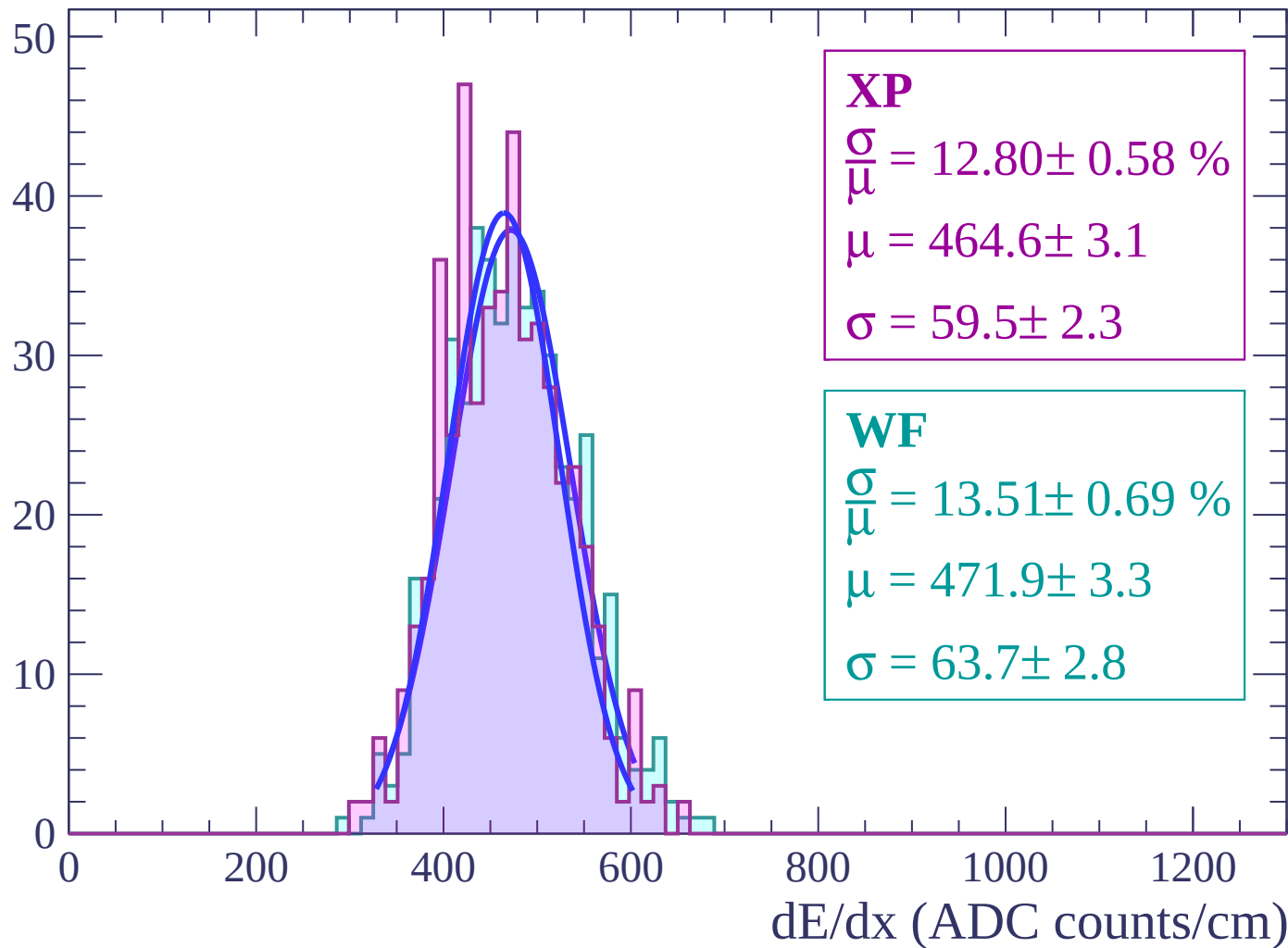
Energy loss | $-12 < \theta < -10$

Count



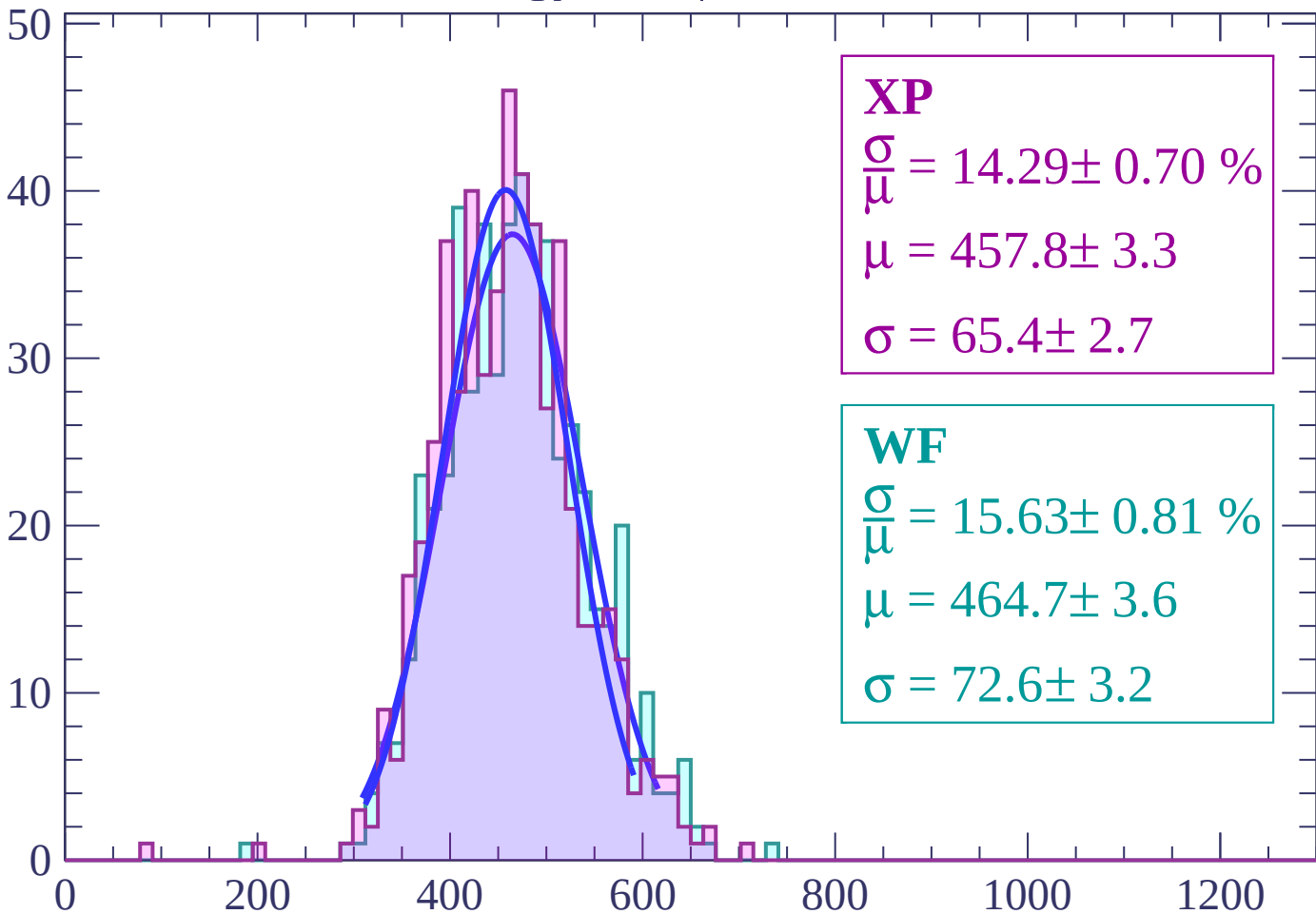
Energy loss | $-10 < \theta < -8$

Count



Energy loss | $-8 < \theta < -6$

Count



XP

$$\frac{\sigma}{\mu} = 14.29 \pm 0.70 \%$$

$$\mu = 457.8 \pm 3.3$$

$$\sigma = 65.4 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 15.63 \pm 0.81 \%$$

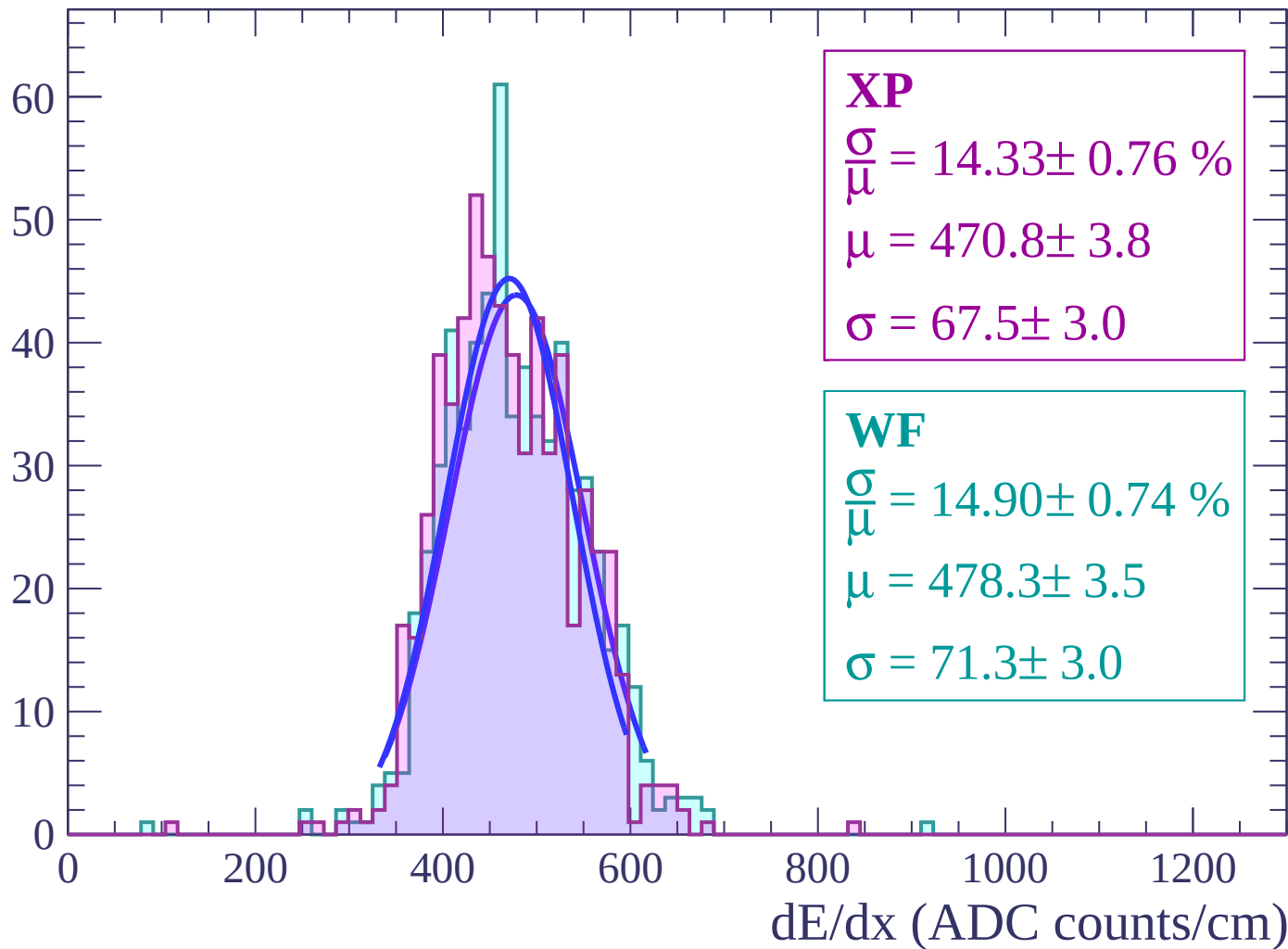
$$\mu = 464.7 \pm 3.6$$

$$\sigma = 72.6 \pm 3.2$$

dE/dx (ADC counts/cm)

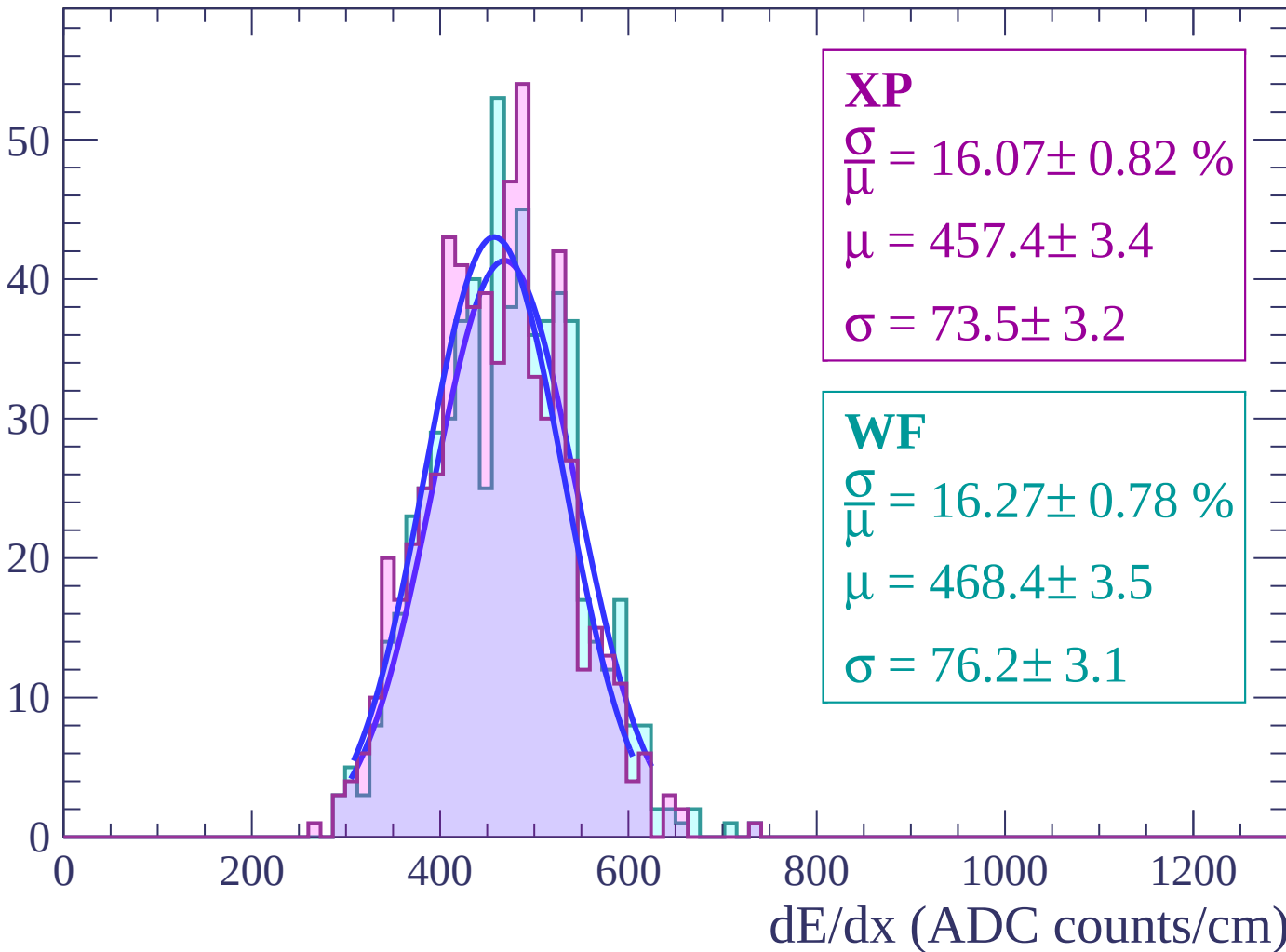
Energy loss | $-6 < \theta < -4$

Count



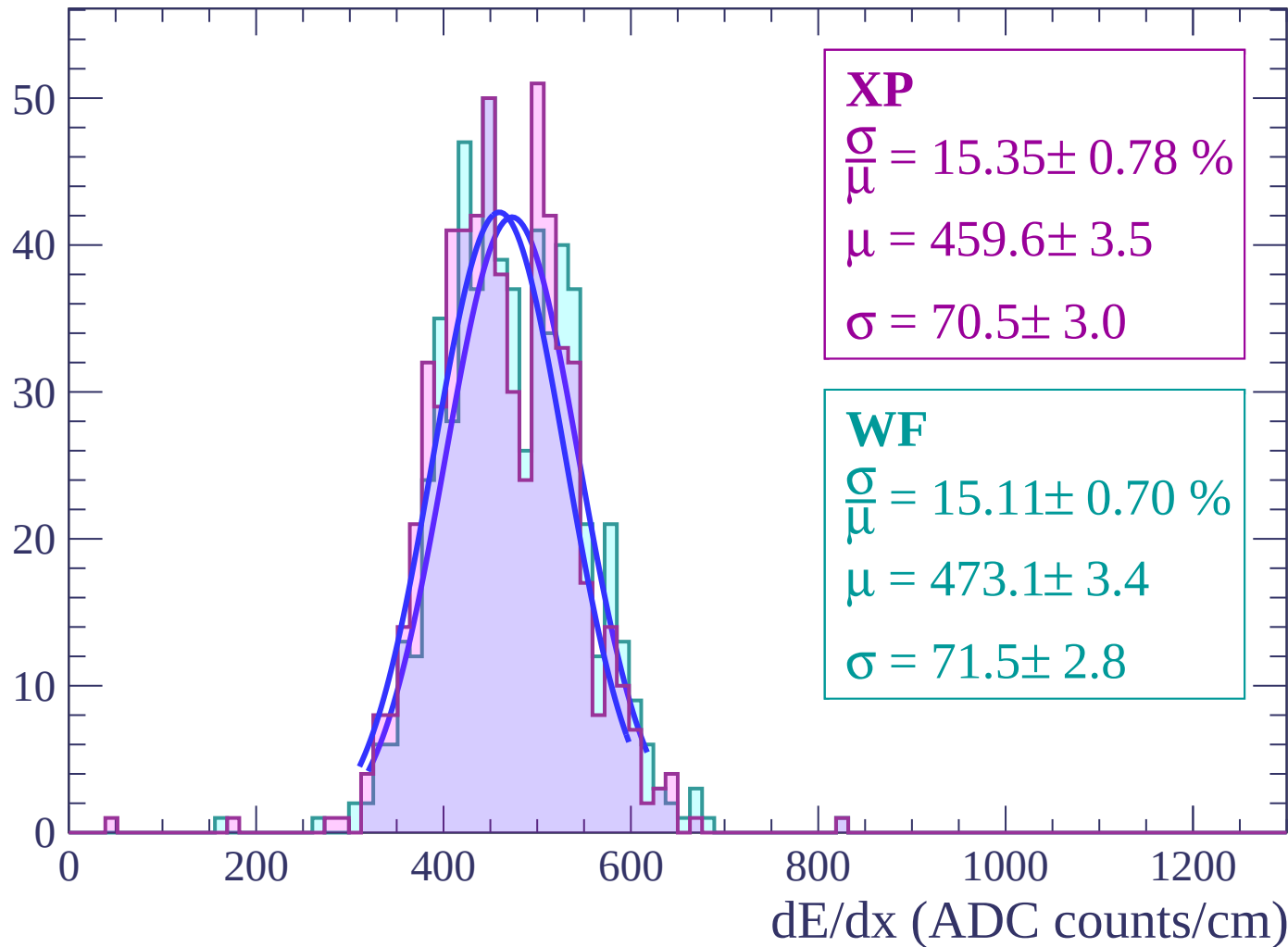
Energy loss | $-4 < \theta < -2$

Count



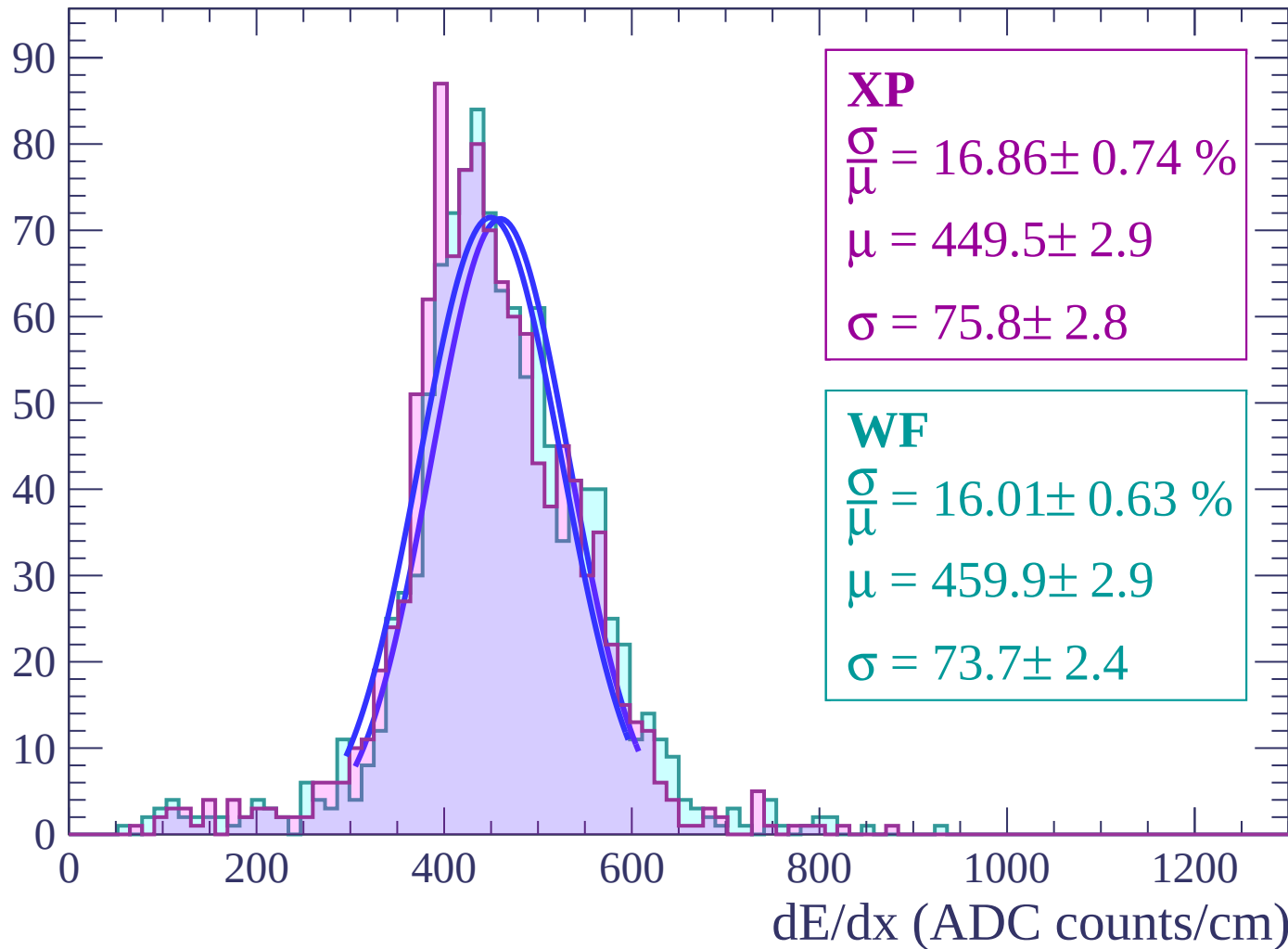
Energy loss $-2 < \theta < 0$

Count



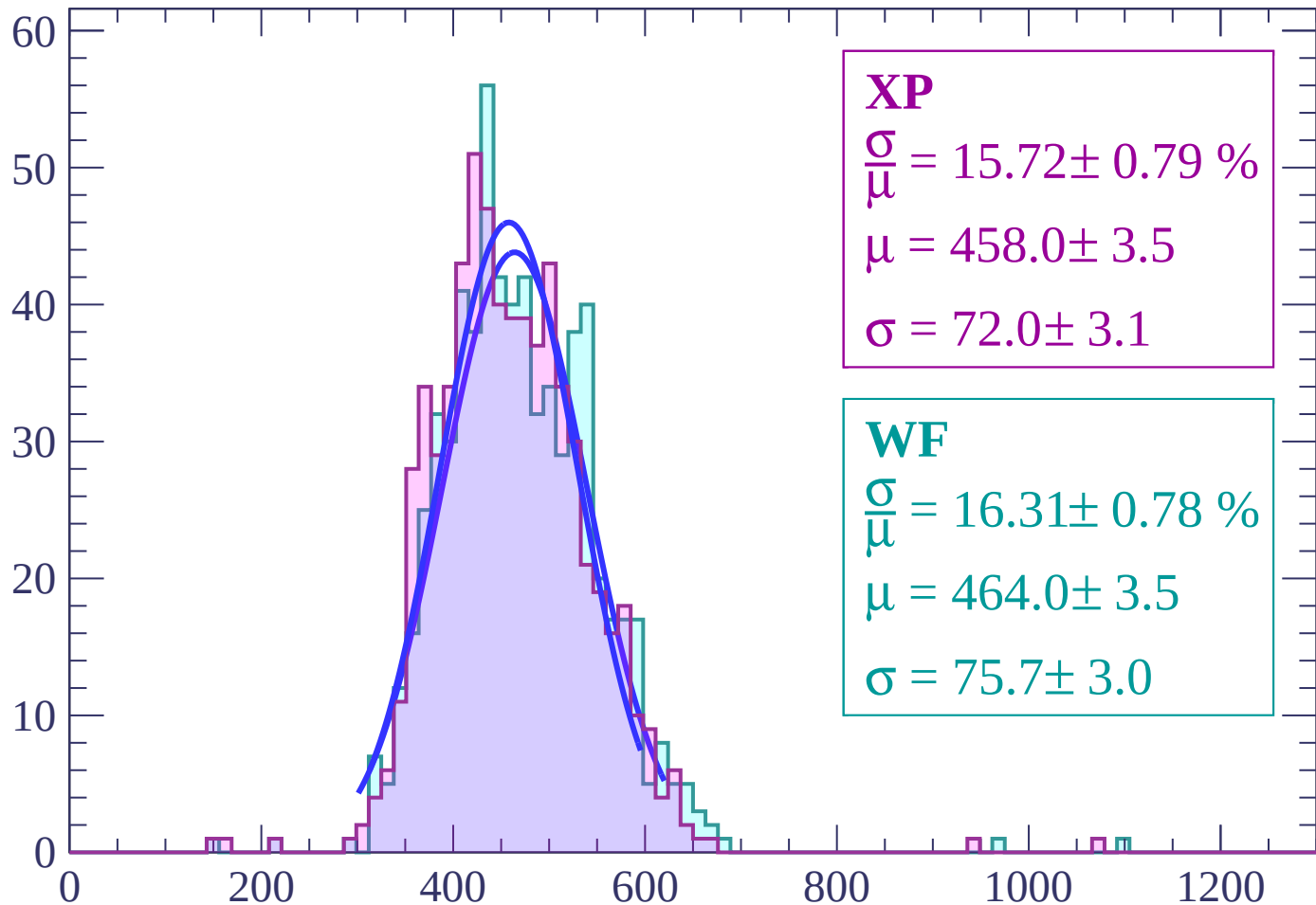
Energy loss | $0 < \theta < 2$

Count



Energy loss | $2 < \theta < 4$

Count



XP

$$\frac{\sigma}{\mu} = 15.72 \pm 0.79 \%$$

$$\mu = 458.0 \pm 3.5$$

$$\sigma = 72.0 \pm 3.1$$

WF

$$\frac{\sigma}{\mu} = 16.31 \pm 0.78 \%$$

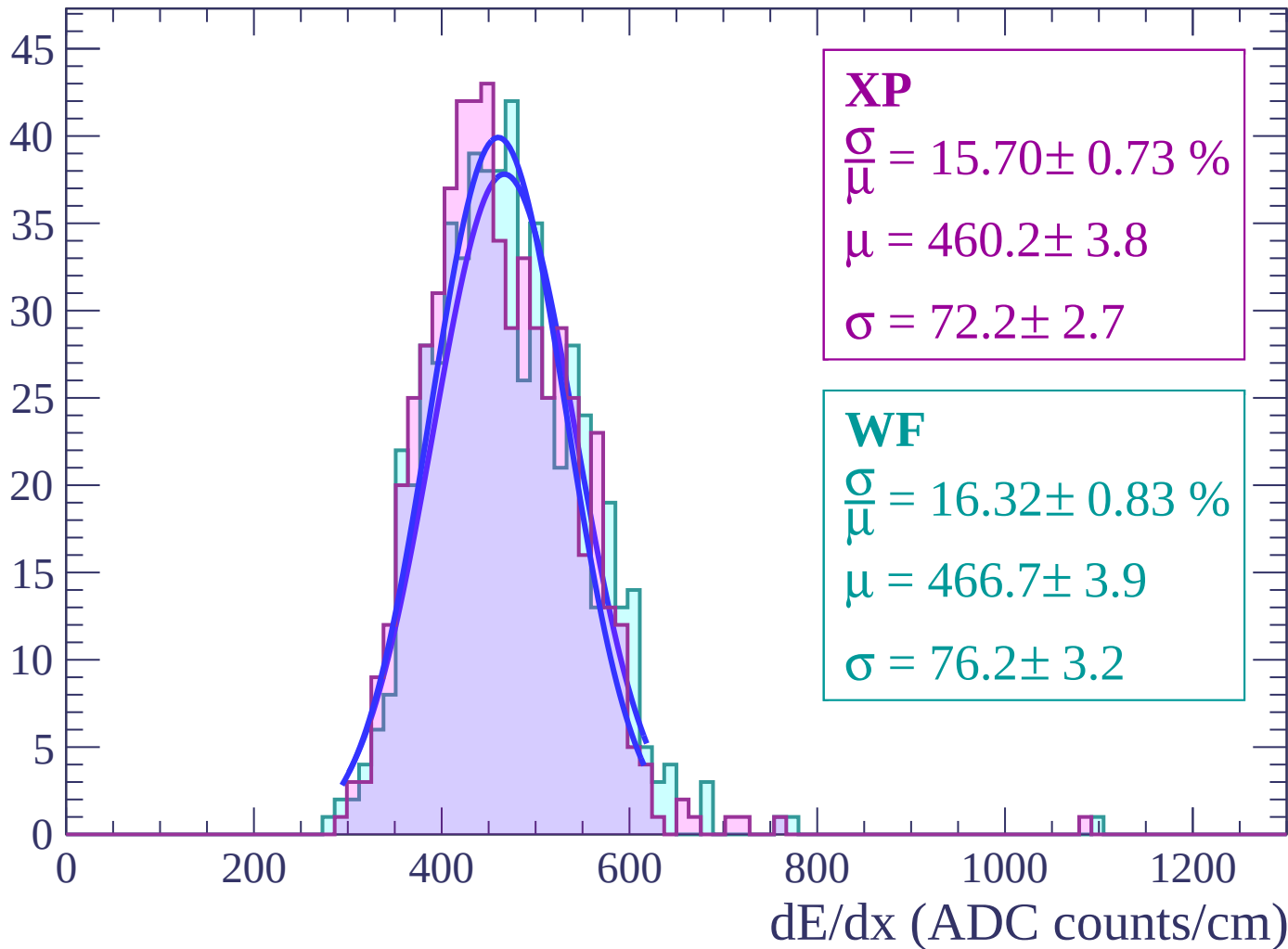
$$\mu = 464.0 \pm 3.5$$

$$\sigma = 75.7 \pm 3.0$$

dE/dx (ADC counts/cm)

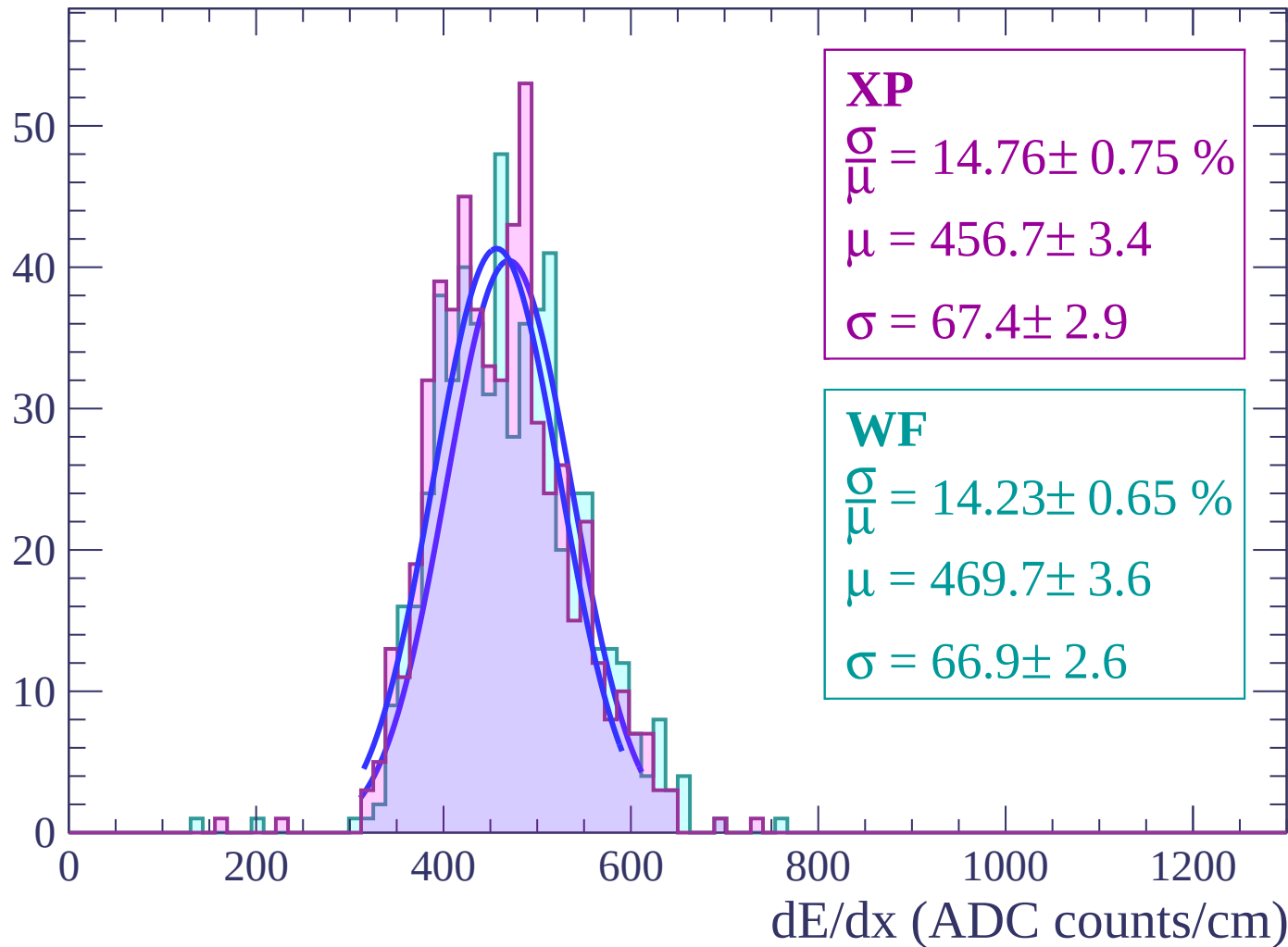
Energy loss | $4 < \theta < 6$

Count



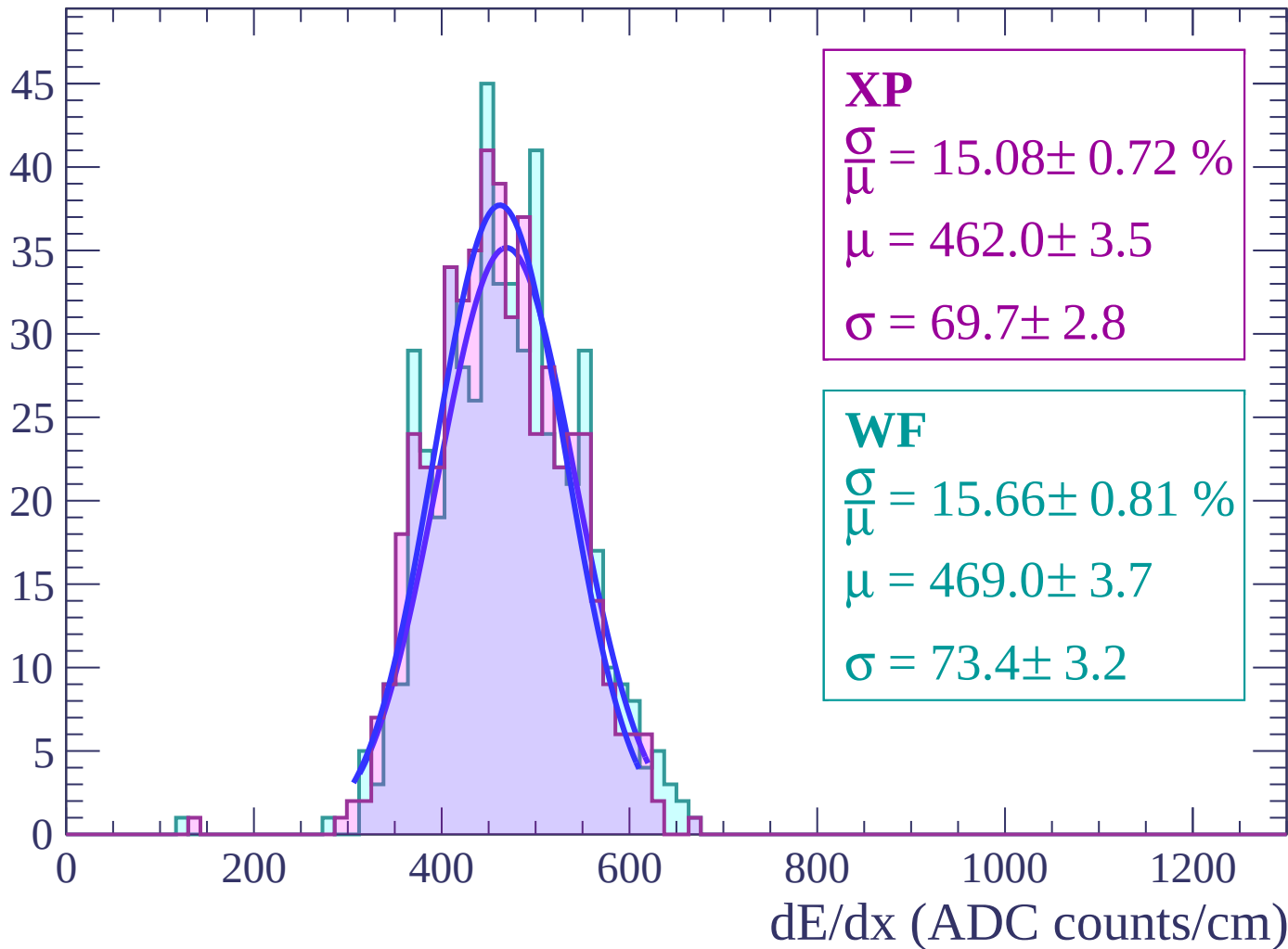
Energy loss | $6 < \theta < 8$

Count



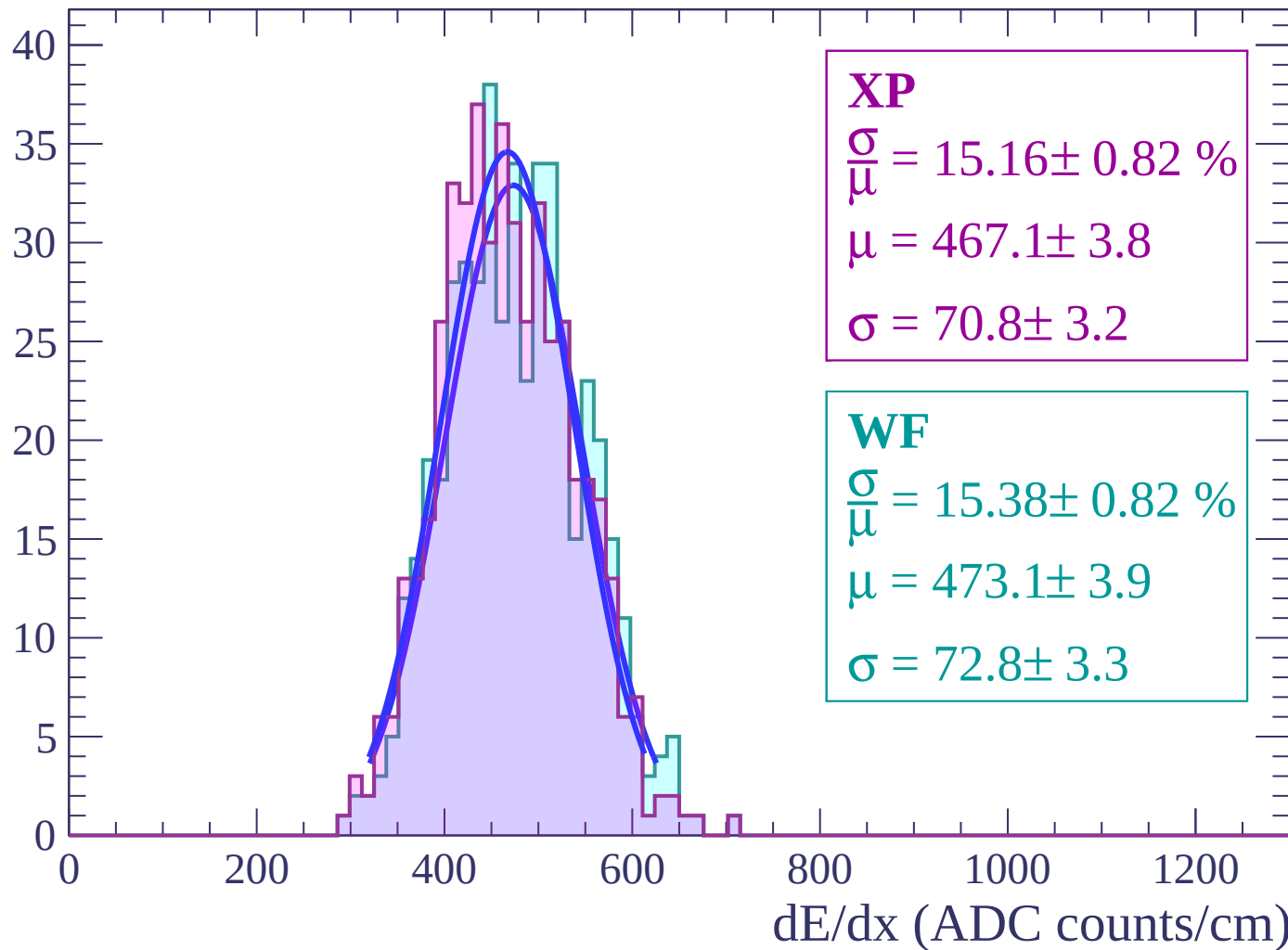
Energy loss | $8 < \theta < 10$

Count



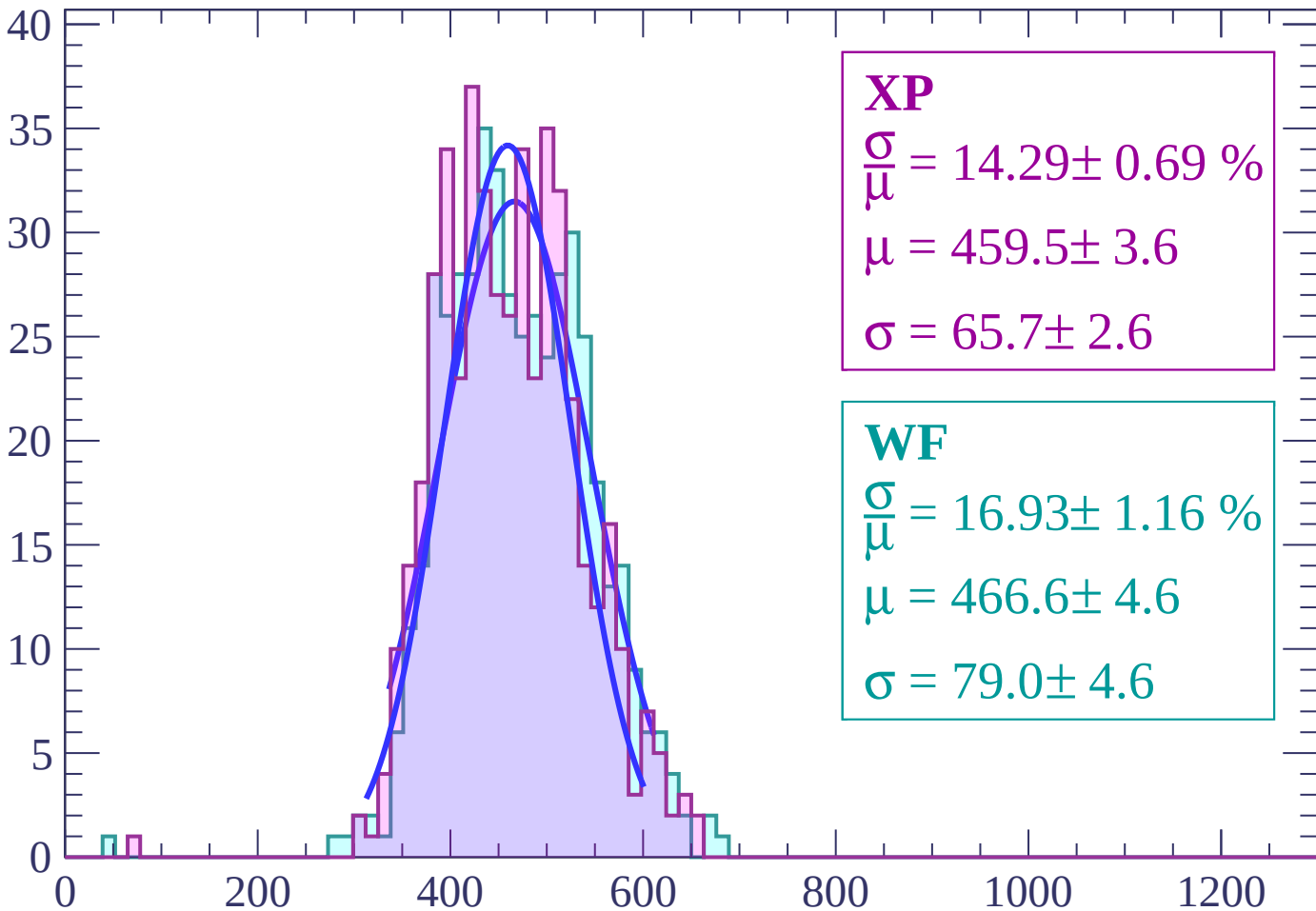
Energy loss | $10 < \theta < 12$

Count



Energy loss | $12 < \theta < 14$

Count



XP

$$\frac{\sigma}{\mu} = 14.29 \pm 0.69 \%$$

$$\mu = 459.5 \pm 3.6$$

$$\sigma = 65.7 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 16.93 \pm 1.16 \%$$

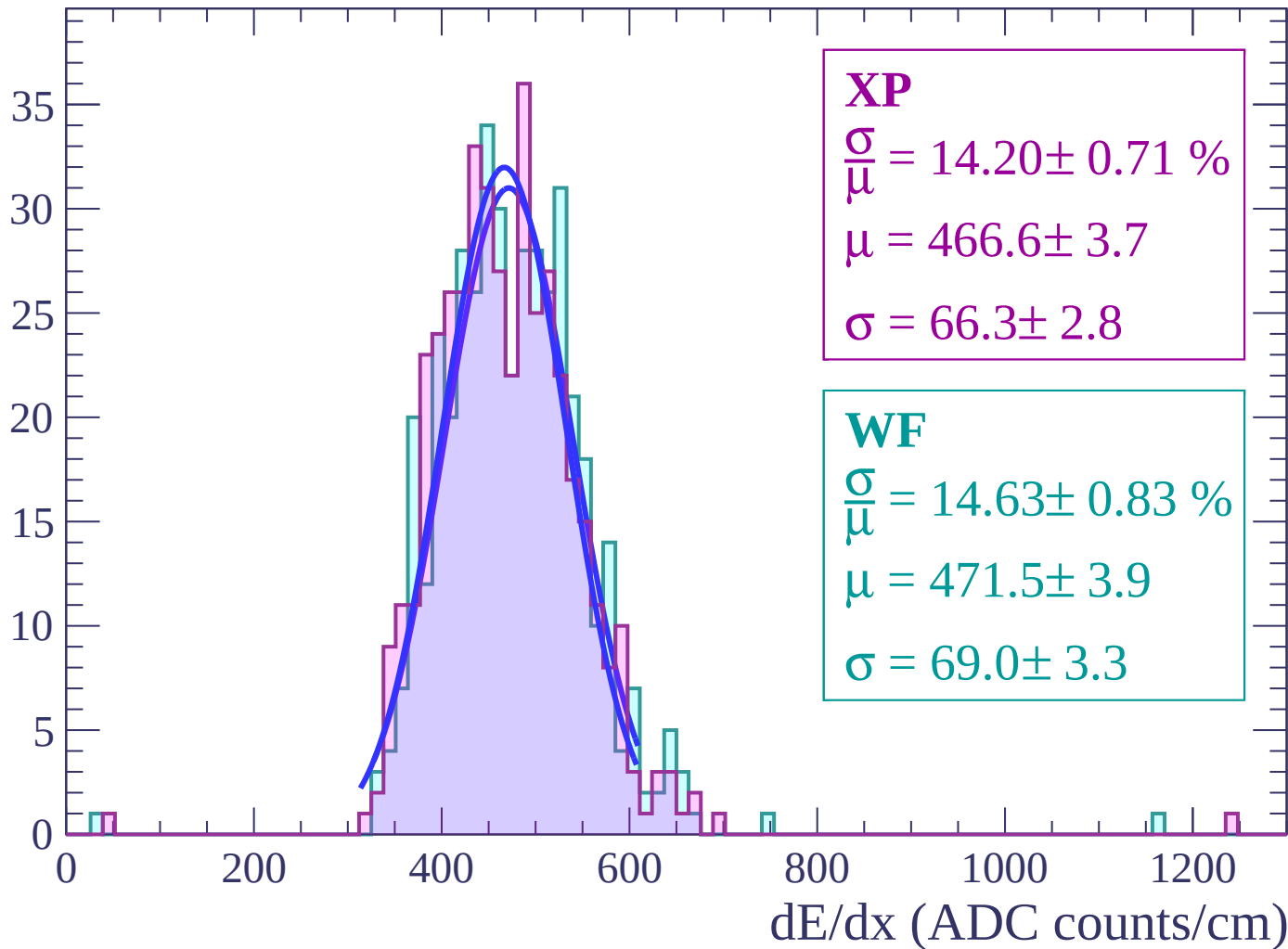
$$\mu = 466.6 \pm 4.6$$

$$\sigma = 79.0 \pm 4.6$$

dE/dx (ADC counts/cm)

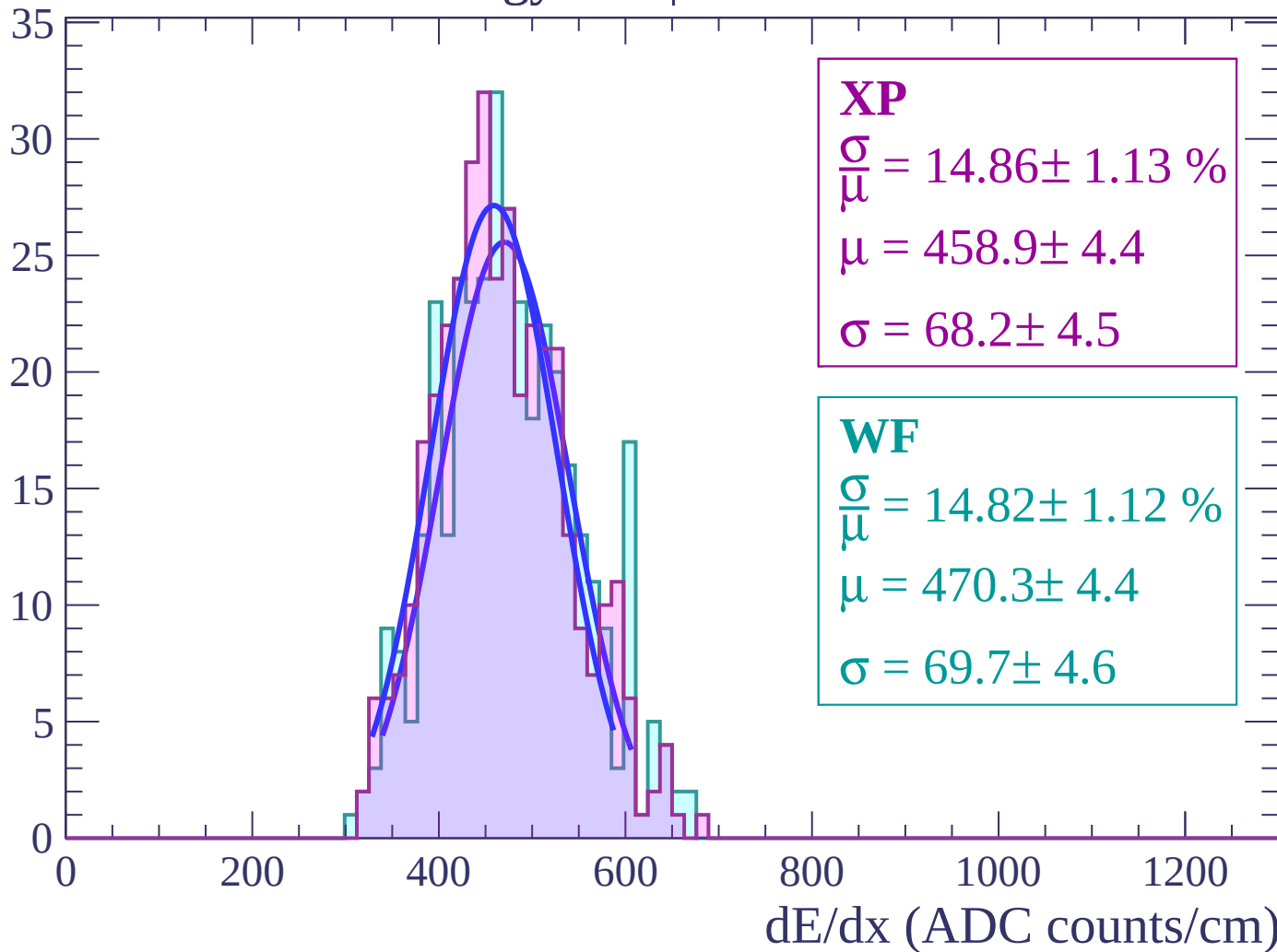
Energy loss | $14 < \theta < 16$

Count



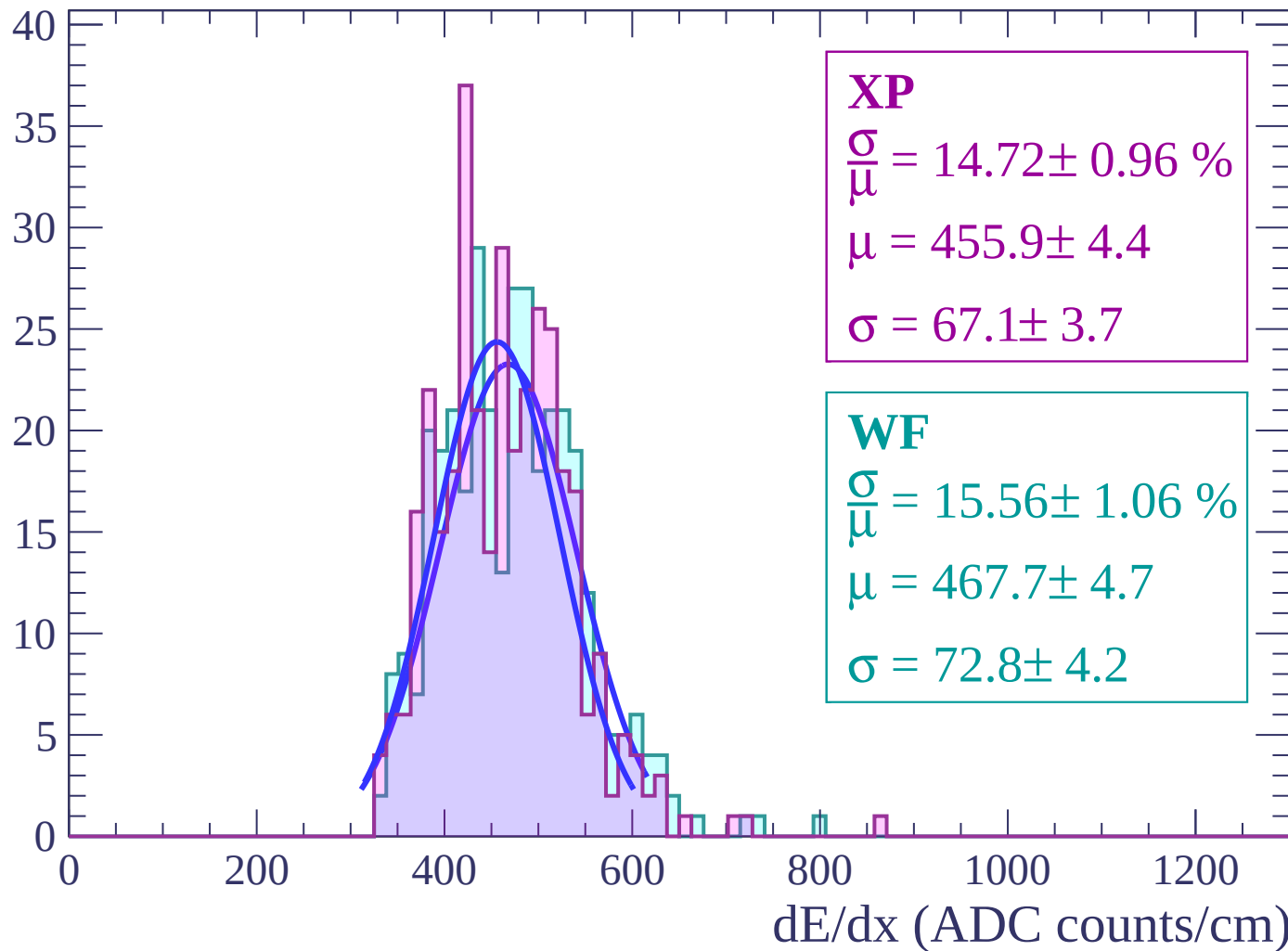
Energy loss | $16 < \theta < 18$

Count



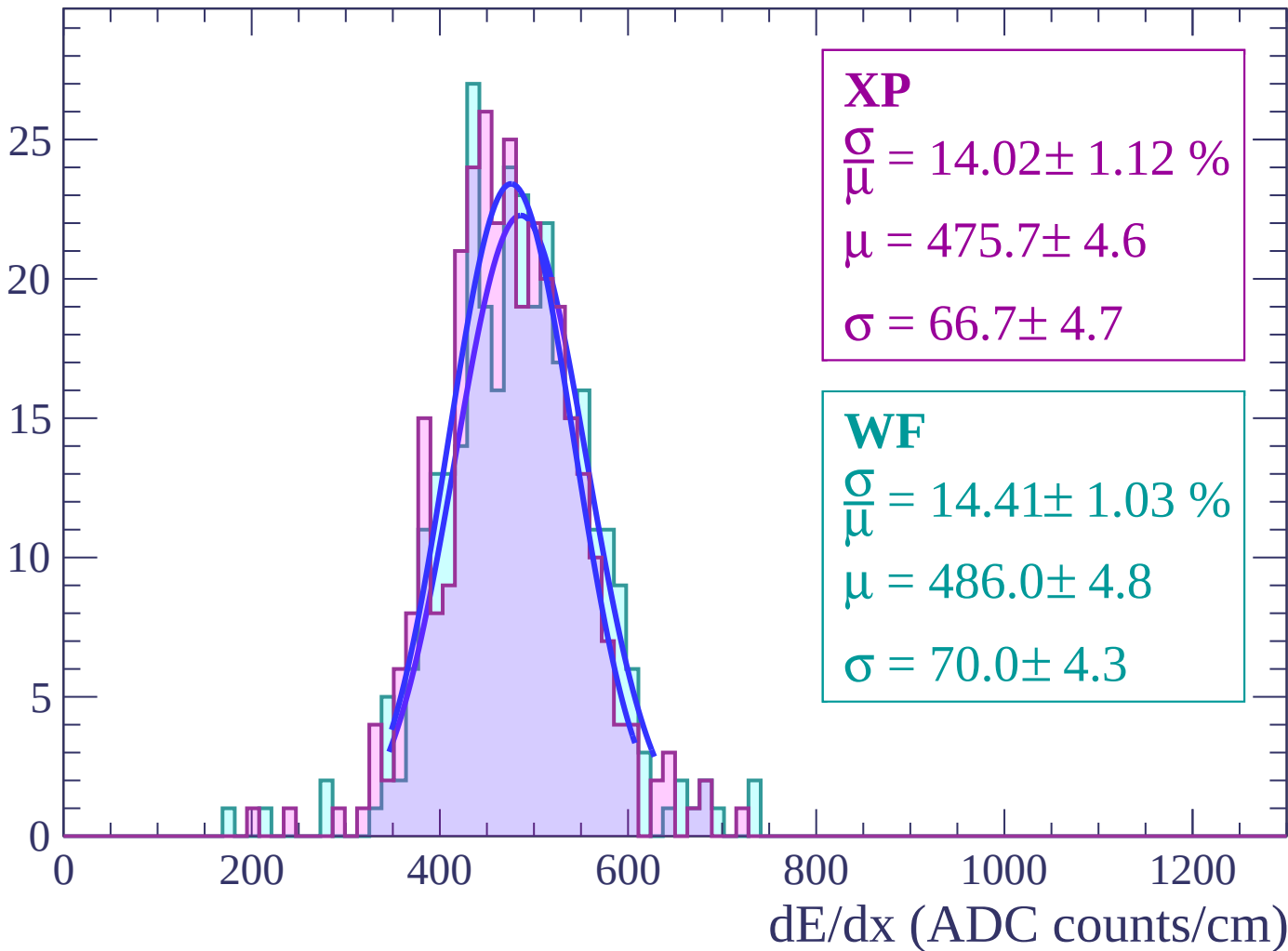
Energy loss | $18 < \theta < 20$

Count



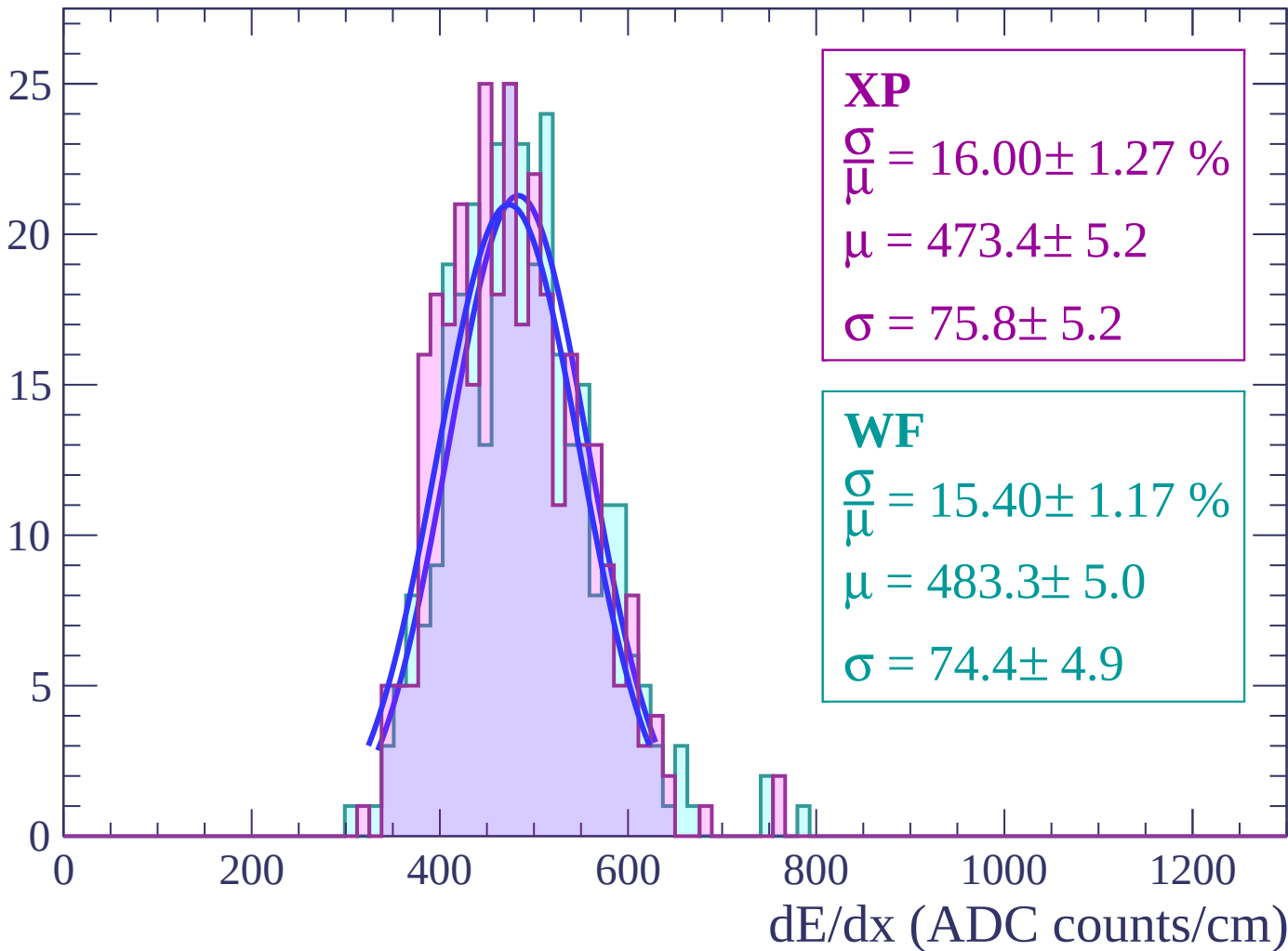
Energy loss | $20 < \theta < 22$

Count



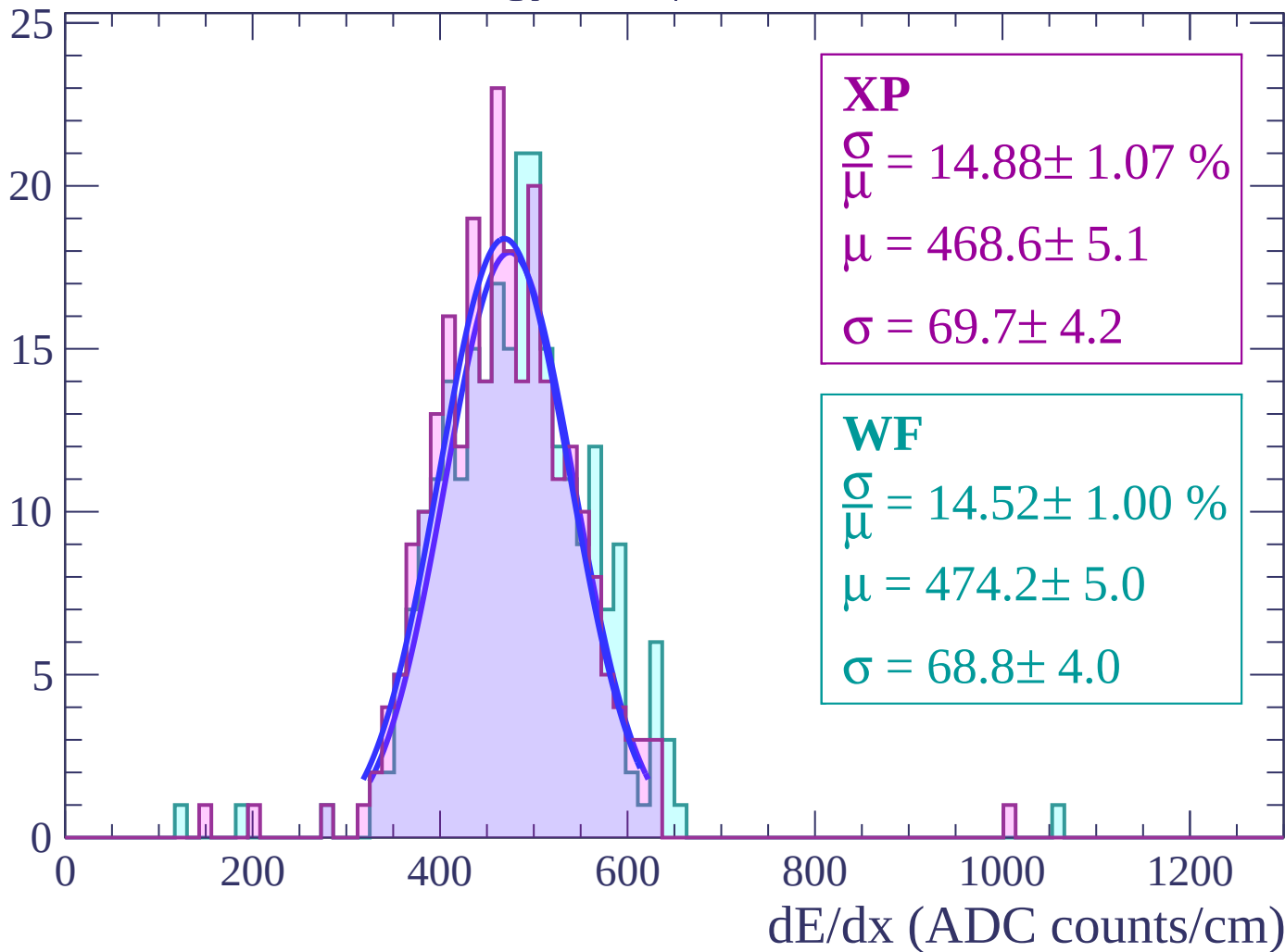
Energy loss | $22 < \theta < 24$

Count



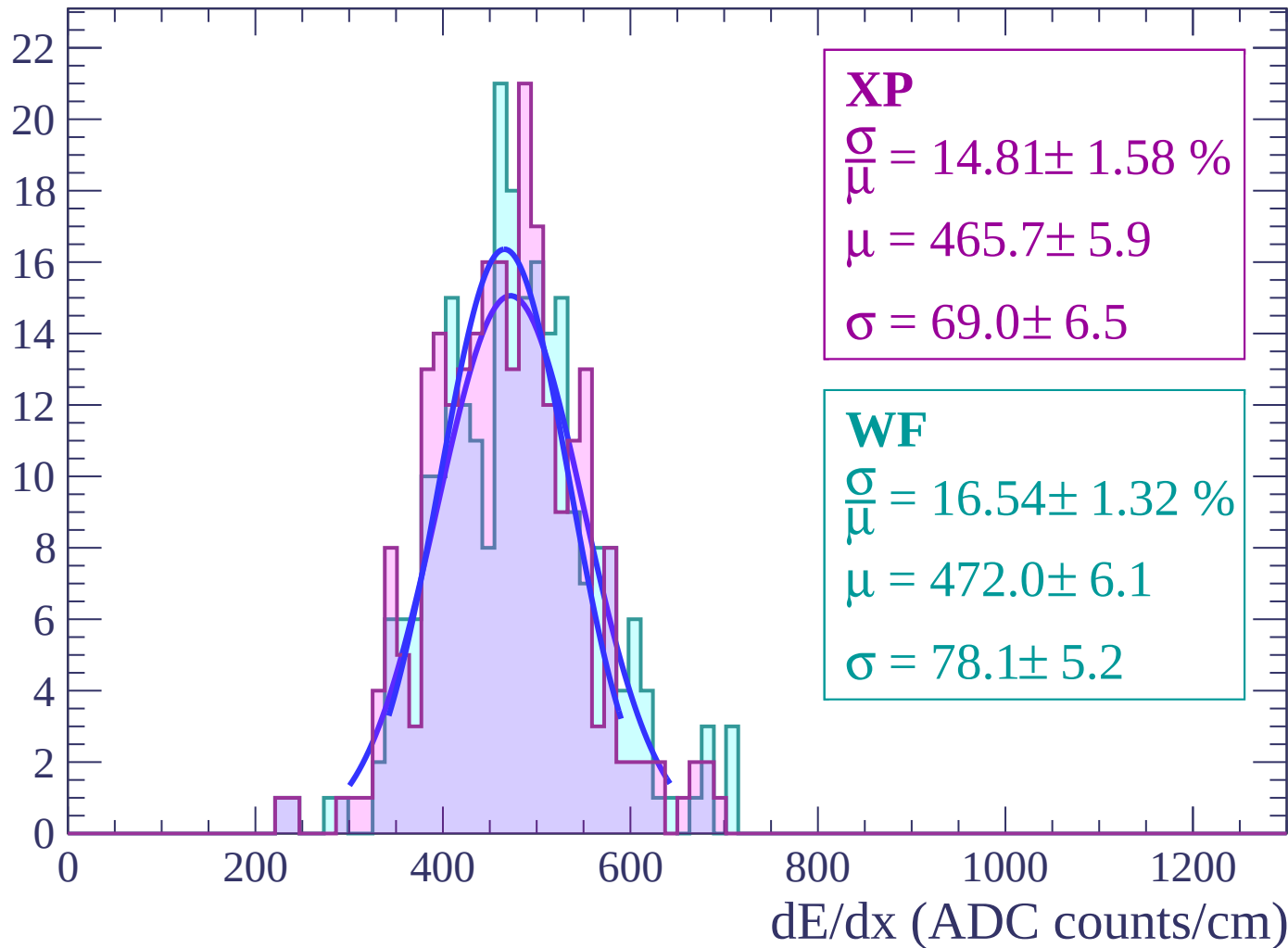
Energy loss | $24 < \theta < 26$

Count



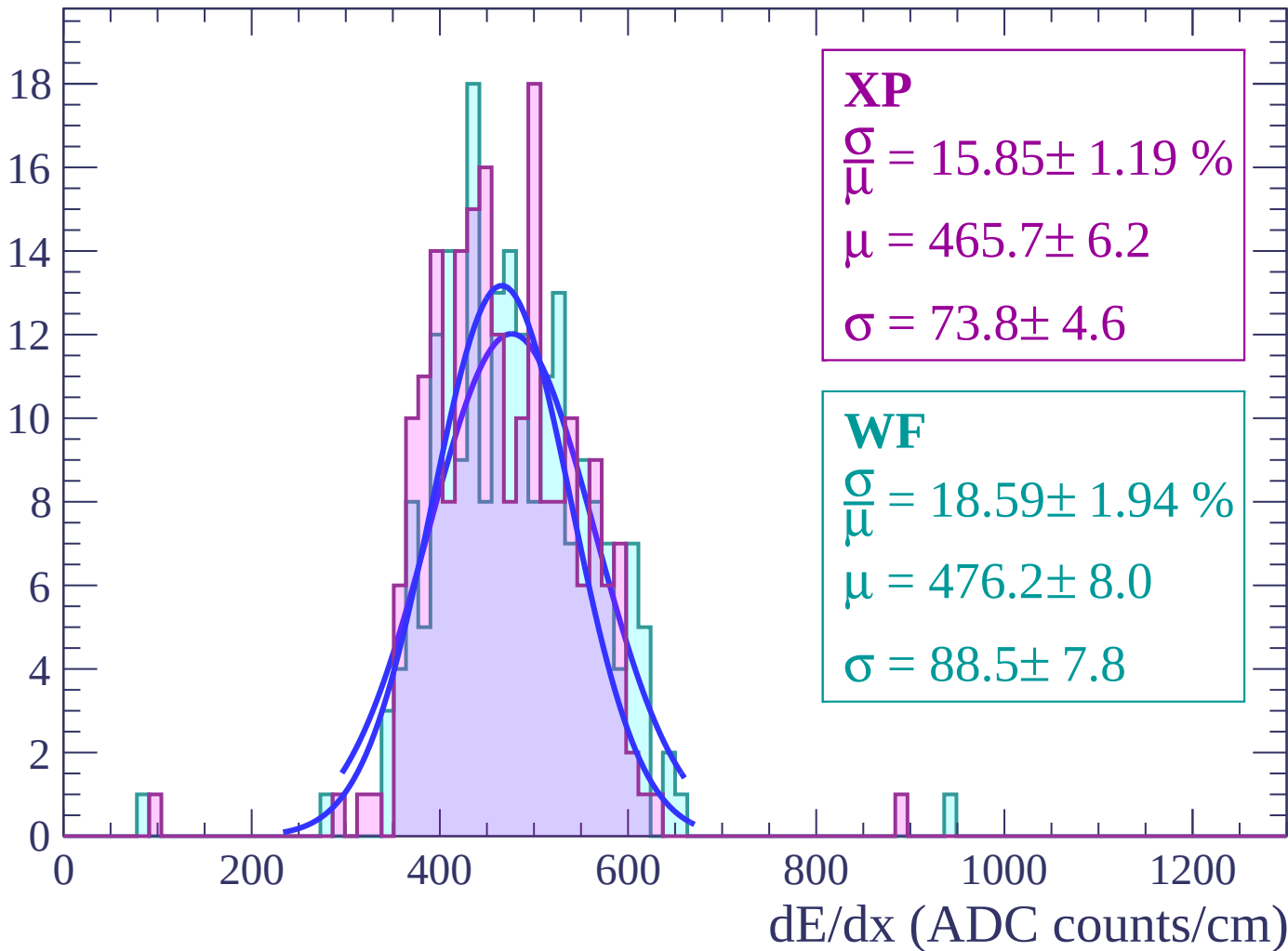
Energy loss | $26 < \theta < 28$

Count



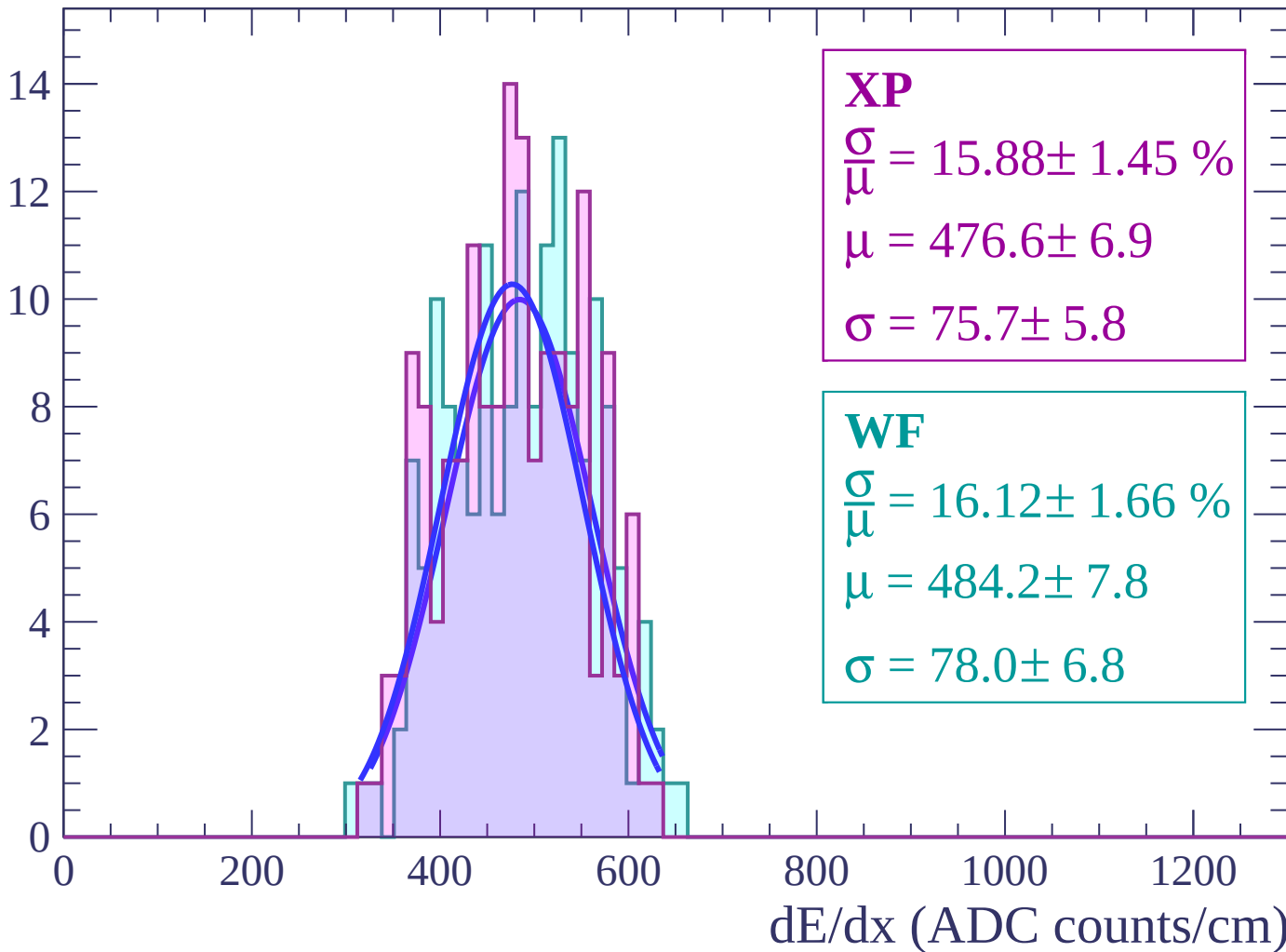
Energy loss | $28 < \theta < 30$

Count



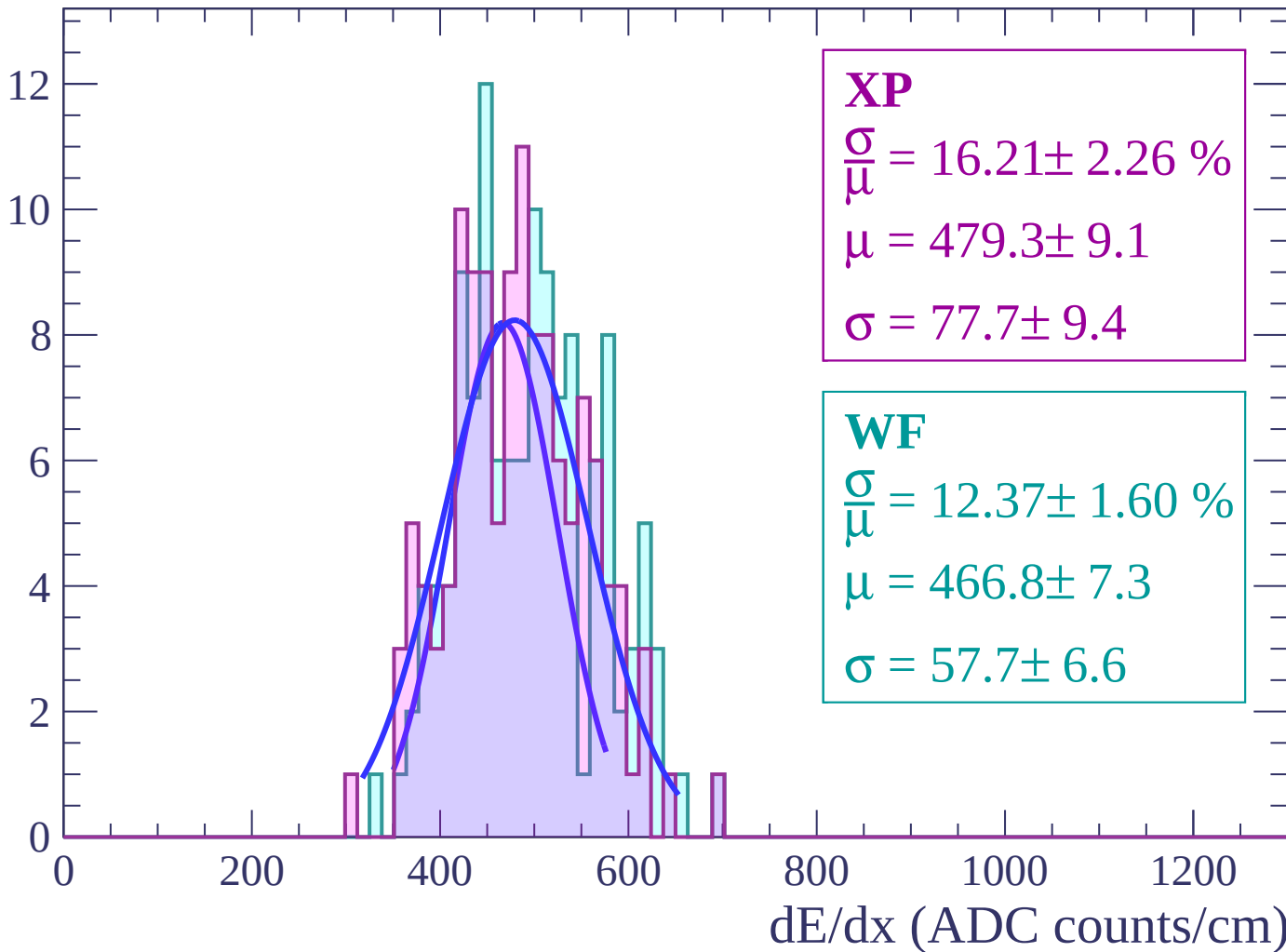
Energy loss | $30 < \theta < 32$

Count



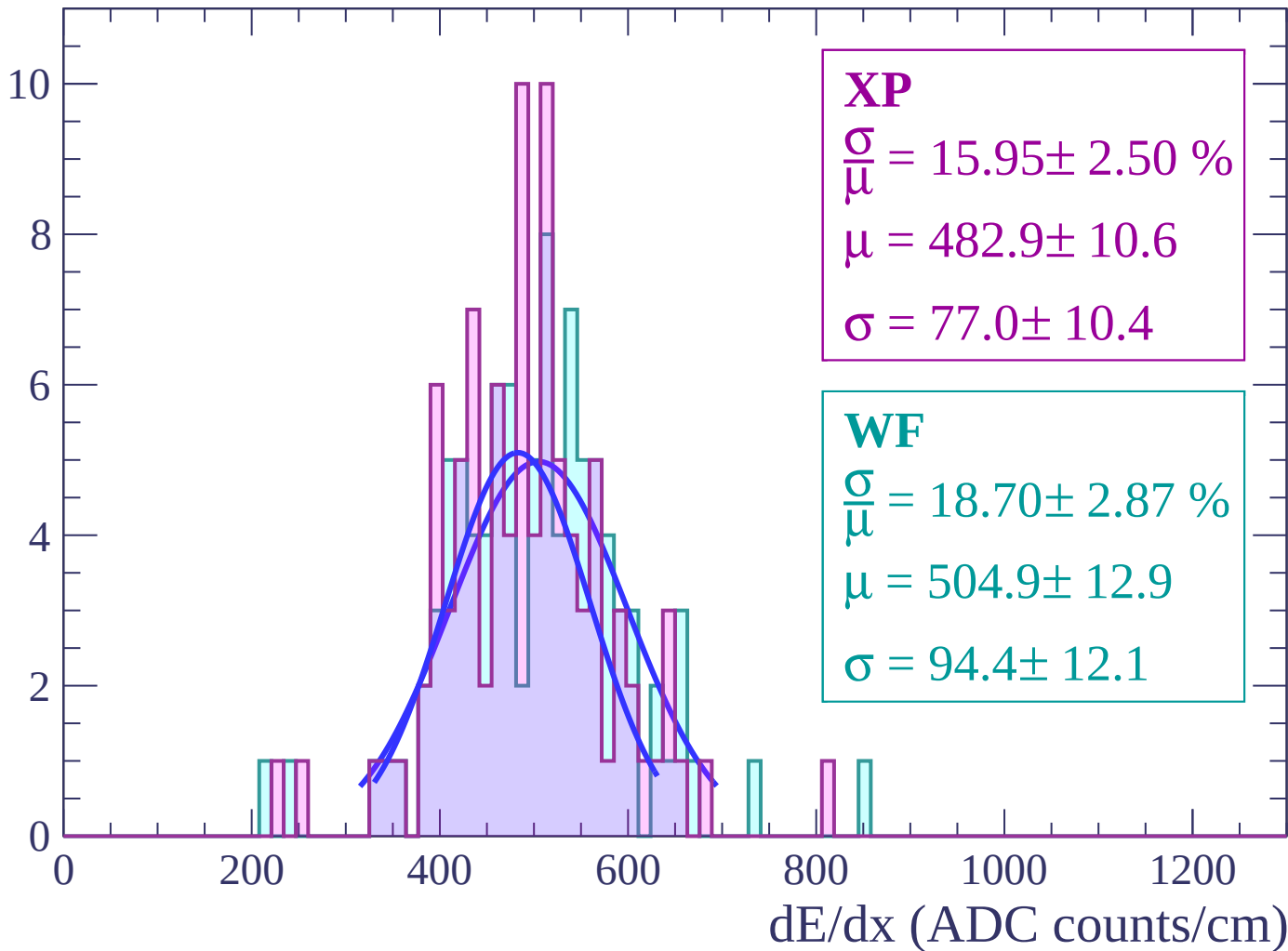
Energy loss | $32 < \theta < 34$

Count



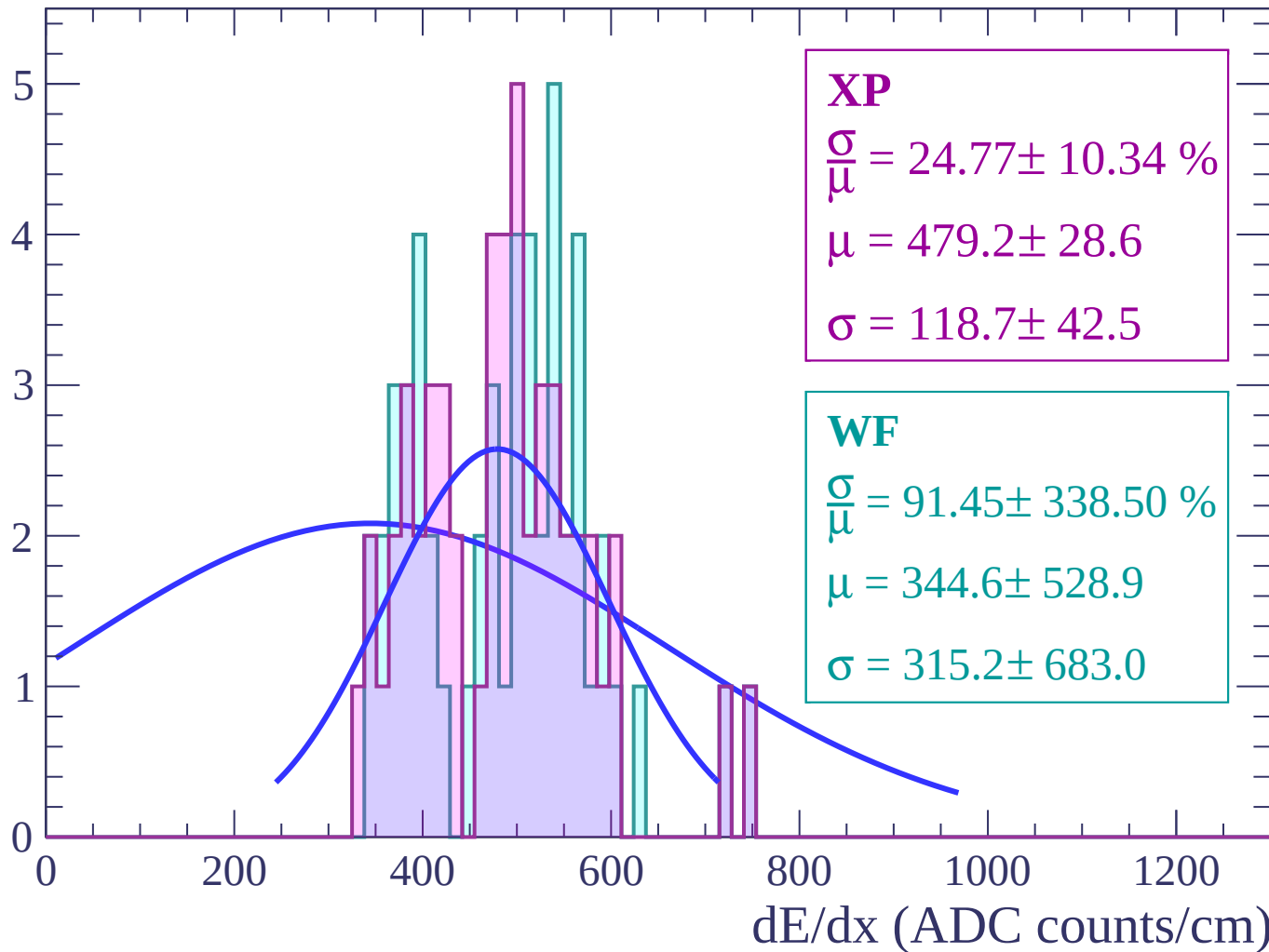
Energy loss | $34 < \theta < 36$

Count



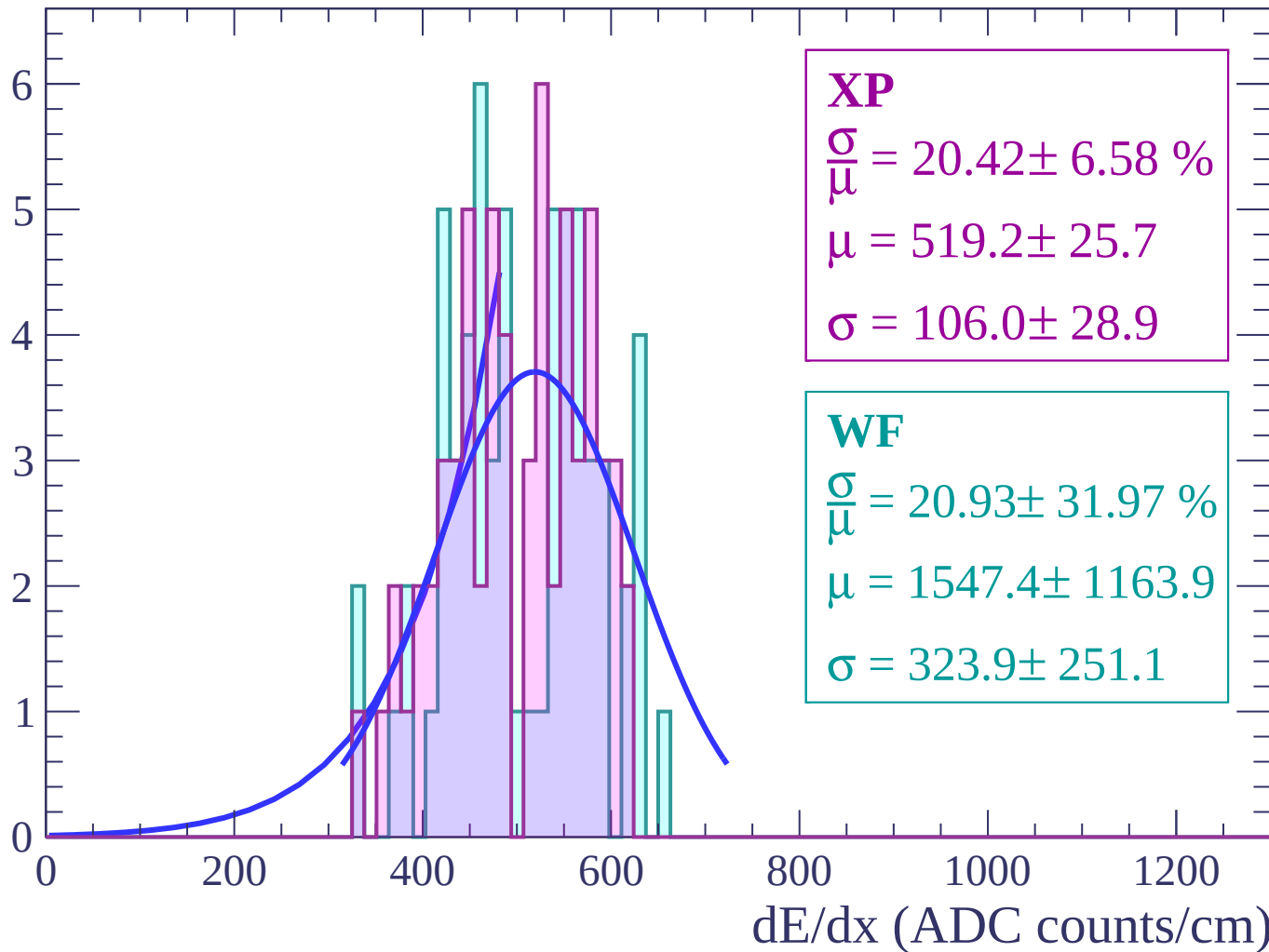
Energy loss | $36 < \theta < 38$

Count



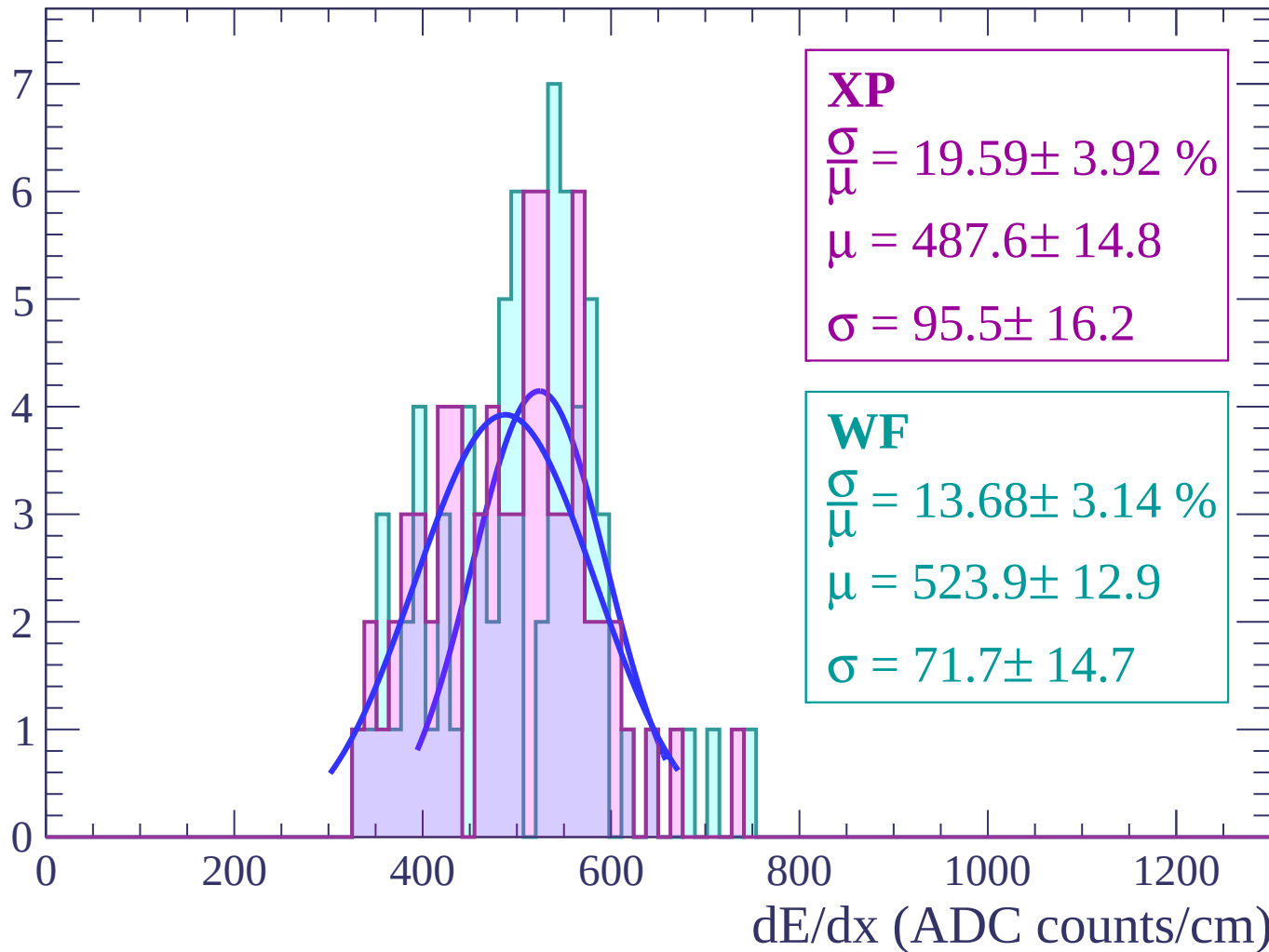
Energy loss | $38 < \theta < 40$

Count



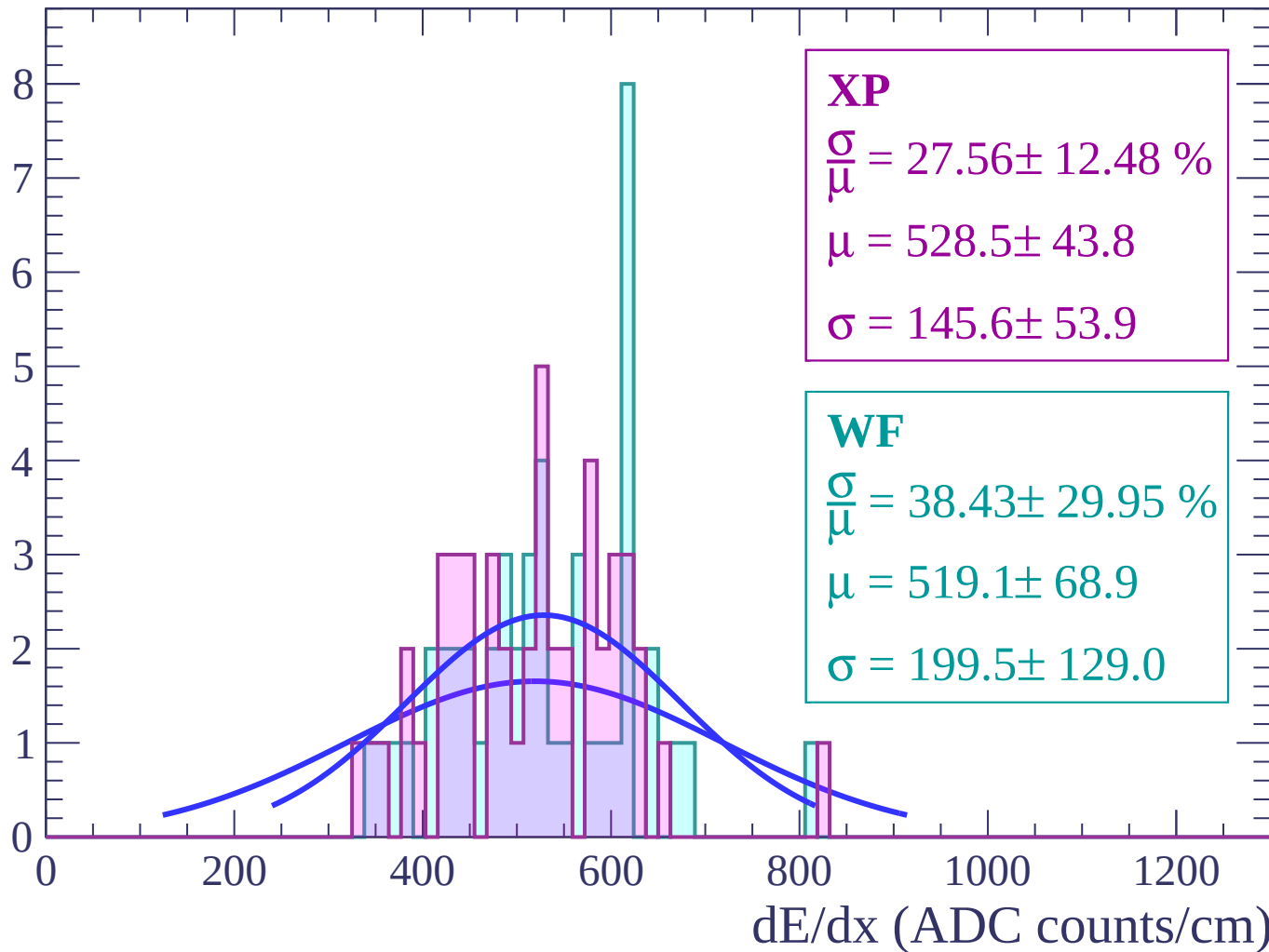
Energy loss | $40 < \theta < 42$

Count



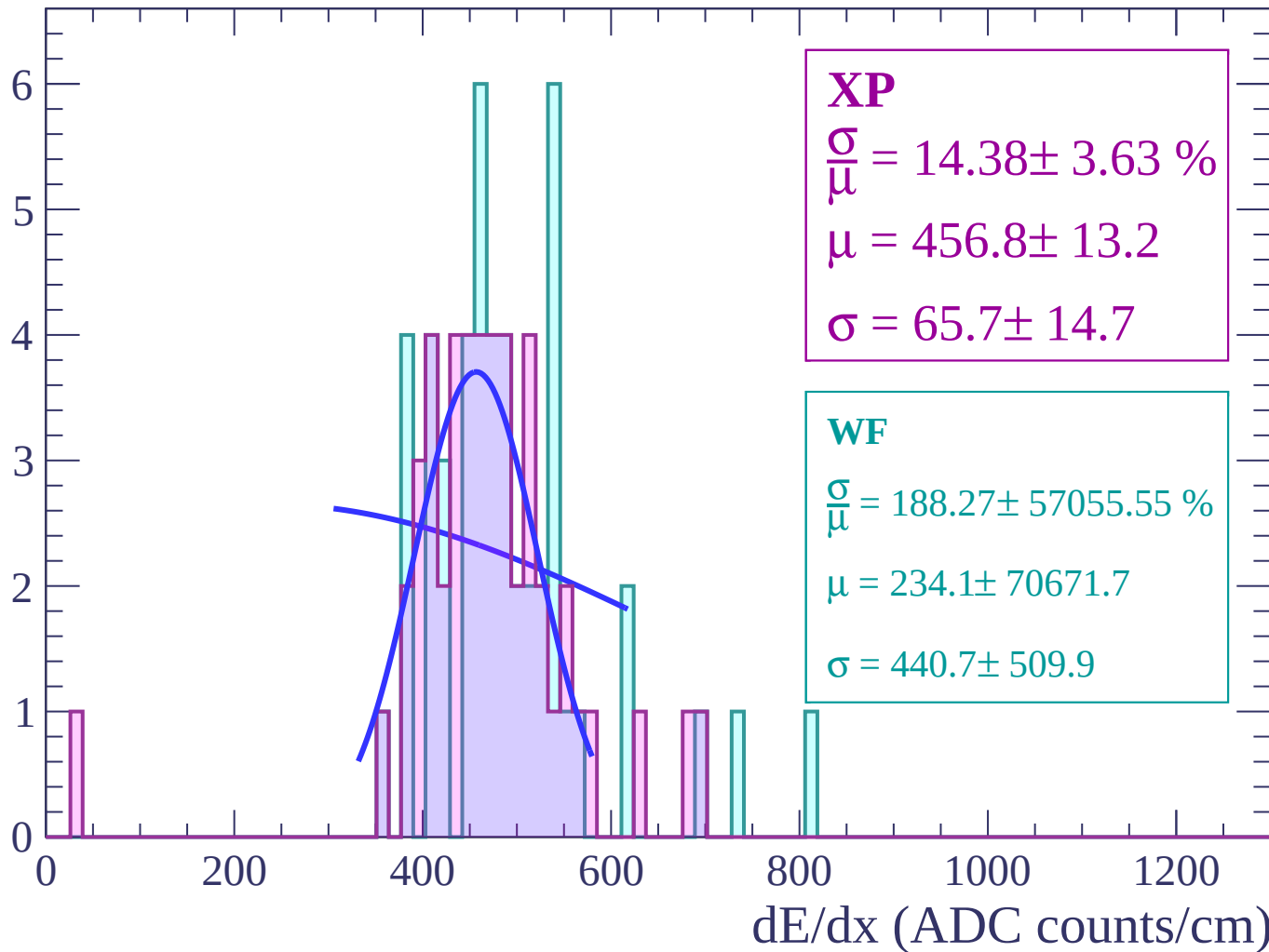
Energy loss | $42 < \theta < 44$

Count



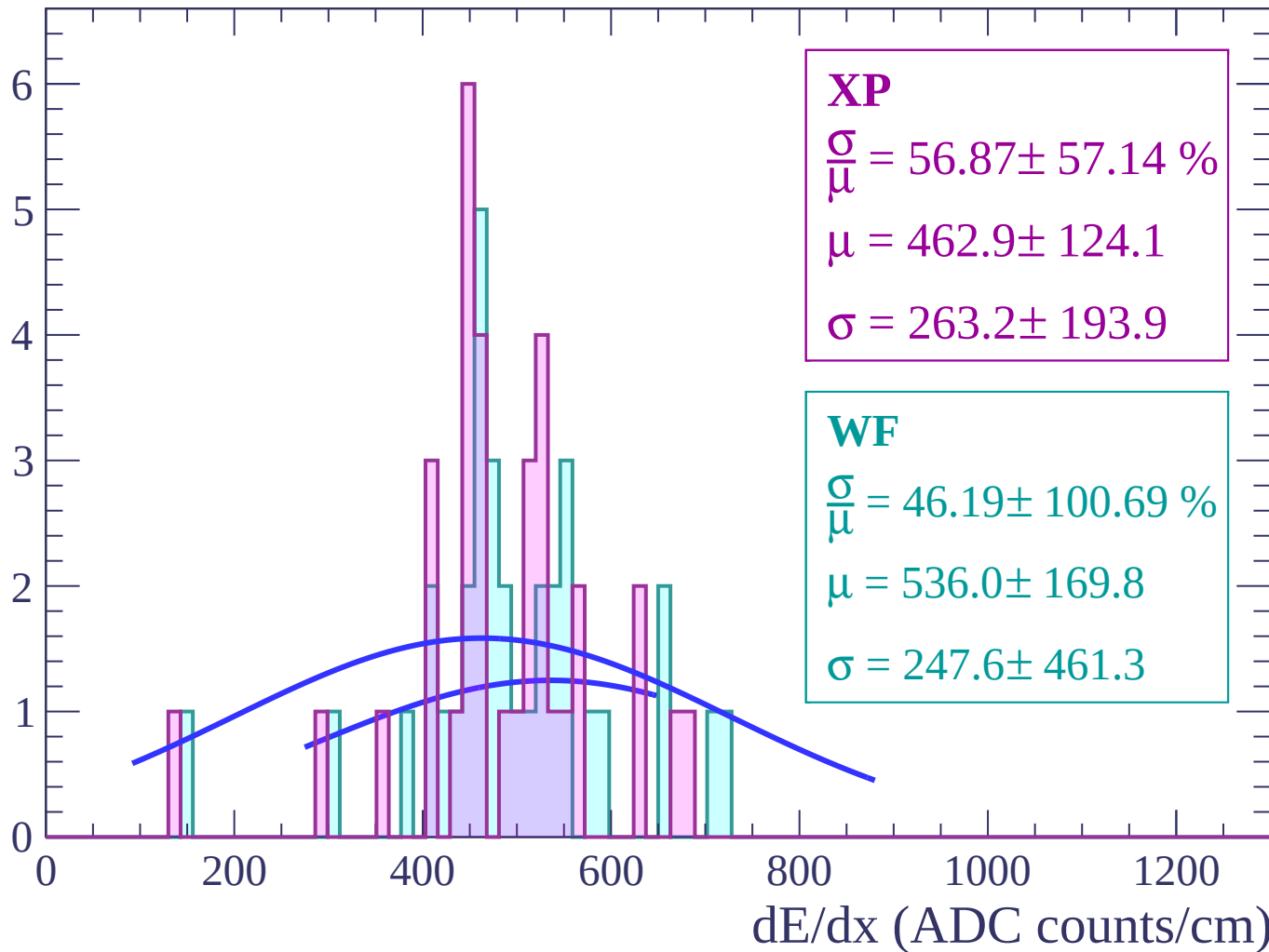
Energy loss | $44 < \theta < 46$

Count



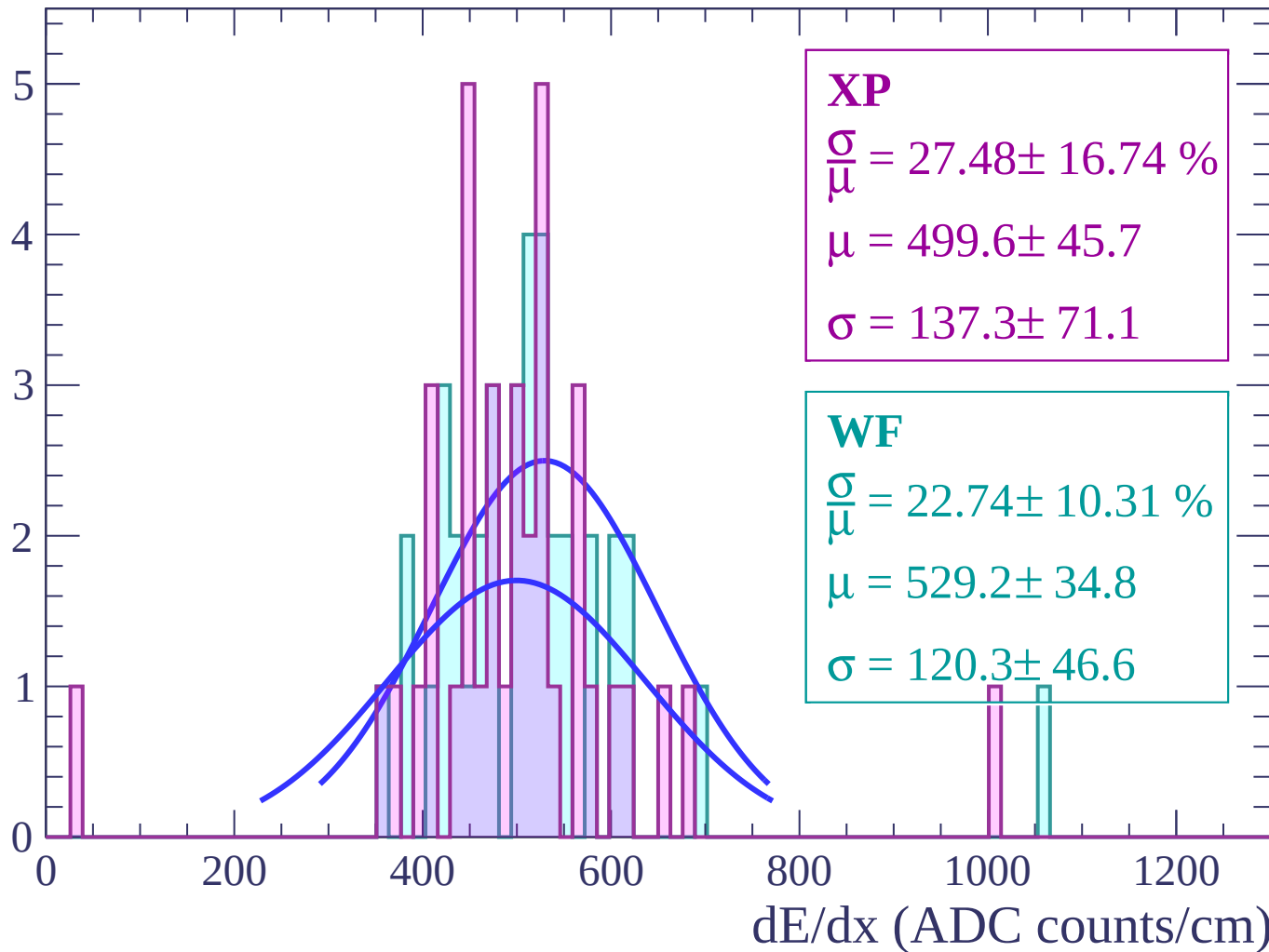
Energy loss | $46 < \theta < 48$

Count



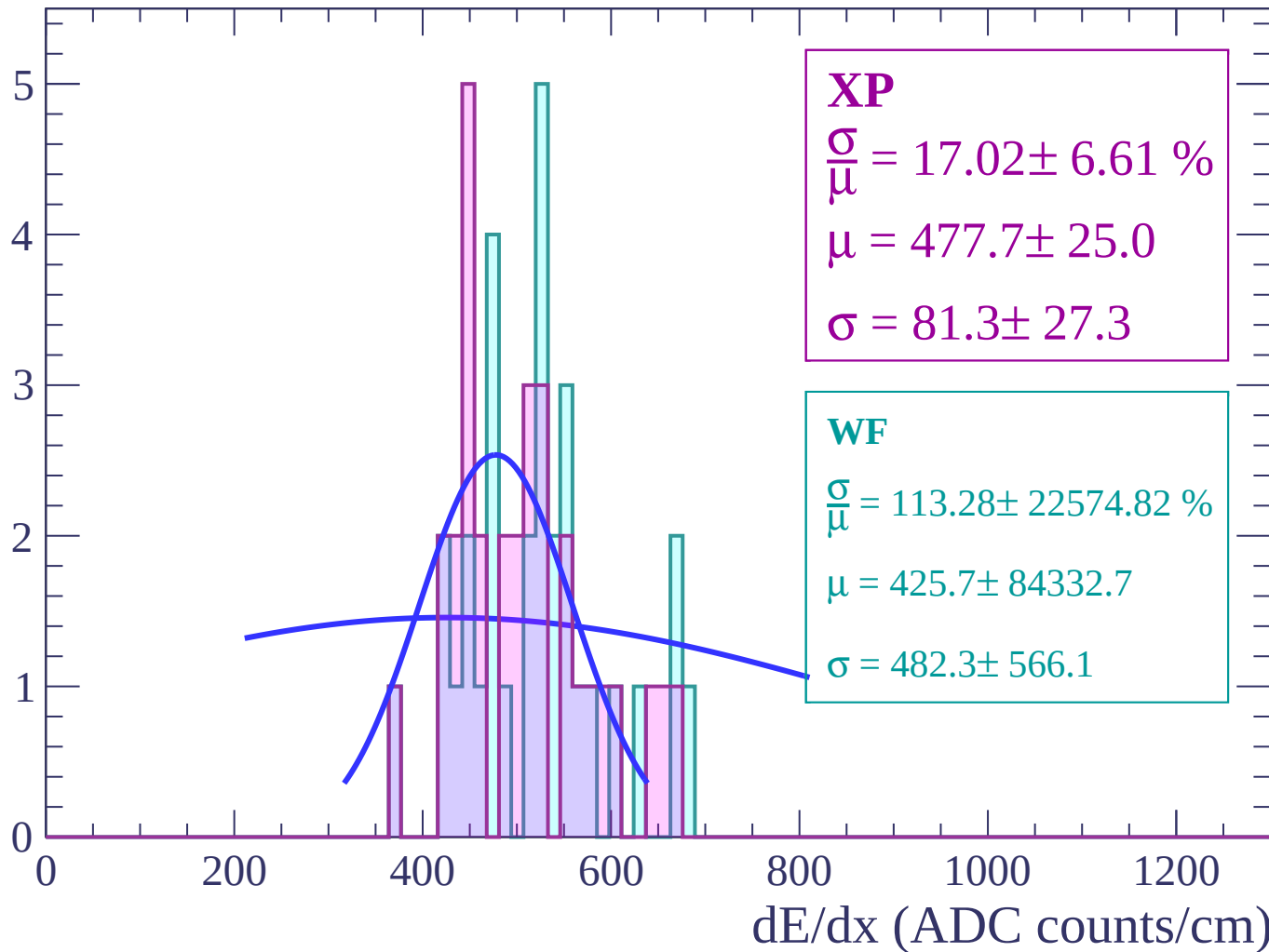
Energy loss | $48 < \theta < 50$

Count



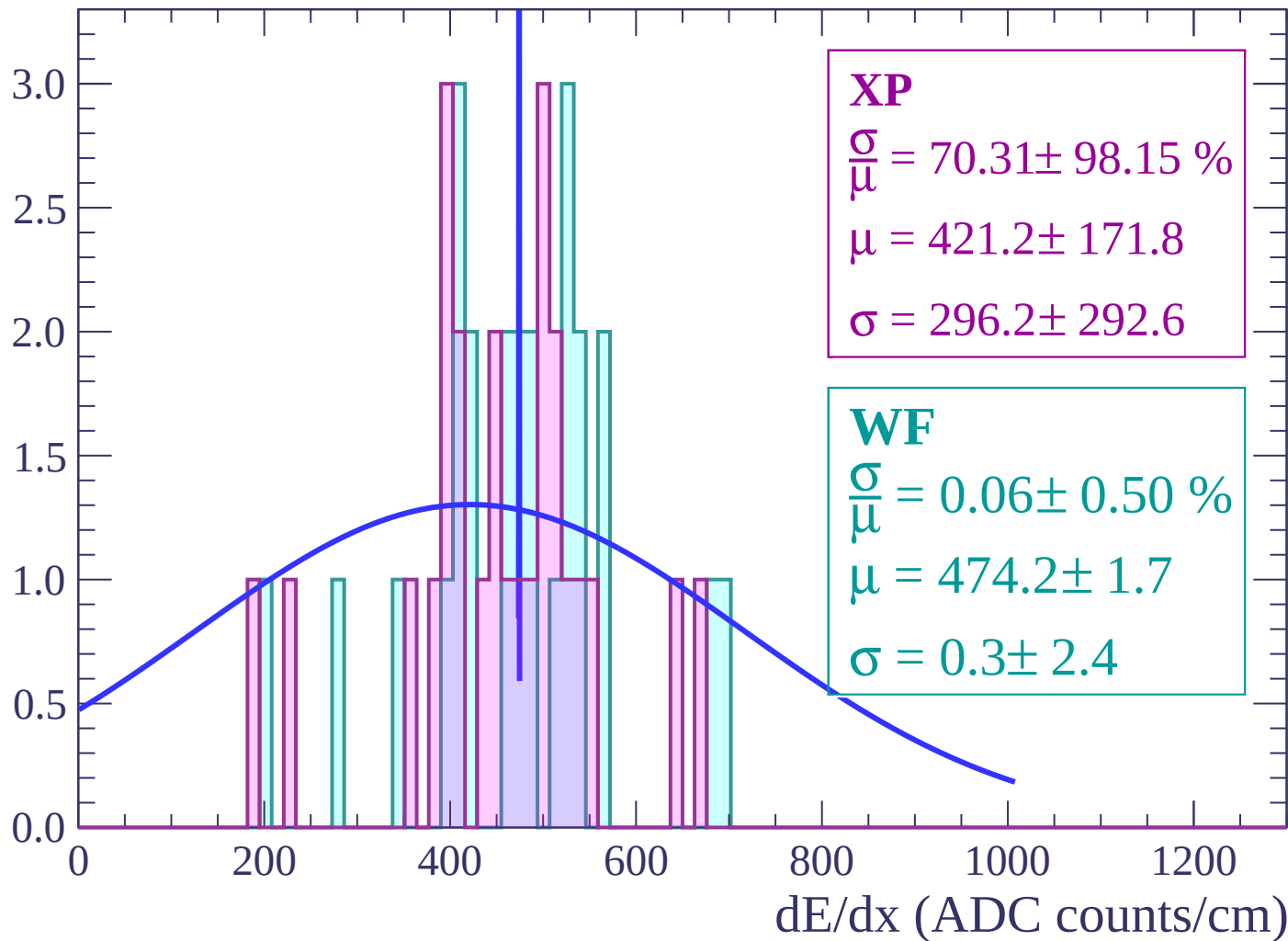
Energy loss | $50 < \theta < 52$

Count



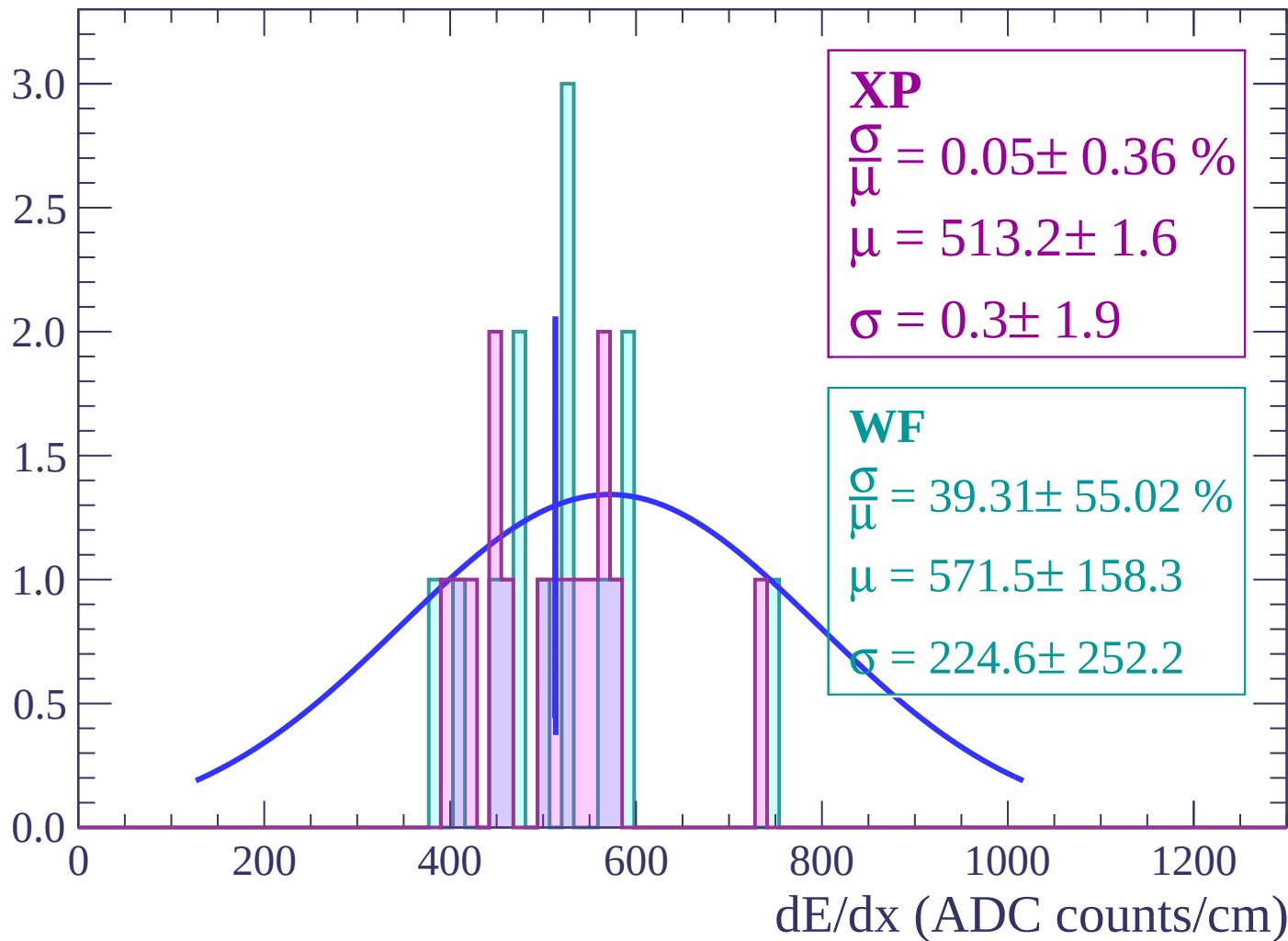
Energy loss | $52 < \theta < 54$

Count



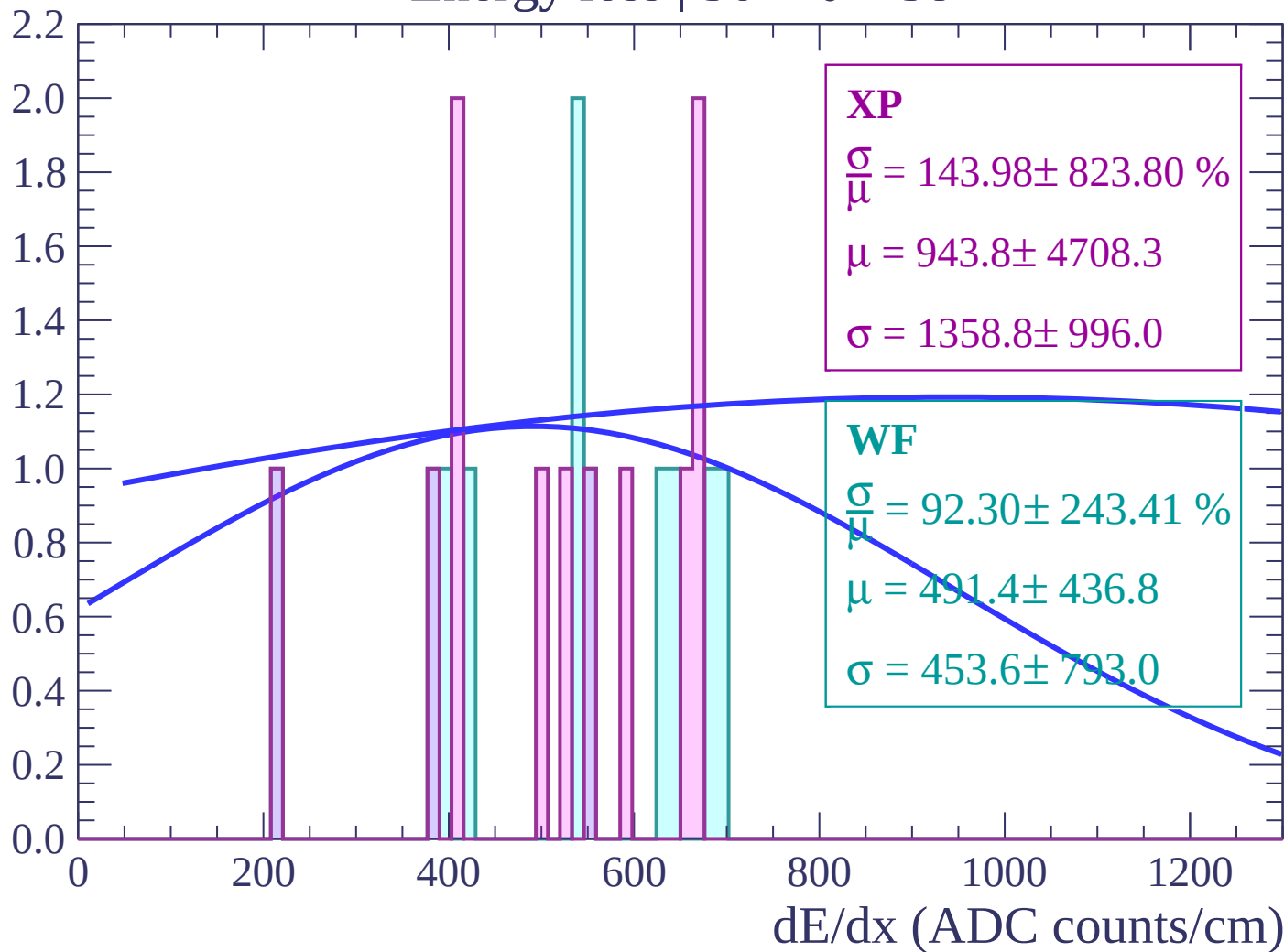
Energy loss | $54 < \theta < 56$

Count



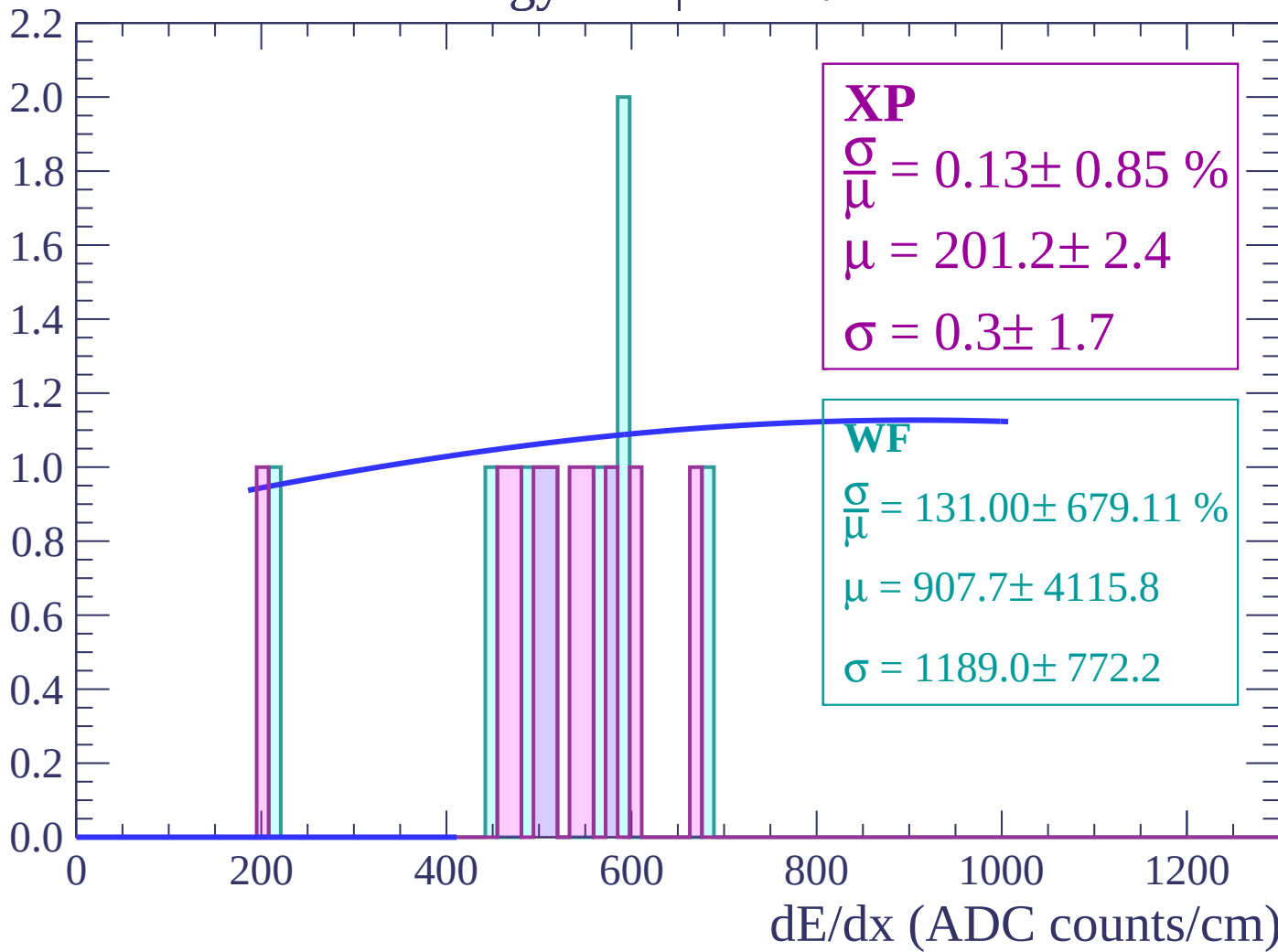
Energy loss | $56 < \theta < 58$

Count



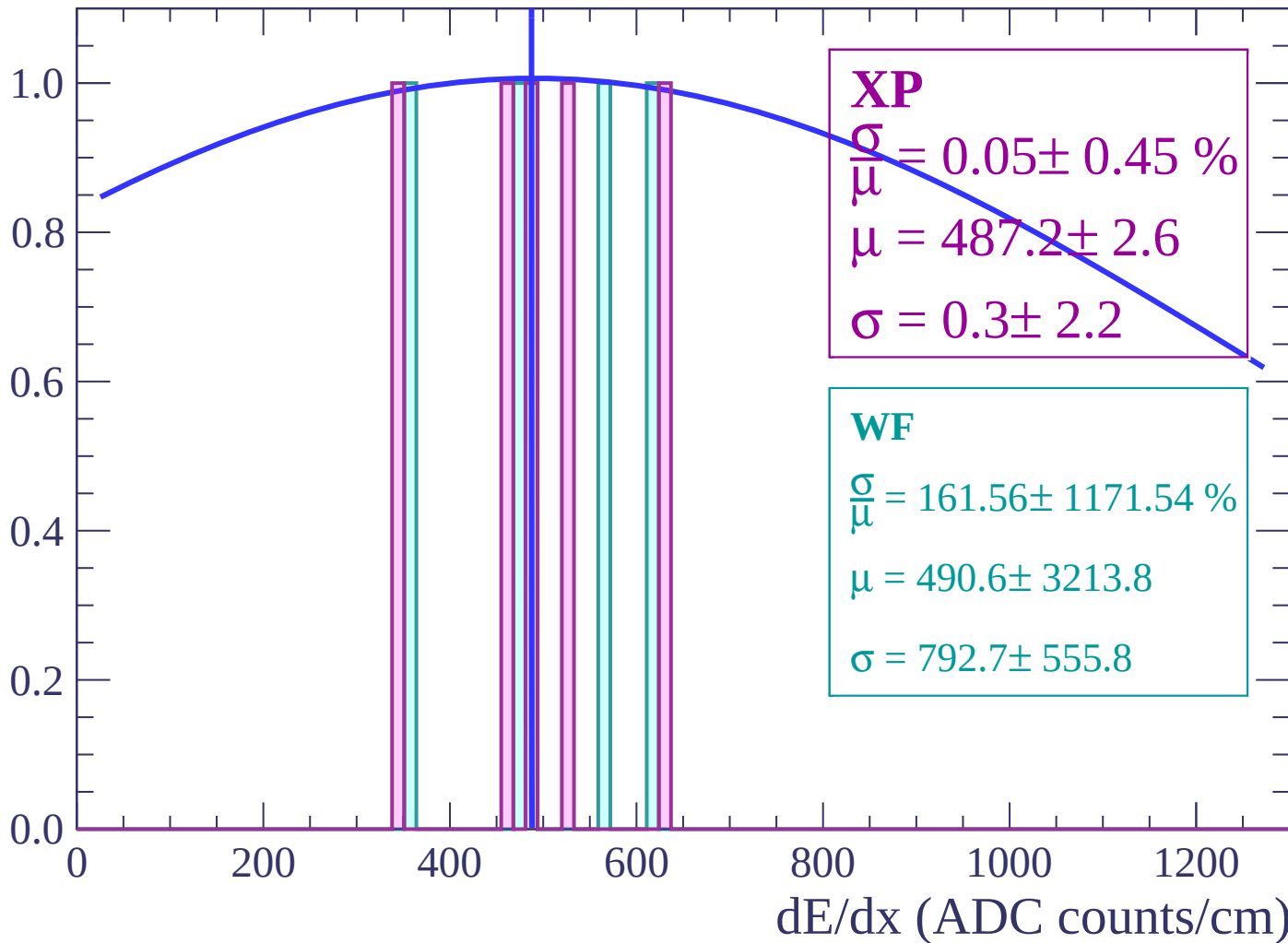
Energy loss | $58 < \theta < 60$

Count



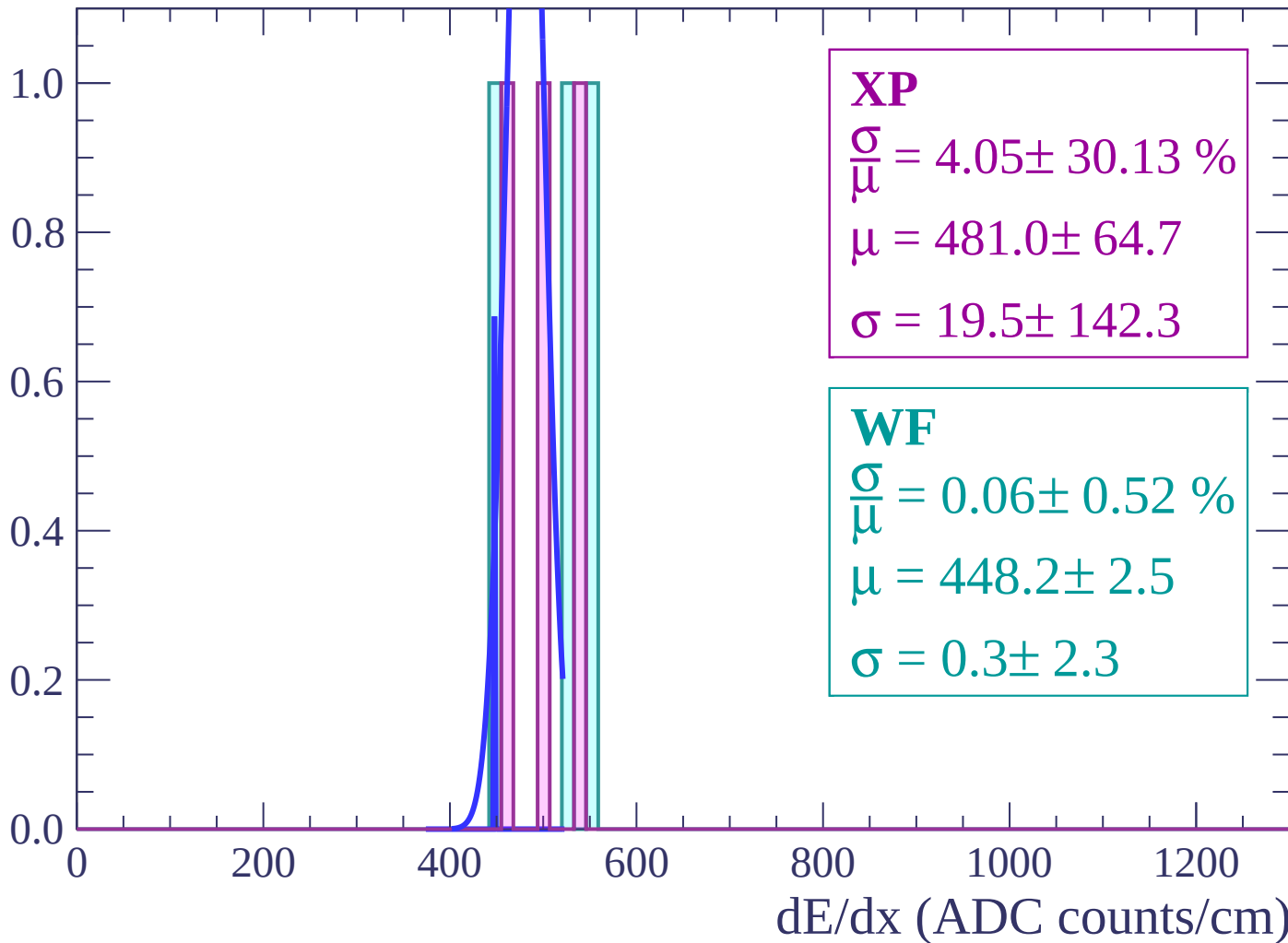
Energy loss | $60 < \theta < 62$

Count



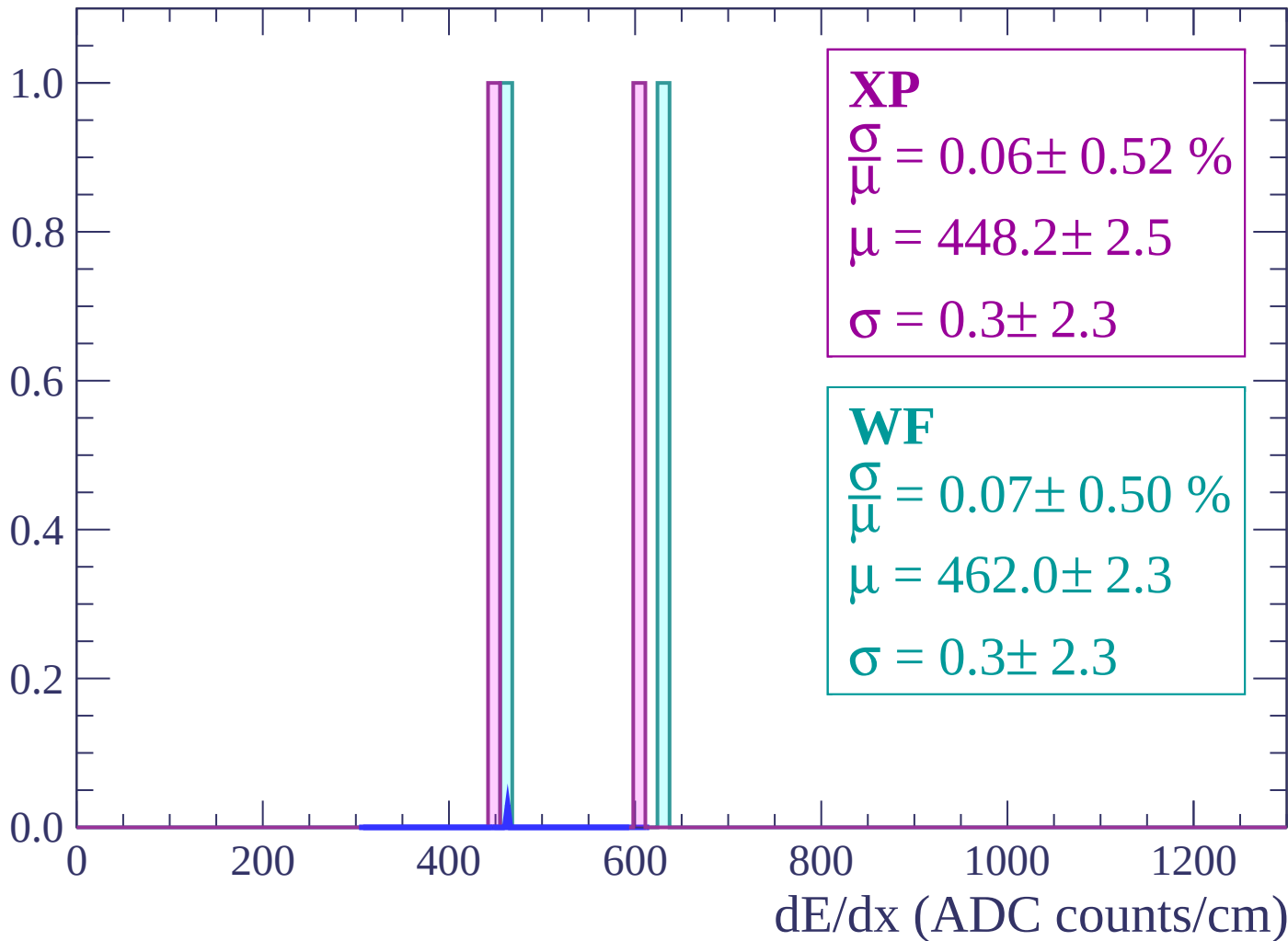
Energy loss | $62 < \theta < 64$

Count



Energy loss | $64 < \theta < 66$

Count



Energy loss | $66 < \theta < 68$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $68 < \theta < 70$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

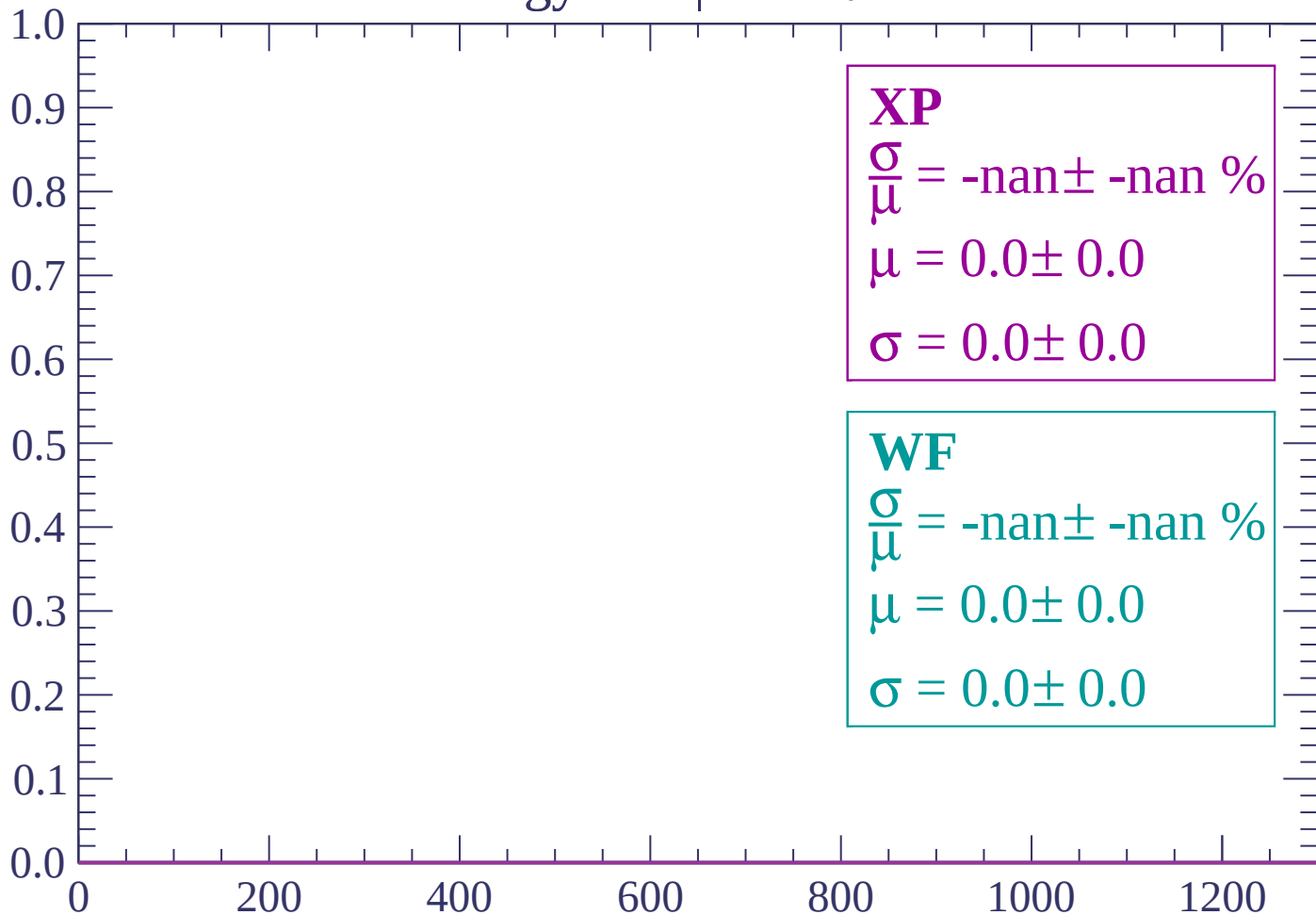
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $70 < \theta < 72$

Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \theta < 74$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

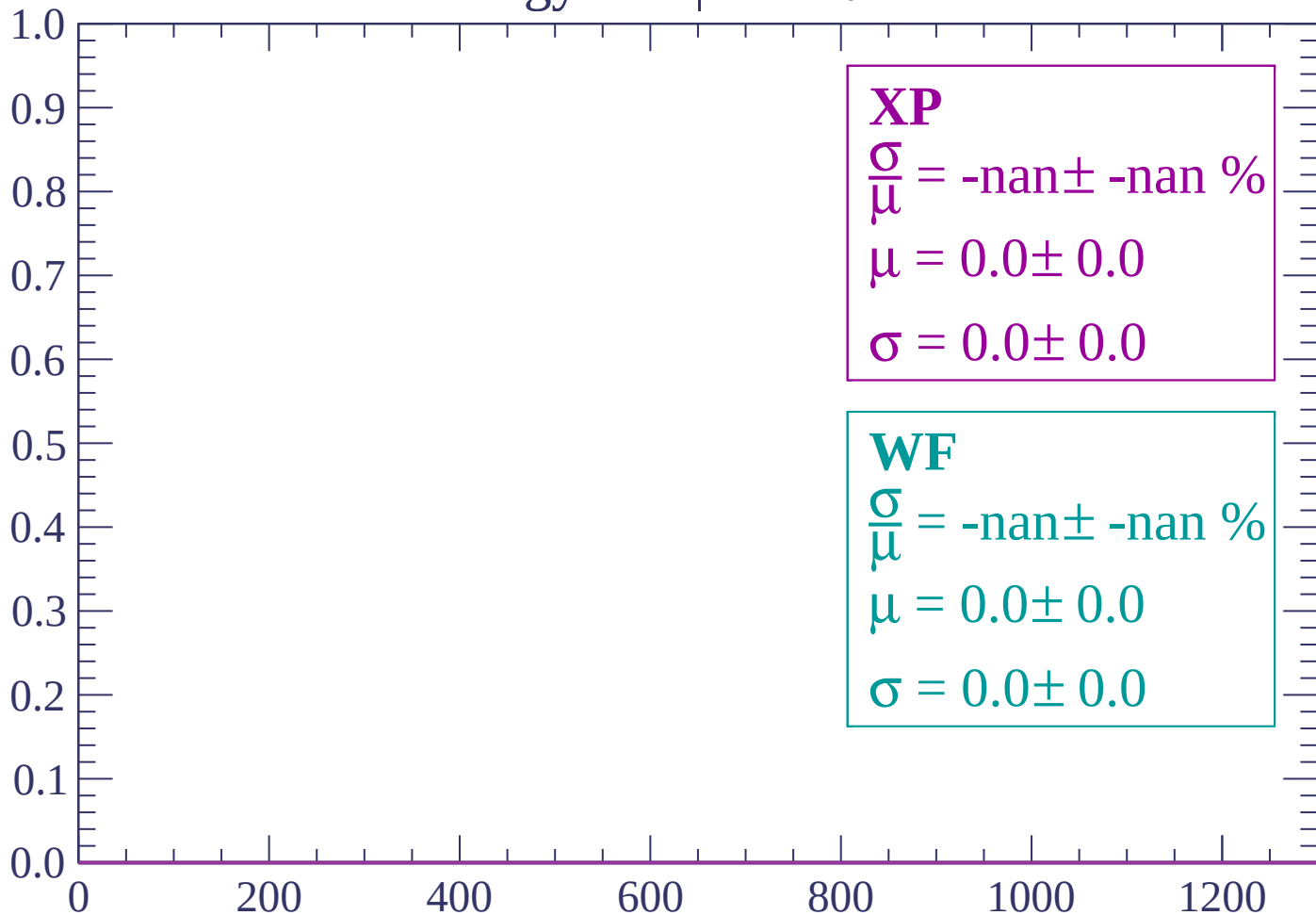
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $74 < \theta < 76$

Count



dE/dx (ADC counts/cm)

Energy loss | $76 < \theta < 78$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

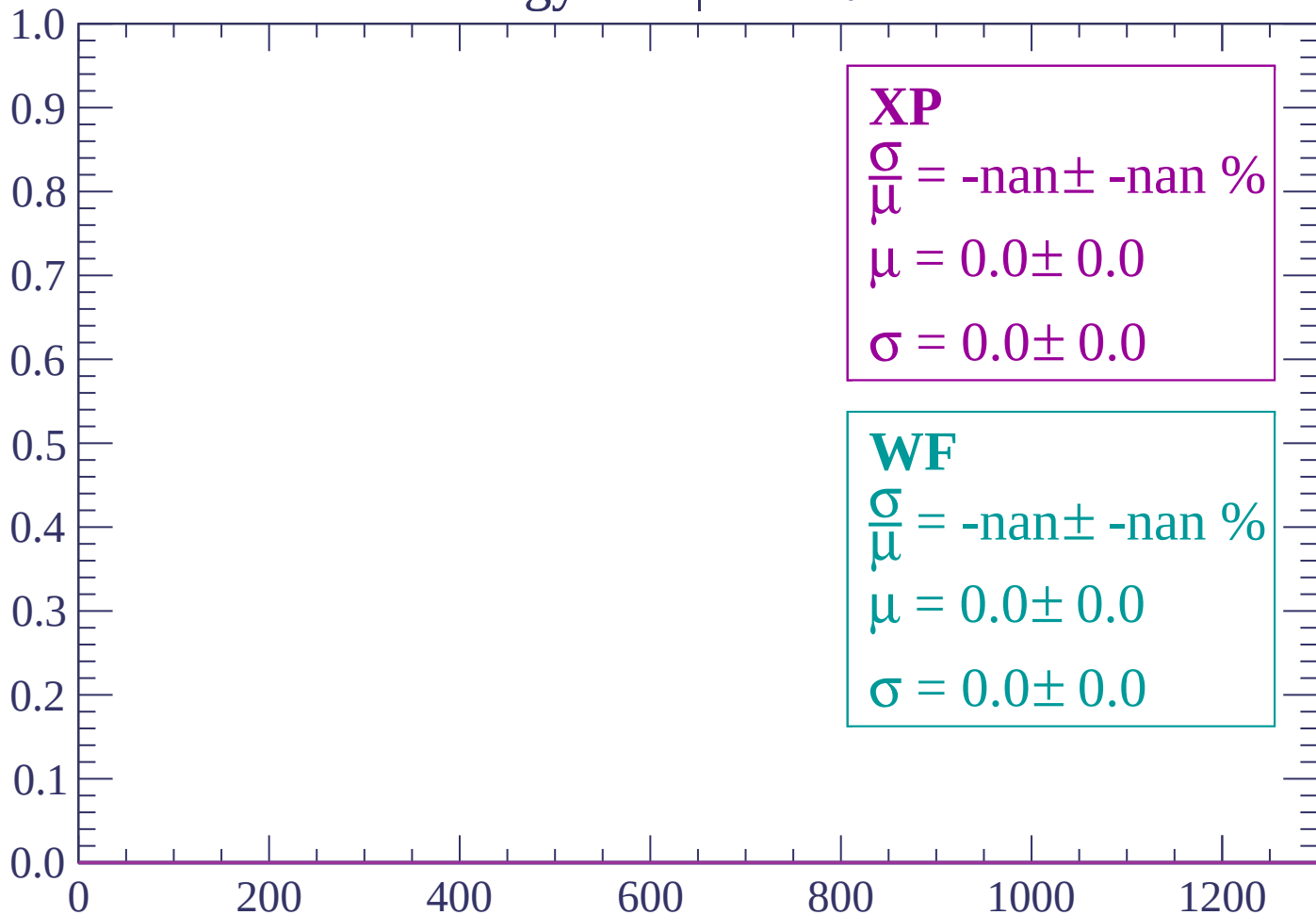
$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $78 < \theta < 80$

Count



dE/dx (ADC counts/cm)

Energy loss | $80 < \theta < 82$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $82 < \theta < 84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

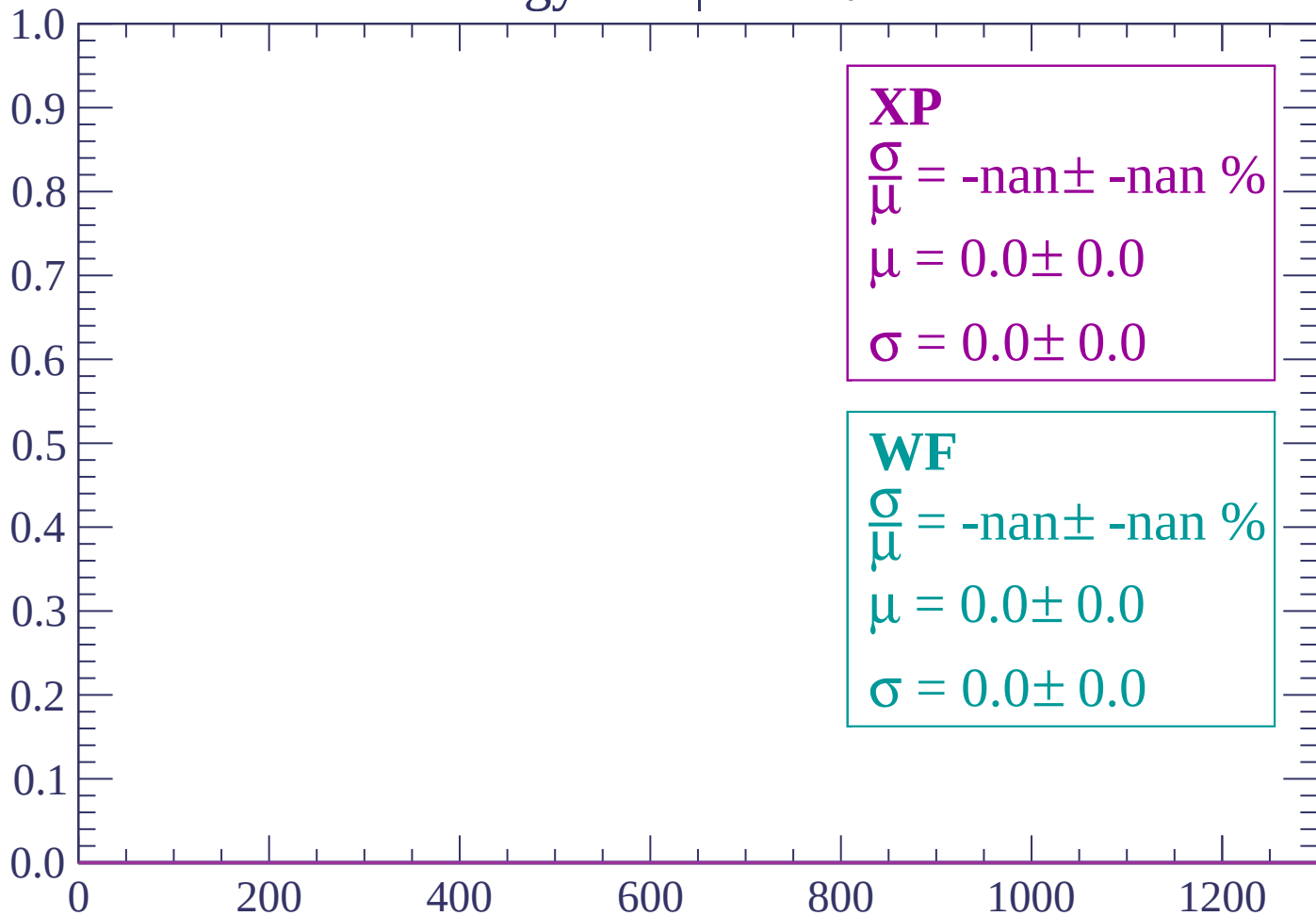
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $84 < \theta < 86$

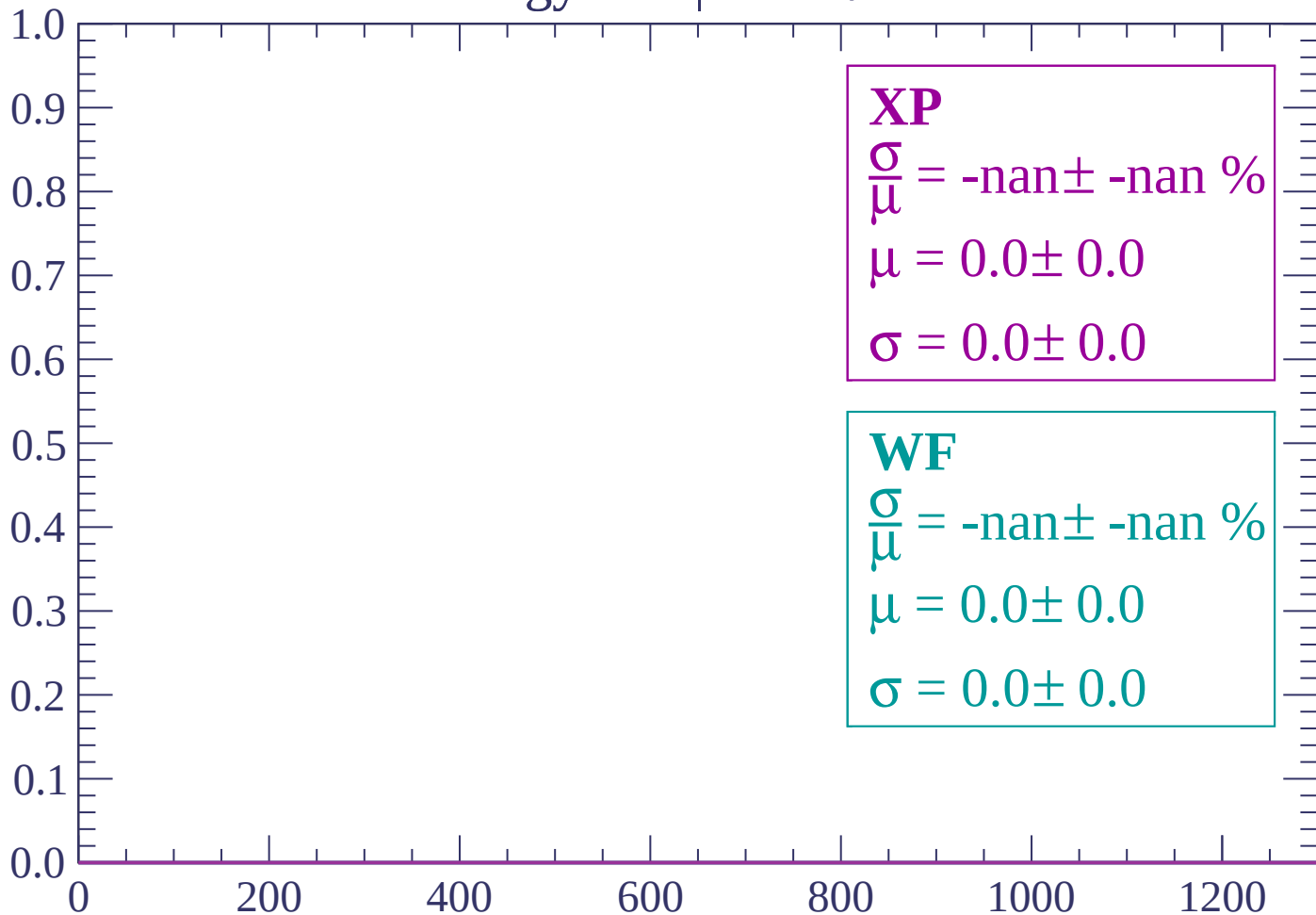
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

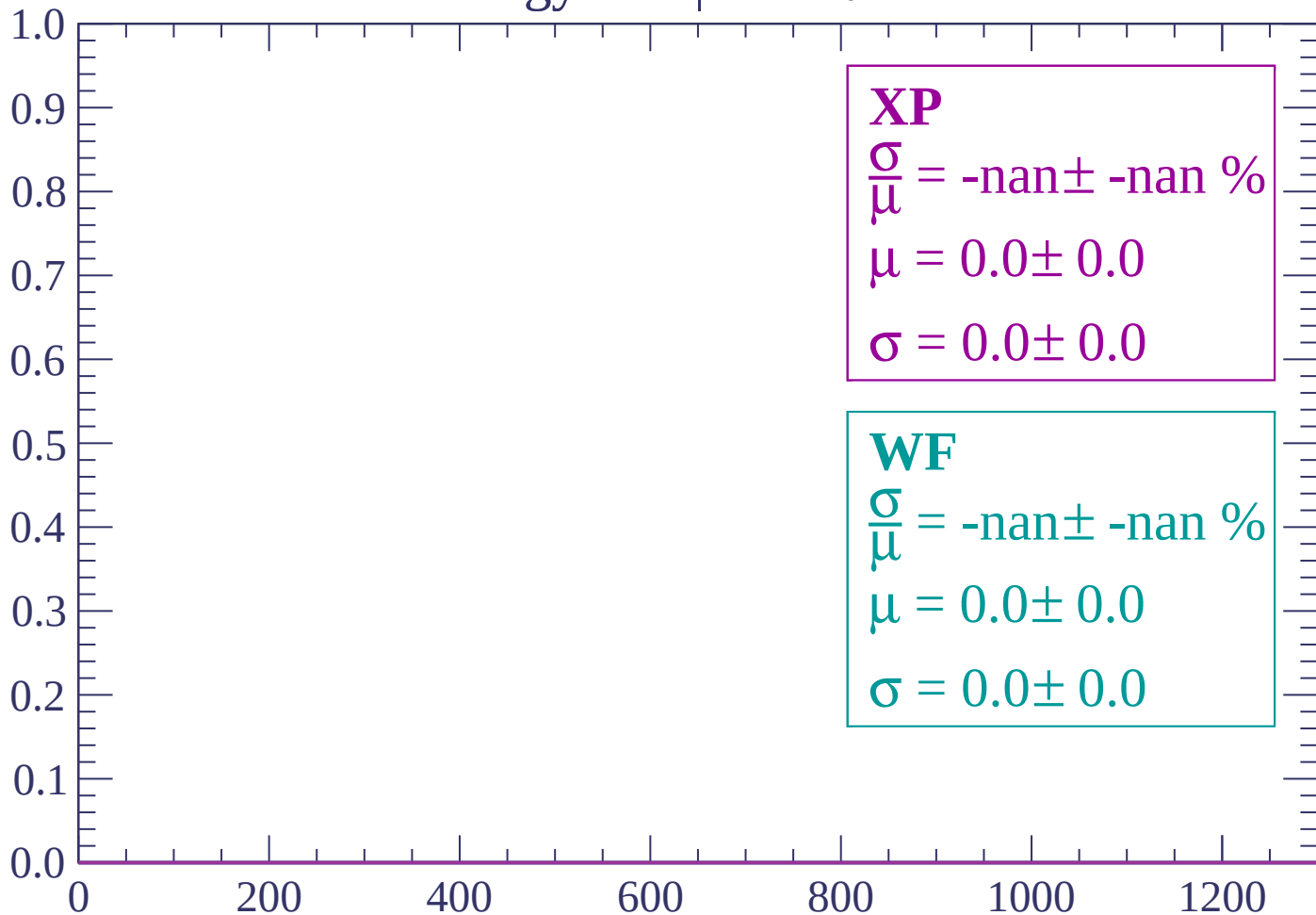
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

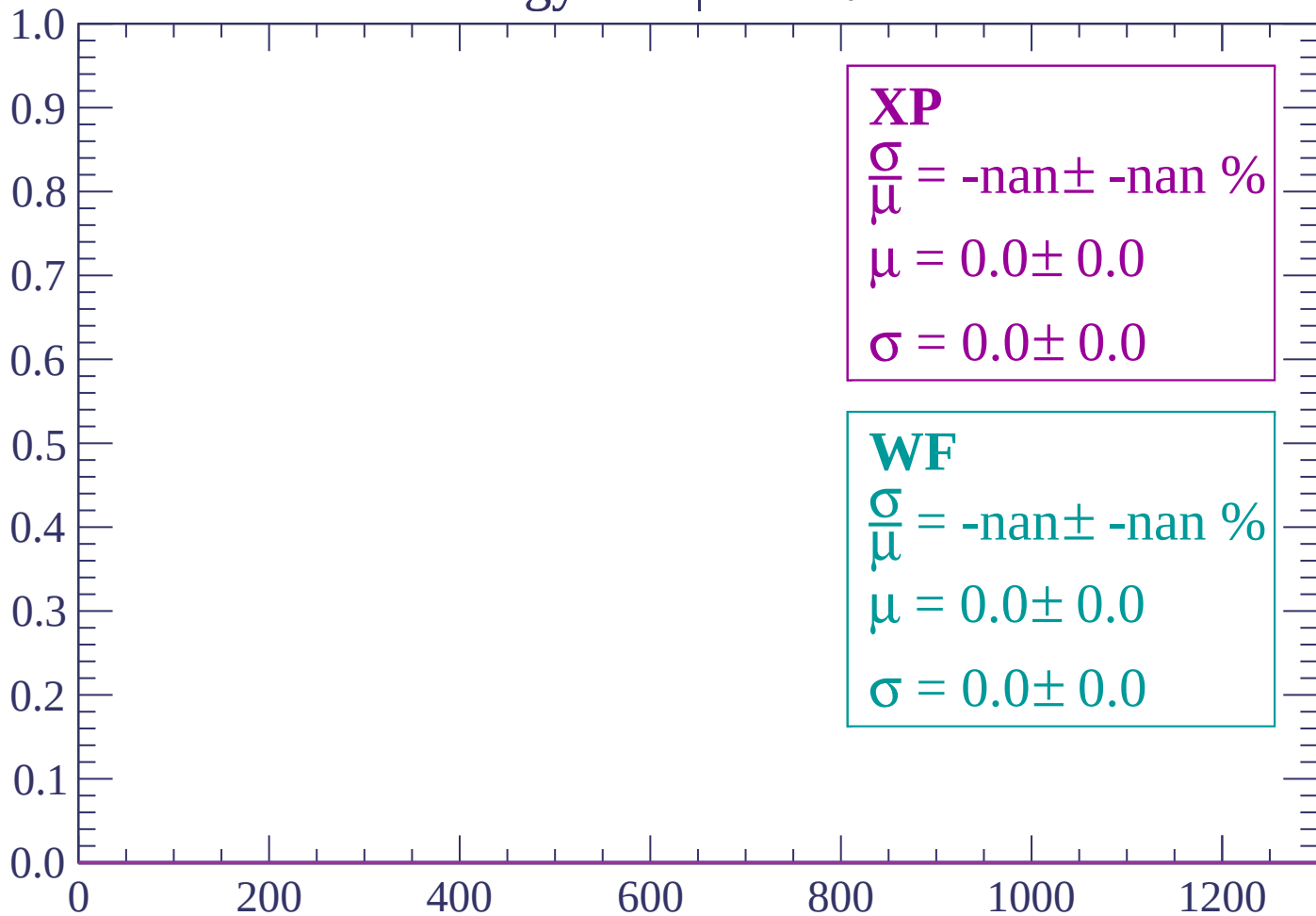
Count



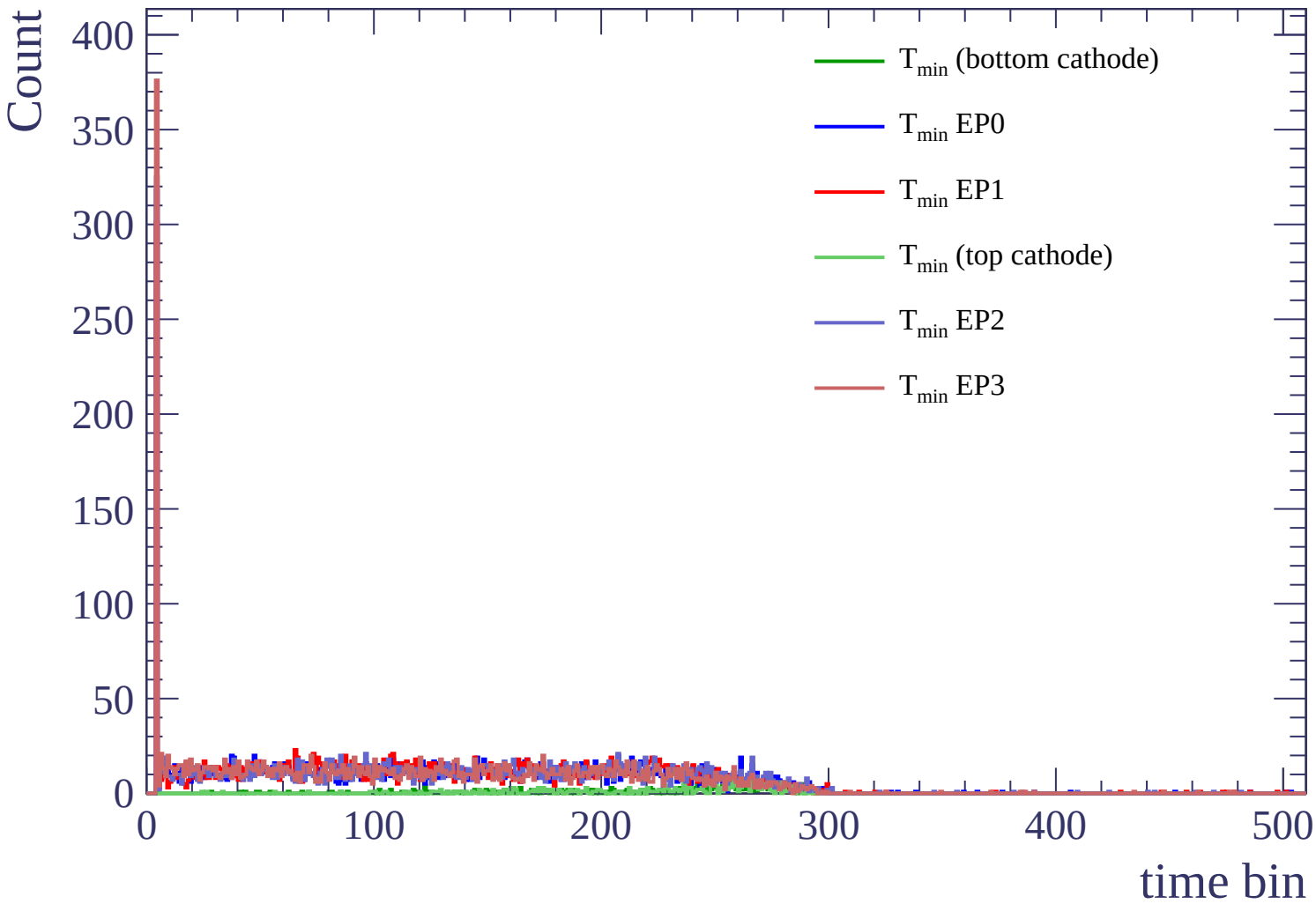
dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)



Count

